

MTH10013

Linear Algebra and Applications

Swinburne University



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1 Matrices

1.1 Definitions

One of the uses of matrices is the solution of simultaneous equations (e.g. to find the electric currents in different parts of a circuit). This application will be discussed later.

Definition 1.1 For any two positive integers $m, n > 0$, an $m \times n$ (pronounced m by n) **matrix** is a rectangular arrangement (array) of nm numbers or variables, arranged in m rows and n columns. The numbers (or variables) are called **elements** or **entries** of the matrix and the array is usually enclosed in brackets. The matrix as a whole is represented by a capital letter.

Example 1.1

$$A = \begin{matrix} & & & \\ & & & \\ & & & \\ 3 \times 3 & \begin{bmatrix} 1 & 2 & -5 \\ 0 & 3 & 2 \\ -1 & 5 & 0 \end{bmatrix} \end{matrix} \quad B = \begin{matrix} & & \\ & & \\ & & \\ 4 \times 1 & \begin{bmatrix} 1 \\ 5 \\ -6 \\ 7 \end{bmatrix} \end{matrix} \quad C = \begin{matrix} & & & & \\ & & & & \\ 2 \times 4 & \begin{bmatrix} 4.3 & -3.15 & 2.7 & 17.5 \\ 0 & 3 & -2.58 & -12.75 \end{bmatrix} \end{matrix}$$

$$D = [0.9 \quad -.98 \quad 1.25 \quad 7.83 \quad 0 \quad 0.02]$$

$$E = [6] \quad F = \begin{bmatrix} 1 & 7 & 3 & 4 \\ 2 & 0 & 8 & 3 \\ -1 & 8 & 9 & 7 \\ -9 & 7 & 6 & 5 \end{bmatrix}$$

Note that a 1×1 matrix (e.g. E) is a scalar quantity. We usually then leave off the brackets.

Thus, in the examples above, A has size 3×3 , B has size 4×1 , C has size 2×4 , D has size 1×6 , E has size 1×1 and F has size 4×4 .

A general matrix with m rows and n columns can be denoted by

$$A = [a_{ik}]_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}.$$

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The elements of the matrix are denoted by double subscripted small letters: a_{ij} , where i is the row number and j is the column number. So the element a_{34} is located in the 3rd row and 4th column.

Definition 1.2 A matrix with only one row is called a **row matrix** (or row vector) and a matrix with only one column is called a **column matrix** (or column vector).

Definition 1.3 A matrix with the same number of rows as columns, i.e. $m = n$, is called a **square matrix**.

The matrix

$$A = \begin{bmatrix} 1 & 2 & -5 \\ 0 & 3 & 2 \\ -1 & 5 & 0 \end{bmatrix}$$

is square and of order 3—it has 3 rows and 3 columns.

Definition 1.4 The **transpose** of a matrix A is a new matrix A^T with rows and columns of the original matrix A interchanged.

Example 1.2 If

$$C = \begin{bmatrix} 4.3 & -3.15 & 2.7 & 17.5 \\ 0 & 3 & -2.58 & -12.75 \end{bmatrix}$$

then

$$C^T = \begin{bmatrix} 4.3 & 0 \\ -3.15 & 3 \\ 2.7 & -2.58 \\ 17.5 & -12.75 \end{bmatrix}.$$

Exercises

1. Write down the order of each of the following matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \\ -1 & 5 \\ 7 & -6 \end{bmatrix},$$

$$B = [1 \quad 2 \quad -3 \quad 21 \quad 15],$$

$$C = \begin{bmatrix} 1.3 & 2.4 & -4 \\ 0.4 & 3.1 & -4.2 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad [1 \ 0 \ -1]$$

$$E = \begin{bmatrix} 1 & -2 & 5 & 11 & -9 \\ 5 & 3 & 8 & 12 & 0 \end{bmatrix},$$

$$F = \begin{bmatrix} 1 & 2 & 4 & -9 & 10 \\ 0 & 3 & -5 & 15 & 21 \\ -1 & 5 & 43 & -31 & 28 \end{bmatrix}.$$

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2. Write down the following elements from the matrices in Question 1:
 $a_{22}, f_{35}, a_{32}, c_{13}, d_{21}, f_{24}, b_{14}, e_{25}$.

3. Find the transpose of each of the following matrices:

$$A = \begin{bmatrix} 5 & -6 & 1 \\ 2 & 4 & 3 \end{bmatrix},$$

$$B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 3 & 4 \\ 2 & 6 \end{bmatrix},$$

$$C = \begin{bmatrix} 5 & 7 & 4 \\ 2 & 4 & -6 \\ -1 & -3 & 5 \end{bmatrix},$$

$$D = \begin{bmatrix} 3.2 \\ 1.7 \\ -2.4 \end{bmatrix}.$$

Answers:

$$1. A = 4 \times 2 \quad B = 1 \times 5 \quad C = 2 \times 3 \quad D = 3 \times 1 \quad E = 2 \times 5 \quad F = 3 \times 5$$

$$2. a_{22} = 3 \quad f_{35} = 28 \quad a_{32} = 5 \quad c_{13} = -4 \quad d_{21} = 0 \quad f_{24} = 15 \quad b_{14} = 21 \quad e_{25} = 0$$

$$3. A^T = \begin{bmatrix} 5 & 2 \\ -6 & 4 \\ 1 & 3 \end{bmatrix}; B^T = \begin{bmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 4 & 6 \end{bmatrix}; C^T = \begin{bmatrix} 5 & 2 & -1 \\ 7 & 4 & -3 \\ 4 & -6 & 5 \end{bmatrix}; \\ D^T = [3.2 \quad 1.7 \quad -2.4]$$

1.2 Operations with matrices

Definition 1.5 Matrices A and B are equal if they have equal size and if their corresponding elements are equal, i.e. $A = B$ if A and B are both $m \times n$ and if $a_{ik} = b_{ik}$ for all $i = 1, \dots, m$ and $k = 1, \dots, n$.

For example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

if and only if $a_{11} = 1, a_{12} = 2, a_{21} = 3, a_{22} = 4$.

Definition 1.6 Let A and B be $m \times n$ matrices. The **sum** $A + B$ is a new $m \times n$ matrix, with its (i, k) entry given by $a_{ik} + b_{ik}$.

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Example 1.3 If

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -5 & 12 \\ 1 & -7 \end{bmatrix}$$

then

$$A + B = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} -5 & 12 \\ 1 & -7 \end{bmatrix} = \begin{bmatrix} 1-5 & 2+12 \\ 0+1 & 3-7 \end{bmatrix} = \begin{bmatrix} -4 & 14 \\ 1 & -4 \end{bmatrix}$$

Matrix addition is commutative, i.e. $A + B = B + A$ and associative, i.e. $(A + B) + C = A + (B + C)$.

Definition 1.7 The **product** of any matrix A by a scalar (or a constant) λ is a new matrix λA , obtained by multiplying every element of the matrix A by λ

Example 1.4 If

$$A = \begin{bmatrix} 5 & -10 & -7 \\ 3 & 11 & -18 \end{bmatrix} \quad \begin{matrix} 3 & -14 \\ 14 & -8 \\ 9 & 1 \end{matrix}$$

then

$$4A = 4 \begin{bmatrix} 5 & -10 & -7 \\ 3 & 11 & -18 \end{bmatrix} = \begin{bmatrix} 20 & -40 & -28 \\ 12 & 44 & -72 \end{bmatrix}.$$

Example 1.5 If

$$B = \begin{bmatrix} 15 & -10 \\ 3 & 11 \\ -17 & 21 \\ 0 & -8 \end{bmatrix}$$

then

$$-2B = -2 \begin{bmatrix} 15 & -10 \\ 3 & 11 \\ -17 & 21 \\ 0 & -8 \end{bmatrix} = \begin{bmatrix} -30 & 20 \\ -6 & -22 \\ 34 & -42 \\ 0 & 16 \end{bmatrix}.$$

Note that the difference of two matrices $A - B$ can be calculated as a sum of A and $-1B$, i.e. $A - B = A + (-B)$.

Example 1.6

$$A - B = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} -5 & 12 \\ 1 & -7 \end{bmatrix} = \begin{bmatrix} 1+5 & 2-12 \\ 0-1 & 3+7 \end{bmatrix} = \begin{bmatrix} 6 & -10 \\ -1 & 10 \end{bmatrix}.$$

Also, if $C = \begin{bmatrix} 12 & -6 \\ 3 & 18 \end{bmatrix}$, we could write this as $3 \begin{bmatrix} 4 & -2 \\ 1 & 6 \end{bmatrix}$, i.e. we can take a common factor out of every element of the matrix.

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Definition 1.8 Assume that A is an $m \times n$ and B is an $n \times p$ matrix. The **product of two matrices** AB is defined as a new $m \times p$ matrix $C = AB$, whose (i, k) entry is given by

$$c_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{in}b_{nk} = \sum_{j=1}^n a_{ij}b_{jk}.$$

Example 1.7 Product of a row matrix and a column matrix.

If

row matrices	column matrices
$A = [5 \quad 7 \quad 9]$	and $B = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$

then

$$AB = [5 \quad 7 \quad 9] \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} = [5 \times 2 + 7 \times 4 + 9 \times 6] = [92] = 92.$$

Notice that each element of the row matrix is multiplied by the corresponding element of the column matrix and the products are added to give a 1×1 matrix. Hence the number of elements (3) in the row of A must be the same as the number of elements (3) in the column of B .

Note: Generally, the product AB exists only if the adjacent dimensions of A and B coincide.

$$\begin{matrix} A & B \\ (m \times n) & (n \times p) \end{matrix} = \begin{matrix} C \\ (m \times p) \end{matrix}$$

Example 1.8 If

$$A = [5 \quad 7 \quad 9] \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 \\ 4 & -3 \\ 6 & 1 \end{bmatrix}$$

then

$$\begin{aligned} AB &= [5 \quad 7 \quad 9] \begin{bmatrix} 2 & 0 \\ 4 & -3 \\ 6 & 1 \end{bmatrix} \\ &= [5 \times 2 + 7 \times 4 + 9 \times 6 \quad 5 \times 0 + 7 \times -3 + 9 \times 1] \\ &= [92 \quad -12]. \end{aligned}$$

BA does not exist, because A is a (1×3) and B is a (3×2) matrix so that the adjacent dimensions of $(3 \times 2)(1 \times 3)$ are not identical.

Definition 1.9 If the product AB exists, then we say that A is **postmultiplied** by B , or B is **premultiplied** by A .

Matrix multiplication is not commutative

Note that generally $AB \neq BA$, even if both, AB and BA exist. When multiplying a product of more than two matrices, work through taking them two at a time.

Example 1.9

$$A = \begin{bmatrix} -1 & 4 \\ 13 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 5 \end{bmatrix}$$

the product

$$\begin{aligned} (CA)B &= \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} -1 & 4 \\ 13 & -3 \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 11 & 5 \\ 39 & -9 \\ 64 & -11 \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 70 & 17 \\ 168 & 87 \\ 287 & 139 \end{bmatrix} \end{aligned}$$

or we could have multiplied the product AB first:

$$\begin{aligned} C(AB) &= \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 5 \end{bmatrix} \left[\begin{bmatrix} -1 & 4 \\ 13 & -3 \end{bmatrix} \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix} \right] \\ &= \begin{bmatrix} 2 & 1 \\ 0 & 3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 7 & -6 \\ 56 & 29 \end{bmatrix} \\ &= \begin{bmatrix} 70 & 17 \\ 168 & 87 \\ 287 & 139 \end{bmatrix}. \end{aligned}$$

It is important to ensure that the order of multiplication, i.e. CAB is not changed.

Remember that generally $CAB \neq ACB$, or any other combination.

Transposition of a product of two matrices

$$(AB)^T = B^T A^T, \text{ for any two matrices } A \text{ and } B.$$

Example 1.10 For the following two matrices A and B

$$A = \begin{bmatrix} -1 & 2 \\ 1 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 2 & 10 \\ 3 & -1 & 4 \end{bmatrix}$$

check that $(AB)^T = B^T A^T$. Indeed we have

$$AB = \begin{bmatrix} 1 & -4 & -2 \\ -4 & 5 & -2 \end{bmatrix} \quad (AB)^T = \begin{bmatrix} 1 & -4 \\ -4 & 5 \\ -2 & -2 \end{bmatrix}$$

On the other hand

$$B^T A^T = \begin{bmatrix} 5 & 3 \\ 2 & -1 \\ 10 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ -4 & 5 \\ -2 & -2 \end{bmatrix} = (AB)^T.$$

Exercises

1. If

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 5 \\ 11 & 7 \\ 6 & -3 \end{bmatrix}, \quad C = \begin{bmatrix} -4 & 3 \\ 2 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} -3 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

calculate the following, if they are defined:

(a) $A + C$; (b) $B + D$; (c) $2A - 3C$; (d) $D + B^T$.

$$\begin{bmatrix} 5 & 2 & 1 \\ -6 & 3 & 1 \end{bmatrix}$$

2. If

$$A = \begin{bmatrix} 5 & -6 \\ 2 & 3 \\ 1 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 3 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 3 \\ -1 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix},$$

$$E = \begin{bmatrix} 5 & -9 & 10 \\ 2 & 1 & 0 \end{bmatrix}, \quad F = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix},$$

calculate the following, if they are defined:

(a) BC ; (b) ED ; (c) AB ; (d) $A^T B$; (e) $D^T A$; (f) DE ; (g) FD ; (h) ACE ; (i) $CB^T A$.

3. If

$$A = \begin{bmatrix} 5 & 7 & -1 \\ 2 & 0 & 3 \\ 4 & -1 & -2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 3 & -10 & 7 \end{bmatrix},$$

$$C = \begin{bmatrix} 1 & -1 \\ 0 & 5 \\ -2 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} w & x \\ y & z \end{bmatrix},$$

calculate the following:

(a) AX ; (b) BX ; (c) BAC ; (d) ACC^T ; (e) CD ; (f) D^2 (i.e. DD).

Answers:

1. (a) $A + C = \begin{bmatrix} -2 & 2 \\ 5 & 1 \end{bmatrix}$ (b) $B + D$ not defined (c) $2A - 3C = \begin{bmatrix} 16 & -11 \\ 0 & 2 \end{bmatrix}$ (d) $D + B^T = \begin{bmatrix} -3 & 13 & 7 \\ 5 & 8 & -3 \end{bmatrix}$

2. (a) $BC = \begin{bmatrix} -1 & -1 \\ 2 & 3 \\ 2 & 5 \end{bmatrix}$ (b) $ED = \begin{bmatrix} 35 \\ 13 \end{bmatrix}$ (c) AB not defined (d) $A^T B = \begin{bmatrix} 5 & 9 \\ 24 & 22 \end{bmatrix}$ (e)

$D^T A = \begin{bmatrix} 36 & 33 \end{bmatrix}$ (f) DE not defined (g) $FD = \begin{bmatrix} 31 \end{bmatrix}$ (h) $ACE = \begin{bmatrix} 122 & -123 & 160 \\ 11 & -6 & 10 \\ -33 & 41 & -50 \end{bmatrix}$

(i) $CB^T A = \begin{bmatrix} 37 & 114 \\ -14 & -46 \end{bmatrix}$

3. (a) $AX = \begin{bmatrix} 5x + 7y - z \\ 2x + 3z \\ 4x - y - 2z \end{bmatrix}$ (b) $BX = \begin{bmatrix} 3x - 10y + 7z \end{bmatrix}$ (c) $BAC = \begin{bmatrix} 117 & -94 \end{bmatrix}$ (d)

$ACC^T = \begin{bmatrix} -20 & 135 & 67 \\ -11 & 35 & 29 \\ 23 & -75 & -61 \end{bmatrix}$ (e) $CD = \begin{bmatrix} w - y & x - z \\ 5y & 5z \\ -2w + 3y & -2x + 3z \end{bmatrix}$ (f) $D^2 = \begin{bmatrix} w^2 + xy & wx + xz \\ wy + yz & xy + z^2 \end{bmatrix}$

1.3 Some special square matrices

Definition 1.10 The **principal diagonal** or the **main diagonal** of a square $(n \times n)$ matrix A consists of the elements from the top left corner to the bottom right corner, i.e. the elements of the main diagonal are a_{ii} , $i = 1, 2, \dots, n$.

Definition 1.11 A square matrix $D = [d_{ik}]_{(n \times n)}$ is called **diagonal** if

identity matrix = ma trận đơn vị

$$d_{ik} = 0, \text{ for } i \neq k.$$

In other words, the only non-zero entries of the matrix D are on the main diagonal. For any two diagonal matrices A and B , their product is commutative, i.e. $AB = BA$.

For example, a (4×4) diagonal matrix is given by

$$D = \begin{bmatrix} d_{11} & 0 & 0 & 0 \\ 0 & d_{22} & 0 & 0 \\ 0 & 0 & d_{33} & 0 \\ 0 & 0 & 0 & d_{44} \end{bmatrix},$$

where d_{ii} , $i = 1, 2, 3, 4$ may or may not be zero.

1 Matrices

Definition 1.12 A square matrix $I = [I_{ik}]_{(n \times n)}$ is called **the unit** (or the **identity**) matrix, if I is diagonal and if $I_{ii} = 1, i = 1, 2, \dots, n$.

identity matrix = ma trận đơn vị

For example, the (4×4) unit matrix is given by

$$I = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad I = 1$$

Note that any $m \times n$ matrix A ($m \neq n$) can be represented as $A = IA$, where I is the unit $m \times m$ matrix, or as $A = AI$, where I is the unit $n \times n$ matrix. For example,

$$\begin{bmatrix} 2 & -3 & 4 \\ 1 & 5 & 8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -3 & 4 \\ 1 & 5 & 8 \end{bmatrix}.$$

Definition 1.13 A square matrix A is **đối xứng** **symmetric** if $A^T = A$. A square matrix A is **anti-symmetric** if $A^T = -A$. This implies that all the elements on the main diagonal of an anti-symmetric matrix must vanish.

get the same result after transpose

Example of a 3×3 symmetric matrix:

$$A = \begin{bmatrix} 1 & 2 & -5 \\ 2 & 3 & 2 \\ -5 & 2 & 0 \end{bmatrix} \quad \text{then} \quad A^T = \begin{bmatrix} 1 & 2 & -5 \\ 2 & 3 & 2 \\ -5 & 2 & 0 \end{bmatrix} = A.$$

Example of a 3×3 anti-symmetric matrix

$$B = \begin{bmatrix} 0 & 2 & -5 \\ -2 & 0 & 2 \\ 5 & -2 & 0 \end{bmatrix} \quad \text{then} \quad B^T = \begin{bmatrix} 0 & -2 & 5 \\ 2 & 0 & -2 \\ -5 & 2 & 0 \end{bmatrix} = -B.$$

Definition 1.14 If all elements of a diagonal matrix are the same then the matrix is called a **scalar matrix**. Any scalar matrix A can be represented as $A = \lambda I$, where λ is a scalar and I is the unit matrix.

Example of a scalar (4×4) matrix

$$A = \begin{bmatrix} 7 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 7 \end{bmatrix} = 7I,$$

where I is the 4×4 identity matrix.

Definition 1.15 A square matrix is **upper-triangular** if each element below the principal diagonal is 0. Similarly, A square matrix is **lower-triangular** if each element **above** the principal diagonal is 0.

For example:

$$A = \begin{bmatrix} 2 & -7 & 8 \\ 0 & 5 & 15 \\ 0 & 0 & 5 \end{bmatrix} \text{ is upper triangular and } B = \begin{bmatrix} 5 & 0 \\ 4 & 5 \end{bmatrix} \text{ is lower triangular.}$$

Definition 1.16 If all elements of a matrix are 0, then the matrix is called a **null matrix**. A null matrix does not have to be square.

Exercises

1. a) Write down the third-order unit matrix.
- b) Write down the (2×4) null matrix.
2. If

$$A = \begin{bmatrix} 4 & 0 \\ 0 & -3 \end{bmatrix} \quad B = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad C = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & -5 \\ 0 & 0 & 6 \end{bmatrix} \quad D = [1 \quad 2 \quad 3]$$

$$E = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad F = \begin{bmatrix} 4 & 0 & 0 \\ 6 & -1 & 0 \end{bmatrix} \quad G = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Calculate:

- a) A^2 ,
 - b) A^3 ,
 - c) CB ,
 - d) EC ,
 - e) EG ,
 - f) GE ,
 - g) DED^T ,
 - h) AF .
3. Which, if any, of the matrices in Question 2 can be described as:

- a) square
- b) upper-triangular
- c) scalar
- d) diagonal
- e) symmetric
- f) lower-triangular

Answers:

1. (a)

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. a) $A^2 = \begin{bmatrix} 16 & 0 \\ 0 & 9 \end{bmatrix}$

b) $A^3 = \begin{bmatrix} 64 & 0 \\ 0 & -27 \end{bmatrix}$. Note that if A is diagonal, then A^n is also diagonal.

c) $CB = \begin{bmatrix} x - y + 2z \\ 3y - 5z \\ 6z \end{bmatrix}$

d) $EC = \begin{bmatrix} 3 & -3 & 6 \\ 0 & 9 & -15 \\ 0 & 0 & 18 \end{bmatrix}$

e) $EG = \begin{bmatrix} -6 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 15 \end{bmatrix}$

f) $GE = \begin{bmatrix} -6 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 15 \end{bmatrix}$

g) $DED^T = [42]$

$$\text{h) } AF = \begin{bmatrix} 16 & 0 & 0 \\ -18 & 3 & 0 \end{bmatrix}$$

3. a) A, C, E, G

b) C

c) E

d) A, E, G

e) A, E, G

f) none

1.4 Determinants

Associated with every **square** matrix is a number called its **determinant**. The determinant occurs commonly in work on transformations and the solution of sets of equations. The determinant of a matrix A can be denoted by $\det(A)$ or $|A|$ or by writing the elements as an array as usual, but between two vertical lines, not brackets.

Definition 1.17 Let A be a 1×1 matrix $A = [a_{11}]$. The determinant of A is then $\det(A) = a_{11}$.

Definition 1.18 Let A be a 2×2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

The determinant of A is given by

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Example 1.11 $\begin{vmatrix} 2 & -9 \\ 5 & 8 \end{vmatrix} = 2 \times 8 - (-9 \times 5) = 61.$

Definition 1.19 For any square matrix $A = [a_{ik}]_{n \times n}$, we define the **minor** of the element a_{ij} (written as M_{ij}), as the determinant obtained by deleting row i and column j of $\det(A)$.

For example for the 3×3 matrix A

M_{11} = delete first column and first line

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 5 & 6 \\ 4 & 8 & 9 \end{bmatrix}$$

we can define 9 minors M_{ik}

$$M_{11} = \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = -3; \quad M_{12} = \begin{vmatrix} 7 & 6 \\ 4 & 9 \end{vmatrix} = 39; \quad M_{32} = \begin{vmatrix} 1 & 3 \\ 7 & 6 \end{vmatrix} = -15, \text{ etc.}$$

Definition 1.20 The **cofactor** C_{ij} of element a_{ij} in $n \times n$ matrix A is defined by $C_{ij} = (-1)^{i+j} M_{ij}$.

Note that the term $(-1)^{i+j}$ has the following pattern:

$$\begin{vmatrix} +1 & -1 & +1 & \dots \\ -1 & +1 & -1 & \dots \\ +1 & -1 & +1 & \dots \\ \dots & & & \end{vmatrix}.$$

This pattern is also written as

$$\begin{vmatrix} + & - & + & \dots \\ - & + & - & \dots \\ + & - & + & \dots \\ \dots & & & \end{vmatrix}.$$

For example, for the 3×3 matrix A above

$$C_{11} = +M_{11} = + \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} = -3; \quad C_{12} = -M_{12} = - \begin{vmatrix} 7 & 6 \\ 4 & 9 \end{vmatrix} = -39;$$

$$C_{32} = -M_{32} = - \begin{vmatrix} 1 & 3 \\ 7 & 6 \end{vmatrix} = 15; \text{ etc.}$$

$\begin{matrix} \text{even} \\ \swarrow \\ \text{no change} \end{matrix}$
 $\begin{matrix} \text{odd} \\ \searrow \\ \text{change} \end{matrix}$

Definition 1.21 The matrix C , which is called the **matrix of cofactors** of A , has the cofactors as its elements.

For example, for a 3×3 matrix A , the matrix of cofactors C is given by

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} M_{11} & -M_{12} & M_{13} \\ -M_{21} & M_{22} & -M_{23} \\ M_{31} & -M_{32} & M_{33} \end{bmatrix}$$

Definition 1.22 The **adjoint matrix** of A , written $\text{adj}(A)$, is defined by $\text{adj}(A) = C^T$, where C is the matrix of cofactors of A .

Definition 1.23 Laplace expansion: The determinant of a $n \times n$ matrix A can be written as a sum of the elements of A in any row or any column, multiplied by the respective cofactor. Thus we have:

$$\det(A) = a_{1i}C_{1i} + a_{2i}C_{2i} + \cdots + a_{ni}C_{ni} = \sum_{j=1}^n a_{ji}C_{ji}$$

for the i -th column of A . Alternatively

$$\det(A) = a_{j1}C_{j1} + a_{j2}C_{j2} + \cdots + a_{jn}C_{jn} = \sum_{i=1}^n a_{ji}C_{ji},$$

for the j -th row of A .

Handwritten note: $a_{11} \cdot C_{11} + a_{12} \cdot C_{12} + a_{13} \cdot C_{13} = \det A$

For example, the determinant of the matrix

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

can be computed by applying the Laplace formula and expanding by the first row of A

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1 \begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}.$$

Example 1.12

$$\begin{aligned}
\begin{vmatrix} 4 & 3 & -1 \\ 4 & -7 & 6 \\ 1 & 3 & -9 \end{vmatrix} &= 4 \begin{vmatrix} -7 & 6 \\ 3 & -9 \end{vmatrix} - 3 \begin{vmatrix} 4 & 6 \\ 1 & -9 \end{vmatrix} + (-1) \begin{vmatrix} 4 & -7 \\ 1 & 3 \end{vmatrix} \\
&= 4(63 - 18) - 3(-36 - 6) - 1(12 + 7) \\
&= 287
\end{aligned}$$

Similarly, a 4×4 determinant is defined in terms of 3×3 determinants:

$$\begin{aligned}
\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix} &= a_1 \begin{vmatrix} b_2 & c_2 & d_2 \\ b_3 & c_3 & d_3 \\ b_4 & c_4 & d_4 \end{vmatrix} - b_1 \begin{vmatrix} a_2 & c_2 & d_2 \\ a_3 & c_3 & d_3 \\ a_4 & c_4 & d_4 \end{vmatrix} \\
&\quad + c_1 \begin{vmatrix} a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \\ a_4 & b_4 & d_4 \end{vmatrix} - d_1 \begin{vmatrix} a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \\ a_4 & b_4 & c_4 \end{vmatrix}
\end{aligned}$$

and then each 3×3 determinant is evaluated as above.

Properties of the determinant

- $\det(AB) = \det(A) \det(B)$, for any $n \times n$ matrices A and B .
- $\det(A^T) = \det(A)$
- $\det(I) = 1$, where I is the unit matrix.
- $\det(\lambda A) = \lambda^n \det(A)$, for any $n \times n$ matrix A and any scalar λ .
- Let B be obtained from A by interchanging any two rows or any two columns. Then $\det(B) = -\det(A)$.
- Hence, if A contains two or more identical rows (or columns), then $\det(A) = 0$.

Example 1.13 *If*

$$A = \begin{bmatrix} 2 & -7 & 3 \\ 1 & 5 & -1 \\ 4 & 0 & 5 \end{bmatrix},$$

find $\det(A)$ and $\det(A^T)$

$$\begin{aligned} \det(A) &= \begin{vmatrix} 2 & -7 & 3 \\ 1 & 5 & -1 \\ 4 & 0 & 5 \end{vmatrix} = 2 \begin{vmatrix} 5 & -1 \\ 0 & 5 \end{vmatrix} - (-7) \begin{vmatrix} 1 & -1 \\ 4 & 5 \end{vmatrix} + 3 \begin{vmatrix} 1 & 5 \\ 4 & 0 \end{vmatrix} \\ &= 2(25 - 0) + 7(5 + 4) + 3(0 - 20) \\ &= 53, \end{aligned}$$

$$A^T = \begin{bmatrix} 2 & 1 & 4 \\ -7 & 5 & 0 \\ 3 & -1 & 5 \end{bmatrix},$$

$$\begin{aligned} \det(A^T) &= \begin{vmatrix} 2 & 1 & 4 \\ -7 & 5 & 0 \\ 3 & -1 & 5 \end{vmatrix} \\ &= 2 \begin{vmatrix} 5 & 0 \\ -1 & 5 \end{vmatrix} - 1 \begin{vmatrix} -7 & 0 \\ 3 & 5 \end{vmatrix} + 4 \begin{vmatrix} -7 & 5 \\ 3 & -1 \end{vmatrix} \\ &= 2(25 + 0) - 1(-35 - 0) + 4(7 - 15) \\ &= 53. \end{aligned}$$

Example 1.14

$$\begin{aligned}
\begin{vmatrix} +1 & -4 & + -3 \\ 6 & 10 & 0 \\ 2 & -9 & 0 \end{vmatrix} &= 1 \begin{vmatrix} 10 & 0 \\ -9 & 0 \end{vmatrix} - 4 \begin{vmatrix} 6 & 0 \\ 2 & 0 \end{vmatrix} + (-3) \begin{vmatrix} 6 & 10 \\ 2 & -9 \end{vmatrix} \\
&= 1(0) - 4(0) - 3(-54 - 20) \\
&= 222
\end{aligned}$$

or using column 3, because there are two zero elements, and following the correct pattern of signs:

$$\begin{aligned}
\begin{vmatrix} +1 & -4 & + -3 \\ 6 & 10 & -0 \\ 2 & -9 & +0 \end{vmatrix} &= -3 \begin{vmatrix} 6 & 10 \\ 2 & -9 \end{vmatrix} - 0 + 0 \\
&= 222
\end{aligned}$$

Exercises

Evaluate the following determinants:

1. $\begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix}$

2. $\begin{vmatrix} -1 & 5 \\ -6 & -11 \end{vmatrix}$

3. $\begin{vmatrix} x & 2 \\ y & -3 \end{vmatrix}$

4. $\begin{vmatrix} 2 & 0 \\ 5 & 7 \end{vmatrix}$

5. $\begin{vmatrix} 5 & 0 & 1 \\ 2 & 1 & -1 \\ 3 & 2 & 0 \end{vmatrix}$

6. $\begin{vmatrix} 1 & 3 & 5 \\ 0 & 0 & 9 \\ -2 & -3 & 1 \end{vmatrix}$

7. $\begin{vmatrix} 1 & -1 & 1 \\ 3 & -5 & 3 \\ 4 & 7 & 4 \end{vmatrix}$

8. $\begin{vmatrix} 5 & 2 & 0 \\ 0 & -6 & -1 \\ 1 & 1 & 3 \end{vmatrix}$

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9.
$$\begin{vmatrix} -2 & 1 & 3 \\ 0 & 4 & 0 \\ 4 & -2 & -6 \end{vmatrix}$$

10. An element of a lamina is subjected to normal and tangential stresses on each of its four sides. The associated stress matrix is $A = \begin{bmatrix} 12 & -8 \\ -8 & 15 \end{bmatrix}$. Calculate the principal stresses, given that each principal stress, λ , satisfies $\det(A - \lambda I) = 0$

Answers:

1. 10
2. 41
3. $-3x - 2y$
4. 14
5. 11
6. -27
7. 0
8. -87
9. 0
10. 21.64; 5.36

1.5 Matrix inversion

Definition 1.24 If A is a square matrix, and there exists a second matrix B of the same size as A which satisfies the matrix equations

$$AB = I \quad \text{and} \quad BA = I,$$

$$1 \cdot \frac{1}{2} = \frac{1}{2} \quad B = A^{-1} : \text{inverse of } A$$

where I is a unit matrix of the same order as A , then B is said to be the **inverse** of A and is written as A^{-1} .

1.5.1 Inverse of a 2×2 matrix

Let A be a 2×2 matrix, with

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad A^{-1} = \begin{bmatrix} e & f \\ g & h \end{bmatrix}.$$

As $AA^{-1} = I$, we have

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

giving

$$\begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Hence,

$$ae + bg = 1, \quad af + bh = 0, \quad ce + dg = 0, \quad cf + dh = 1.$$

Solving for e, f, g and h gives

$$e = \frac{d}{ad - bc}, \quad f = \frac{-b}{ad - bc}, \quad g = \frac{-c}{ad - bc}, \quad h = \frac{a}{ad - bc}.$$

As $\det(A) = ad - bc$, we find

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$$

where it was assumed that $ad - bc \neq 0$.

When $\det A \neq 0$ we can use matrix multiplication to check that

$$\frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} A = I.$$

Definition 1.25 If $\det A = 0$ then A^{-1} does not exist and the matrix A is said to be **singular**. In fact, for any square matrix A we have $\det(A) = 0 \iff A^{-1}$ does not exist $\iff A$ is singular.

Example 1.15 If $A = \begin{bmatrix} 8 & 2 \\ 4 & 3 \end{bmatrix}$ find A^{-1} .

Solution.

$$A^{-1} = \frac{1}{(8)(3) - (2)(4)} \begin{bmatrix} 3 & -2 \\ -4 & 8 \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 3 & -2 \\ -4 & 8 \end{bmatrix}.$$

1.5.2 Inverse of a $n \times n$ matrix

Calculating the inverse from the adjoint matrix. Let A be a non-singular square matrix, i.e. $\det(A) \neq 0$. The inverse A^{-1} is then given by

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A),$$

where $\text{adj}(A)$ is the adjoint matrix, as defined in Def.1.22.

Example 1.16 Given that

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 5 & 6 \\ 4 & 8 & 9 \end{bmatrix}$$

find C , $\text{adj}(A)$, $\det(A)$ and A^{-1} .

Solution.

$$C = \begin{bmatrix} + \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} & - \begin{vmatrix} 7 & 6 \\ 4 & 9 \end{vmatrix} & + \begin{vmatrix} 7 & 5 \\ 4 & 8 \end{vmatrix} \\ - \begin{vmatrix} 2 & 3 \\ 8 & 9 \end{vmatrix} & + \begin{vmatrix} 1 & 3 \\ 4 & 9 \end{vmatrix} & - \begin{vmatrix} 1 & 2 \\ 4 & 8 \end{vmatrix} \\ + \begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} & - \begin{vmatrix} 1 & 3 \\ 7 & 6 \end{vmatrix} & + \begin{vmatrix} 1 & 2 \\ 7 & 5 \end{vmatrix} \end{bmatrix}$$

$$\Rightarrow C = \begin{bmatrix} -3 & -39 & 36 \\ 6 & -3 & 0 \\ -3 & 15 & -9 \end{bmatrix}$$

$$\text{adj}A = C^T = \begin{bmatrix} -3 & 6 & -3 \\ -39 & -3 & 15 \\ 36 & 0 & -9 \end{bmatrix}$$

$$\begin{aligned} \det A &= 1 \begin{vmatrix} 5 & 6 \\ 8 & 9 \end{vmatrix} - 2 \begin{vmatrix} 7 & 6 \\ 4 & 9 \end{vmatrix} + 3 \begin{vmatrix} 7 & 5 \\ 4 & 8 \end{vmatrix} \\ &= (5)(9) - (6)(8) - 2\{(7)(9) - (4)(6)\} + 3\{(7)(8) - (5)(4)\} \\ &= 45 - 48 - 2\{63 - 24\} + 3\{56 - 20\} \\ &= 30 - 2(39) + 3(36) \\ &= -3 - 78 + 108 \\ &= 27 \end{aligned}$$

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$$A^{-1} = \frac{\text{adj}A}{\det A} = \frac{1}{27} \begin{bmatrix} -3 & 6 & -3 \\ -39 & -3 & 15 \\ 36 & 0 & -9 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{3}{27} & \frac{6}{27} & -\frac{3}{27} \\ -\frac{39}{27} & -\frac{3}{27} & \frac{15}{27} \\ \frac{36}{27} & \frac{0}{27} & -\frac{9}{27} \end{bmatrix} = \begin{bmatrix} -\frac{1}{9} & \frac{2}{9} & -\frac{1}{9} \\ -\frac{13}{9} & -\frac{1}{9} & \frac{5}{9} \\ \frac{4}{3} & 0 & -\frac{1}{3} \end{bmatrix}.$$

Exercise 1.1 Find the inverses of the following matrices and in each case show that $AA^{-1} = I$ and $A^{-1}A = I$:

1. $A = \begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix}$

2. $A = \begin{bmatrix} u & 3 \\ 5 & v \end{bmatrix}$

3. $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & -1 & -4 \\ -2 & 5 & 1 \end{bmatrix}$

4. $A = \begin{bmatrix} 1 & 6 & 9 \\ 7 & 5 & 2 \\ 3 & 4 & 8 \end{bmatrix}$

Answers:

1. $\begin{bmatrix} 2 & -3 \\ -5 & 8 \end{bmatrix}$

2. $\frac{1}{uv-15} \begin{bmatrix} v & -3 \\ -5 & u \end{bmatrix}$

3. $\frac{1}{7} \begin{bmatrix} 19 & -17 & -11 \\ 6 & -5 & -2 \\ 8 & -9 & -5 \end{bmatrix}$

4. $\frac{1}{151} \begin{bmatrix} -32 & 12 & 33 \\ 50 & 19 & -61 \\ -13 & -14 & 37 \end{bmatrix}$

1.6 Laws of matrix algebra

For any $n \times n$ invertible matrices A and B

- If $AB = I$ then $B = A^{-1}$ and $A = B^{-1}$
- $(A^{-1})^{-1} = A$
- $(kA)^{-1} = k^{-1}A^{-1}$, for any nonzero scalar k
- $(A^T)^{-1} = (A^{-1})^T$
- $\det(A^{-1}) = (\det A)^{-1}$
- $(AB)^{-1} = B^{-1}A^{-1}$

Example 1.17 Let A, B, C and D be some invertible matrices so that

$$BACD = I.$$

Express the matrix A in terms of B^{-1}, C^{-1} and D^{-1} .

Solution

Solving for A in three steps

$$B^{-1}BACD = ACD = B^{-1}I = B^{-1}$$

next

$$ACDD^{-1} = AC = B^{-1}D^{-1}$$

finally

$$A = ACC^{-1} = B^{-1}D^{-1}C^{-1}.$$

Example 1.18 Using the laws of matrix algebra, show that

$$(ABC^T)^{-1} = (C^{-1})^T B^{-1} A^{-1}$$

Solution

The proof follows in two steps

$$(ABC^T)^{-1} = (C^T)^{-1}(AB)^{-1} = (C^{-1})^T B^{-1} A^{-1}.$$

1.7 Linear equations and applications

1.7.1 Solving linear equations by matrix inversion

Consider a system of n linear equations for n unknown variables x_i , $i = 1, 2, \dots, n$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n &= b_n, \end{aligned}$$

where a_{ik} , $i, k = 1, 2, \dots, n$ and b_i , $i = 1, 2, \dots, n$ are given numbers.

Introducing the following three matrices

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix},$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix} \text{ and } B = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_n \end{bmatrix},$$

the system of equations can be written in the matrix form

$$AX = B.$$

Let us now assume that the matrix A is invertible so that A^{-1} exists. Then the matrix of the unknowns X is found as

$$X = A^{-1}B.$$

1.7.2 Electric circuits

Loop current analysis can be applied to any electric circuit that contains resistors and voltage sources. The currents in all branches of the circuit can be found using the

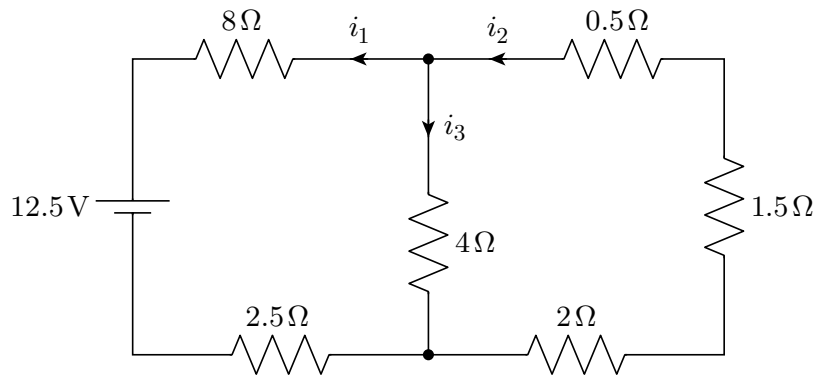


Figure 1.1: Electric circuit in example 1.19

- Around any closed path in an electric circuit, the sum of the voltage drops through the resistors equals the sum of the voltage rises through the voltage sources.
- At any junction, the sum of the currents flowing in equals the sum of the currents flowing out.
- Voltage drop V through a resistor with resistance Ω is given by Ohm's law

$$V = I\Omega,$$

where I is the current that flows through the resistor. Note that I is taken as positive, if the direction of the path coincides with the direction of the current I . Otherwise, the current is taken as negative.

Note that an extension of the Kirchhoff's Voltage Law for electric circuits that contain capacitors and inductors will be discussed in Module 6.

Example 1.19 Applying the Kirchhoff's Voltage law to the circuit shown in Figure 1.1, we find that the currents i_1 , i_2 , and i_3 (in amperes) satisfy the following coupled equations:

$$-10.5i_1 + 4i_3 = 12.5,$$

$$i_1 - i_2 + i_3 = 0,$$

$$4i_2 + 4i_3 = 0.$$

Solve these equations using matrix inversion.

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Solution. These equations can be written in matrix form thus:

$$\begin{bmatrix} -10.5i_1 + 0i_2 + 4i_3 \\ 1i_1 - 1i_2 + 1i_3 \\ 0i_1 + 4i_2 + 4i_3 \end{bmatrix} = \begin{bmatrix} 12.5 \\ 0 \\ 0 \end{bmatrix}.$$

The left-hand matrix can be written as a product of two matrices, giving

$$\underbrace{\begin{bmatrix} -10.5 & 0 & 4 \\ 1 & -1 & 1 \\ 0 & 4 & 4 \end{bmatrix}}_{\text{Matrix A}} \underbrace{\begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix}}_{\text{Matrix X}} = \underbrace{\begin{bmatrix} 12.5 \\ 0 \\ 0 \end{bmatrix}}_{\text{Matrix B}}$$

i.e. $AX = B$.

To solve for X, premultiply both sides by A^{-1}

$$\begin{aligned} \Rightarrow A^{-1}AX &= A^{-1}B \\ \Rightarrow IX &= A^{-1}B \\ \Rightarrow X &= A^{-1}B \\ \Rightarrow \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} &= \underbrace{\begin{bmatrix} -0.08 & 0.16 & 0.04 \\ -0.04 & -0.42 & 0.145 \\ 0.04 & 0.42 & 0.105 \end{bmatrix}}_{\substack{A^{-1}, \text{ as obtained by} \\ \text{the adj-det method}}} \begin{bmatrix} 12.5 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -0.5 \\ 0.5 \end{bmatrix} \\ \Rightarrow i_1 &= -1 \text{ amp}, \quad i_2 = -0.5 \text{ amp} \quad \text{and} \quad i_3 = 0.5 \text{ amp}. \end{aligned}$$

Exercise 1.2 1. Using matrix inversion, solve

a) $2x + 3y = 8; \quad 4x + 5y = 14.$

b) $2x - y = 5; \quad 3x + 2y = 4.$

2. For the electric circuit shown in Figure 1.2, the equations for the mesh currents (i_1 , i_2 , and i_3 , in amperes) are given by

$$\begin{aligned} 14i_1 - 4i_2 - 5i_3 &= 6, \\ -4i_1 + 19i_2 - 3i_3 &= -15, \\ -5i_1 - 3i_2 + 20i_3 &= 3. \end{aligned}$$

Solve the equations using matrix inversion, and give your answers correct to 3 significant figures.

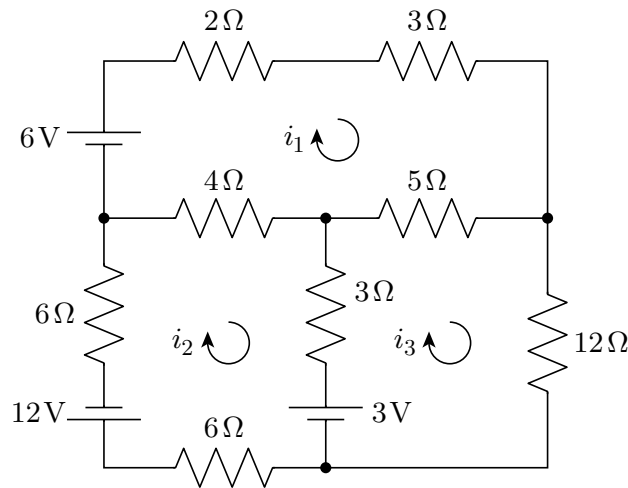


Figure 1.2: Electric circuit in exercise 1.2 (b).

3. For the circuit shown in Figure 1.3, the currents i_1 , i_2 , and i_3 (in amperes) satisfy these equations:

$$i_1 + i_2 + i_3 = 0,$$

$$6i_1 - 8i_3 = 24,$$

$$7i_2 - 2i_3 = 9.$$

Solve the equations using matrix inversion, giving all answers correct to 3 decimal places.

Answers:

1. a) $x = 1, y = 2$
b) $x = 2, y = -1$
2. 0.262 amps, -0.717 amps, 0.108 amps.
3. 1.309 amps, 0.709 amps, -2.018 amps.

1.7.3 Cramer's rule

A system of n linear equations for n unknowns from the previous Section 1.7 can also be solved using the so-called **Cramer's rule**.

Theorem 1.1 Consider the system of n linear equations in the matrix form

$$AX = B,$$

with the $n \times n$ coefficient matrix A , the $n \times 1$ column matrix of the right-hand side B and the $n \times 1$ solution matrix X . The entries x_i of the solution matrix X can be found from

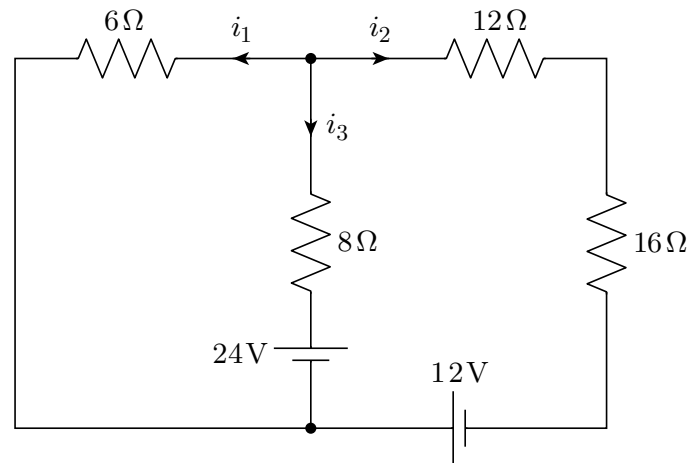


Figure 1.3: Electric circuit in exercise 1.2(c).

$$x_i = \frac{\Delta_i}{\Delta} = \frac{\det(A_i)}{\det(A)},$$

where for brevity of notations we denote $\Delta_i = \det(A_i)$ and $\Delta = \det(A)$. Here, A_i is the matrix formed by replacing the i -th column of A by the column matrix of the right-hand side B .

Cramer's rule for two unknowns.

Example 1.20 For the electric circuit shown in Figure 1.4, the equations for the electric currents (i_1 and i_2 in amperes) are given:

$$8i_1 - 3i_2 = -2 \quad -3i_1 + 10i_2 = 6.$$

Solve the equations using Cramer's rule for two unknowns.

Solution

$$\begin{aligned} i_1 &= \frac{\Delta_1}{\Delta} = \frac{\begin{vmatrix} -2 & -3 \\ 8 & 10 \end{vmatrix}}{\begin{vmatrix} 8 & -3 \\ -3 & 10 \end{vmatrix}} \\ &= \frac{(-2)(10) - (-3)(6)}{(8)(10) - (-3)(-3)} \\ &= \frac{-20 + 18}{80 - 9} \\ &= \frac{-2}{71} \text{ amps.} \end{aligned}$$

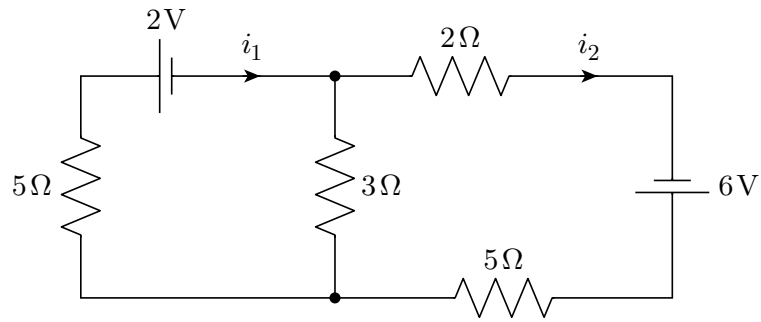


Figure 1.4: Electric circuit in example 1.20.

$$i_2 = \frac{\Delta_2}{\Delta} = \frac{\begin{vmatrix} 8 & -2 \\ -3 & 6 \end{vmatrix}}{\begin{vmatrix} 8 & -3 \\ -3 & 10 \end{vmatrix}} = \frac{42}{71} \text{ amps}$$

Cramer's rule for three unknowns.

Example 1.21 For the electric circuit shown in Figure 1.5, the equations for the mesh currents (i_1 , i_2 , and i_3 , in amperes) are given:

$$\begin{aligned} 10i_1 - 3i_2 - i_3 &= 3 \\ -3i_1 + 14i_2 - 2i_3 &= -6 \\ -i_1 - 2i_2 + 6i_3 &= 4. \end{aligned}$$

Solve the equations using Cramer's rule.

Solution

$$i_1 = \frac{\Delta_1}{\Delta}, \quad i_2 = \frac{\Delta_2}{\Delta}, \quad i_3 = \frac{\Delta_3}{\Delta},$$

where

$$\Delta = \begin{vmatrix} 10 & -3 & -1 \\ -3 & 14 & -2 \\ -1 & -2 & 6 \end{vmatrix} = 720$$

$$\Delta_1 = \begin{vmatrix} 3 & -3 & -1 \\ -6 & 14 & -2 \\ 4 & -2 & 6 \end{vmatrix} = 200$$

$$\Delta_2 = \begin{vmatrix} 10 & 3 & -1 \\ -3 & -6 & -2 \\ -1 & 4 & 6 \end{vmatrix} = -202$$

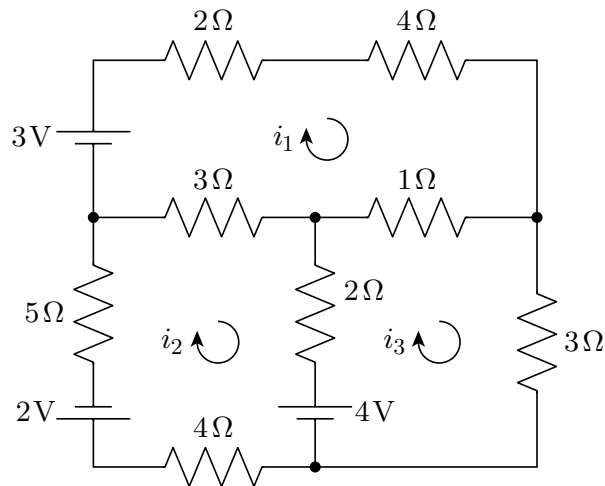


Figure 1.5: Electric circuit in example 1.21.

and

$$\Delta_3 = \begin{vmatrix} 10 & -3 & 3 \\ -3 & 14 & -6 \\ -1 & -2 & 4 \end{vmatrix} = 446$$

Hence,

$$\begin{aligned} i_1 &= \frac{\Delta_1}{\Delta} = \frac{200}{720} \approx 0.278 \text{ amps,} \\ i_2 &= \frac{\Delta_2}{\Delta} = \frac{-202}{720} \approx -0.281 \text{ amps,} \\ i_3 &= \frac{\Delta_3}{\Delta} = \frac{446}{720} \approx 0.619 \text{ amps.} \end{aligned}$$

Exercise 1.3 1. Using Cramer's rule for two unknowns solve

a) $2x + 3y = 8; \quad 4x + 5y = 14.$

b) $2x - y = 5; \quad 3x + 2y = 4.$

2. For the electric circuit shown in Figure 1.6, the equations for the mesh currents (i_1 , i_2 , and i_3 , in amperes) are given by

$$\begin{aligned} 14i_1 - 4i_2 - 5i_3 &= 6, \\ -4i_1 + 19i_2 - 3i_3 &= -15, \\ -5i_1 - 3i_2 + 20i_3 &= 3. \end{aligned}$$

Solve the equations using Cramer's rule, giving your answers correct to 3 significant figures.

Answers.

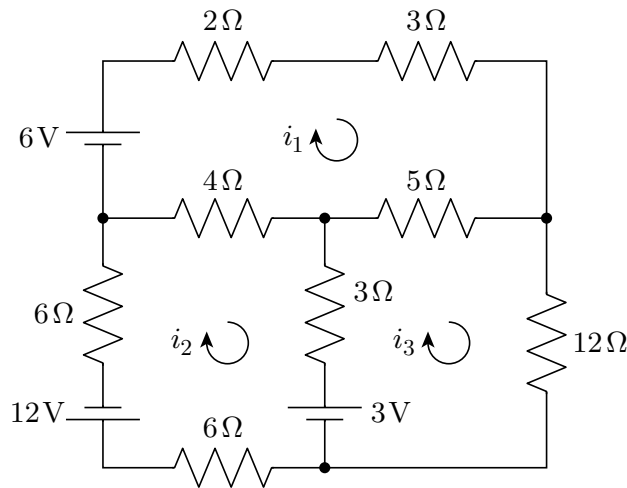


Figure 1.6: Electric circuit in exercise 1.3(b).

1. a) $x = 1, y = 2$;
b) $x = 2, y = -1$.
2. 0.262 amps, -0.717 amps, 0.108 amps.

1.7.4 Gaussian elimination

Another very efficient algorithm for solving systems of linear equations is called **Gaussian elimination**, named after the German mathematician Carl Friedrich Gauss (1777–1855). This algorithm is computationally much more efficient than the previously discussed methods of matrix inversion and Cramer's rule. For comparison, to solve a system of n equations for n unknowns, the Gaussian elimination method requires only $O(n^3)$ arithmetic operations, as compared with $O(n!)$ operations, needed by other methods. In addition, the Gaussian elimination method can also be used to solve **overdetermined** or **underdetermined** systems, when the number of unknowns does not coincide with the number of equations.

Consider a general system of m linear equations for n unknowns. It is important to emphasize that in the general case, the number of equations m does not necessarily coincide with the number of unknowns n . Such a system of equations can also be written in the matrix form, similar to how it was done in Section 1.7

$$AX = B,$$

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where A is the $m \times n$ matrix of the left-hand side coefficients

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix},$$

X is the $n \times 1$ matrix of the unknowns and B is the $m \times 1$ matrix of the right-hand side coefficients

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}, \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{bmatrix}.$$

Definition 1.26 In the cases when $n > m$ ($n < m$), the system of equations $AX = B$ is called **underdetermined** (**overdetermined**).

Definition 1.27 Depending on the coefficient matrices A and B , we generally distinguish between the following three cases

- The system $AX = B$ has a single (**unique**) solution. In this case, the system is called **exactly determined and consistent**.
- The system $AX = B$ has infinitely many solutions. In this case, the system is called **consistent and indeterminate**.
- The system $AX = B$ has no solutions. In this case the system is called **inconsistent**.

Definition 1.28 Combination of the matrices A and B written in the compact form

$$[A|B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right]$$

is called the **augmented matrix**.

Definition 1.29 Elementary row operations: Let R_i denotes the i -th row of the augmented matrix of a certain system of equations. The augmented matrix can be transformed by applying the following types of operations on its rows

- Interchange rows R_i and R_j , $i \neq j$.
- Replace R_i by λR_i , where λ is any nonzero constant.
- Replace R_i by $R_i + \lambda R_j$, with $i \neq j$.

Definition 1.30 Row echelon form. The augmented matrix is said to be in row echelon form, if

- any zero row lies below the non-zero rows
- the leading entry in each non-zero row (i.e. the left-most non-zero entry) is 1. Note that the leading entry is called .
- the leading entry in the $i + 1$ -th row is strictly to the right of the leading entry in the i -th row

Definition 1.31 Reduced row echelon form. The augmented matrix is said to be in reduced row echelon form, if it is in the row echelon form and if additionally the following property is satisfied

- if a column contains a leading entry of some row, then all other entries in this column are zero.

In the example below the matrix A is in reduced row echelon form. The matrix B is in row echelon form, however, it is not in reduced row echelon form, because the 5-th column in B contains the leading entry of the 3-rd row and another non-zero entry $b_{15} = 10$. If we set $b_{15} = 0$ (by applying $R_1 \rightarrow R_1 - 10R_3$), then the matrix B is in reduced row echelon form

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 10 & 20 \\ 0 & 1 & 7 & 0 & -6 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 & 0 & 10 & 20 \\ 0 & 1 & 7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}.$$

Theorem 1.2 Gaussian elimination algorithm. Any augmented matrix can be transformed to a new matrix in reduced row echelon form by a finite sequence of elementary row operations. The solution can be read out from the transformed augmented matrix in the reduced row echelon form.

However, it is important to note that

In practice we often read out the solution from the augmented matrix in row echelon form

as explained in examples below.

1.7.5 Examples of Gaussian elimination: unique solution

Example 1.22 Solve the following equations using Gaussian elimination:

$$x + 2y + 3z = 14$$

$$2x + 3y + z = 11$$

$$3x + y + 4z = 17.$$

Solution.

The augmented matrix is

$$M = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 2 & 3 & 1 & 11 \\ 3 & 1 & 4 & 17 \end{array} \right]$$

The first step is to change m_{21} to 0 by adding a multiple of row 2 to a multiple of row 1. The row operation $R_2 \rightarrow R_2 - 2R_1$ will do this. The second step is to alter m_{31} to zero by adding a multiple of row 3 to a multiple of row 1. The row operation $R_3 \rightarrow R_3 - 3R_1$ will do this. The result is:

$$\begin{aligned} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 2 & 3 & 1 & 11 \\ 3 & 1 & 4 & 17 \end{array} \right] & \xRightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 2 - 2(1) & 3 - 2(2) & 1 - 2(3) & 11 - 2(14) \\ 3 - 3(1) & 1 - 3(2) & 4 - 3(3) & 17 - 3(14) \end{array} \right] \\ & = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & -1 & -5 & -17 \\ 0 & -5 & -5 & -25 \end{array} \right]. \end{aligned}$$

The next step is to change m_{32} to 0 using a row operation involving rows 2 and 3. A suitable row operation is $R_3 \rightarrow R_3 - 5R_2$. The result of this operation is:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & -1 & -5 & -17 \\ 0 & -5 & -5 & -25 \end{array} \right] \xRightarrow{R_3 \rightarrow R_3 - 5R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & -1 & -5 & -17 \\ 0 & 0 & 20 & 60 \end{array} \right]$$

In principle it is now possible to read out the solution by solving the equations sequentially from the bottom of the augmented matrix to the top.

Thus, the equations corresponding to the preceding augmented matrix are written down:

$$x + 2y + 3z = 14$$

$$-y - 5z = -17$$

$$20z = 60$$

$$\Rightarrow z = \frac{60}{20} = 3.$$

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Substituting $z = 3$ in $-y - 5z = -17$ yields $y = 2$. Finally, substituting $z = 3$ and $y = 2$ in $x + 2y + 3z = 14$, we find $x = 1$.

Note that the same answer can also be obtained by further reducing the augmented matrix to **reduced row echelon form**

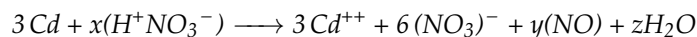
$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & -1 & -5 & -17 \\ 0 & 0 & 20 & 60 \end{array} \right] \xRightarrow{\begin{array}{l} R_1 \rightarrow 2R_2 + R_1 \\ R_2 \rightarrow -R_2 \\ R_3 \rightarrow R_3/20 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & -7 & -20 \\ 0 & 1 & 5 & 17 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

and, finally,

$$\left[\begin{array}{ccc|c} 1 & 0 & -7 & -20 \\ 0 & 1 & 5 & 17 \\ 0 & 0 & 1 & 3 \end{array} \right] \xRightarrow{\begin{array}{l} R_1 \rightarrow R_1 + 7R_3 \\ R_2 \rightarrow R_2 - 5R_3 \end{array}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

From the last augmented matrix, we arrive at the same solution $x = 1$, $y = 2$, $z = 3$, as before.

Example 1.23 As the next example, consider the reaction of oxidation of Cadmium Cd. After balancing the electrical charge for the reactants and products, the reaction can be written as



where x , y and z are called the of the reaction.

Solution

In order to balance this chemical reaction, one equates the number of atoms of each chemical element before and after the reaction.

Noticing that Cd is already balanced, we need to balance the number of atoms of H, N and O. Thus we obtain

$$\begin{array}{ll} \text{H:} & x = 2z \\ \text{N:} & x = 6 + y \\ \text{O:} & 3x = 18 + y + z. \end{array} \quad (1.1)$$

The corresponding augmented matrix is given by

$$M = \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 1 & -1 & 0 & 6 \\ 3 & -1 & -1 & 18 \end{array} \right].$$

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First, we reshuffle the rows

$$M = \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 1 & -1 & 0 & 6 \\ 3 & -1 & -1 & 18 \end{array} \right] \xRightarrow{\substack{R_3 \rightarrow R_1 \\ R_1 \rightarrow R_3}} \left[\begin{array}{ccc|c} 3 & -1 & -1 & 18 \\ 1 & -1 & 0 & 6 \\ 1 & 0 & -2 & 0 \end{array} \right]$$

Next, we transform the augmented matrix to the row echelon form in three steps

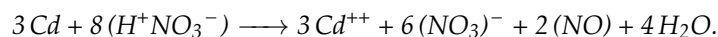
$$\left[\begin{array}{ccc|c} 3 & -1 & -1 & 18 \\ 1 & -1 & 0 & 6 \\ 1 & 0 & -2 & 0 \end{array} \right] \xRightarrow{\substack{R_2 \rightarrow 3R_2 - R_1 \\ R_3 \rightarrow 3R_3 - R_1}} \left[\begin{array}{ccc|c} 3 & -1 & -1 & 18 \\ 0 & -2 & 1 & 0 \\ 0 & 1 & -5 & -18 \end{array} \right]$$

$$\xRightarrow{R_3 \rightarrow 2R_3 + R_2} \left[\begin{array}{ccc|c} 3 & -1 & -1 & 18 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & -9 & -36 \end{array} \right] \xRightarrow{\substack{R_3 \rightarrow -R_3/9 \\ R_1 \rightarrow R_1/3 \\ R_2 \rightarrow -R_2/2}} \left[\begin{array}{ccc|c} 1 & -1/3 & -1/3 & 6 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 1 & 4 \end{array} \right].$$

The solution can now be written as

$$z = 4, \quad y = z/2 = 2, \quad x = 6 + y/3 + z/3 = 8$$

so that the balanced chemical reaction is



1.7.6 Examples of Gaussian elimination: infinitely many solutions

Example 1.24 Solve the following equations by Gaussian elimination:

$$x + y + z = 3, \tag{1.2}$$

$$x - y + 2z = 1 \tag{1.3}$$

$$3x - y + 5z = 5 \tag{1.4}$$

Solution.

The augmented matrix is

$$M = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & -1 & 2 & 1 \\ 3 & -1 & 5 & 5 \end{array} \right].$$

The first step is to change m_{21} to 0 by adding a multiple of row 2 to a multiple of row 1. The row operation $R_2 \rightarrow R_2 - R_1$ will do this. The second step is to alter m_{31} to 0 by adding a multiple of row 3 to a multiple of row 1. The row operation $R_3 \rightarrow R_3 - 3R_1$ will do this. The result is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 1 & -1 & 2 & 1 \\ 3 & -1 & 5 & 5 \end{array} \right] \xRightarrow{\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -2 & 1 & -2 \\ 0 & -4 & 2 & -4 \end{array} \right]$$

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The next step is to change m_{32} to 0 using a row operation involving rows 2 and 3. A suitable row operation is $R_3 \rightarrow R_3 - 2R_2$. The result of this operation is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -2 & 1 & -2 \\ 0 & -4 & 2 & -4 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 3 \\ 0 & -2 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

The resulting equations are:

$$x + y + z = 3 \quad (1.5)$$

$$-2y + z = -2 \quad (1.6)$$

$$0 = 0. \quad (1.7)$$

As the last equation says nothing about x , y and z , and is fulfilled by any x, y, z , only the first two equations are of use. As there are two equations containing three unknowns, any two of the variables can be expressed in terms of the remaining variable, e.g. x and y in terms of z .

$$(1.6) \quad \Rightarrow \quad -2y = -2 - z \quad (1.8)$$

$$\Rightarrow \quad y = \frac{-2 - z}{-2} \quad (1.9)$$

$$\Rightarrow \quad y = 1 + \frac{z}{2}. \quad (1.10)$$

Substituting (1.10) into (1.5)

$$x + 1 + \frac{z}{2} + z = 3 \quad (1.11)$$

$$\Rightarrow \quad x = 2 - \frac{3z}{2}. \quad (1.12)$$

It is sometimes useful (e.g. in vectors, see Section 2) to let

$$z = t. \quad (1.13)$$

Here, t can have any value and is called a **parameter** (i.e. a variable which x , y and z can be written in terms of). Hence the solution of the equations is

$$x = 2 - \frac{3t}{2} \quad (1.14)$$

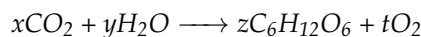
$$y = 1 + \frac{t}{2} \quad (1.15)$$

$$z = t. \quad (1.16)$$

So, for example, if $t=0$, $x = 2$, $y = 1$ and $z=0$; if $t=-2$, $x = 5$, $y = 0$, and $z = -2$, and so on.

Example 1.25 Balance the reaction of photosynthesis

1 Matrices



Solution

The balancing conditions are

$$x = 6z, \quad 2x + y = 6z + 2t, \quad 2y = 12z.$$

In this case, we have four variables x, y, z, t and three equations. This system is underdetermined. Rewriting the balance equations in the form

$$2x + y - 6z - 2t = 0, \quad x - 6z = 0, \quad 2y - 12z = 0,$$

we obtain the 3×5 augmented matrix

$$\left[\begin{array}{cccc|c} 2 & 1 & -6 & -2 & 0 \\ 1 & 0 & -6 & 0 & 0 \\ 0 & 2 & -12 & 0 & 0 \end{array} \right].$$

Transformation to the row echelon form occurs in three steps

$$\left[\begin{array}{cccc|c} 2 & 1 & -6 & -2 & 0 \\ 1 & 0 & -6 & 0 & 0 \\ 0 & 2 & -12 & 0 & 0 \end{array} \right] \xRightarrow[R_3 \rightarrow R_3/2]{R_2 \rightarrow 2R_2 - R_1} \left[\begin{array}{cccc|c} 2 & 1 & -6 & -2 & 0 \\ 0 & -1 & -6 & 2 & 0 \\ 0 & 1 & -6 & 0 & 0 \end{array} \right]$$

$$\left[\begin{array}{cccc|c} 2 & 1 & -6 & -2 & 0 \\ 0 & -1 & -6 & 2 & 0 \\ 0 & 1 & -6 & 0 & 0 \end{array} \right] \xRightarrow[R_1 \rightarrow R_1/2]{R_3 \rightarrow R_3 + R_2} \left[\begin{array}{cccc|c} 1 & 1/2 & -3 & -1 & 0 \\ 0 & -1 & -6 & 2 & 0 \\ 0 & 0 & -12 & 2 & 0 \end{array} \right].$$

Finally

$$\left[\begin{array}{cccc|c} 1 & 1/2 & -3 & -1 & 0 \\ 0 & -1 & -6 & 2 & 0 \\ 0 & 0 & -12 & 2 & 0 \end{array} \right] \xRightarrow[R_3 \rightarrow -R_3/12]{R_2 \rightarrow -R_2} \left[\begin{array}{cccc|c} 1 & 1/2 & -3 & -1 & 0 \\ 0 & 1 & 6 & -2 & 0 \\ 0 & 0 & 1 & -1/6 & 0 \end{array} \right].$$

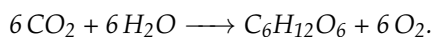
Because the system is underdetermined, we need to introduce one (or several) free parameters. Let us take t as such parameter

$$t \in (-\infty, +\infty).$$

Then the remaining three coefficients x, y, z can be expressed in terms of t as follows

$$z = t/6, \quad y = 2t - 6z = 2t - t = t, \quad x = t + 3z - y/2 = t + t/2 - t/2 = t.$$

From the mathematical point of view, this solution cannot be simplified any further. However, since x, y, z, t are the Stoichiometric Coefficients of a chemical reaction, they can only take integer values. Thus, we conclude that the balanced reaction of photosynthesis with the minimal number of molecules corresponds to $t = 6$



1.7.7 Example of Gaussian elimination: no solution exists

Example 1.26 Solve the following equations using Gaussian elimination:

$$x + 3y + z = 4, \quad (1.17)$$

$$x + y - z = 1, \quad (1.18)$$

$$2x + 4y = 0. \quad (1.19)$$

Solution. The augmented matrix is:

$$M = \left[\begin{array}{ccc|c} 1 & 3 & 1 & 4 \\ 1 & 1 & -1 & 1 \\ 2 & 4 & 0 & 0 \end{array} \right].$$

The first step is to change m_{21} to 0 by adding a multiple of row 2 to a multiple of row 1. The row operation $R_2 \rightarrow R_2 - R_1$ will do this. The second step is to alter m_{31} to 0 by adding a multiple of row 3 to a multiple of row 1. The row operation $R_3 \rightarrow R_3 - 2R_1$ will do this. The result is:

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 4 \\ 1 & 1 & -1 & 1 \\ 2 & 4 & 0 & 0 \end{array} \right] \xRightarrow{\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1}} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 4 \\ 0 & -2 & -2 & -3 \\ 0 & -2 & -2 & -8 \end{array} \right].$$

The next step is to change m_{32} to 0 using a row operation involving rows 2 and 3. A suitable row operation is $R_3 - R_2$. The result of this operation is:

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 4 \\ 0 & -2 & -2 & -3 \\ 0 & -2 & -2 & -8 \end{array} \right] \xRightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 3 & 1 & 4 \\ 0 & -2 & -2 & -3 \\ 0 & 0 & 0 & -5 \end{array} \right].$$

The resulting equations are:

$$\begin{aligned} x + 3y + z &= 4 \\ -2y - 2z &= -3 \\ 0 &= -5. \end{aligned}$$

As the last equation is impossible, this means that the original equations have no solution. There must be some inconsistency between the equations. If the original equations are thought of as representing the equations of three different planes, then as the equations have no solution, this means that the planes have no common point of intersection.

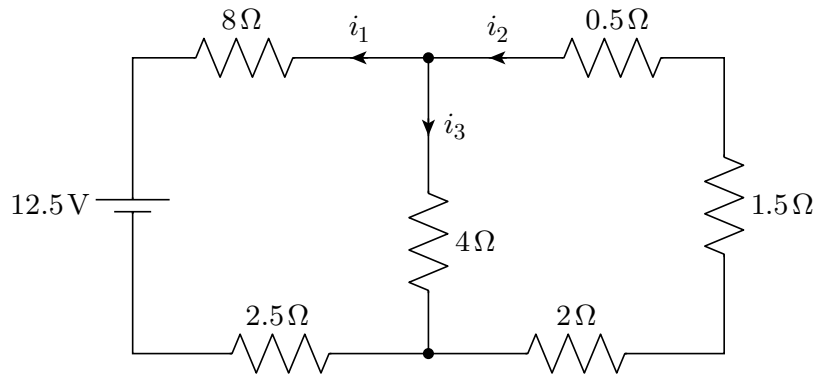


Figure 1.7: Circuit in the example 1.27.

1.7.8 Examples of Gaussian elimination: overdetermined system of equations

Example 1.27 For the circuit shown in Figure 1.7 the currents i_1 , i_2 , and i_3 (in amperes) satisfy the equations:

$$\begin{aligned} i_1 - i_2 + i_3 &= 0 \\ -10.5i_1 + 4i_3 &= 12.5 \\ 10.5i_1 + 4i_2 &= -12.5 \\ 4i_2 + 4i_3 &= 0 \end{aligned}$$

Solve these equations using Gaussian elimination.

Solution. The augmented matrix is:

$$M = \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ -10.5 & 0 & 4 & 12.5 \\ 10.5 & 4 & 0 & -12.5 \\ 0 & 4 & 4 & 0 \end{array} \right].$$

The next step is to change the values of m_{21} , m_{31} and m_{41} to 0s using row operations involving R_1 (i.e. row 1), i.e. $R_2 \rightarrow R_2 + 10.5R_1$ and $R_3 \rightarrow R_3 - 10.5R_1$. In this particular example m_{41} is already 0, so no row operations are necessary on R_4 (i.e. Row 4). The resulting augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ -10.5 & 0 & 4 & 12.5 \\ 10.5 & 4 & 0 & -12.5 \\ 0 & 4 & 4 & 0 \end{array} \right] \xRightarrow{\substack{R_2 \rightarrow R_2 + 10.5R_1 \\ R_3 \rightarrow R_3 - 10.5R_1}} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & -10.5 & 14.5 & 12.5 \\ 0 & 14.5 & -10.5 & -12.5 \\ 0 & 4 & 4 & 0 \end{array} \right]$$

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The next step is to change the values of m_{32} and m_{42} to 0 using row operations involving R_2 , i.e. $R_3 \rightarrow R_3 + 14.5/10.5R_2$, and $R_4 \rightarrow R_4 + 4/10.5R_2$, giving:

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & -10.5 & 14.5 & 12.5 \\ 0 & 0 & 9.5238 & 4.7619 \\ 0 & 0 & 9.5328 & 4.7619 \end{array} \right]$$

The next step is to change the value of m_{43} to a 0 using a row operation involving R_3 , i.e. $R_4 \rightarrow R_4 - R_3$, giving:

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & -10.5 & 14.5 & 12.5 \\ 0 & 0 & 9.5238 & 4.7619 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

The augmented matrix can then be written in equation form, i.e.

$$i_1 - i_2 + i_3 = 0 \quad (1.20)$$

$$-10.5i_2 + 14.5i_3 = 12.5 \quad (1.21)$$

$$9.5238i_3 = 4.7619. \quad (1.22)$$

Equation (1.22) gives

$$i_3 = \frac{4.7619}{9.5238} = 0.5 \text{ amps.} \quad (1.23)$$

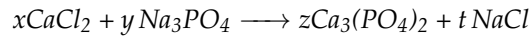
Substituting (1.23) in (1.21) gives

$$\begin{aligned} \Rightarrow -10.5i_2 &= 12.5 - 7.25 = 5.25 \\ \Rightarrow i_2 &= -\frac{5.25}{10.5} \\ \Rightarrow i_2 &= -0.5 \text{ amps.} \end{aligned} \quad (1.24)$$

Substituting (1.24) and (1.23) in (1.20)

$$\begin{aligned} \Rightarrow i_1 - (-0.5) + 0.5 &= 0 \\ \Rightarrow i_1 &= -1 \text{ amp.} \end{aligned}$$

Example 1.28 Using Gaussian elimination, balance the chemical reaction



and thus, determine the minimal values of x, y, z, t .

Solution

1 Matrices

Balancing the reaction, we obtain

$$x = 3z, 3y = t, y = 2z, 4y = 8z, 2x = t.$$

Ordering the coefficients according to x, y, z, t , we obtain the augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ 0 & 3 & 0 & -1 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 4 & -8 & 0 & 0 \\ 2 & 0 & 0 & -1 & 0 \end{array} \right].$$

Transformation to the row echelon form occurs in three steps:

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ 0 & 3 & 0 & -1 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 4 & -8 & 0 & 0 \\ 2 & 0 & 0 & -1 & 0 \end{array} \right] \xRightarrow{\substack{R_5 \rightarrow R_5 - 2R_1 \\ R_3 \rightarrow 3R_3 - R_2 \\ R_4 \rightarrow 3R_4/4 - R_2}} \left[\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ 0 & 3 & 0 & -1 & 0 \\ 0 & 0 & -6 & 1 & 0 \\ 0 & 0 & -6 & 1 & 0 \\ 0 & 0 & 6 & -1 & 0 \end{array} \right],$$

next

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ 0 & 3 & 0 & -1 & 0 \\ 0 & 0 & -6 & 1 & 0 \\ 0 & 0 & -6 & 1 & 0 \\ 0 & 0 & 6 & -1 & 0 \end{array} \right] \xRightarrow{\substack{R_5 \rightarrow R_5 + R_3 \\ R_4 \rightarrow R_4 - R_3}} \left[\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ 0 & 3 & 0 & -1 & 0 \\ 0 & 0 & -6 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Finally

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ 0 & 3 & 0 & -1 & 0 \\ 0 & 0 & -6 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xRightarrow{\substack{R_2 \rightarrow R_2/3 \\ R_3 \rightarrow -R_3/6}} \left[\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ 0 & 1 & 0 & -1/3 & 0 \\ 0 & 0 & 1 & -1/6 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

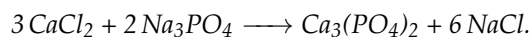
This is the case of an overdetermined system of five equations for four unknown coefficients. However, because the augmented matrix contains two zero rows, the effective number of equations is reduced to three. The solution can be obtained from the following augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 0 \\ 0 & 1 & 0 & -1/3 & 0 \\ 0 & 0 & 1 & -1/6 & 0 \end{array} \right].$$

Thus we have

$$z = t/6, y = t/3, x = 3z = t/2, t \in (-\infty, \infty)$$

The minimal number of molecules $\text{Ca}_3(\text{PO}_4)_2$ is one, so that we choose $t = 6$. The balanced reaction is given by



1.7.9 Examples of Gaussian elimination: underdetermined system of equations

If an underdetermined system is consistent, it has infinitely many solutions. The set of all solutions can be described using one or several free parameters.

Example 1.29 Consider the following augmented matrix

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 2 & -2 & 4 \end{array} \right]$$

It appears that the second and the third equations are equivalent

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 2 & -2 & 4 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

Now we are left with two consistent equations for four unknown variables. Therefore, the solution set contains $4 - 2 = 2$ free parameters. The choice of the free parameters is usually arbitrary, however, in practice it is convenient to choose these parameters as follows

In the augmented matrix reduced to row echelon form, determine the columns that do not contain pivots. Choose the corresponding variables as free parameters.

In our case, the second and the fourth columns do not contain pivots. Consequently, we choose

$$x_4, x_2 \in (-\infty, +\infty).$$

The solution can now be written in terms of x_2 and x_4

$$x_3 = 2 + x_4, \quad x_1 = 5 - 4x_4 - 3x_3 - 2x_2 = 5 - 4x_4 - 3(2 + x_4) - 2x_2 = -1 - 7x_4 - 2x_2.$$

Solution in the vector form is given by

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -7 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad x_4, x_2 \in (-\infty, +\infty).$$

1.8 Rank of a matrix

Definition 1.32 The **rank** of a matrix A , denoted by $\text{Rank}(A)$, is given by the number of nonzero rows of A when it is transformed to reduced row echelon form, or simply, to row echelon form.

Note that the precise definition of rank involves setting up the concepts of row and column subspaces of a matrix. We sidestep this and simply point out that the rank turns out to be a measure of the number of linearly independent rows (regarded as vectors) of the matrix. This property is preserved through the Gaussian elimination process and so the rank can be decided after applying row operations. The concept of linear dependence/independence of vectors will be explained in the next Chapter.

Example 1.30 Let

$$M = \begin{bmatrix} -1 & 6 & -2 \\ 3 & 1 & -1 \\ 2 & -10 & 3 \end{bmatrix}.$$

To find $\text{Rank}(M)$, we reduce M to row echelon form

$$\begin{aligned} \begin{bmatrix} -1 & 6 & -2 \\ 3 & 1 & -1 \\ 2 & -10 & 3 \end{bmatrix} &\xRightarrow{\substack{R_2 \rightarrow R_2 + 3R_1 \\ R_3 \rightarrow R_3 + 2R_1}} \begin{bmatrix} 1 & -6 & 2 \\ 0 & 19 & -7 \\ 0 & 2 & -1 \end{bmatrix} \xRightarrow{\substack{R_3 \rightarrow R_3/2 \\ R_3 \leftrightarrow R_2}} \begin{bmatrix} 1 & -6 & 2 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 19 & -7 \end{bmatrix} \\ &\xRightarrow{\substack{R_1 \rightarrow R_1 + 6R_2 \\ R_3 \rightarrow R_3 - 19R_2}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & \frac{5}{2} \end{bmatrix} \xRightarrow{\substack{R_1 \rightarrow R_1 + (2/5)R_3 \\ R_2 \rightarrow R_2 + (1/5)R_3 \\ R_3 \rightarrow (2/5)R_3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

Consequently,

$$\text{Rank}(M) = \text{Rank} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 3$$

because no row can be completely eliminated.

Example 1.31 On the other hand let

$$M = \begin{bmatrix} -1 & 6 & -6 \\ 3 & 1 & -20 \\ 2 & -10 & 8 \end{bmatrix}$$

Reducing M to the row echelon form, we obtain

$$\begin{bmatrix} -1 & 6 & -6 \\ 3 & 1 & -20 \\ 2 & -10 & 8 \end{bmatrix} \xRightarrow{\substack{R_2 \rightarrow R_2 + 3R_1 \\ R_3 \rightarrow R_3 + 2R_1}} \begin{bmatrix} -1 & 6 & -6 \\ 0 & 19 & -38 \\ 0 & 2 & -4 \end{bmatrix} \xRightarrow{\substack{R_3 \rightarrow R_3/2 \\ R_2 \rightarrow R_2/19}} \begin{bmatrix} -1 & 6 & -6 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix}$$

Finally, we conclude

$$\text{Rank}(M) = \text{Rank} \begin{bmatrix} -1 & 6 & -6 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \text{Rank} \begin{bmatrix} -1 & 6 & -6 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix} = 2.$$

1.8.1 Fundamental theorem of linear algebra

Theorem 1.3 *The rank of a $m \times n$ matrix A , coincides with the rank of the transposed matrix A^T*

$$\text{Rank}(A) = \text{Rank}(A^T).$$

It can further be shown that for a $m \times n$ matrix A , the rank $\text{Rank}(A)$ cannot be greater than either m or n

$$\text{Rank}(A) \leq \min(m, n).$$

Definition 1.33 *In case when*

$$\text{Rank}(A) = \min(m, n),$$

the matrix A is said to have . Otherwise, the matrix A is .

Example 1.32 *The rank of the 2×3 matrix*

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \end{bmatrix}$$

is obviously

$$\text{Rank}(A) = 2.$$

Row echelon form reduction of the transposed matrix A^T

$$A^T = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & -2 \end{bmatrix} \xrightarrow{\begin{matrix} R_1 \rightarrow R_3 \\ R_3 \rightarrow R_1 \end{matrix}} \begin{bmatrix} 3 & -2 \\ 2 & 1 \\ 1 & 0 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 \rightarrow 3R_2/2 - R_1 \\ R_3 \rightarrow 3R_3 - R_1 \end{matrix}} \begin{bmatrix} 3 & -2 \\ 0 & 5/2 \\ 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -2 \\ 0 & 5/2 \\ 0 & 2 \end{bmatrix} \xrightarrow{R_3 \rightarrow 5R_3/4 - R_2} \begin{bmatrix} 3 & -2 \\ 0 & 5/2 \\ 0 & 0 \end{bmatrix}.$$

proves that

$$\text{rank}(A^T) = 2.$$

1.9 Null space of a matrix

Definition 1.34 The **null space** (or the **kernel**) of a $m \times n$ matrix A , denoted by $\text{Null}(A)$, is a set of all solutions of the **homogeneous** matrix equation $AX = 0$, where X is the $n \times 1$ solution matrix and 0 is an $m \times 1$ zero matrix.

$$\text{Null}(A) = \{X \in (n \times 1) \text{ matrices} \mid AX = 0\}.$$

Note that if the right-hand-side B of the matrix equation $AX = B$ is not a zero matrix, i.e. if $B \neq 0$, then such system is called **inhomogeneous** or **non-homogeneous**.

Definition 1.35 The dimension of the null space of A is called the **nullity**, denoted by $\text{nullity}(A)$.

$$\text{nullity}(A) = \text{dimension}\{X \mid AX = 0\}.$$

In practice, the dimension of the null space is determined as the number of free parameters in X .

Definition 1.36 Any homogeneous system

$$AX = 0$$

has a solution

$$X = 0$$

and thus, $AX = 0$ is always consistent.

Note that if the trivial solution $X = 0$ is the only solution of $AX = 0$, then the dimension of the null space of the matrix A is zero, i.e. $\text{nullity}(A) = 0$.

Properties of a homogeneous system:

- A system of equations $AX = 0$ either has a unique trivial solution $X = 0$, or it has infinitely many solutions.
- If A is a square $n \times n$ matrix, the system $AX = 0$ has a unique trivial solution only if $\det(A) \neq 0$. Otherwise, the system has infinitely many solutions.
- If X_1 and X_2 are any two solutions of the homogeneous system $AX = 0$, then $X = \alpha X_1 + \beta X_2$ with some scalars α and β is also a solution.

- If Y is any specific solution of an inhomogeneous system

$$AY = B$$

and if X is any solution of the homogeneous system

$$AX = 0$$

then $X + Y$ is a solution of the inhomogeneous system $A(X + Y) = B$.

Theorem 1.4 For any $m \times n$ matrix A , the rank and nullity are related via the so-called **rank-nullity theorem**

$$\text{nullity}(A) + \text{Rank}(A) = n.$$

In order to exemplify the null space of a matrix, consider the matrices from the examples (1.30) and (1.31). The set of solutions of the homogeneous matrix equation $AX = 0$ can be found using Gaussian elimination method. For the matrix A from (1.30) we solve

$$\begin{bmatrix} -1 & 6 & -2 \\ 3 & 1 & -1 \\ 2 & -10 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

Using the same elementary row operations as in (1.30), we reduce the augmented matrix to the row echelon form

$$\left[\begin{array}{ccc|c} -1 & 6 & -2 & 0 \\ 3 & 1 & -1 & 0 \\ 2 & -10 & 3 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right].$$

The solution X is given by a zero matrix

$$X = \begin{bmatrix} x \\ y \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Because the solution matrix does not contain any free parameter, the dimension of X is zero, i.e. $\text{nullity}(A) = \text{dimension}(X) = 0$. As it was shown in (1.30), the rank of A is $\text{Rank}(A) = 3$. Therefore, the nullity theorem is fulfilled, i.e.

$$\text{nullity}(A) + \text{Rank}(A) = 0 + 3 = 3.$$

Now let us take the matrix A from (1.31) and solve

$$\begin{bmatrix} -1 & 6 & -6 \\ 3 & 1 & -20 \\ 2 & -10 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0.$$

Using the same elementary row operations as in (1.31), we reduce the augmented matrix to the row echelon form

$$\left[\begin{array}{ccc|c} -1 & 6 & -6 & 0 \\ 3 & 1 & -20 & 0 \\ 2 & -10 & -8 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} -1 & 6 & -6 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

In this case, the last row of the augmented matrix contains all zeros and thus, can be deleted. The solution X can be written in the parametric form

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6t \\ 2t \\ t \end{bmatrix},$$

where t is any real number. Because the solution matrix contains one free parameter, the dimension of X is $\text{nullity}(A) = \text{dimension}(X) = 1$. As shown in (1.31), the rank of A is $\text{Rank}(A) = 2$. Therefore, the nullity theorem is fulfilled, i.e.

$$\text{nullity}(A) + \text{Rank}(A) = 1 + 2 = 3.$$

1.10 Rank of an augmented matrix and solvability of the inhomogeneous systems

Consider an inhomogeneous system of m equations for n unknowns $AX = B$, where A is an $(m \times n)$ matrix of the left hand side coefficients, X is the $n \times 1$ column matrix of unknowns and B is an $m \times 1$ column matrix of the right-hand-side coefficients. Recall that an augmented matrix $[A|B]$ that corresponds to system $AX = B$ is written as

$$[A|B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right].$$

The rank of $[A|B]$ is determined using the standard definition 1.32, namely, $\text{rank}([A|B])$ is given by the number of nonzero rows in the row echelon form. We say that the row i of $[A|B]$ is nonzero if at least one of the numbers a_{ik} , $k = 1, \dots, n$, or b_i is nonzero. For example,

$$\text{rank} \left(\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 7 \end{array} \right] \right) = 3, \quad \text{rank} \left(\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right] \right) = 2$$

The solvability of $AX = B$ can be determined by comparing the rank of the augmented matrix $[A|B]$ with the rank of the matrix A . More specifically, the following theorem, known as the **Rouche-Capelli theorem** applies

Theorem 1.5 *The inhomogeneous system $AX = B$ is*

- *inconsistent, if*

$$\text{rank}([A|B]) > \text{rank}(A)$$

- *consistent and has a unique solution if*

$$\text{rank}([A|B]) = \text{rank}(A) = n$$

- *consistent and has infinitely many solutions if*

$$\text{rank}([A|B]) = \text{rank}(A) < n$$

Example 1.33 *The system*

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

is consistent and has infinitely many solutions, because

$$\text{rank}\left(\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 5 & 4 \end{array}\right]\right) = \text{rank}\left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 5 \end{bmatrix}\right) = 2 < 3.$$

Example 1.34 *The following system*

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

is inconsistent, because

$$\text{rank}\left(\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 0 & 0 & 4 \end{array}\right]\right) = 2 \neq \text{rank}\left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}\right) = 1.$$

Example 1.35 *The system*

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 5 \end{bmatrix}$$

is consistent and has a unique solution, because the number of unknowns is 3 and this number coincides with

$$\text{rank}\left(\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 5 & 4 \\ 0 & 0 & 1 & 5 \end{array}\right]\right) = \text{rank}\left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}\right) = 3.$$

The following example illustrates the usage of the Rouché-Capelli theorem 1.5, when the entries of the matrices A and B are unknown.

Example 1.36 *Decide if the system $AX = B$ is consistent or inconsistent and comment on the properties of its solution, if it is known that $\text{rank}([A|B]) = 3$, $\text{rank}(A) = 3$ and A is a 3×4 matrix.*

Solution

Since A is a 3×4 matrix, the number of unknowns in the system $AX = B$ is four. Using the Rouché-Capelli theorem 1.5, we conclude that $AX = B$ is consistent and indeterminate, because $\text{rank}([A|B]) = \text{rank}(A) = 3 < 4$. The set of solutions has one free parameter.

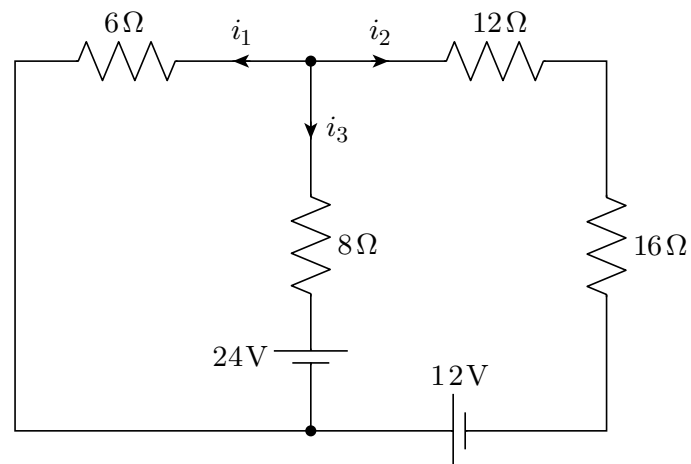


Figure 1.8: Circuit in the exercise 3.

Exercises

1. Using Gaussian elimination, solve the following equations for x and y :

(a) $x + 7y = 11$
 $2x + 16y = 26$

(b) $2x + 4y = -2$
 $-x + 3y = -7$

(c) $2x + y = 7$
 $3x - 15y = 27$

2. Use Gaussian elimination to solve each of the following systems of equations:

(a) $a - b + c = -2$
 $2a + 2b + c = 3$
 $3a + b - 2c = 5$

(b) $a + 2b + c = 6$
 $2a - b - c = -1$
 $3a + 2b + c = 8$

3. For the circuit shown in Figure 1.8 the currents i_1 , i_2 , and i_3 (in amperes) satisfy these equations:

$$i_1 + i_2 + i_3 = 0,$$

$$6i_1 - 8i_3 = 24,$$

$$7i_2 - 2i_3 = 9.$$

Solve these equations using Gaussian elimination.

4. Solve the following overdetermined system of equations

$$x + y = 1$$

$$x - y = 3$$

$$2x + 3y = 1$$

5. Solve the following set of equations

1 Matrices

$$\begin{array}{ll} \text{(a)} & x + y + z = 3 \\ & 2x + 4y + z = 8 \\ & 6x + 10y + 4z = 22 \\ \text{(b)} & x + 2y + 3z = 8 \\ & 2x + 4y + 3z = 7 \\ & 6x + 12y + 12z = 30 \end{array}$$

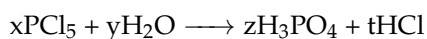
6. Using Gaussian elimination solve the following underdetermined system

$$\begin{array}{l} x + 2y + z = 3 \\ 2x + y - z = 4 \end{array}$$

7. Rewrite the following systems of equations in the matrix form $AX = B$. Determine the rank and the nullity of the matrix A and find the solution matrix X .

$$\begin{array}{ll} \text{(a)} & a + b - c = 2, \\ & 2a + 3b - c = 0, \\ & 3a + 4b - 2c = 1. \\ \text{(b)} & x + y + z = 1, \\ & -x + y + z = 1, \\ & y + z = 2. \end{array}$$

8. Using Gaussian elimination, balance the chemical reaction



and thus, determine the minimal possible values of x, y, z, t .

Answers:

1. (a) $x = -3, y = 2$; (b) $x = 2.2, y = -1.6$; (c) $x = 4, y = -1$

2. (a) $a = 0.5, b = 1.5, c = -1$; (b) $a = 1, b = 2, c = 1$

3. 1.309 amps; 0.709 amps; -2.018 amps

4. $x = 2, y = -1$;

5. a) $x = 2 - \frac{3t}{2}; \quad y = \frac{t}{2} + 1; \quad z = t.$

b) $x = -1 - 2t; \quad y = t; \quad z = 3.$

6. $x = \frac{1}{3}(5 + 3t); \quad y = \frac{1}{3}(2 - 3t); \quad z = t.$

7. (a) $\text{Rank}(A) = 2, \text{nullity}(A) = 1$, no solutions; (b) $\text{Rank}(A) = 2, \text{nullity}(A) = 1$, no solutions.



2 Vectors

2.1 The geometric approach

Definition 2.1 A **vector** is a quantity that has both **magnitude (size)** and **direction**. Any vector can be represented graphically by an arrow, the length of the arrow representing its magnitude and the direction of the arrowhead being the direction of the vector.

In this study guide, vectors that give a certain direction in space will be represented using bold type **a**, e.g. a wind velocity $\vec{v}$ of 6 m/s in a south-east direction, a force $\vec{f}$ of 10 kN acting upwards on a beam, an acceleration $\vec{a}$ of 9.8 m/s² downwards, etc.

Definition 2.2 Sometimes a vector is associated with two selected points in space. The point *A* where the vector starts is called the **initial** point and the point *B* where the vector stops is called the **terminal** point. The corresponding vector is denoted as $\overline{AB}$, or as $\overrightarrow{AB}$. The arrowhead can be at the terminal point or between the initial and terminal points, as shown in Figure 2.1.

Definition 2.3 The **magnitude** (length) of a vector is written as $|\overline{AB}|$, or AB , or $|\vec{a}|$, or $|\vec{a}|$, or a .

Definition 2.4 Two vectors are **equal** if they have the same length (magnitude) and the same direction.

For example, $\overline{AB}$ and $\overline{CD}$ in Figure 2.2 both point north-west and both have length 10. Hence, $\overline{AB} = \overline{CD}$.

Note that two equal vectors may have different initial and terminal points. In other words, any vector $\vec{a}$ can be shifted parallel to self in space.

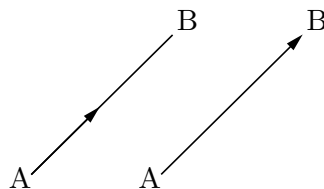


Figure 2.1: Vector $\overline{AB}$ connecting the initial point *A* with the terminal point *B*.

2 Vectors

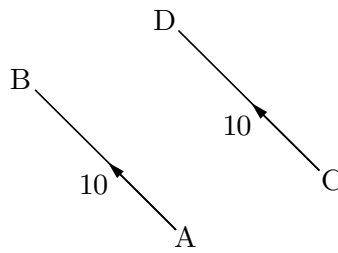


Figure 2.2: Two equal vectors.

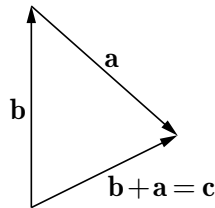


Figure 2.3: Sum of two vectors: the triangle rule. $\vec{a} + \vec{b} = \vec{c}$.

Definition 2.5 Triangle rule for vector addition. The sum of two vectors $\vec{a}$ and $\vec{b}$ is a new vector $\vec{c}$ that is obtained geometrically by adding vectors $\mathbf{a}$ and $\mathbf{b}$ simply **head to tail**, as shown in Figure 2.3.

Example 2.1 During a voyage of the old Bass Strait ferry, the Abel Tasman, the rip at the entrance to Port Phillip Bay caused the navigator some concern. He needed to sail due north, but the current was flowing east at 20 km/h. In still water, the Abel Tasman had a maximum speed of 30 km/h.

Calculate:

1. the actual direction of travel if he had simply pointed the boat towards north,
2. the course necessary to travel in the required direction,
3. the ship's velocity relative to Point Lonsdale.

Solution

(a) Let $\vec{b}$ represents the velocity of the ship relative to the water and $\vec{a}$ is the velocity of the water (see Figure 2.4(a)). The magnitude of the velocity $\vec{b}$, i.e. the speed, is $|\vec{b}| = 30$ km/h and $|\vec{a}| = 20$ km/h.

To find the actual direction of travel we can 'combine' vectors $\vec{a}$ and $\vec{b}$ to get vector $\vec{c}$. This is simply adding vectors $\vec{a}$ and $\vec{b}$ 'head to tail'. The resulting velocity of the ship with respect to the land is $\vec{c} = \vec{a} + \vec{b}$, as shown in Figure 2.4(a).

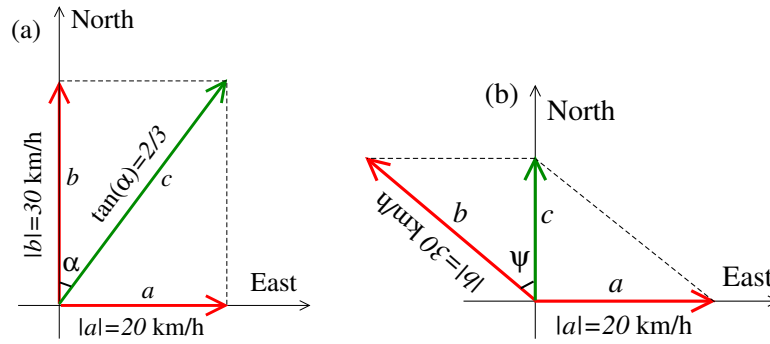


Figure 2.4: (a) Velocity of the ship travelling due north $\vec{b}$ in waters, where the current flows east $\vec{a}$. The sum of two vectors $\vec{c} = \vec{a} + \vec{b}$ is found using the triangle rule. (b) Velocity of the ship $\vec{b}$ with respect to the water compensates for the water current $\vec{a}$ so that the ship travels due north.

To calculate the direction of travel (let it be an angle α measured clockwise from north, as shown in Figure 2.4(a)) we obtain

$$\alpha = \arctan(2/3) = 33.7^\circ$$

(b) Now let us find out in which direction the boat needs to travel so that it is moving due north with respect to the land. Clearly the boat needs to compensate for the water current $\vec{a}$ that flows east. Let $\vec{b}$ be the velocity vector of the boat with respect to the water. Then, we require that the sum $\vec{b} + \vec{a}$ is a vector that points due north, as shown in Figure 2.4(b).

Let Ψ be the angle measured anticlockwise from north giving the direction in which the boat must be pointed, i.e. giving the direction of the vector $\vec{b}$. Then Ψ can be found as

$$\Psi = \arcsin(2/3) = 41.8^\circ$$

(c) The boat's *speed* relative to Point Lonsdale is the magnitude of the vector $\vec{c}$ in Figure 2.4(b)

$$c = \sqrt{30^2 - 20^2} = 22.36 \text{ km/h.}$$

2.1.1 Cartesian coordinates and components of a vector

Definition 2.6 In three-dimensional space, the **Cartesian coordinate system** is formed by three perpendicular lines (**coordinate axis**) that intersect at one point (the **origin**). Usually O denotes the origin and the coordinate axes are denoted by (x, y, z) , as shown in Figure 2.5.

2 Vectors

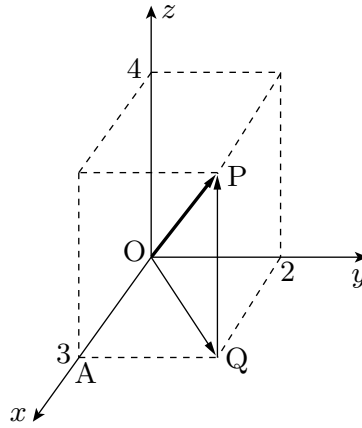


Figure 2.5: Position vector $\overline{OP}$ in a Cartesian coordinate system.

Definition 2.7 When the unit of length is chosen, any vector $\vec{u}$, whose length is equal to one in the chosen units $|\vec{u}| = u = 1$ is called a **unit vector**. For an arbitrary vector $\vec{a}$, the unit vector $\hat{a}$ in the direction of $\vec{a}$ is given by $\hat{a} = \vec{a}/a$. If the length of the vector is zero, such a vector is called **zero vector** and is written as $\vec{0}$.

Note that the unit vector $\vec{u}$ that points in the opposite direction to vector $\vec{a}$ is found as $\vec{u} = -\vec{a}/a$.

Definition 2.8 Let $\vec{i}$, $\vec{j}$ and $\vec{k}$ be unit vectors in the directions of the x -, y - and z -axis of the Cartesian coordinate system, respectively. The triplet of mutually perpendicular unit vectors $(\vec{i}, \vec{j}, \vec{k})$ is called a **basis** of the Cartesian coordinate system.

Theorem 2.1 Let P be any point in space. Then the vector $\overline{OP}$ that connects the origin O with the point P , is called the **position vector** of the point P . Let point A denote a **perpendicular projection** of point P onto the x -axis (see Figure 2.5). Then the vector $\overline{OA}$ can be written as

$$\overline{OA} = x\vec{i},$$

where x is the **signed** distance from the origin to the perpendicular projection of point P onto the x -axis. x is positive if the direction of $\overline{OA}$ coincides with the x -axis and negative otherwise.

Theorem 2.2 Any position vector $\overline{OP}$ can be represented as

$$\overline{OP} = x\vec{i} + y\vec{j} + z\vec{k},$$

2 Vectors

where x , y and z correspond to **signed** distances from the origin to the perpendicular projections of point P onto the x -, y - and z -axis, respectively.

The vectors $x\vec{i}$, $y\vec{j}$ and $z\vec{k}$ are called the **components** of the position vector $\overline{OP}$.

The **length** of the vector $\overline{OP}$ is found from the Pythagoras theorem and is given by

$$OP = |\overline{OP}| = \sqrt{x^2 + y^2 + z^2}.$$

Example 2.2 For example, if a rectangular box of height 4, length 3, and width 2 is drawn as in Figure 2.5 below, it follows that

$$\begin{aligned}\overline{OP} &= \overline{OQ} + \overline{QP} \\ \Rightarrow \overline{OP} &= (\overline{OA} + \overline{AQ}) + \overline{QP} \\ \Rightarrow \overline{OP} &= 3\vec{i} + 2\vec{j} + 4\vec{k},\end{aligned}$$

where $\vec{k}$ is the unit vector in the z -direction.

The length OP is found as

$$OP = \sqrt{3^2 + 2^2 + 4^2} = \sqrt{29}.$$

Direction cosines

The directions of a 3D vector can be specified by the angles (α, β, γ) it makes with the x -, y - and z -axes respectively.

Let $\vec{r} = a\vec{i} + b\vec{j} + c\vec{k}$ (see Figure 2.6). Then $r = OP$

In triangle OCP , $\cos \gamma = c/r$.

Similarly, in triangle OQP $\cos \alpha = a/r$, and in triangle OSP $\cos \beta = b/r$.

The direction cosines of $\vec{r}$ are l , m and n , defined by $l = \cos \alpha$, $m = \cos \beta$ and $n = \cos \gamma$.

Note. $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$.

For any two points $A = (x_A, y_A, z_A)$ and $B = (x_B, y_B, z_B)$, the vector $\overline{AB}$ that starts at point A and terminates at point B is given by

$$\overline{AB} = (x_B - x_A)\vec{i} + (y_B - y_A)\vec{j} + (z_B - z_A)\vec{k}$$

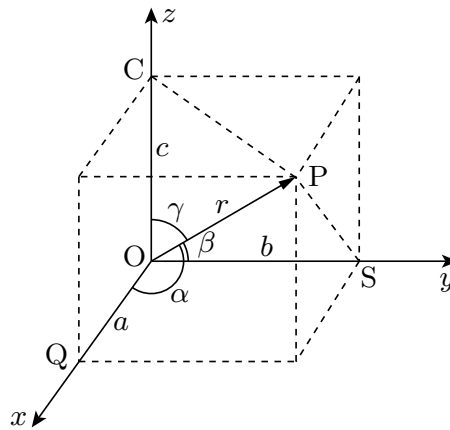


Figure 2.6: Direction cosines $\cos \alpha$, $\cos \beta$ and $\cos \gamma$ of the position vector $\overline{OP}$

Example 2.3 Find the direction cosines of $\overline{AB}$, given that A has coordinates (4,3,2) and B has coordinates (7,8,9) and hence find the angles the vector makes with the coordinate axes.

Solution.

$$\begin{aligned}
 \overline{OA} + \overline{AB} &= \overline{OB} \\
 \Rightarrow \overline{AB} &= \overline{OB} - \overline{OA} \\
 \Rightarrow \overline{AB} &= 7\vec{i} + 8\vec{j} + 9\vec{k} - (4\vec{i} + 3\vec{j} + 2\vec{k}) \\
 \Rightarrow \overline{AB} &= 3\vec{i} + 5\vec{j} + 7\vec{k} \\
 AB &= \sqrt{3^2 + 5^2 + 7^2} = \sqrt{83} \\
 \Rightarrow l = \cos \alpha &= \frac{3}{\sqrt{83}}, \quad m = \cos \beta = \frac{5}{\sqrt{83}}, \quad n = \cos \gamma = \frac{7}{\sqrt{83}} \\
 \Rightarrow \alpha &= 70^\circ 46', \quad \beta = 56^\circ 43', \quad \gamma = 39^\circ 47'.
 \end{aligned}$$

Example 2.4 Given $\vec{a} = 3\vec{i} + 4\vec{j} + 2\vec{k}$, find a unit vector in the direction of $\vec{a}$.

Solution.

$$\begin{aligned}
 a &= \sqrt{3^2 + 4^2 + 2^2} = \sqrt{29} \\
 \Rightarrow \hat{a} &= \frac{\vec{a}}{a} = \frac{3\vec{i} + 4\vec{j} + 2\vec{k}}{\sqrt{29}} \\
 \Rightarrow \hat{a} &= \frac{3}{\sqrt{29}}\vec{i} + \frac{4}{\sqrt{29}}\vec{j} + \frac{2}{\sqrt{29}}\vec{k}.
 \end{aligned}$$

Exercises

Exercises 2.1

A boat is rowed north with a velocity 4 km/h w.r.t. the water across a river which is flowing east at 3 km/h. Find the magnitude and direction of the resultant velocity of the boat w.r.t. the river bank.

A stream is running at 3 km/h eastwards and its breadth is 150 m. If a woman can row a boat at 5 km/h, find the direction in which she must row in order to go north across the stream, and the time it takes to cross.

During a voyage of the old Bass Strait ferry, the Abel Tasman, the rip at the entrance to Port Phillip Bay caused the navigator some concern. He needed to sail due north, but the current was flowing northwest at 10 km/h. In still water, the Abel Tasman has a maximum speed of 25 km/h.

Find the following:

- (a) the actual direction of travel if he had simply pointed the boat towards north;
- (b) the course necessary to travel in the required direction;
- (c) the ship's velocity relative to Point Lonsdale (a fixed point on land).

Write the position vectors of the following points in the form $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$:

- (a) $(1, -2, -3)$; (b) $(2, 0, -5)$; (c) $(1.5, 1, -5)$.

If $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j} - \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}$, $\mathbf{c} = 2\mathbf{i} + \mathbf{k}$, find

- (a) $\mathbf{a} + \mathbf{b}$, (b) $\mathbf{a} + 3\mathbf{c}$, (c) $3\mathbf{a} - \mathbf{b} - 2\mathbf{c}$, (d) $1.5\mathbf{c} - \mathbf{b} + 0.5\mathbf{a}$.

Find (a) the magnitude and (b) the direction cosines of each of the following vectors:

- (i) $\mathbf{i} + 2\mathbf{j} + \mathbf{k}$; (ii) $3\mathbf{i} + 5\mathbf{k}$; (iii) $2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$; (iv) $3\mathbf{j} - \mathbf{k}$.

For each of the following vectors, find (a) the magnitude and (b) the unit vector parallel to the given vector:

- (i) $3\mathbf{i} + 4\mathbf{j}$; (ii) $-5\mathbf{i} + 12\mathbf{j}$; (iii) $-7\mathbf{i} - 24\mathbf{j}$; (iv) $9\mathbf{i} - 12\mathbf{j}$.

Find

- (a) an $\mathbf{i}, \mathbf{j}, \mathbf{k}$ representation for the position vector $\overline{OP}$;
- (b) the magnitude of $\overline{OP}$;
- (c) a unit vector in the direction of $\overline{OP}$;
- (d) the angle which $\overline{OP}$ makes with the x -axis;

for each of the following points P with coordinates:

- (i) $(-1, 4)$, (ii) $(6, 8)$, (iii) $(2, 2, 1)$, (iv) $(-3, 4, 5)$.

The position vectors of points P and Q are $2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$ and $3\mathbf{i} - 7\mathbf{j} + 12\mathbf{k}$ respectively. Find the length of $\overline{PQ}$. What are the direction cosines of $\overline{PQ}$?

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Given the vector $3\vec{i} + 2\vec{j} - 6\vec{k}$, find

- (a) its direction cosines; and
(b) the angles it makes with the coordinate axes.

Answers

1. 5 km/h, $36^\circ 52'$ E of N
2. $36^\circ 52'$ W of N, 2 min 15 s
- 3.(a) $12^\circ 26'$ W of N (b) $16^\circ 26'$ E of N
(c) 31.05 km/h due north
- 4.(a) $\vec{i} - 2\vec{j} - 3\vec{k}$
(b) $2\vec{i} - 5\vec{k}$
(c) $\frac{3}{2}\vec{i} + \vec{j} - \frac{1}{2}\vec{k}$
- 5.(a) $6\vec{i} - \vec{j} + 2\vec{k}$
(b) $10\vec{i} + 3\vec{j} + 2\vec{k}$
(c) $6\vec{i} + 13\vec{j} - 8\vec{k}$
(d) $3\vec{i} + (\frac{11}{2})\vec{j} - 2\vec{k}$
- 6.(i) (a) $\sqrt{6}$ (b) $1/\sqrt{6}, 2/\sqrt{6}, 1/\sqrt{6}$
(ii) (a) $\sqrt{34}$ (b) $3/\sqrt{34}, 0, 5/\sqrt{34}$
(iii) (a) $\sqrt{14}$ (b) $2/\sqrt{14}, -3/\sqrt{14}, 1/\sqrt{14}$
(iv) (a) $\sqrt{10}$ (b) $0, 3/\sqrt{10}, -1/\sqrt{10}$
- 7.(i) (a) 5 (b) $(3\vec{i} + 4\vec{j})/5$
(ii) (a) 13 (b) $(-5\vec{i} + 12\vec{j})/13$
(iii) (a) 25 (b) $(-7\vec{i} - 24\vec{j})/25$
(iv) (a) 15 (b) $(3\vec{i} - 4\vec{j})/5$
- 8.(i) (a) $-\vec{i} + 4\vec{j}$ (b) $\sqrt{17}$ (c) $(-\vec{i} + 4\vec{j})/\sqrt{17}$ (d) 1.816
(ii) (a) $6\vec{i} + 8\vec{j}$ (b) 10 (c) $0.2(3\vec{i} + 4\vec{j})$ (d) 0.927
(iii) (a) $2\vec{i} + 2\vec{j} + \vec{k}$ (b) 3 (c) $(2\vec{i} + 2\vec{j} + \vec{k})/3$ (d) 0.841
(iv) (a) $-3\vec{i} + 4\vec{j} + 5\vec{k}$ (b) $5\sqrt{2}$ (c) $(-3\vec{i} + 4\vec{j} + 5\vec{k})/(5\sqrt{2})$ (d) 2.009
9. 9 $1/9, -4/9, 8/9$
- 10.(a) $(3/7, 2/7, -6/7)$ (b) $\alpha = 64^\circ 37', \beta = 73^\circ 24', \gamma = 149^\circ 00'$

2.2 Vector spaces, linear independence and basis

Up to now, we have talked loosely about vectors in 2 and 3 dimensions. When introducing a new entity (such as vectors) in mathematics, it is important to set up a rigorous formal structure in which to embed that entity. For the recently introduced concept of vectors, we introduce the idea of a *vector space*. In engineering mathematics and most areas of physics and economics, we use the notion of a vector space defined over the real (or possibly complex) (see below) numbers. Quantum mechanics in particular, requires use of vector spaces over the complex numbers. (We will stick to the reals here however.)

We will not go very deeply into the formal description, but rather, we give some brief sketches of how things work. For a deeper treatment the reader is referred to any good book on linear algebra.

Definition 2.9 *A vector space over the real numbers is a set V containing elements known as vectors, $\vec{u}$, $\vec{v}$, $\vec{w}$ etc., along with rules for combining those vectors with each other using also elements from the real number set. The space is closed under application of operations of addition and multiplication by a scalar. That means*

For any $\vec{v}$, $\vec{u} \in V$ and any scalar k

$$(\vec{v} + \vec{u}) \in V \quad \text{and} \quad k\vec{v} \in V$$

For V to be a vector space, the means by which these rules act must satisfy a number of restrictions. For all $\vec{u}, \vec{v}, \vec{w} \in V$, all $k, l \in \mathbb{R}$ we require the following eight rules.

Four are connected with the addition:

1. Associativity: $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$.
2. Existence of null vector: there exists $\vec{0} \in V$ such that $\vec{u} + \vec{0} = \vec{0} + \vec{u} = \vec{u}$.
3. Existence of inverse for addition: for every $\vec{u} \in V$, there exists another vector $-\vec{u} \in V$ such that $\vec{u} + (-\vec{u}) = -\vec{u} + \vec{u} = \vec{0}$.
4. Commutativity: $\vec{u} + \vec{v} = \vec{v} + \vec{u}$.

Four more are connected with the scalar multiplication:

1. Distributivity: for $\vec{u}, \vec{v} \in V$ and $k \in \mathbb{R}$, $k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}$.
2. Distributivity: $(k + l)\vec{u} = k\vec{u} + l\vec{u}$.
3. Associativity: $(kl)\vec{u} = k(l\vec{u})$.

2 Vectors

4. Existence of scalar multiplicative unit ($1 \in \mathbb{R}$), such that $1\vec{u} = \vec{u}1 = \vec{u}$.

Example 2.5 Show that a set of solutions X to a homogeneous system of linear equations

$$AX = 0$$

is a vector space.

Solution

We need to show that the set is closed under the operations of addition and multiplication by a scalar. Let X_1 and X_2 be any two solutions of $AX = 0$, then $X_1 + X_2$ and λX_1 , where λ is a scalar, also satisfy the same homogeneous equation

$$A(X_1 + X_2) = AX_1 + AX_2 = 0 + 0 = 0, \text{ and } A(\lambda X_1) = \lambda AX_1 = \lambda \times 0 = 0.$$

However, the set of solutions X of a non-homogeneous system

$$AX = B$$

is **not** a vector space, because

$$A(X_1 + X_2) = AX_1 + AX_2 = B + B = 2B \neq B.$$

Example 2.6 The set of polynomials $P_n(x) = \{a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n\}$ of degree n is a vector space.

Solution

Indeed, adding two such polynomials $P_n(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots + a_nx^n$ and $Q_n(x) = b_0 + b_1x + b_2x^2 + b_3x^3 + \cdots + b_nx^n$ we have

$$P_n(x) + Q_n(x) = (a_0 + b_0) + (a_1 + b_1)x + (a_2 + b_2)x^2 + (a_3 + b_3)x^3 + \cdots + (a_n + b_n)x^n$$

and multiplying by a scalar λ

$$\lambda P_n(x) = \lambda a_0 + (\lambda a_1)x + (\lambda a_2)x^2 + (\lambda a_3)x^3 + \cdots + (\lambda a_n)x^n.$$

Example 2.7 Determine for which values of scalar a , the functions $f(x)$ that satisfy the equation

$$f''(x) + f(x) = a$$

form a vector space.

Solution

Let $f(x)$ and $g(x)$ be such that $f''(x) + f = a$ and $g''(x) + g = a$. Then adding the two equations, we see that the sum $f + g$ satisfies the equation

$$(f + g)'' + (f + g) = a + a = 2a$$

Thus, the sum $f + g$ satisfies the same equation if $a = 0$.

We now check that if $a = 0$ then $\lambda f(x)$ satisfies the equation

$$(\lambda f)'' + (\lambda f) = 0.$$

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2.2.1 Linear combination of vectors

Definition 2.10 A set of m vectors $\vec{u}_i, i = 1 \dots m$ is a sum

$$k_1\vec{u}_1 + k_2\vec{u}_2 + k_3\vec{u}_3 + \dots + k_m\vec{u}_m,$$

with some scalar coefficients k_i .

Example 2.8 Determine if the vector

$$\vec{v} = \mathbf{i} - \mathbf{k}$$

can be represented as a linear combination of the following three vectors

$$\vec{u}_1 = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}, \quad \vec{u}_2 = -\mathbf{i} + \mathbf{j} + \mathbf{k}, \quad \vec{u}_3 = \mathbf{j} + \mathbf{k}.$$

Solution We need to show if it is possible to find three scalars a, b and c such that

$$\vec{v} = a\vec{u}_1 + b\vec{u}_2 + c\vec{u}_3.$$

Any two vectors are equal only if their respective coordinates are identical. Thus we arrive at the following set of three equations for three unknowns

$$\begin{aligned} 1 &= a - b \\ 0 &= 2a + b + c \\ -1 &= 3a + b + c. \end{aligned}$$

The solution follows using Gaussian elimination

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 2 & 1 & 1 & 0 \\ 3 & 1 & 1 & -1 \end{array} \right] \begin{array}{l} \\ R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \Rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 3 & 1 & -2 \\ 0 & 4 & 1 & -4 \end{array} \right]$$

Finally

$$\left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 3 & 1 & -2 \\ 0 & 4 & 1 & -4 \end{array} \right] \begin{array}{l} \\ \\ R_3 \rightarrow 3R_3/4 - R_2 \end{array} \Rightarrow \left[\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & -1/4 & -1 \end{array} \right].$$

The system has a unique solution

$$a = -1, \quad b = -2, \quad c = 4.$$

Consequently, the vector $\vec{v}$ can be represented as a linear combination of $\vec{u}_1, \vec{u}_2$ and $\vec{u}_3$.

Note that in what follows we will be representing a vector

$$\vec{v} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$$

as a 3×1 matrix

$$\vec{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}.$$

Definition 2.11 Vectors $\vec{u}_i, i = 1, \dots, m$ are called *linearly dependent*, if one of these vectors, e.g. $\vec{u}_k$ can be represented as a linear combination of the remaining vectors $\vec{u}_i, i = 1, \dots, m, (i \neq k)$. Otherwise, the vectors are called *linearly independent*.

Technically, we say that the vectors $\vec{u}_i, i = 1, \dots, m$ are linearly dependent if the system of linear equations

$$k_1\vec{u}_1 + k_2\vec{u}_2 + k_3\vec{u}_3 + \dots + k_m\vec{u}_m = \vec{0}$$

has a non-trivial solution $k_i, i = 1, \dots, m$.

Example 2.9 Determine if the vectors

$$\vec{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{u}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \vec{u}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix},$$

are linearly dependent or independent.

Solution:

We need to solve the system of linear equations

$$x\vec{u}_1 + y\vec{u}_2 + z\vec{u}_3 = \vec{0}$$

with respect to the scalar coefficients (x, y, z) . If $x = y = z = 0$ is the **only** solution then we conclude that $\vec{u}_i, i = 1, 2, 3$ are linearly independent.

To find x, y, z we set up the system of three equations

$$x \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + y \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

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and solve it, using the method of Gaussian elimination:

$$\begin{aligned}
 \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 3 & 1 & -1 & 0 \end{array} \right] & \xRightarrow{R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 4 & -1 & 0 \end{array} \right] \xRightarrow{R_3 \rightarrow R_3 - 2R_2} \\
 \left[\begin{array}{ccc|c} 1 & -1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -3 & 0 \end{array} \right] & \Rightarrow \left[\begin{array}{l} x = 0 \\ y = 0 \\ z = 0 \end{array} \right]
 \end{aligned}$$

Consequently, the vectors $\vec{u}_i$ are linearly independent.

Example 2.10 In 2 dimensions let $\vec{u} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$.

Solution. Clearly $5\vec{u} - \vec{v} = \vec{0}$ and so $\vec{u}$ and $\vec{v}$ are linearly dependent.

In Section 1.9 we have established that a homogeneous system of m linear equations for m unknowns has a non-trivial solution only if the determinant of the matrix of left-hand-side coefficients is zero. This theorem allows us to decide if any set of m vectors with m components each are linearly dependent or independent. More specifically we may use the following *method of determinants* to decide about linear dependence or independence of vectors:

Consider m vectors $\vec{u}_i$, $i = 1, 2, \dots, m$, where each vector is represented as an $m \times 1$ column matrix. Let $A = [\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m]$ be an $m \times m$ matrix, whose columns are formed by the vectors $\vec{u}_i$. The vectors $(\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m)$ are linearly dependent (independent) if $\det(A) = 0$ ($\det(A) \neq 0$).

Example 2.11 Determine if the vectors

$$\vec{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{u}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad \vec{u}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix},$$

are linearly dependent or independent.

Solution

Using the method of determinants

$$\begin{vmatrix} 1 & -1 & 0 \\ 2 & 0 & 1 \\ 3 & 1 & -1 \end{vmatrix} = 1 \times (-1 \times 0 - 1 \times 1) - (-1) \times (2 \times (-1) - 1 \times 3) = -1 - 5 = -6 \neq 0.$$

Consequently, the vectors $\vec{u}_i$, ($i = 1, 2, 3$) are linearly independent.

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In the more general situation of m vectors

$$\vec{u}_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ \dots \\ a_{n1} \end{bmatrix}, \quad \vec{u}_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ \dots \\ a_{n2} \end{bmatrix}, \dots, \vec{u}_m = \begin{bmatrix} a_{1m} \\ a_{2m} \\ \dots \\ a_{nm} \end{bmatrix}$$

where each vector is represented as an $n \times 1$ column matrix with $m < n$, the decision about linear dependence or independence of the vectors can be made based on the rank of the $n \times m$ matrix $A = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m]$.

The rank of the matrix $A = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m]$ gives the number of linearly independent vectors amongst $(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m)$. Therefore,

- $\text{Rank}(A) < m$, the vectors $(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m)$ are linearly dependent.
- $\text{Rank}(A) = m$, the vectors $(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m)$ are linearly independent.

Note that for $m < n$, $\max(\text{Rank}(A)) \leq \min(m, n) = m$.

From the above result it follows that any m vectors with n components are necessarily linearly dependent if $m > n$. Indeed, for $m > n$, the maximal possible value of the rank of matrix A is

$$\max(\text{Rank}(A)) = \min(m, n) = n < m.$$

Then, according to the Rank-Nullity theorem 1.4, the minimal possible value of the Nullity of A is

$$\text{Nullity}(A) = m - \text{Rank}(A) = m - n > 0.$$

This implies that the set of solutions of the system of n equations

$$AX = 0$$

has $m - n > 0$ unknown parameters.

This proves that the vectors $\vec{u}_i$ are linearly dependent.

Example 2.12 Decide if the vectors $\vec{u} = 2\vec{i} + \vec{j} - \vec{k}$, $\vec{v} = 6\vec{i} + 2\vec{j}$ and $\vec{w} = \vec{i} + \vec{k}$. can be used as a basis in three dimensional space.

Solution.

Three vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ can be used as a basis in three dimensional space only if they are linearly independent. To check this, we use the method of determinants and construct a 3×3 matrix with columns given by the coordinates of the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$

$$\begin{vmatrix} 2 & 6 & 1 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 2 \times 2 - 6 \times 1 + 1 \times 2 = 0.$$

Consequently, the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ are linearly dependent and thus, cannot be used as a basis in three dimensional space.

Definition 2.12 The set V of all vectors $\vec{v}$ of the form

$$\vec{v} = k_1\vec{u}_1 + k_2\vec{u}_2 + k_3\vec{u}_3 + \dots + k_m\vec{u}_m,$$

is called the by the set of m vectors $\vec{u}_i$, $i = 1 \dots m$.

Note: can you prove that V is a vector space?

Formally, we denote V as

$$V = \text{Span}(\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m).$$

Example 2.13 Given vector

$$\vec{v} = -\vec{i} + \vec{j} + 3\vec{k},$$

determine if $\vec{v} \in \text{Span}(\vec{u}, \vec{w})$, where

$$\vec{u} = 5\vec{i} + \vec{j} + \vec{k}, \quad \vec{w} = \vec{i} - 2\vec{j} - \vec{k}.$$

Solution

For $\vec{v}$ to be in $\text{Span}(\vec{u}, \vec{w})$, we require

$$\vec{v} = a\vec{u} + b\vec{w},$$

where a and b are some scalars. Thus, we have

$$-\vec{i} + \vec{j} + 3\vec{k} = a[5\vec{i} + \vec{j} + \vec{k}] + b[\vec{i} - 2\vec{j} - \vec{k}]$$

Comparing the respective coordinates in the left- and the right-hand sides, we obtain

$$\begin{aligned} -1 &= 5a + b \\ 1 &= a - 2b \\ 3 &= a - b. \end{aligned}$$

The augmented matrix is

$$\left[\begin{array}{cc|c} 5 & 1 & -1 \\ 1 & -2 & 1 \\ 1 & -1 & 3 \end{array} \right] \xRightarrow{\substack{R_2 \rightarrow 5R_2 - R_1 \\ R_3 \rightarrow 5R_3 - R_1}} \left[\begin{array}{cc|c} 5 & 1 & -1 \\ 0 & -11 & 6 \\ 0 & -6 & 16 \end{array} \right]$$

Finally

$$\left[\begin{array}{cc|c} 5 & 1 & -1 \\ 0 & -11 & 6 \\ 0 & -6 & 16 \end{array} \right] \xRightarrow{R_3 \rightarrow 11R_3/6 - R_2} \left[\begin{array}{cc|c} 5 & 1 & -1 \\ 0 & -11 & 6 \\ 0 & 0 & 16 \times 11/6 - 6 \neq 0 \end{array} \right]$$

The system is inconsistent. Consequently, the vector $\vec{v}$ is not in the vector space spanned by $\vec{u}$ and $\vec{w}$, i.e.

$$\vec{v} \notin \text{Span}(\vec{u}, \vec{w}).$$

2 Vectors

Definition 2.13 We define the of a vector space to be the smallest number of linearly independent vectors needed to span the entire space.

Definition 2.14 If V is an n -dimensional vector space, then a for V is a set of n linearly independent vectors in V which span V .

Example 2.14 Decide if the vectors $\vec{u} = 2\vec{i} + \vec{j} - \vec{k}$, $\vec{v} = 6\vec{i} + 2\vec{j}$ and $\vec{w} = \vec{i} + \vec{k}$. can be used as a basis in three dimensional space.

Solution.

Three vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ can be used as a basis in three dimensional space only if they are linearly independent. To check this, we use the method of determinants and construct a 3×3 matrix with columns given by the coordinates of the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$

$$\begin{vmatrix} 2 & 6 & 1 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 2 \times 2 - 6 \times 1 + 1 \times 2 = 0.$$

Consequently, the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ are linearly dependent and thus, cannot be used as a basis in three dimensional space.

Exercises

1. Determine which sets of vectors of the following form a linearly independent set.

a) $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$,

b) $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$,

c) $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$,

d) $\vec{u} = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} -4 \\ 10 \\ 2 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} 8 \\ 7 \\ 8 \end{bmatrix}$,

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$$\text{e) } \vec{u} = \begin{bmatrix} -3 \\ 0 \\ 4 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 5 \\ -1 \\ 2 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix},$$

$$\text{f) } \vec{u} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 6 \\ -1 \\ 1 \end{bmatrix}, \quad \vec{t} = \begin{bmatrix} 7 \\ 0 \\ -2 \end{bmatrix},$$

$$\text{g) } \vec{u} = \begin{bmatrix} 3 \\ 8 \\ 7 \\ -3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 1 \\ 5 \\ 3 \\ -1 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 2 \\ -1 \\ 2 \\ 6 \end{bmatrix}, \quad \vec{t} = \begin{bmatrix} 1 \\ 4 \\ 0 \\ 3 \end{bmatrix},$$

$$\text{h) } \vec{u} = \begin{bmatrix} 0 \\ 0 \\ 2 \\ 2 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 3 \\ 3 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{w} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ -1 \end{bmatrix}.$$

2. Given $\vec{u} = \vec{i} + 2\vec{k}$, $\vec{v} = \vec{i} - \vec{j} + 3\vec{k}$, $\vec{w} = \vec{i} + 4\vec{j}$ and $\vec{x} = \vec{i} + \vec{j} - \vec{k}$, show that $\vec{u}$, $\vec{v}$ and $\vec{w}$ could be used as a basis for 3-dimensional space and express $\vec{x}$ in terms of $\vec{u}$, $\vec{v}$ and $\vec{w}$.

3. Determine the dimension n of the vector space V given by

$$V = \text{Span} \left(\begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} -4 \\ -7 \\ 6 \end{bmatrix} \right).$$

Answers

1. The vectors are linearly independent:

a) yes

b) no $\vec{w} = \vec{v} + 2\vec{u}$

c) no $\vec{w} - \vec{v} - \vec{u} = 0$

d) no $-3\vec{u} - \vec{v} + \vec{w} = 0$

e) yes

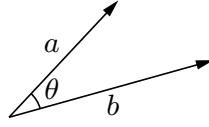
f) no

g) yes

h) yes

2. Hint: one needs to show that $\vec{u}$, $\vec{v}$ and $\vec{w}$ are linearly independent.

3. $n = 2$.

Figure 2.7: Angle θ between two vectors $\vec{a}$ and $\vec{b}$.

2.3 Scalar (dot) product of two vectors

Definition 2.15 If $\vec{a}$ and $\vec{b}$ are two vectors, with θ the angle between them (see Figure 2.7), then the scalar (dot) product of $\vec{a}$ and $\vec{b}$ is written as $\vec{a} \cdot \vec{b}$ and is defined by

$$\vec{a} \cdot \vec{b} = ab \cos \theta. \quad (2.25)$$

If $\vec{a}$ and $\vec{b}$ are at right angles to each other, then

$$\vec{a} \cdot \vec{b} = ab \cos 90^\circ = ab(0) = 0.$$

Hence,

$$\vec{i} \cdot \vec{j} = 0, \quad \vec{j} \cdot \vec{i} = 0, \quad \vec{i} \cdot \vec{k} = 0, \quad \vec{k} \cdot \vec{i} = 0, \quad \vec{j} \cdot \vec{k} = 0 \quad \text{and} \quad \vec{k} \cdot \vec{j} = 0. \quad (2.26)$$

If $\vec{a}$ and $\vec{b}$ are unit vectors and $\vec{a} = \vec{b}$, then

$$\vec{a} \cdot \vec{b} = (1)(1) \cos 0^\circ = (1)(1)(1) = 1$$

and, hence,

$$\vec{i} \cdot \vec{i} = 1, \quad \vec{j} \cdot \vec{j} = 1, \quad \text{and} \quad \vec{k} \cdot \vec{k} = 1. \quad (2.27)$$

If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$, then

$$\begin{aligned} \vec{a} \cdot \vec{b} &= (a_1\vec{i} + a_2\vec{j} + a_3\vec{k}) \cdot (b_1\vec{i} + b_2\vec{j} + b_3\vec{k}) \\ \Rightarrow \vec{a} \cdot \vec{b} &= a_1b_1\vec{i} \cdot \vec{i} + a_1b_2\vec{i} \cdot \vec{j} + a_1b_3\vec{i} \cdot \vec{k} + a_2b_1\vec{j} \cdot \vec{i} + a_2b_2\vec{j} \cdot \vec{j} + a_2b_3\vec{j} \cdot \vec{k} \\ &\quad + a_3b_1\vec{k} \cdot \vec{i} + a_3b_2\vec{k} \cdot \vec{j} + a_3b_3\vec{k} \cdot \vec{k} \end{aligned}$$

Using the results in (2.26) and (2.27), this simplifies to

$$\vec{a} \cdot \vec{b} = a_1b_1 + 0 + 0 + 0 + a_2b_2 + 0 + 0 + 0 + a_3b_3.$$

Hence,

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3. \quad (2.28)$$

The dot product of $\vec{a}$ and $\vec{b}$ is useful both in the form given by (2.25) and in the form given by (2.28).

Example 2.15 Given that $\vec{a} = 2\vec{i} + 3\vec{j} + 5\vec{k}$ and $\vec{b} = 4\vec{i} + 1\vec{j} + 6\vec{k}$, find

- (a) $\vec{a} \cdot \vec{b}$ (b) $a = |\vec{a}|$ (c) $b = |\vec{b}|$
 (d) $\cos \theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$
 (e) θ

Solution.

(a) Using (2.28), $\vec{a} \cdot \vec{b} = (2)(4) + (3)(1) + (5)(6) = 8 + 3 + 30 = 41$

(b) $a = |\vec{a}| = \sqrt{2^2 + 3^2 + 5^2} = \sqrt{38}$ (c) $b = |\vec{b}| = \sqrt{4^2 + 1^2 + 6^2} = \sqrt{53}$

(d) (2.25) $\Rightarrow \cos \theta = \frac{\vec{a} \cdot \vec{b}}{ab} = \frac{41}{\sqrt{38}\sqrt{53}}$ (e) $\Rightarrow \theta = \cos^{-1} \left(\frac{41}{\sqrt{38}\sqrt{53}} \right) = 24^\circ 00'$

2.3.1 Dot product in higher dimensions

In dimensions higher than three, the dot product between any two vectors can be defined similar to Eq. (2.28). Let V be an n -dimensional vector space. Any element $\vec{v}$ from V can be represented as a $n \times 1$ matrix

$$\vec{v} \in V : \Rightarrow \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix}.$$

The dot product between any two vectors $\vec{v}$ and $\vec{w}$ from V is given by

$$\vec{v} \cdot \vec{w} = \vec{v}^T \vec{w} = \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \dots \\ w_n \end{bmatrix} = v_1 w_1 + v_2 w_2 + \dots + v_n w_n.$$

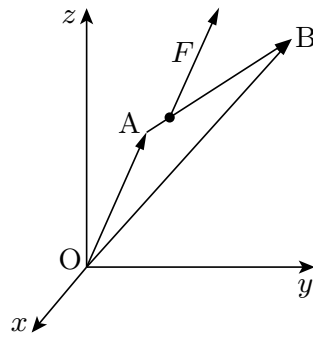


Figure 2.8: Work done by force $\vec{F}$ moving a particle between A and B

2.3.2 Work done by a force

An application of the dot (scalar) product of two vectors is to find the work done by a force acting on a particle which moves between two points.

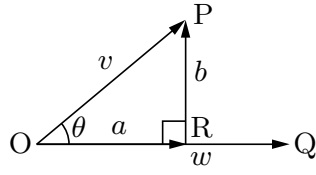
Example 2.16 A force $\vec{F} = 5\vec{i} + 2\vec{j} + 3\vec{k}$ newtons acts on a particle which moves from $A(1, 7, 11)$ to $B(4, 16, 18)$ (see Figure 2.8), with the coordinates in metres. Find W , the work done by the force on the particle. Use $W = \vec{F} \cdot \overline{AB}$.

Solution.

$$\begin{aligned}
 \overline{AB} &= \overline{OB} - \overline{OA} \\
 &= 4\vec{i} + 16\vec{j} + 18\vec{k} - (1\vec{i} + 7\vec{j} + 11\vec{k}) \\
 &= 3\vec{i} + 9\vec{j} + 7\vec{k} \\
 W &= \vec{F} \cdot \overline{AB} = (5\vec{i} + 2\vec{j} + 3\vec{k}) \cdot (3\vec{i} + 9\vec{j} + 7\vec{k}) \\
 &= (5)(3) + (2)(9) + (3)(7) \\
 &= 15 + 18 + 21 \\
 &= 54 \text{ newton metres} \\
 &= 54 \text{ joule}
 \end{aligned}$$

2.3.3 Projection of a vector

Consider the vectors $\vec{v}$ and $\vec{w}$ as in Figure 2.9, with $\vec{v} = \overline{OP}$ and $\vec{w} = \overline{OQ}$. The vectors $\vec{a}$ and $\vec{b}$ are defined by $\vec{a} = \overline{OR}$ and $\vec{b} = \overline{RP}$.

Figure 2.9: Component of $\vec{v}$ in the $\vec{w}$ direction.

- The vector $\vec{a}$ is called the vector component (or vector projection) of $\vec{v}$ in the $\vec{w}$ -direction

$$\vec{a} = \text{proj}_{\vec{w}}(\vec{v}).$$

- The vector $\vec{b}$ is called the vector component of $\vec{v}$ perpendicular to the $\vec{w}$ -direction.

From the right-angled triangle OPR it can be seen that

$$OR = a = v \cos \theta. \quad (2.29)$$

But, using the dot product definition,

$$vw \cos \theta = \vec{v} \cdot \vec{w} \quad (2.30)$$

$$\Rightarrow a = v \cos \theta = \frac{\vec{v} \cdot \vec{w}}{w} = \vec{v} \cdot \left(\frac{\vec{w}}{w} \right). \quad (2.31)$$

As $\vec{a} = a \frac{\vec{w}}{w}$, we finally obtain

$$\vec{a} = \text{proj}_{\vec{w}}(\vec{v}) = (\vec{v} \cdot \vec{w}) \frac{\vec{w}}{w^2}. \quad (2.32)$$

Using this result we can now find the vector component of $\vec{v}$ perpendicular to the $\vec{w}$ -direction

$$\vec{b} = \vec{v} - \vec{a} = \vec{v} - (\vec{v} \cdot \vec{w}) \frac{\vec{w}}{w^2}. \quad (2.33)$$

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Example 2.17 Given that $\vec{v} = 3\vec{i} + 2\vec{j} - \vec{k}$ and $\vec{w} = 4\vec{i} + 4\vec{j}$, find
 (a) the vector component of $\vec{v}$ in the direction of $\vec{w}$, and
 (b) the vector component of $\vec{v}$ perpendicular to the direction of $\vec{w}$.

Solution.

(a) The vector $\vec{a}$ is found using

$$\vec{a} = (\vec{v} \cdot \vec{w}) \frac{\vec{w}}{w^2},$$

where

$$w = \sqrt{4^2 + 4^2} = 4\sqrt{2}.$$

Hence,

$$\vec{a} = (3\vec{i} + 2\vec{j} - \vec{k}) \cdot (4\vec{i} + 4\vec{j}) \frac{(4\vec{i} + 4\vec{j})}{32} = (3 \times 4 + 2 \times 4) \frac{(4\vec{i} + 4\vec{j})}{32} = \frac{5}{2}(\vec{i} + \vec{j}).$$

(b)

$$\begin{aligned} \vec{b} &= \vec{v} - \vec{a} = 3\vec{i} + 2\vec{j} - \vec{k} - \frac{5}{2}(\vec{i} + \vec{j}) \\ \Rightarrow \vec{b} &= \frac{\vec{i}}{2} - \frac{\vec{j}}{2} - \vec{k}. \end{aligned}$$

2.3.4 Orthogonal and orthonormal basis

Definition 2.16 Any n linearly independent vectors $\vec{u}_1, \vec{u}_2, \dots, \vec{u}_n$ form an orthogonal basis in an n -dimensional vector space, if $\vec{u}_i$ are mutually, i.e. if for any $i \neq k$

$$\vec{u}_i \cdot \vec{u}_k = 0.$$

Definition 2.17 If additionally to orthogonality, all basis vectors $\vec{u}_i$ have unit length, i.e. if

$$u_i = |\vec{u}_i| = 1, \quad i = 1, 2, \dots, n$$

then the basis is called .

For example, the unit vectors of the Cartesian coordinate system $(\vec{i}, \vec{j}, \vec{k})$ form an orthonormal basis in three dimensional vector space, because they are linearly independent, have unit length

$$|\vec{i}| = |\vec{j}| = |\vec{k}| = 1$$

and are mutually orthogonal

$$\vec{i} \cdot \vec{j} = 0, \quad \vec{i} \cdot \vec{k} = 0, \quad \vec{j} \cdot \vec{k} = 0.$$

2.3.5 The Gram-Schmidt Process

Definition 2.18 *The Gram-Schmidt process is an algorithm for constructing an orthogonal (orthonormal) basis in n -dimensional vectors space from any set of n linearly independent vectors $\vec{u}_i$, $i = 1, 2, 3, \dots, n$.*

In order to exemplify the algorithm, we consider a two-dimensional vector space V , spanned by

$$V = \text{Span}(\vec{u}_1 = \vec{i} + \vec{j}, \vec{u}_2 = 2\vec{i} + 2\vec{j} + 3\vec{k}).$$

In order to find an orthogonal (orthonormal) basis in V , we may choose $\vec{u}_1$ as the first vector of the basis. Note that we can always normalize $\vec{u}_1$ according to

$$\hat{\vec{u}}_1 = \frac{\vec{u}_1}{u_1} = \frac{\vec{i}}{\sqrt{2}} + \frac{\vec{j}}{\sqrt{2}}.$$

Next, we find the vector component $\vec{w}_\perp$ of $\vec{u}_2$ in the direction, perpendicular to $\vec{u}_1$. This is done using Eq. (2.33)

$$\vec{w}_\perp = \vec{u}_2 - \text{proj}_{\vec{u}_1}(\vec{u}_2) = \vec{u}_2 - \vec{u}_1 \cdot \vec{u}_2 \frac{\vec{u}_1}{u_1^2}.$$

Explicitly we obtain

$$\vec{w}_\perp = 2\vec{i} + 2\vec{j} + 3\vec{k} - (2 \times 1 + 2 \times 1) \frac{(\vec{i} + \vec{j})}{2} = 3\vec{k}.$$

As the result of this construction, we have found the orthogonal basis of V

$$V = \text{Span}(\vec{i} + \vec{j}, 3\vec{k}).$$

The orthonormal basis is obtained by normalizing the basis vectors

$$V = \text{Span}\left(\frac{\vec{i}}{\sqrt{2}} + \frac{\vec{j}}{\sqrt{2}}, \vec{k}\right).$$

Note that in the case when an n -dimension vector space V is spanned by n linearly independent n -dimensional vectors, the Gram-Schmidt process becomes redundant. Indeed, in this case, one can always choose the standard orthonormal basis. For example, if V is an n -dimensional vectors space, then any $\vec{v} \in V$ can be represented as an $n \times 1$ matrix

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \dots \\ v_n \end{bmatrix}.$$

2 Vectors

The standard basis is given by

$$\begin{bmatrix} 1 \\ 0 \\ \dots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \dots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \dots \\ 1 \end{bmatrix}.$$

The vector space spanned by the standard basis is

$$V = \text{Span} \left(\begin{bmatrix} 1 \\ 0 \\ \dots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \dots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \dots \\ 1 \end{bmatrix} \right).$$

Example 2.18 Construct an orthonormal basis for

$$V = \text{Span} \left(\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \vec{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \vec{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 2 \end{bmatrix} \right).$$

Solution We choose

$$\vec{w}_1 = \vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

as the first vector of the new basis. The second vector $\vec{w}_2$ of the new basis is

$$\vec{w}_2 = \vec{v}_2 - (\vec{v}_2 \cdot \vec{v}_1) \frac{\vec{v}_1}{v_1^2} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 2 \\ -1/2 \\ 0 \end{bmatrix}.$$

In order to find the third vector $\vec{w}_3$ of the new basis, we calculate the projections of $\vec{v}_3$ onto $\vec{w}_1$ and $\vec{w}_2$

$$\text{proj}_{\vec{w}_1}(\vec{v}_3) = (\vec{v}_3 \cdot \vec{w}_1) \frac{\vec{w}_1}{w_1^2} = \frac{2}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

and

$$\text{proj}_{\vec{w}_2}(\vec{v}_3) = (\vec{v}_3 \cdot \vec{w}_2) \frac{\vec{w}_2}{w_2^2} = \frac{2}{9} \begin{bmatrix} 1/2 \\ 2 \\ -1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/9 \\ 4/9 \\ -1/9 \\ 0 \end{bmatrix}.$$

2 Vectors

Consequently, the vector $\vec{v}_3$ can be represented as

$$\vec{v}_3 = \text{proj}_{\vec{w}_2}(\vec{v}_3) + \text{proj}_{\vec{w}_1}(\vec{v}_3) + \vec{w}_3,$$

where $\vec{w}_3$ is orthogonal to both $\vec{w}_1$ and $\vec{w}_2$. Thus we find

$$\vec{w}_3 = \vec{v}_3 - \text{proj}_{\vec{w}_2}(\vec{v}_3) - \text{proj}_{\vec{w}_1}(\vec{v}_3).$$

Or explicitly

$$\vec{w}_3 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/9 \\ 4/9 \\ -1/9 \\ 0 \end{bmatrix} = \begin{bmatrix} 8/9 \\ -4/9 \\ -8/9 \\ 2 \end{bmatrix}.$$

This implies that the new orthogonal basis is given by

$$V = \text{Span} \left(\vec{w}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \vec{w}_2 = \begin{bmatrix} 1/2 \\ 2 \\ -1/2 \\ 0 \end{bmatrix}, \vec{w}_3 = \begin{bmatrix} 8/9 \\ -4/9 \\ -8/9 \\ 2 \end{bmatrix} \right).$$

Exercises

- If $\mathbf{a} = 3\mathbf{i} - \mathbf{j}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j}$ find:
 - $\mathbf{a} \cdot \mathbf{b}$;
 - the angle between $\mathbf{a}$ and $\mathbf{b}$.
- If $\mathbf{u} = 2\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$ and $\mathbf{v} = \mathbf{i} - \mathbf{j} - \mathbf{k}$, find:
 - $\mathbf{u} \cdot \mathbf{v}$;
 - $\mathbf{u} \cdot \mathbf{u}$;
 - $\mathbf{v} \cdot \mathbf{v}$;
 - a unit vector in the direction of $\mathbf{v}$.
- A force $\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ newtons moves a particle from the position $3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ to the position $-\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$, where the displacement is measured in metres. Find the work done.
- In moving a crate of electrical fittings, a force $\mathbf{F}$ of magnitude 84 newton is applied in the direction of $\mathbf{b}$, where $\mathbf{b} = 3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$.
If the crate is moved from point A $(-2, -1, 1)$ to the point B $(1, 1, 2)$, measured in metres, find the work done.
- The points A $(1, 2)$, B $(4, -1)$, C $(5, -3)$ and D $(2, -6)$ are the vertices of a quadrilateral. Show, using vector methods, that $\overline{AB}$ and $\overline{CD}$ are at right angles.
- Find
 - the vector component of $\mathbf{u}$ in the direction of $\mathbf{v}$
 - the vector component of $\mathbf{u}$ perpendicular to $\mathbf{v}$
 for each of the following pairs of vectors:
 - $\mathbf{u} = \mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$, $\mathbf{v} = \mathbf{i} + \mathbf{j}$;
 - $\mathbf{u} = 4\mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $\mathbf{v} = -2\mathbf{i} + 3\mathbf{j} - \mathbf{k}$;
 - $\mathbf{u} = 6\mathbf{i} + 12\mathbf{j}$, $\mathbf{v} = -3\mathbf{i} - 2\mathbf{j}$.

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7. Given that $\vec{v} = 2\vec{i} + 2\vec{j} + \vec{k}$ and $\vec{w} = 3\vec{i} - \vec{j} + \vec{k}$ find
 (i) the vector component of $\vec{v}$ in the direction of $\vec{w}$;
 (ii) the vector component of $\vec{v}$ perpendicular to the the direction of $\vec{w}$.
8. The angle between $\mathbf{x}$ and $\mathbf{y}$ is $\arccos(4/21)$. Find the scalar t given that $\mathbf{x} = 6\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ and $\mathbf{y} = -2\mathbf{i} + t\mathbf{j} - 4\mathbf{k}$.
9. Using the Gram-Schmidt process, construct orthonormal basis for V given by

$$V = \text{Span} \left(\begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix} \right).$$

Answers 1. (a) 1 (b) $1.429^\circ = 81.9^\circ$

2. (a) 1 (b) 29 (c) 3 (d) $\frac{\mathbf{i} - \mathbf{j} - \mathbf{k}}{\sqrt{3}}$

3. 1 joule

4. 132 joule

5. Hint: Calculate the scalar product between $\overline{AB}$ and $\overline{CD}$.

6.

- | | | | |
|--------|--|------|--|
| (a)(i) | $2(\mathbf{i} + \mathbf{j});$ | (ii) | $-\mathbf{i} + \mathbf{j} - 4\mathbf{k}$ |
| (b)(i) | $-\frac{3}{14}(-2\mathbf{i} + 3\mathbf{j} - \mathbf{k})$ | (ii) | $\frac{1}{14}(50\mathbf{i} + 37\mathbf{j} + 11\mathbf{k})$ |
| (c)(i) | $\frac{42}{13}(3\mathbf{i} + 2\mathbf{j})$ | (ii) | $-(\frac{24}{13})(-2\mathbf{i} + 3\mathbf{j})$ |

7. (i) $\frac{5}{11}(3\vec{i} - \vec{j} + \vec{k});$ (ii) $\frac{1}{11}(7\vec{i} + 27\vec{j} + 6\vec{k})$

8. 4

9.

$$V = \text{Span} \left(\frac{1}{\sqrt{29}} \begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}, \frac{1}{\sqrt{11310}} \begin{bmatrix} 95 \\ 46 \\ -13 \end{bmatrix} \right)$$

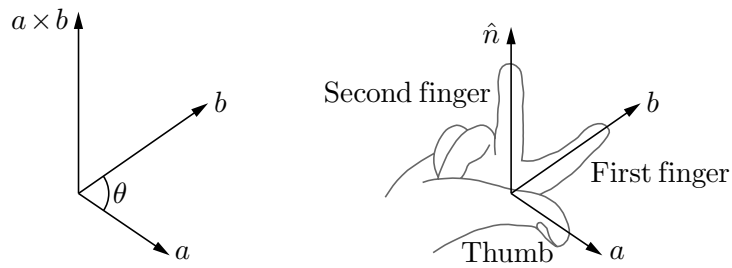


Figure 2.10: Right-hand rule to determine the direction of a cross product $\vec{a} \times \vec{b}$.

2.4 The vector (cross) product

Definition 2.19 If $\vec{a}$ and $\vec{b}$ are two vectors, with θ the angle between them (see Figure 2.7), then the vector (cross) product of $\vec{a}$ and $\vec{b}$ is written as $\vec{a} \times \vec{b}$ and is defined by

$$\vec{a} \times \vec{b} = (ab \sin \theta) \hat{n}, \quad (2.34)$$

where $\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, and a and b are the magnitudes of $\vec{a}$ and $\vec{b}$ respectively. The direction of $\hat{n}$ is given by the right-hand rule; i.e. if we take $\vec{a}$ to be pointing in the direction of the thumb on the right hand, and $\vec{b}$ to be pointing in the direction of the first finger on the right hand, then $\hat{n}$ is pointing in the direction of the second finger on the right hand (see Figure 2.10).

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Geometrically, the magnitude of the cross product between $\vec{a}$ and $\vec{b}$ is equal to the area S of the parallelogram, built on $\vec{a}$ and $\vec{b}$.

$$|\vec{a} \times \vec{b}| = ab |\sin \theta| = S.$$

Next, we define the so-called right-handed coordinate systems

Consider a Cartesian coordinate system, formed by three mutually orthogonal axes (x, y, z) that intersect in the origin O . The system is called if the unit vectors $\vec{i}, \vec{j}$ and $\vec{k}$ that point in the positive directions of the x, y and z axes, respectively, are related via

$$\vec{i} \times \vec{j} = \vec{k}.$$

Using the right-hand rule, it can further be shown that

$$\vec{j} \times \vec{k} = \vec{i}, \quad \vec{k} \times \vec{i} = \vec{j}, \quad \vec{i} \times \vec{j} = \vec{k}, \quad \vec{k} \times \vec{j} = -\vec{i}, \quad \vec{i} \times \vec{k} = -\vec{j},$$

and

$$\vec{i} \times \vec{i} = \vec{0} = \vec{j} \times \vec{j} = \vec{k} \times \vec{k}.$$

If

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}, \text{ and } \vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k},$$

then

$$\begin{aligned} \vec{a} \times \vec{b} &= (a_1\vec{i} + a_2\vec{j} + a_3\vec{k}) \times (b_1\vec{i} + b_2\vec{j} + b_3\vec{k}) \\ &= a_1b_1\vec{i} \times \vec{i} + a_1b_2\vec{i} \times \vec{j} + a_1b_3\vec{i} \times \vec{k} \\ &\quad + a_2b_1\vec{j} \times \vec{i} + a_2b_2\vec{j} \times \vec{j} + a_2b_3\vec{j} \times \vec{k} \\ &\quad + a_3b_1\vec{k} \times \vec{i} + a_3b_2\vec{k} \times \vec{j} + a_3b_3\vec{k} \times \vec{k} \end{aligned}$$

Using the previous results for the cross products of the unit vectors, this simplifies to

$$\begin{aligned} \vec{a} \times \vec{b} &= a_1b_1\vec{0} + a_1b_2\vec{k} + a_1b_3(-\vec{j}) + a_2b_1(-\vec{k}) + a_2b_2\vec{0} + a_2b_3(\vec{i}) \\ &\quad + a_3b_1\vec{j} + a_3b_2(-\vec{i}) + a_3b_3\vec{0} \\ &= (a_2b_3 - a_3b_2)\vec{i} + (a_3b_1 - a_1b_3)\vec{j} + (a_1b_2 - a_2b_1)\vec{k} \end{aligned}$$

This can be rewritten in determinant form as

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$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \quad (2.35)$$

Note that here we used $(a_1b_3 - a_3b_1) = -(a_3b_1 - b_3a_1)$. It is usually more useful to use the cross product in the form given in (2.35) than in the form given in (2.34).

Taking into account the above result, it is now possible to show that the determinant $\det(A)$ of a 2×2 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

is equal to the signed area of the parallelogram, built on two vectors

$$\vec{v} = a_{11}\vec{i} + a_{12}\vec{j}, \quad \vec{w} = a_{21}\vec{i} + a_{22}\vec{j}.$$

Indeed, the cross product $\vec{v} \times \vec{w}$ is given by

$$\vec{v} \times \vec{w} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \end{vmatrix} = \vec{k}(a_{11}a_{22} - a_{12}a_{21}) = \vec{k} \det(A).$$

Recalling that the area S of the parallelogram built on $\vec{v}$ and $\vec{w}$ is the magnitude of $\vec{v} \times \vec{w}$, we conclude

$$S = |\vec{v} \times \vec{w}| = |\det(A)|.$$

The properties of the cross product follow from the properties of determinants

- $\vec{a} \times \vec{a} = 0$
- $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$
- If any three unit vectors $\vec{a}, \vec{b}$ and $\vec{c}$ ($a = b = c = 1$) are mutually orthogonal, and if $\vec{a} = \vec{b} \times \vec{c}$ then

$$\vec{b} = \vec{c} \times \vec{a}, \quad \vec{c} = \vec{a} \times \vec{b}.$$

Example 2.19 Given that $\vec{a} = 2\vec{i} + 3\vec{j} + 4\vec{k}$ and $\vec{b} = \vec{i} + 5\vec{j} + 6\vec{k}$, find

1. $\vec{a} \times \vec{b}$ and
2. a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

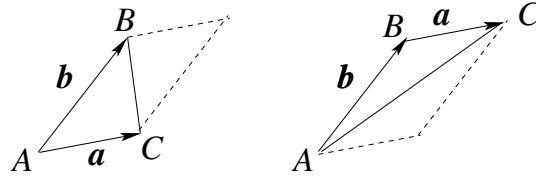


Figure 2.11: Two different triangles that can be built on two given vectors $\vec{a}$ and $\vec{b}$.

Solution.

1.

$$\begin{aligned}\vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 3 & 4 \\ 1 & 5 & 6 \end{vmatrix} \\ &= \vec{i} \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} - \vec{j} \begin{vmatrix} 2 & 4 \\ 1 & 6 \end{vmatrix} + \vec{k} \begin{vmatrix} 2 & 3 \\ 1 & 5 \end{vmatrix} \\ &= \vec{i}(3 \times 6 - 4 \times 5) - \vec{j}(2 \times 6 - 1 \times 4) + \vec{k}(2 \times 5 - 3 \times 1) \\ &= -2\vec{i} - 8\vec{j} + 7\vec{k}\end{aligned}$$

2. As $\vec{a} \times \vec{b} = |\vec{a} \times \vec{b}| \hat{n}$ and $\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, $\hat{n}$ is an answer ($-\hat{n}$ is also an answer, as it is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$), where

$$\hat{n} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{-2\vec{i} - 8\vec{j} + 7\vec{k}}{\sqrt{(-2)^2 + (-8)^2 + (7)^2}} = \frac{-2\vec{i} - 8\vec{j} + 7\vec{k}}{\sqrt{117}}.$$

Example 2.20 Determine the area of a triangle, whose two sides are given by vectors $\vec{a} = \vec{i} - \vec{j} + 3\vec{k}$ and $\vec{b} = 2\vec{i} + \vec{j} - 5\vec{k}$.

Solution

Let A , B and C be the vertices of the triangle. With two given vectors $\vec{a}$ and $\vec{b}$, one can construct two different triangles Δ_1 and Δ_2 as shown in Figure 2.11

$$\Delta_1 : \overline{AB} = \vec{b}, \overline{AC} = \vec{a}$$

and

$$\Delta_2 : \overline{AB} = \vec{b}, \overline{BC} = \vec{a}$$

The areas S_i of these two triangles are identical and equal to half of the area S of the parallelogram, built on $\vec{a}$ and $\vec{b}$. The area S is found as the magnitude of the cross product between $\vec{a}$ and $\vec{b}$

$$S_1 = S_2 = \frac{S}{2} = \frac{1}{2} |\vec{a} \times \vec{b}|,$$

where

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 3 \\ 2 & 1 & -5 \end{vmatrix} = 2\vec{i} + 11\vec{j} + 3\vec{k}.$$

Thus,

$$S_1 = S_2 = \frac{S}{2} = \frac{1}{2} |\vec{a} \times \vec{b}| = \frac{1}{2} \sqrt{2^2 + 11^2 + 3^2} = \frac{1}{2} \sqrt{134}.$$

2.4.1 Scalar triple product

The scalar triple product of three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is defined as the scalar product of one vector with the cross product of the other two vectors

$$\vec{a} \cdot (\vec{b} \times \vec{c})$$

Using Eq. (2.35) the scalar triple product can be found as

$$\vec{a} \cdot \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

From the geometric point of view, the scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c})$ gives the signed volume of the parallelepiped, built on (or spanned by) the three vectors $\vec{a}, \vec{b}, \vec{c}$, as shown in Figure 2.12.

Indeed, the area S of the parallelogram, built on $\vec{b}$ and $\vec{c}$ is given by the absolute value of the cross product

$$S = |\vec{b} \times \vec{c}|.$$

The volume V of the parallelepiped can be represented as the area of the base S times the height h . Clearly, the height h is the projection of $\vec{a}$ onto the direction, orthogonal to the parallelogram S , i.e $h = a \cos(\theta)$. Recalling that $\vec{b} \times \vec{c}$ is orthogonal to S , we conclude that

$$V = Sh = Sa \cos(\theta) = |\vec{a} \cdot (\vec{b} \times \vec{c})|$$

.

Using the properties of the determinants, one can show that

$$\begin{aligned} \vec{a} \cdot (\vec{b} \times \vec{c}) &= -\vec{b} \cdot (\vec{a} \times \vec{c}) \\ \vec{a} \cdot (\vec{b} \times \vec{c}) &= -\vec{a} \cdot (\vec{c} \times \vec{b}) \\ \vec{a} \cdot (\vec{b} \times \vec{c}) &= \vec{b} \cdot (\vec{c} \times \vec{a}). \end{aligned}$$

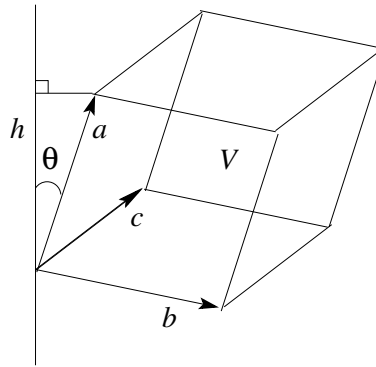


Figure 2.12: Parallelepiped with volume $V = |\vec{a} \cdot (\vec{b} \times \vec{c})|$ built on the vectors $\vec{a}, \vec{b}$ and $\vec{c}$.

Example 2.21 Find the volume of the parallelepiped built on $\vec{a} = \vec{i} + \vec{j}$, $\vec{b} = \vec{j} + \vec{k}$ and $\vec{c} = -\vec{i} + 2\vec{k}$

Solution

The volume V is given by

$$V = |\vec{a} \cdot (\vec{b} \times \vec{c})|.$$

Thus we have

$$V = \text{modulus} \left(\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ -1 & 0 & 2 \end{vmatrix} \right) = |2 - 1| = 1.$$

Using the scalar triple product one can easily verify if a given triplet of vectors is right-handed, left-handed or co-planar.

Triplet of vectors $\vec{a}, \vec{b}, \vec{c}$ (in this particular order) is

- right-handed if $\vec{a} \cdot (\vec{b} \times \vec{c}) > 0$
- left-handed if $\vec{a} \cdot (\vec{b} \times \vec{c}) < 0$
- vectors are co-planar, if $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$.

2.4.2 Vector triple product

The vector triple product of three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is defined as the cross product of one vector with the cross product of the other two vectors

$$\vec{a} \times (\vec{b} \times \vec{c})$$

The following reduction rule, also known as Lagrange formula, holds for any three vectors $\vec{a}, \vec{b}$ and $\vec{c}$

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}). \quad (2.36)$$

The proof follows by comparing the respective coordinates of the left- and the right-hand sides. For example for the x -component we have

$$(\vec{a} \times (\vec{b} \times \vec{c}))_x = a_y(b_x c_y - b_y c_x) - a_z(b_z c_x - b_x c_z)$$

and

$$\begin{aligned} (\vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}))_x &= b_x(a_x c_x + a_y c_y + a_z c_z) - c_x(a_x b_x + a_y b_y + a_z b_z) \\ &= b_x(a_y c_y + a_z c_z) - c_x(a_y b_y + a_z b_z) \\ &= a_y(b_x c_y - b_y c_x) - a_z(b_z c_x - b_x c_z). \end{aligned}$$

Similarly, one can show that y and z coordinates of the vectors in the left- and the right-hand sides of Eq. (2.36) are identical.

Note that the position of brackets in the vector triple product is not arbitrary

$$\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$$

Indeed, using the properties of the vector product and the Lagrange reduction formula we obtain

$$\begin{aligned} (\vec{a} \times \vec{b}) \times \vec{c} &= -\vec{c} \times (\vec{a} \times \vec{b}) \\ &= -(\vec{a}(\vec{c} \cdot \vec{b}) - \vec{b}(\vec{a} \cdot \vec{c})) \\ &= \vec{b}(\vec{a} \cdot \vec{c}) - \vec{a}(\vec{c} \cdot \vec{b}) \\ &\neq \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}). \end{aligned}$$

Example 2.22 Show that for any two orthogonal vectors $\vec{a}$ and $\vec{b}$

$$\vec{a} \times (\vec{a} \times \vec{b}) = -a^2 \vec{b}.$$

Solution Using Lagrange formula, we obtain

$$\vec{a} \times (\vec{a} \times \vec{b}) = \vec{a}(\vec{a} \cdot \vec{b}) - \vec{b}(\vec{a} \cdot \vec{a}).$$

Because $\vec{a} \cdot \vec{b} = 0$, we conclude

$$\vec{a} \times (\vec{a} \times \vec{b}) = -\vec{b} |\vec{a}|^2 = -a^2 \vec{b}.$$

Exercises

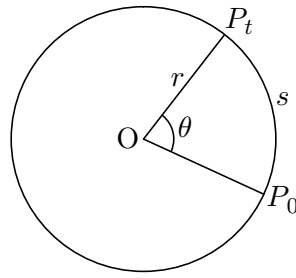
1. Find the vector product $\vec{a} \times \vec{b}$ where
 - a) $\vec{a} = 2\vec{i} - \vec{j}$ $\vec{b} = 3\vec{i} + 4\vec{j}$
 - b) $\vec{a} = 3\vec{i} - \vec{j} - \vec{k}$ $\vec{b} = \vec{i} + \vec{j} - \vec{k}$
2. Noting that the area of a triangle is $0.5(ab \sin C)$, which is $0.5|\overline{CA} \times \overline{CB}|$, find the area of a triangle whose vertices have position vectors:
 - a) $\vec{0}, \vec{i} + \vec{j} + \vec{k}, 2\vec{i} - \vec{j} + 3\vec{k}$
 - b) $\vec{i} + \vec{j} + \vec{k}, 2\vec{i} - \vec{j} - 3\vec{k}, 3\vec{i} + 2\vec{j} + 4\vec{k}$
3. Simplify:
 - a) $\vec{a} \times (2\vec{a} + \vec{b})$
 - b) $(\vec{a} + \vec{b}) \times (2\vec{a} - \vec{b})$
 - c) $\vec{a} \cdot (\vec{a} \times \vec{b})$
4. If $\vec{a} = -2\vec{i} + 4\vec{j} + 3\vec{k}$, $\vec{b} = 3\vec{i} - 6\vec{j} - 4.5\vec{k}$, $\vec{c} = 1.5\vec{i} + 9\vec{j} + 3\vec{k}$, and $\vec{d} = \vec{i} + 6\vec{j} + 2\vec{k}$, show that $(\vec{a} \times \vec{c}) + (\vec{b} \times \vec{d}) = \vec{0}$
5. Find a unit vector perpendicular to the plane containing the vectors $2\vec{i} + 4\vec{j} - 3\vec{k}$ and $\vec{i} - \vec{j} - \vec{k}$
6. If $\vec{a} = 3\vec{i} + 6\vec{j} + 2\vec{k}$ and $\vec{b} = -6\vec{i} + 2\vec{j} + 3\vec{k}$, find
 - a) a unit vector parallel to $2\vec{a} + \vec{b}$, and
 - b) a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$
7. P, Q and R are points in space with coordinates $(6, 3, -4)$, $(7, 5, -5)$ and $(8, 4, -3)$, respectively. Find the area of the triangle PQR.
8. a) Given that $\vec{a} = 2\vec{i} + 4\vec{j} + 4\vec{k}$, $\vec{b} = 5\vec{i} + 2\vec{j} + 6\vec{k}$, and $\vec{c} = 7\vec{i} + 2\vec{j}$, find
 - i. $\vec{a} \cdot \vec{b}$
 - ii. a
 - iii. b
 - iv. θ (the angle between $\vec{a}$ and $\vec{b}$)
 - v. $\vec{a} \times \vec{b}$
 - vi. $(\vec{a} \times \vec{b}) \cdot \vec{c}$

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- b) Given that $\vec{v} = 2\vec{i} + 2\vec{j} + \vec{k}$ and $\vec{w} = 3\vec{i} - \vec{j} + \vec{k}$, find a unit vector perpendicular to both $\vec{v}$ and $\vec{w}$
9. A charge $q = 15 \times 10^{-20}$ coulombs moves in a magnetic field $\vec{B} = 300\vec{i} + 100\vec{j} - 500\vec{k}$ tesla with a velocity $\vec{v} = \vec{i} + \vec{j} - 2\vec{k}$ m/s. Find the resulting force $\vec{F}$ on that charge given that $\vec{F} = q\vec{v} \times \vec{B}$.

Answers

1. (a) $11\vec{k}$ (b) $2\vec{i} + 2\vec{j} + 4\vec{k}$
2. (a) $0.5\sqrt{26}$ (b) $2.5\sqrt{6}$
3. (a) $\vec{a} \times \vec{b}$ (b) $-3(\vec{a} \times \vec{b})$ (c) 0
4. Calculate and compare $\vec{a} \times \vec{c}$ and $\vec{b} \times \vec{b}$.
5. $\pm \frac{1}{\sqrt{86}}(7\vec{i} + \vec{j} + 6\vec{k})$
6. (a) $\frac{1}{\sqrt{5}}(2\vec{j} + \vec{k})$ (b) $\pm \frac{1}{7}(2\vec{i} - 3\vec{j} + 6\vec{k})$
7. $1.5\sqrt{3}$
8. a) (i) 42 (ii) 6 (iii) $\sqrt{65}$ (iv) $29^\circ 45'$ (v) $16\vec{i} + 8\vec{j} - 16\vec{k}$ (vi) 128
 b) $\frac{1}{\sqrt{74}}(-3\vec{i} - \vec{j} + 8\vec{k})$
9. $-15 \times 10^{-18}(3\vec{i} + \vec{j} + 2\vec{k})$ newtons

Figure 2.13: Point P is moving along a circle of radius r .

2.4.3 Circular motion and angular velocity

Suppose that a point P travels along a circular path with point O the centre of the circle of radius r , as shown in Fig. 2.13. Suppose that at time $t = 0$ seconds, P is at position P_0 and that at time t , P is at position P_t . Then the distance covered by P is $s = r\theta$, where θ is measured in radians.

Definition 2.20 The ω is defined as the time derivative of the angle θ

$$\omega = \frac{d\theta}{dt}.$$

Differentiating $s = r\theta$ with respect to time, we obtain the relation between the angular speed ω and the linear speed $v = ds/dt$

$$v = \frac{ds}{dt} = r \frac{d\theta}{dt} = r\omega$$

Example 2.23 Find the speeds and angular speeds for the second and minute hands of a clock, their lengths being 5 and 8 cm respectively.

Solution. The second hand travels 2π radians in 1 minute, i.e. $\frac{2\pi}{60}$ radians in 1 second, i.e. $\omega = \frac{\pi}{30}$ radians/s.

Using $v = \omega r$,

$$v = \left(\frac{\pi}{30}\right)5 = \frac{\pi}{6} \text{ cm/s.}$$

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The minute hand travels 2π radians in 1 hour, i.e. $\frac{2\pi}{60}$ radians in 1 minute, i.e. $\frac{2\pi}{60 \times 60}$ radians in 1 second, i.e. $\omega = \frac{2\pi}{60 \times 60} = \frac{\pi}{1800}$ radians/s. Using $v = \omega r$,

$$v = \left(\frac{\pi}{1800} \right) 8 = \frac{\pi}{225} \text{ cm/s.}$$

2.4.4 Angular velocity as a vector

Consider a rigid body (e.g. a wheel) rotating with a constant angular speed of ω radians/s about a fixed axis LM, which passes through a fixed point A, as shown in Fig. 2.14.

The point P on the rigid body, with position vector $\vec{r}$ relative to A, will move in a circle (perpendicular to LM) whose centre N is on LM, with speed $v = \omega(NP) = \omega(AP) \sin \theta$, i.e.

$$v = \omega r \sin \theta \quad (2.37)$$

The velocity is a vector $\vec{v}$ with magnitude v and direction that of the tangent to the circle at P, with the tangent pointing in the direction of travel.

Definition 2.21 The angular velocity is a vector ω with magnitude ω and direction along the axis of rotation such that

$$\vec{v} = \omega \times \vec{r}, \quad (2.38)$$

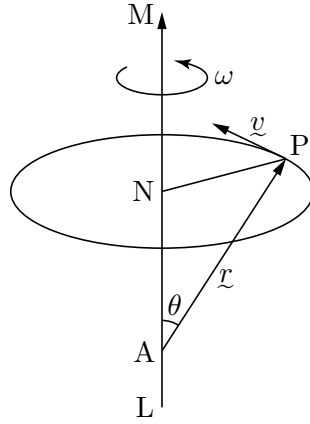
i.e. ω is in the direction of the vector $\overline{LM}$.

Note that the magnitude of $\vec{v}$ given by (2.35) is consistent with the definition (2.34).

Example 2.24 A rigid body rotates with an angular velocity of 4 radians/s about an axis in the direction of $\vec{i} + 2\vec{j} + 4\vec{k}$ and passing through the point A(3, 5, 8). Find the (linear) velocity $\vec{v}$ at the point P(4, 6, 10) of the body.

Solution.

$$\begin{aligned} \vec{r} &= \overline{AP} = \overline{OP} - \overline{OA} \\ &= 4\vec{i} + 6\vec{j} + 10\vec{k} - (3\vec{i} + 5\vec{j} + 8\vec{k}) \\ &= (4 - 3)\vec{i} + (6 - 5)\vec{j} + (10 - 8)\vec{k} \\ &= \vec{i} + \vec{j} + 2\vec{k} \end{aligned}$$

Figure 2.14: Rotation of a rigid body about the LM -axis.

$$\begin{aligned}\omega &= \omega \hat{\omega} = 4 \frac{\vec{i} + 2\vec{j} + 4\vec{k}}{\sqrt{1^2 + 2^2 + 4^2}} \\ &= \frac{4}{\sqrt{21}} (\vec{i} + 2\vec{j} + 4\vec{k}).\end{aligned}$$

Hence,

$$\vec{v} = \omega \times \vec{r} = \frac{4}{\sqrt{21}} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 4 \\ 1 & 1 & 2 \end{vmatrix} = \frac{4}{\sqrt{21}} (2\vec{j} - \vec{k}).$$

Example 2.25 The position vector $\vec{r}$ of a point particle changes with time according to

$$\vec{r}(t) = (\cos(t) + 5)\vec{i} + \sqrt{2}\vec{j}\sin(t) + (5 - \cos(t))\vec{k}.$$

1. Determine the velocity vector $\vec{v} = \dot{\vec{r}}$ and calculate its magnitude.
2. Assume that the angular velocity ω of the particle is given by $\omega = w(\vec{i} + \vec{k})$, where w is some scalar. Determine scalar w .
3. Determine the radius of the circular trajectory of the particle.

Solution

1. The velocity of the particle is

$$\vec{v} = \dot{\vec{r}} = -\sin(t)\vec{i} + \sqrt{2}\vec{j}\cos(t) + \sin(t)\vec{k}.$$

The magnitude of $\vec{v}$ is

$$|\vec{v}| = v = \sqrt{\sin^2(t) + 2\cos^2(t) + \sin^2(t)} = \sqrt{2(\cos^2(t) + \sin^2(t))} = \sqrt{2}$$

2 Vectors

2. We need to show that $\vec{v} = \boldsymbol{\omega} \times \vec{r}$

$$\begin{aligned}\boldsymbol{\omega} \times \vec{r} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ w & 0 & w \\ \cos(t) + 5 & \sqrt{2} \sin(t) & 5 - \cos(t) \end{vmatrix} \\ &= -w\sqrt{2} \sin(t) \vec{i} + w(\cos(t) + 5 - 5 + \cos(t)) \vec{j} + w\sqrt{2} \sin(t) \vec{k} \\ &= w\sqrt{2}(-\sin(t) \vec{i} + \sqrt{2} \cos(t) \vec{j} + \sin(t) \vec{k}) = w\sqrt{2} \vec{v}\end{aligned}$$

Thus, we have

$$w = \frac{1}{\sqrt{2}}, \Rightarrow, \boldsymbol{\omega} = \frac{1}{\sqrt{2}}(\vec{i} + \vec{k}).$$

3. The radius R of the circular trajectory is

$$R = \frac{|\vec{v}|}{|\boldsymbol{\omega}|} = \sqrt{2}.$$

Example 2.26 The Coriolis force.

The so-called Coriolis force is a type of inertia force that acts on any object that moves with respect to a rotating reference frame. It can be shown that if the angular velocity of the rotating reference frame is $\boldsymbol{\omega}$ and if the instantaneous velocity of an object with respect to the rotating reference frame is $\vec{v}$, then this object moves with the instantaneous acceleration

$$\vec{a} = \dot{\vec{v}} = -2\vec{\omega} \times \vec{v}.$$

Show that any moving body in the Northern (Southern) Hemisphere will deflect to the right (to the left) of its instantaneous direction of motion.

Solution

The Earth rotates anti-clockwise (clockwise), as seen from the north pole (south pole). Consider the motion in the Northern Hemisphere. Let z -axis be directed upwards along the vector of Earth's angular velocity $\boldsymbol{\omega} = w\vec{k}$, where $w > 0$ is the magnitude of the angular velocity. The velocity $\vec{v}$ of an object in the Northern Hemisphere can always be represented as

$$\vec{v} = \vec{v}_{\parallel} + \vec{v}_{\perp},$$

where $\vec{v}_{\parallel} = v_x \vec{i} + v_y \vec{j}$ is the velocity in the equatorial plane and $\vec{v}_{\perp} = v_z \vec{k}$ is the velocity along z -axis.

Then the Coriolis acceleration is

$$\vec{a} = -2\boldsymbol{\omega} \times \vec{v} = -2\boldsymbol{\omega} \times (\vec{v}_{\parallel} + \vec{v}_{\perp}) = -2\boldsymbol{\omega} \times \vec{v}_{\parallel},$$

because $\boldsymbol{\omega} \times \vec{v}_{\perp} = 0$.

2 Vectors

Thus, we obtain

$$\vec{a} = -2 \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & w \\ v_x & v_y & 0 \end{vmatrix} = 2wv_y\vec{i} - 2wv_x\vec{j}.$$

Finally, we notice that the Coriolis acceleration $\vec{a}$ is directed to the right from $\vec{v}_{\parallel}$, when seen from the north pole. This can be shown by checking that the triplet of vectors $(\vec{a}, \vec{v}_{\parallel}, \vec{k})$ is right-handed. Indeed

$$\vec{a} \cdot (\vec{v}_{\parallel} \times \vec{k}) = \begin{vmatrix} 2wv_y & -2wv_x & 0 \\ v_x & v_y & 0 \\ 0 & 0 & 1 \end{vmatrix} = 2w(v_y^2 + v_x^2) > 0.$$

Exercises

1. Find the angular speed and speed of the hour hand of a clock, its length being 6 cm
2. The blade of a lawnmower has a rotation rate of 3600 revolutions per minute. Given that the blade is 0.3 m. in radius, what is the angular speed and speed at the tip of the blade?
3. A rigid body is rotating with an angular velocity 5 radians/s about an axis in the direction of vector $\vec{i} + 4\vec{j} + 2\vec{k}$ and passing through the point A(3, 5, 9). Dimensions are in metres. Find the linear velocity $\vec{v}$ of the point P(4, 8, 11) on the body (use $\vec{v} = \boldsymbol{\omega} \times \vec{r}$).

Answers

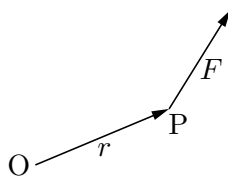
1. 1.454×10^{-4} radians/s, $\frac{\pi}{3600}$ cm/s
2. 120π radians/s, 113 m/s
3. $\frac{10}{\sqrt{21}}\vec{i} - \frac{5}{\sqrt{21}}\vec{k}$ m/s

2.4.5 Moment of a force (torque)

Definition 2.22 If a force $\vec{F}$ acts through a point P, which has position vector $\vec{r}$, then the moment of the force $\vec{F}$ about the origin O is given by

$$\vec{M} = \vec{r} \times \vec{F},$$

as illustrated in Fig. 2.15.

Figure 2.15: Force $\vec{F}$ acting at a point P .

The moment of force about a point is a measure of the tendency of the force to rotate an object about the point. If a body in equilibrium is acted upon by several forces, then: (i) the sum of the forces is 0; and (ii) the sum of the moments of the forces about any point is 0. These relationships can be used to calculate unknown forces (e.g. the forces on the supports of the body) from known forces (e.g. gravity).

Example 2.27 A force of 4 newtons acts through $P(2,3,5)$ in the direction of $\vec{a} = 4\vec{i} + 7\vec{j} + 9\vec{k}$. Find the moment of the force about $A(6,10,3)$. Dimensions are in metres.

Solution.

$$\begin{aligned}\vec{r} &= \overline{AP} = \overline{OP} - \overline{OA} \\ &= 2\vec{i} + 3\vec{j} + 5\vec{k} - (6\vec{i} + 10\vec{j} + 3\vec{k}) \\ &= -4\vec{i} - 7\vec{j} + 2\vec{k}\end{aligned}$$

$$\vec{F} = 4\hat{a} = 4 \frac{4\vec{i} + 7\vec{j} + 9\vec{k}}{\sqrt{4^2 + 7^2 + 9^2}} = \frac{4}{\sqrt{146}}(4\vec{i} + 7\vec{j} + 9\vec{k}).$$

Hence,

$$\vec{M} = \vec{r} \times \vec{F} = \frac{4}{\sqrt{146}} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -4 & -7 & 2 \\ 4 & 7 & 9 \end{vmatrix} = \frac{4}{\sqrt{146}}(-77\vec{i} + 44\vec{j}).$$

Example 2.28 The gyroscope problem.

Consider a gyroscope that spins at a certain angular velocity around the x -axis, as shown in Figure 2.16. The direction of spinning is such that the vector of angular velocity is given by

$$\omega = w\vec{i},$$

where w is some positive scalar.

The gyroscope is supported at point O at the distance $OA = l$ from its centre. We assume that the mass of the axis OA is negligibly small.

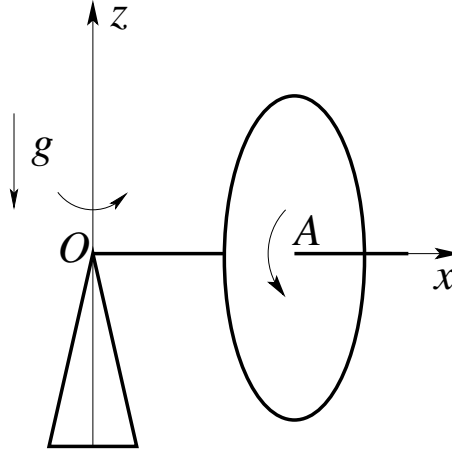


Figure 2.16: Gyroscope supported at O spins with the angular velocity $\omega = \omega \vec{i}$ directed along the x -axis. Distance from the centre of the gyroscope A to the point of support O is $OA = l$. Precession of the gyroscope is anti-clockwise, as seen from above.

The angular momentum of the gyroscope is the vector

$$\vec{L} = I\omega,$$

where $I > 0$ is the so-called moment of inertia with respect to the x -axis.

According to the second Newton's law of rotational motion, the change in the angular momentum of a solid body is given by the moment of forces, acting on the body.

$$\dot{\vec{L}} = \vec{r} \times \vec{F},$$

where $\vec{r}$ is the position vector of the point, at which the force $\vec{F}$ is applied.

Determine the direction of precession (i.e. change in the orientation of the rotational axis) of the gyroscope.

Solution

Let m be the mass of the gyroscope. We calculate the moment of the gravity force $\vec{F} = -mg\vec{k}$ that acts on the gyroscope about point O

$$\vec{r} \times m\vec{g} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ l & 0 & 0 \\ 0 & 0 & -mg \end{vmatrix} = lmg\vec{j}.$$

This shows that the direction of $\dot{\vec{L}}$ is along the y -axis. Since the triplet (x, y, z) forms a right-handed system, the y -axis is directed perpendicular to the plane (x, z) away from the reader.

2.4. Equation of a straight line in three dimensions

Thus, at any moment of time, the vectors $\vec{L} = I\omega$ and $\dot{\vec{L}}$ are orthogonal.

It is now easy to realize that the vector of angular momentum is rotating about the z-axis in the anti-clockwise direction, as seen from above. Indeed, if we assume

$$\vec{L}(t) = I\omega(\cos(\Omega t)\vec{i} + \sin(\Omega t)\vec{j}),$$

where Ω is the precession frequency about the z-axis, then

$$\dot{\vec{L}} = I\omega\Omega(-\sin(\Omega t)\vec{i} + \cos(\Omega t)\vec{j})$$

Setting $t = 0$, which corresponds to the situation in Figure 2.16, we obtain

$$I\omega\Omega = lmg$$

so that the precession frequency is

$$\Omega = \frac{lmg}{I\omega}.$$

Exercises

1. A force $\vec{F}$ with magnitude 4 newtons acts through $P(1, 2, 4)$ in the direction of $\vec{a} = 2\vec{i} + 3\vec{j} + 5\vec{k}$. Find the moment of the force about $A(0, 7, 4)$. Dimensions are in metres.
2. A cable AB helps support a tower. The point A is at the top of the tower, while B is the point on the ground to which the cable is attached. The origin is the centre of the base of the tower. Given that A has the coordinates $(0, 0, 80)$ and B has the coordinates $(0, 40, 0)$, with both coordinates in metres, and the tension in the cable is 4 kN, find
 - a) the tension in vector form, and
 - b) the moment about O due to the force exerted by the cable at A.

Answers

1. $-\frac{4}{\sqrt{38}}(25\vec{i} + 5\vec{j} - 13\vec{k})$ Nm
2. a) $\frac{4}{\sqrt{5}}(\vec{j} - 2\vec{k})$ kN
 b) $-\frac{320}{\sqrt{5}}\vec{i}$ kNm

3 Linear Geometry

3.1 Equation of a straight line in three dimensions

Let a straight line in the direction of $\vec{a}$, where $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ pass through a fixed point $C(c_1, c_2, c_3)$, and let $P(x, y, z)$ be any point on the line (see Fig. 3.1).

By the triangle rule for vector addition,

$$\overrightarrow{OP} = \overrightarrow{OC} + \overrightarrow{CP}.$$

That is,

$$\begin{aligned}\vec{r} &= \vec{c} + (\text{a multiple of } \vec{a}) \\ &= \vec{c} + t\vec{a}.\end{aligned}\tag{3.39}$$

Equation (3.39) is the of a straight line in three dimensions. The term t is called a parameter. As $\overrightarrow{CP} = t\vec{a}$, if, for example, $\overrightarrow{CP} = 3\vec{a}$, then $t = 3$.

If the vectors $\vec{c}$ and $\vec{a}$ in (3.39) are each expanded out as a sum of vector components, (3.39) becomes

$$\vec{r} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k} + t(a_1\vec{i} + a_2\vec{j} + a_3\vec{k}).\tag{3.40}$$

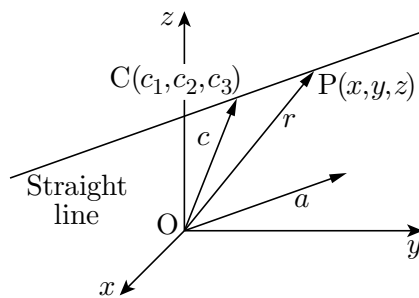


Figure 3.1: Line passes through point C in the direction $\vec{a}$.

3 Linear Geometry

Equation (3.40) is also the vector equation of a straight line in three dimensions. Similarly, if $\vec{r}$ is expanded out as a sum of vector components, then (3.40) becomes

$$x\vec{i} + y\vec{j} + z\vec{k} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k} + t(a_1\vec{i} + a_2\vec{j} + a_3\vec{k}). \quad (3.41)$$

If the left-hand and right-hand coefficients of $\vec{i}$, $\vec{j}$, and $\vec{k}$ in (3.40) are respectively equated, we obtain

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + t \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \text{ or } \begin{bmatrix} x = c_1 + ta_1 \\ y = c_2 + ta_2 \\ z = c_3 + ta_3 \end{bmatrix} \quad (3.42)$$

Equations (3.42) are together called the of the straight line. They can be rewritten as

$$\begin{aligned} t &= \frac{x - c_1}{a_1} \\ t &= \frac{y - c_2}{a_2} \\ t &= \frac{z - c_3}{a_3}, \end{aligned} \quad (3.43)$$

as long as the components of $\vec{a}$ are non-zero. Equations (3.43) imply that

$$\frac{x - c_1}{a_1} = \frac{y - c_2}{a_2} = \frac{z - c_3}{a_3} \quad (3.44)$$

If any component of $\vec{a}$ is zero, then the corresponding coordinate x , y or z is constant (given by the corresponding component of $\vec{c}$). The equations in (3.44) are known as the equations without parameter of the straight line (or the symmetric equations).

Example 3.1 Find the equation of the straight line that passes through the point $C(4, 2, 3)$ and which is in the direction of the vector $\vec{a} = 4\vec{i} - 2\vec{j} + 3\vec{k}$ in

1. vector form
2. scalar parametric form, and
3. in Cartesian form (without parameter).

Solution.

As C has the coordinates $(4, 2, 3)$, $\vec{c} = 4\vec{i} + 2\vec{j} + 3\vec{k}$.

1. Using (3.39), the vector equation is

$$\vec{r} = 4\vec{i} + 2\vec{j} + 3\vec{k} + t(4\vec{i} - 2\vec{j} + 3\vec{k}).$$

2. Using (3.42), the scalar parametric equations are

$$\begin{aligned} x &= 4 + 4t \\ y &= 2 - 2t \\ z &= 3 + 3t. \end{aligned} \quad (3.45)$$

3. Using (3.44), the Cartesian form is

$$\frac{x - 4}{4} = \frac{y - 2}{-2} = \frac{z - 3}{3}.$$

3.1.1 Distance between lines in three dimensions

Example 3.2 Given two lines

$$l_1 : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad l_2 : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix} + s \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}, \quad t, s \in \mathbb{R}$$

Find such points A and B on l_1 and on l_2 , respectively so that the distance AB is the smallest possible. Thus determine the distance between the lines l_1 and l_2 .

Solution

When the two lines are given by the respective parametric equations, it is important to use different letters to denote the free parameters. Here we use t and s for lines l_1 and l_2 , respectively.

Let A and B be the closest points on lines l_1 and l_2 , respectively. Their coordinates can be written in terms of t and s

$$A = (-1 + t, 1 + 2t, 3 + t), \quad B = (3 + s, -1 - s, 3 - s).$$

From the geometrical point of view, the vector $\overline{AB}$ is orthogonal to the direction vectors of the two lines. We use this condition to find the values of t and s . Thus, we require

$$\overline{AB} \perp \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \text{and} \quad \overline{AB} \perp \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix},$$

where

$$\begin{aligned} \overline{AB} &= (3 + s - (-1 + t))\vec{i} + (-1 - s - (1 + 2t))\vec{j} + (3 - s - (3 + t))\vec{k} \\ &= (4 + s - t)\vec{i} + (-2 - s - 2t)\vec{j} + (-s - t)\vec{k}. \end{aligned}$$

The dot product between any two orthogonal vectors vanishes

$$\begin{aligned} ((4 + s - t)\vec{i} + (-2 - s - 2t)\vec{j} + (-s - t)\vec{k}) \cdot (\vec{i} + 2\vec{j} + \vec{k}) &= 0 \\ ((4 + s - t)\vec{i} + (-2 - s - 2t)\vec{j} + (-s - t)\vec{k}) \cdot (\vec{i} - \vec{j} - \vec{k}) &= 0. \end{aligned}$$

3 Linear Geometry

Thus we obtain a system of two linear equations for t and s

$$\begin{aligned}(4 + s - t) + 2(-2 - s - 2t) + (-s - t) &= -6t - 2s = 0 \\ (4 + s - t) - (-2 - s - 2t) - (-s - t) &= 2t + 3s = -6.\end{aligned}$$

This system can be solved using any method (e.g. the Cramer's rule) to yield the solution

$$t = \frac{6}{7}, \quad s = -\frac{18}{7}.$$

Consequently,

$$\overline{AB} = \frac{4}{7}\mathbf{i} - \frac{8}{7}\mathbf{j} + \frac{12}{7}\mathbf{k}$$

The distance d between the lines is

$$d = |\overline{AB}| = \frac{1}{7}\sqrt{16 + 64 + 144} = \frac{\sqrt{224}}{7}$$

3.1.2 Lines in two dimensions

Any line in a two-dimensional space can be written as

$$y = kx + b,$$

where b is the y -intercept of the line and $k = \tan \alpha$. Here α is the angle between the line and the x -axis. In what follows we restrict α to the interval $[0, \pi]$.

Next we rewrite the equation of the line in the parametric form. Taking into account that the choice of the parameter is not specified, we choose x as the line parameter and set

$$t = x, \quad t \in (-\infty, \infty).$$

The position vector $\overline{OP}$ of an arbitrary point on the line $P = (x, y)$ can be written as

$$\overline{OP} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ kx + b \end{bmatrix} = \begin{bmatrix} 0 \\ b \end{bmatrix} + t \begin{bmatrix} 1 \\ k \end{bmatrix}, \quad t \in (-\infty, \infty). \quad (3.46)$$

The last equation shows that the direction vector of the line is $\boldsymbol{\tau} = \mathbf{i} + k\mathbf{j}$.

3 Linear Geometry

Note that Eq. (3.46) can only be used if $k \neq (0, \infty)$. However, if the line is parallel to one of the coordinate axes, then Eq. (3.46) is modified as follows

$$x = b : \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} b \\ t \end{bmatrix}, \quad t \in (-\infty, \infty).$$

and

$$y = b : \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} t \\ b \end{bmatrix}, \quad t \in (-\infty, \infty).$$

Example 3.3 Determine the Cartesian equation of the line that passes through $P = (3, 3)$ and is perpendicular to the line $y = 1 - 2x$.

Solution. The parametric equation of the line $y = 1 - 2x$ is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad t \in (-\infty, \infty).$$

This shows that the direction vector of the line $y = 1 - 2x$ is $\tau = \mathbf{i} - 2\mathbf{j}$. First, we find a vector $\mathbf{a}$ that is perpendicular to τ . Because the length of $\mathbf{a}$ can be arbitrary, we may set $\mathbf{a} = \mathbf{i} + a_y\mathbf{j}$. The value of a_y is determined from

$$\tau \cdot \mathbf{a} = 1 - 2a_y = 0.$$

Thus we obtain $a_y = 1/2$.

Consequently, the vector equation of the line that passes through $P = (3, 3)$ and is perpendicular to $y = 1 - 2x$ can be written in the form

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} + s \begin{bmatrix} 1 \\ 1/2 \end{bmatrix}, \quad s \in (-\infty, \infty).$$

Finally, we transform the parametric equation of the line into the Cartesian equation by eliminating parameter s

$$x = 3 + s, \quad y = 3 + s/2, \rightarrow y = 3 + (x - 3)/2 = 3/2 + x/2.$$

3.1.3 The angle between two lines

Two lines have vector equations $\vec{r} = \vec{c} + t\vec{a}$ and $\vec{r} = \vec{d} + t\vec{b}$ (see Figure 3.2). Depending on $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ the two lines may or may not intersect in one point.

The angle θ between the two lines is found using the dot product, i.e.

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{ab} \right)$$

Note that the angle can also be determined as $(\pi - \theta)$.

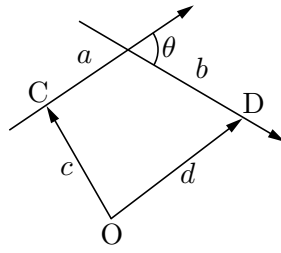


Figure 3.2: Angle θ between two lines with direction vectors $\vec{a}$ and $\vec{b}$. Alternatively, the angle between the lines is $\pi - \theta$.

Example 3.4 Given two straight lines with equations

$$\vec{r} = \vec{i} + 2\vec{j} - \vec{k} + t(\vec{i} + 3\vec{j} + 2\vec{k})$$

and

$$\vec{r} = 3\vec{i} + 4\vec{j} + 7\vec{k} + t(\vec{i} + 2\vec{j} + 3\vec{k}),$$

find θ , the angle between the two lines.

Solution.

$$\begin{aligned} \cos \theta &= \frac{(\vec{i} + 3\vec{j} + 2\vec{k}) \cdot (\vec{i} + 2\vec{j} + 3\vec{k})}{\sqrt{1^2 + 3^2 + 2^2} \sqrt{1^2 + 2^2 + 3^2}} \\ &= \frac{(1)(1) + (3)(2) + (2)(3)}{\sqrt{14}\sqrt{14}} \\ &= \frac{13}{14} \\ \Rightarrow \theta &= \cos^{-1} \frac{13}{14} \\ &\approx 21^\circ 47'. \end{aligned}$$

Example 3.5 Determine the angle between the lines $y = 2x + 1$ and $y = -x$ in two dimensions.

Solution

The direction vectors $\vec{v}$ and $\vec{w}$ of the line $y = 2x + 1$ and $y = -x$ are given by

$$\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \text{ and } \vec{w} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix},$$

respectively

The angle θ between the lines is

$$\theta = \cos^{-1} \left(\frac{\vec{v} \cdot \vec{w}}{|\vec{v}| |\vec{w}|} \right) = \cos^{-1} \left(\frac{-1}{\sqrt{10}} \right) = 1.8925 \text{ (rads)}$$

3.1.4 Intersection of two lines

Any two lines in three-dimensional space can intersect in one point, coincide, or have no common points of intersection. The following examples cover these three possibilities.

Example 3.6 An underground pipeline, P_1 , passes through the points $A(0, 8, -10)$ and $B(2, 10, -12)$. A second pipeline, P_2 , yet to be constructed, is designed to pass through the points $C(1, 8, -11)$ and $D(7, 16, -17)$.

1. Find the equation of the line P_1 (use t as the parameter) in vector and parametric form.
2. Find the equation of the second pipeline (use s as the parameter) in vector and parametric form.
3. The designers are concerned that the two pipelines might intersect. Is their concern justified, and if so, where would they intersect?

Solution.

1. Using (3.39), P_1 has vector equation

$$\begin{aligned}\vec{r} &= \overline{OA} + t(\overline{AB}) \\ &= \overline{OA} + t(\overline{OB} - \overline{OA}) \\ &= 0\vec{i} + 8\vec{j} - 10\vec{k} + t(2\vec{i} + 10\vec{j} - 12\vec{k} - (0\vec{i} + 8\vec{j} - 10\vec{k})) \\ &= 0\vec{i} + 8\vec{j} - 10\vec{k} + t(2\vec{i} + 2\vec{j} - 2\vec{k}).\end{aligned}\tag{3.47}$$

Equation (3.47) implies that

$$x\vec{i} + y\vec{j} + z\vec{k} = 2t\vec{i} + (8 + 2t)\vec{j} + (-10 - 2t)\vec{k}.\tag{3.48}$$

Equating respectively the $\vec{i}$, $\vec{j}$, and $\vec{k}$ components on the left- and right-hand sides of (3.48) gives

$$x = 2t\tag{3.49}$$

$$y = 8 + 2t\tag{3.50}$$

$$z = -10 - 2t,\tag{3.51}$$

which are the parametric equations of P_1 .

2. From (3.39), P_2 has vector equation

$$\begin{aligned}\vec{r} &= \overline{OC} + s(\overline{CD}) \\ &= \overline{OC} + s(\overline{OD} - \overline{OC}) \\ &= 1\vec{i} + 8\vec{j} - 11\vec{k} + s(7\vec{i} + 16\vec{j} - 17\vec{k} - (1\vec{i} + 8\vec{j} - 11\vec{k})) \\ &= 1\vec{i} + 8\vec{j} - 11\vec{k} + s(6\vec{i} + 8\vec{j} - 6\vec{k}).\end{aligned}\tag{3.52}$$

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Equation (3.52) implies that

$$x\vec{i} + y\vec{j} + z\vec{k} = (1 + 6s)\vec{i} + (8 + 8s)\vec{j} + (-11 - 6s)\vec{k} \quad (3.53)$$

Equating respectively the $\vec{i}$, $\vec{j}$, and $\vec{k}$ components on the left- and right-hand sides of (3.53) gives

$$x = 1 + 6s \quad (3.54)$$

$$y = 8 + 8s \quad (3.55)$$

$$z = -11 - 6s, \quad (3.56)$$

which are the parametric equations of P_2 .

3. The two pipelines will intersect if there is a point where their x , y , and z coordinates are the same. Equating (3.49) and (3.54), (3.50) and (3.55), and (3.51) and (3.56), we obtain

$$2t = 1 + 6s$$

$$8 + 2t = 8 + 8s$$

$$-10 - 2t = -11 - 6s.$$

These equations can be rewritten as

$$-6s + 2t = 1 \quad (3.57)$$

$$-8s + 2t = 0 \quad (3.58)$$

$$6s - 2t = -1. \quad (3.59)$$

These are three equations with two unknowns, which, if they are consistent, can be solved using Gaussian elimination. We row reduce the matrix of coefficients from the equations

$$\left[\begin{array}{cc|c} -6 & 2 & 1 \\ -8 & 2 & 0 \\ 6 & -2 & -1 \end{array} \right]$$

to get

$$\left[\begin{array}{cc|c} 1 & 0 & \frac{1}{2} \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right]$$

which tells us that there is a consistent solution $s = \frac{1}{2}$, $t = 2$ and so the pipelines do indeed intersect.

Substituting this value for s into (3.54), (3.55), and (3.56) (or the value of t into (3.49), (3.50), and (3.51)) gives $x = 4$, $y = 12$, $z = -14$, the coordinates of the point of intersection of the two lines. If the pipelines had not intersected, then we would have found no consistent solution for s and t .

Example 3.7 The lines l_1 and l_2 are given by the equations

$$l_1 : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + t \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}, t \in (-\infty, \infty) \text{ and } l_2 : \frac{x-a}{4} = \frac{1-y}{2} = \frac{z-5}{4},$$

where a is some parameter. Determine for which values of a the lines l_1 and l_2

1. coincide,
2. have no common points of intersection.

Solution

First, we rewrite the equation of line l_2 in the vector form

$$l_2 : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ 1 \\ 5 \end{bmatrix} + s \begin{bmatrix} 4 \\ -2 \\ 4 \end{bmatrix}, s \in (-\infty, \infty)$$

The points of intersection between l_1 and l_2 are found from

$$\begin{aligned} 1 + 2t &= a + 4s \\ 2 - t &= 1 - 2s \\ 3 + 2t &= 5 + 4s. \end{aligned}$$

We rewrite the equations in the form

$$\begin{aligned} 2t - 4s &= a - 1 \\ -t + 2s &= -1 \\ 2t - 4s &= 2 \end{aligned}$$

and then row reduce the augmented matrix

$$\left[\begin{array}{cc|c} 2 & -4 & a-1 \\ -1 & 2 & -1 \\ 2 & -4 & 2 \end{array} \right]$$

to get

$$\left[\begin{array}{cc|c} 2 & -4 & a-1 \\ 0 & 0 & a-3 \\ 0 & 0 & 3-a \end{array} \right].$$

Thus we conclude

1. If $a = 3$, there are infinitely many points of intersections so that the lines l_1 and l_2 coincide. Indeed, for any value of the parameter s , there exist the corresponding value of t

$$t = 1 + 2s.$$

2. If $a \neq 3$, the lines l_1 and l_2 have no common points of intersection. Note that in this case l_1 and l_2 are parallel, since their directions coincide.

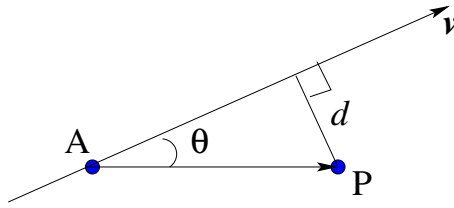


Figure 3.3: Distance d between the point P and the line $\vec{v}$.

3.1.5 Distance from a point to a line

Consider the line l that passes through point $A = (a_x, a_y, a_z)$ in the direction of $\vec{v} = v_x\vec{i} + v_y\vec{j} + v_z\vec{k}$. The vector equation of l is

$$l : \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}, t \in (-\infty, \infty).$$

The distance d from a point $P = (p_x, p_y, p_z)$ to the line l is determined from the orthogonal projection of P onto l , as shown in Figures 3.3. Let θ be the angle between the vector $\overline{AP}$ and the direction of the line $\vec{v}$. Then

$$d = |\overline{AP} \sin \theta| = \frac{|\overline{AP} \times \vec{v}|}{|\vec{v}|},$$

where we have used the definition of the cross product

$$|\overline{AP} \times \vec{v}| = |\overline{AP}| |\vec{v}| |\sin \theta|.$$

Example 3.8 Determine the distance from point $P = (1, 2)$ to the line $y = 2x + 3$ in two dimensions.

Solution

The vector equation of the line $y = 2x + 3$ is obtained from Eq. (3.46)

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, t \in (-\infty, \infty).$$

Next, we take an arbitrary point on the line, e.g. $A = (0, 3, 0)$ and calculate the vector $\overline{AP} = \vec{i} - \vec{j}$. Consequently, the distance d from P to the line is

$$d = \frac{|\overline{AP} \times \vec{v}|}{|\vec{v}|},$$

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where $\vec{v} = \vec{i} + 2\vec{j}$ gives the direction of the line. Finally we obtain

$$d = \frac{1}{\sqrt{1+4}} \text{mod} \left(\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 0 \\ 1 & 2 & 0 \end{vmatrix} \right) = \frac{1}{\sqrt{5}} |3\vec{k}| = \frac{3}{\sqrt{5}}.$$

Exercises

- Find the equation of the straight line through $(1, 2, 3)$ with direction $4\vec{i} - \vec{j} + \vec{k}$ in
 - vector form,
 - scalar parametric form, and
 - without parameter (i.e. Cartesian form).

- Find the vector (parametric) equation of the pipeline passing through $(0, 3, 4)$ and $(1, 5, 0)$.

- Convert the straight line equation

$$\frac{x-3}{1} = \frac{y+2}{2} = \frac{z-3}{4}$$

to vector form.

- Find the angle between the lines

$$\vec{r} = \vec{i} - 2\vec{j} - \vec{k} + t(\vec{i} + 3\vec{j} + 2\vec{k})$$

and

$$\vec{r} = -\vec{i} - \vec{j} - 3\vec{k} + t(2\vec{i} + \vec{j} + 3\vec{k})$$

- An underground pipeline passes through the points $(0, 16, -30)$ and $(20, 20, -32)$ with the x - and y -axes in the horizontal surface of the ground. Find the equation of the pipeline.

A second pipeline, yet to be constructed, has been designed to pass through the points $(10, 10, -32)$ and $(25, 17, -33)$. Find the equation of the second pipeline (use s as the parameter). The designers are concerned that the two pipelines might intersect. Is their concern justified, and if so, where would they intersect?

Answers

- $\vec{r} = (1 + 4t)\vec{i} + (2 - t)\vec{j} + (3 + t)\vec{k}$
 - $x = 1 + 4t, \quad y = 2 - t, \quad z = 3 + t$

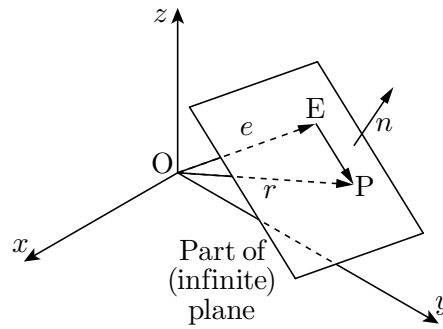


Figure 3.4: A plane passes through point E . $\vec{n}$ is the normal vector to the plane.

c) $\frac{x-1}{4} = \frac{y-2}{-1} = \frac{z-3}{1}$

2. $\vec{r} = t\vec{i} + (2t+3)\vec{j} + (-4t+4)\vec{k}$

3. $\vec{r} = (t+3)\vec{i} + (2t-2)\vec{j} + (4t+3)\vec{k}$

4. $38^\circ 12' 48''$

5. $\vec{r} = 16\vec{j} - 30\vec{k} + t(20\vec{i} + 4\vec{j} - 2\vec{k})$, $\vec{r} = 10\vec{i} + 10\vec{j} - 32\vec{k} + s(15\vec{i} + 7\vec{j} - \vec{k})$,
Yes, at $(40, 24, -34)$

3.2 Equation of a plane

Suppose that vector $\vec{n}$ is normal (i.e. perpendicular) to a plane, with $\vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$. Suppose also that $E(e_1, e_2, e_3)$ is a fixed point on the plane, with $\overrightarrow{OE} = \vec{e}$, and let $P(x, y, z)$ be any point on the plane, with $\overrightarrow{OP} = \vec{r}$. The plane is shown in Fig. 3.4.

The vector $\overrightarrow{EP} = \vec{r} - \vec{e}$.

As $\overrightarrow{EP}$ is perpendicular to $\vec{n}$, then $\overrightarrow{EP} \cdot \vec{n} = 0$, i.e.

$$\begin{aligned} & (\vec{r} - \vec{e}) \cdot \vec{n} = 0 \\ \implies & \vec{r} \cdot \vec{n} - \vec{e} \cdot \vec{n} = 0 \\ \implies & \vec{r} \cdot \vec{n} = \vec{e} \cdot \vec{n} \\ \implies & \vec{r} \cdot \vec{n} = d, \end{aligned}$$

where $d = \vec{e} \cdot \vec{n}$.

Expanding $\vec{r} \cdot \vec{n}$ in terms of the vector components gives

$$(x\vec{i} + y\vec{j} + z\vec{k}) \cdot (a\vec{i} + b\vec{j} + c\vec{k}) = d,$$

i.e.

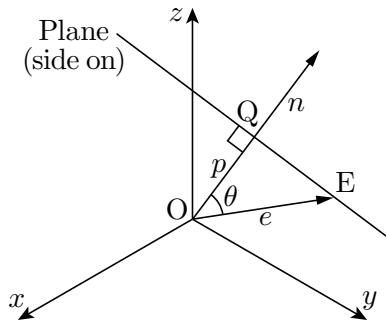


Figure 3.5: Distance p from the plane to the origin.

$$ax + by + cz = d.$$

(3.60)

Equation (3.60) is the equation of the plane.

3.2.1 Distance from a plane to the origin

Suppose that p is the perpendicular distance of the plane from the origin, with Q the foot of the perpendicular from O to the plane, i.e. $p = OQ$.

There are two possibilities: (i) $\vec{n}$ is in the same direction as $\overrightarrow{OQ}$, or (ii) $\vec{n}$ is in the opposite direction to $\overrightarrow{OQ}$.

The first possibility is illustrated in Figure 3.5, with the plane shown side-on.

$$\begin{aligned} p &= OQ \\ &= e \cos \theta \\ &= e \frac{\vec{e} \cdot \vec{n}}{|\vec{n}|} \\ &= \frac{\vec{e} \cdot \vec{n}}{|\vec{n}|} \\ &= \vec{e} \cdot \hat{n} \\ &= (e_1 \vec{i} + e_2 \vec{j} + e_3 \vec{k}) \cdot \left(\frac{a\vec{i} + b\vec{j} + c\vec{k}}{\sqrt{a^2 + b^2 + c^2}} \right) \\ &= \frac{ae_1 + be_2 + ce_3}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{d}{\sqrt{a^2 + b^2 + c^2}} \quad (\text{using (3.60)}). \end{aligned}$$

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If $\vec{n}$ is in the opposite direction to $\overrightarrow{OQ}$, then d turns out to be negative and it can be shown that

$$p = -\frac{d}{\sqrt{a^2 + b^2 + c^2}}.$$

As p is always positive or 0, the last two formula can be summarized in one formula, i.e.

$$p = \frac{|d|}{\sqrt{a^2 + b^2 + c^2}}. \quad (3.61)$$

Note. Equation (3.61) implies

that ℓ , the distance between two parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$, is given by

$$\ell = \frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}.$$

Example 3.9 Given that $2\vec{i} + 3\vec{j} + \vec{k}$ is normal to a plane which passes through the point $(4, 7, 8)$, find the equation of the plane and its distance from the origin.

Solution.

Using (3.60), the equation of the plane is

$$2x + 3y + z = d \quad (3.62)$$

As $(4, 7, 8)$ is on the plane, substituting $x = 4$, $y = 7$, and $z = 8$ into (3.62) gives

$$d = (2)(4) + (3)(7) + (1)(8) = 8 + 21 + 8 = 37.$$

Hence, the equation of the plane is $2x + 3y + 1z = 37$.

Using (3.61), the distance of the plane from the origin is

$$p = \frac{|37|}{\sqrt{2^2 + 3^2 + 1^2}} = \frac{37}{\sqrt{14}}.$$

Example 3.10 1. Find

- the equation of a plane normal to $4\vec{i} + 3\vec{j} + 12\vec{k}$ and a distance 4 from the origin, and
- the intercepts of the plane with the three axes.

2. Sketch the plane.

Solution.

1. a) Using (3.60), the equation of the plane is $4x + 12y + 3z = d$. Using (3.61),

$$\begin{aligned} 4 &= \frac{|d|}{\sqrt{4^2 + 3^2 + 12^2}} = \frac{|d|}{13} \\ \implies |d| &= 52 \\ \implies d &= \pm 52. \end{aligned}$$

Hence there are two planes with a normal $4\vec{i} + 3\vec{j} + 12\vec{k}$ and a distance 4 from the origin. The equation of one plane is

$$4x + 3y + 12z = 52 \quad (3.63)$$

and the equation of the other plane is

$$4x + 3y + 12z = -52.$$

b) For the plane defined by (3.63), the plane intercepts (cuts) the x -axis where $y = 0$ and $z = 0$, and hence, by (3.63),

$$4x = 52 \implies x = 13.$$

The plane cuts the y -axis where $x = 0$ and $z = 0$, and hence, by (3.63),

$$3y = 52 \implies y = \frac{52}{3}.$$

The plane cuts the z -axis where $x = 0$ and $y = 0$, and hence, by (3.63),

$$12z = 52 \implies z = \frac{13}{3}.$$

2. The intercepts can be sketched on a 3D diagram as shown in Fig. 3.6. These three points can be joined together (two at a time) by straight lines. The three straight lines form a triangle, which is part of the plane.

3.2.2 Distance from a given point to a plane

The result of the previous section allows us to find the distance from any given point $P = (p_x, p_y, p_z)$ to the plane

$$ax + by + cz = d.$$

Let $E = (e_x, e_y, e_z)$ be an arbitrary point on the plane, i.e.

$$e_x a + e_y b + e_z c = d.$$

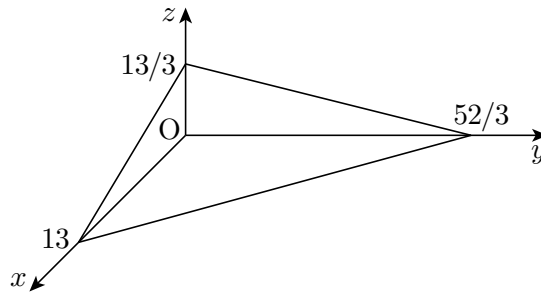


Figure 3.6: Intercepts of the plane with the coordinate axes.

Then the distance p from P to the plane is given by

$$\begin{aligned} p &= \frac{\text{mod}(\overline{EP} \cdot \vec{n})}{|\vec{n}|} = \frac{|a(p_x - e_x) + b(p_y - e_y) + c(p_z - e_z)|}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{|ap_x + bp_y + cp_z - d|}{\sqrt{a^2 + b^2 + c^2}}. \end{aligned} \quad (3.64)$$

Example 3.11 Determine the distance from $P = (1, 2, 3)$ to the plane $x = y$.

Solution

The normal $\vec{n}$ to the plane is

$$\vec{n} = \vec{i} - \vec{j}.$$

Using Eq. (3.64) we find the distance p

$$p = \frac{|1 - 2|}{\sqrt{1 + 1}} = \frac{1}{\sqrt{2}}.$$

3.2.3 The angle between two planes

Example 3.12 Find the angle between the planes $4x + 5y + 12z = 3$ and $x + 3y + 2z = -8$.

Solution.

The angle θ is defined as the angle between the normals to the planes. The angle between the

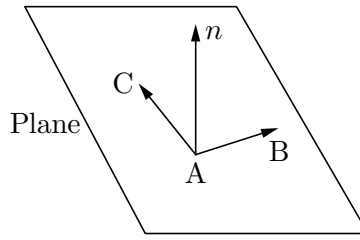


Figure 3.7: Plane that passes through three points A B and C.

normals can be found using the dot product, i.e.

$$\begin{aligned}
 \cos \theta &= \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|} = \frac{(4\vec{i} + 5\vec{j} + 12\vec{k}) \cdot (\vec{i} + 3\vec{j} + 2\vec{k})}{\sqrt{4^2 + 5^2 + 12^2}\sqrt{1^2 + 3^2 + 2^2}} \\
 &= \frac{(4)(1) + (5)(3) + (12)(2)}{\sqrt{185}\sqrt{14}} \\
 &= \frac{4 + 15 + 24}{\sqrt{185}\sqrt{14}} \\
 &= \frac{43}{\sqrt{185}\sqrt{14}} \\
 \Rightarrow \quad \theta &= \cos^{-1}\left(\frac{43}{\sqrt{185}\sqrt{14}}\right) \\
 &\approx 32^\circ 20'.
 \end{aligned}$$

3.2.4 Equation of a plane given three points on the plane

A plane can be specified by any three points that do not belong to one line.

Example 3.13 Find the equation of the plane which passes through the points A(1, 1, 1), B(0, 1, 2), and C(-1, 4, -1).

Solution.

A diagram of the plane is shown in Figure 3.7, with $\vec{n}$ being defined by

$$\vec{n} = \overline{AB} \times \overline{AC}.$$

Thus, $\vec{n}$ is a vector that is normal to the plane, and

$$\begin{aligned}
 \overline{AB} &= \overline{OB} - \overline{OA} = 0\vec{i} + 1\vec{j} + 2\vec{k} - (1\vec{i} + 1\vec{j} + 1\vec{k}) \\
 &= -1\vec{i} + 0\vec{j} + 1\vec{k} \\
 \overline{AC} &= \overline{OC} - \overline{OA} = -1\vec{i} + 4\vec{j} - 1\vec{k} - (1\vec{i} + 1\vec{j} + 1\vec{k}) \\
 &= -2\vec{i} + 3\vec{j} - 2\vec{k}.
 \end{aligned}$$

Hence,

$$\begin{aligned}\vec{n} &= \overline{\text{AB}} \times \overline{\text{AC}} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 0 & 1 \\ -2 & 3 & -2 \end{vmatrix} \\ &= \vec{i} \begin{vmatrix} 0 & 1 \\ 3 & -2 \end{vmatrix} - \vec{j} \begin{vmatrix} -1 & 1 \\ -2 & -2 \end{vmatrix} + \vec{k} \begin{vmatrix} -1 & 0 \\ -2 & 3 \end{vmatrix} \\ &= -3\vec{i} - 4\vec{j} - 3\vec{k}.\end{aligned}$$

Hence, the equation of the plane is $-3x - 4y - 3z = d$. As $(1, 1, 1)$ is on the plane,

$$\begin{aligned}d &= -3(1) - 4(1) - 3(1) \\ &= -10.\end{aligned}$$

Hence, the equation of the plane is $-3x - 4y - 3z = -10$.

3.2.5 Intersection of a line and a plane

In three dimensions, any line may either intersect with a plane at one point, or have no common points of intersection, or the line may be contained in the plane.

Example 3.14 Given a plane

$$x + y + 2z = 37 \quad (3.65)$$

and the straight line

$$x = 2 + 3t, \quad y = 1 + t, \quad z = 1 + 2t, \quad (3.66)$$

find

1. the coordinates of the point of intersection, and
2. the angle between the plane and the line.

Solution.

The plane and straight line are shown in Fig. 3.8.

1. Substituting (3.66) into (3.65) gives

$$\begin{aligned}(2 + 3t) + (1 + t) + 2(1 + 2t) &= 37 \\ \implies 5 + 8t &= 37 \\ \implies 8t &= 32 \\ \implies t &= 4\end{aligned} \quad (3.67)$$

Substituting (3.67) into (3.66) gives

$$x = 14, \quad y = 5, \quad z = 9.$$

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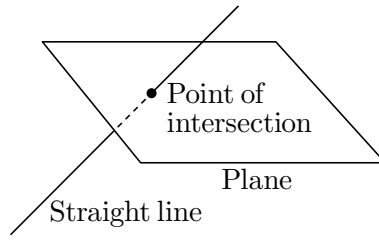


Figure 3.8: Intersection of a line and a plane

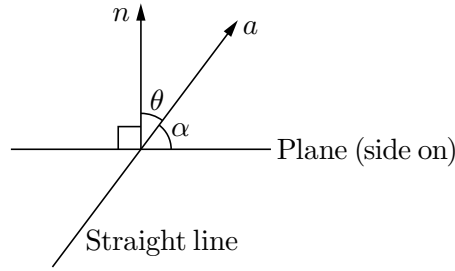


Figure 3.9: Side view of a plane, intersecting with a line.

2. A side-on view of the plane is shown in Fig. 3.9. The angle between the plane and the straight line is α and the angle between the normal to the plane and the straight line is θ . As the two angles add up to a right angle, $\alpha = \pi/2 - \theta$. Given that the straight line is in the direction of the vector $\vec{a}$, and $\vec{n}$ is a normal to the plane, θ is found using the dot product, i.e.

$$\cos \theta = \frac{\vec{n} \cdot \vec{a}}{|\vec{n}| |\vec{a}|}. \quad (3.68)$$

To find $\vec{a}$ (the direction of the line), substitute (3.66) into the vector equation of a point on the line, $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$, which gives

$$\begin{aligned} \vec{r} &= (2 + 3t)\vec{i} + (1 + t)\vec{j} + (1 + 2t)\vec{k} \\ &= (2\vec{i} + \vec{j} + \vec{k}) + t(3\vec{i} + \vec{j} + 2\vec{k}). \end{aligned} \quad (3.69)$$

As the equation of the line is of the form $\vec{r} = \vec{c} + t\vec{a}$, comparison with (3.69) tells us that

$$\vec{a} = 3\vec{i} + \vec{j} + 2\vec{k}. \quad (3.70)$$

As the equation of the plane is $x + y + 2z = 37$, the normal is

$$\vec{n} = \vec{i} + \vec{j} + 2\vec{k}. \quad (3.71)$$

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Substituting (3.70) and (3.71) into (3.68) gives

$$\begin{aligned}\cos \theta &= \frac{(\vec{i} + \vec{j} + 2\vec{k}) \cdot (3\vec{i} + \vec{j} + 2\vec{k})}{\sqrt{1^2 + 1^2 + 2^2} \sqrt{3^2 + 1^2 + 2^2}} \\ &= \frac{8}{\sqrt{84}} \\ \Rightarrow \quad \theta &= \cos^{-1} \frac{8}{\sqrt{84}} \\ &\approx 29^\circ 12' 21''\end{aligned}$$

and

$$\alpha \approx 90^\circ - 29^\circ 12' 21'' = 60^\circ 47' 39''.$$

Example 3.15 Given a plane

$$x + y = 1$$

and the straight line

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \quad t \in (-\infty, \infty),$$

determine for which values of a the line is contained in the plane and for which values of a the line and the plane have no common points of intersection.

Solution

To find the points of intersection between the line and the plane, we set

$$x = a + t, \quad y = 1 - t, \quad z = -1 + 2t$$

and substitute into the equation of the plane

$$a + t + 1 - t = 1, \Rightarrow a = 0.$$

Consequently, if $a = 0$, the line is contained in the plane. Otherwise, the line and the plane are parallel and have no common points of intersection.

3.2.6 Parametric equation of a plane

Next we construct a parametric equation of a plane. Assume that a plane is passing through a point $E = (e_x, e_y, e_z)$. In addition, assume that two linearly independent vectors

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$\vec{v}$ and $\vec{w}$ are parallel to the plane. The position vector $\overline{OP}$ of an arbitrary point $P = (x, y, z)$ on a plane can be represented as

$$\overline{OP} = \overline{OE} + t\vec{v} + s\vec{w}, \quad t, s \in (-\infty, \infty).$$

or, equivalently

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} e_x \\ e_y \\ e_z \end{bmatrix} + t \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} + s \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix}, \quad t, s \in (-\infty, \infty). \quad (3.72)$$

This equation is known as the parametric equation of the plane.

Example 3.16 A plane passes through $(1, 2, 3)$ and is parallel to vectors $\vec{i}$ and $\vec{j} + \vec{k}$. Determine the parameteric equation of the plane and reduce it to the equation of the plane without parameters.

Solution The parameteric equation of the plane is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad t, s \in (-\infty, \infty).$$

In order to obtain the equation of the same plane without parameters, we calculate the vector $\vec{n}$ that is normal to the plane

$$\vec{n} = \vec{i} \times (\vec{j} + \vec{k}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{vmatrix} = -\vec{j} + \vec{k}.$$

Now the equation of the plane can be written as

$$-y + z = d.$$

If we set $x = 1$, $y = 2$ and $z = 3$, we obtain $d = 1$.

3.2.7 Intersection of planes

In three dimensions, any three planes may either intersect at one point, or intersect along a line, or coincide, or have no common points of intersection. Any two planes may either intersect along a line, or coincide, or have no common points of intersection.

Common points of intersection of any two (three) planes can be found by solving a system of two (three) simultaneous linear equations. One method of solution is the Gaussian elimination, as discussed earlier in Section 1.7.4.

Example 3.17 In each of the problems given below, find whether and how the given planes intersect.

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1. $x + y - 2z = -2$, $2x - y + z = 9$, $x + 4y + z = 1$
2. $x + 3y + z = 6$, $x + 2y - z = 5$, $3x + 8y + z = 10$
3. $x + 3y + z = 4$, $2x + 2y - z = 6$,
4. $x + y - z = 3$, $2x - y - z = 0$, $3x + 9y - 5z = 21$

Solution.

1. The equations are:

$$x + y - 2z = -2$$

$$2x - y + z = 9$$

$$x + 4y + z = 1.$$

We use Gaussian elimination on the matrix of coefficients

$$M = \left[\begin{array}{ccc|c} 1 & 1 & -2 & -2 \\ 2 & -1 & 1 & 9 \\ 1 & 4 & 1 & 1 \end{array} \right]$$

We give the row operations here just in this example and leave the reader to work them out in parts (b) to (d).

$R2 \rightarrow R2 - 2R1$, along with $R3 \rightarrow R3 - R1$.

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & -2 \\ 0 & -3 & 5 & 13 \\ 0 & 3 & 3 & 3 \end{array} \right].$$

Now $R3 \rightarrow R3/3$, then swap $R2$ and $R3$:

$$\left[\begin{array}{ccc|c} 1 & 1 & -2 & -2 \\ 0 & 1 & 1 & 1 \\ 0 & -3 & 5 & 13 \end{array} \right].$$

$R1 \rightarrow R1 - R2$, along with $R3 \rightarrow R3 + 3R2$:

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -3 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 8 & 16 \end{array} \right]$$

$R3 \rightarrow R3/8$, followed by $R2 \rightarrow R2 - R3$ and $R1 \rightarrow R1 + 3R3$:

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right],$$

from which we can read the solution $x = 3$, $y = -1$, $z = 2$. Hence, the planes intersect at the point $(3, -1, 2)$. Check: Substitute these coordinates into all three of the original equations to make sure that the point $(3, -1, 2)$ is on all three planes.

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2. The equations are

$$\begin{aligned}x + 3y + z &= 6, \\x + 2y - z &= 5, \\3x + 8y + z &= 10.\end{aligned}$$

$$M = \left[\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ 1 & 2 & -1 & 5 \\ 3 & 8 & 1 & 10 \end{array} \right]$$

Two simple row operations along with multiplication of rows by -1 bring this to the form

$$\left[\begin{array}{ccc|c} 1 & 3 & 1 & 6 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 2 & 8 \end{array} \right]$$

from which we see that $y + 2z = 1$ as well as $y + 2z = 8$. This is impossible, so the equations are inconsistent. There is no solution, and the three planes do not intersect at a single point. This can happen for a number of reasons. The planes might be (i) all parallel and non-coincident, (ii) two of them parallel and non-coincident, with the third intersecting both in lines or (iii) a roof-like shape consisting of two sloping sides and a base, i.e. three sides of a triangular cylinder, in which each pair intersects in a line but there is no common intersection of all three planes.

3. Number the equations as follows:

$$x + 3y + z = 4 \tag{3.73}$$

$$2x + 2y - z = 6. \tag{3.74}$$

In this case, we have only two equations, each representing a plane in 3D space.

If (3.73) was exactly the same as (3.74) (or a multiple of (3.74)), then we would really have only one plane, and the intersection would consist of all the points in the plane; but that is not what we have here.

If the left-hand side of (3.73) was a constant k times the left-hand side of (3.74), but the right-hand side of (3.73) was not k times the right-hand side of (3.74), we would have two parallel but unequal planes, which would never meet, and there would be no points of intersection; but, again, that is not what we have here.

So we must have two non-parallel planes. Any two non-parallel planes in 3D space must intersect along a straight line. We now proceed to find the parametric equations of that line. Here

$$M = \left[\begin{array}{ccc|c} 1 & 3 & 1 & 4 \\ 2 & 2 & -1 & 6 \end{array} \right]$$

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Row reduction turns this into

$$\left[\begin{array}{ccc|c} 1 & 0 & -\frac{5}{4} & \frac{5}{2} \\ 0 & 1 & \frac{3}{4} & \frac{1}{2} \end{array} \right]$$

which can only be solved to the extent that

$$x = \frac{5}{2} + \frac{5z}{4}, \quad y = \frac{1}{2} - \frac{3z}{4}.$$

We rename z as the parameter t to obtain the result that the planes intersect in the straight line whose parametric equations are

$$x = \frac{5}{2} + \frac{5t}{4}, \quad y = \frac{1}{2} - \frac{3t}{4}, \quad z = t.$$

4. Proceeding as before,

$$M = \left[\begin{array}{ccc|c} 1 & 1 & -1 & 3 \\ 2 & -1 & -1 & 0 \\ 3 & 9 & -5 & 21 \end{array} \right]$$

A few row operations bring M to the form

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 3 \\ 0 & 3 & -1 & 6 \\ 0 & 3 & -1 & 6 \end{array} \right]$$

Note the second and third rows contain the same information. This means that one of our original equations was redundant, i.e. it was a combination of the other two equations.

Further reduction gives

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 3 \\ 0 & 3 & -1 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

In the language of the final section of this chapter, the rank of the matrix is only 2. By continuing the reduction as far as possible to

$$\left[\begin{array}{ccc|c} 1 & 0 & -\frac{2}{3} & 1 \\ 0 & 1 & -\frac{1}{3} & 2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

or by immediately solving then back-substituting, we obtain a solution (with z chosen as the parameter) $x = 1 + \frac{2t}{3}$, $y = 2 + \frac{t}{3}$, $z = t$, i.e. we find that the planes intersect in the straight line whose parametric equations are:

$$x = 1 + \frac{2t}{3}, \quad y = 2 + \frac{t}{3}, \quad z = t.$$

Exercises

1. Find the equation of the plane which passes through $(4, 7, 8)$ and is normal to $2\vec{i} + 3\vec{j} + \vec{k}$; hence find the plane's (perpendicular) distance from the origin.
2. Find the equation of the plane normal to the vector $2\vec{i} + \vec{j} - 3\vec{k}$ and a distance 5 from the origin.
3. For the plane defined by $4x + 2y + 4z = 9$, find
 - a) a unit vector normal to the plane,
 - b) the perpendicular distance from the plane to the origin, and
 - c) the intercepts of the plane with the axes.
 - d) the distance between the plane in (a) and the plane $4x + 2y + 4z = -6$.
4. Repeat Question 3(a)–(c) for the plane $x - y + z = 0$.
5. Which of the planes P_1 and P_2 below passes closer to the origin?
 $P_1 = \{(x, y, z) : -4x + y - 3z = 9\};$
 $P_2 = \{(x, y, z) : x + y + z = 3\}.$
6. Find the angle between the planes $5x + y + z = 1$ and $3x + 3y + 3z = 5$.
7. Find the equation of the plane which passes through the points $(1, 6, 1)$, $(3, 5, 2)$, and $(3, 2, 1)$.
8. Find the equation of the plane which passes through $(3, 0, 4)$ and is normal to $2\vec{i} + \vec{j} - 3\vec{k}$.
9. Find the equation of the plane which passes through the points $(1, 1, 0)$, $(1, 0, 1)$, and $(0, 0, 1)$.
10. For the line $x = 2 + t$, $y = 3 - t$, $z = 1 + 2t$, and the plane $x + y + 3z = 4$, find
 - a) the point of intersection, and
 - b) the angle between the line and the plane.
11. In each of the problems given below, find where the given planes intersect.
 - a) $x - 3y + z = -2$ $2x + y + z = 7$ $x - 4y + 2z = -1$

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b) $x + 4y - 3z = 2$ $x + 2y + z = 5$ $x + 3y - z = 4$

c) $x + 2y + z = 3$ $2x + y - z = 4$

d) $2x - y - z = -1$ $x + 3y + z = 2$

Give the line in two dimensions $l_1 : y = 1 - 2x$,

12. a) determine the distance from $(1, 1)$ to the line l_1 ,
 b) find a vector $\vec{v}$, orthogonal to the line l_1
 c) determine the equation of a line that passes through $(1, 1)$ in the direction orthogonal to the line l_1 .

Answers

1. $2x + 3y + z = 37$, $\frac{37}{\sqrt{14}}$

2. $2x + y - 3z = 5\sqrt{14}$ or $2x + y - 3z = -5\sqrt{14}$

3. a) $\frac{1}{3}(2\vec{i} + \vec{j} + 2\vec{k})$

b) $\frac{3}{2}$

c) $\frac{9}{4}, \quad \frac{9}{2}, \quad \frac{9}{4}$

d) $\frac{5}{2}$

4. a) $\frac{1}{\sqrt{3}}(\vec{i} - \vec{j} + \vec{k})$

b) 0

c) $(0, 0, 0)$

5. The distance of P_1 is about 1.77. The distance of P_2 is about 1.73. Hence, P_2 passes closer to the origin.

6. $38^\circ 56' 33''$

7. $2x + y - 3z = 5$

8. $2x + y - 3z = -6$

9. $y + z = 1$

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10. (a) When $t = -\frac{2}{3}$, $(\frac{4}{3}, \frac{11}{3}, -\frac{1}{3})$ (b) $47^\circ 36' 29''$

11. a) $(1, 2, 3)$

b) do not intersect

c) $x = \frac{1}{3}(5 + 3t)$, $y = \frac{1}{3}(2 - 3t)$, $z = t$

d) $x = \frac{1}{7}(2t - 1)$, $y = \frac{1}{7}(5 - 3t)$, $z = t$

12. a) $2/\sqrt{5}$

b) $\vec{v} = 2\vec{i} + \vec{j}$

c) $1 + x = 2y$

4 Linear transformations

Any linear transformation can be defined as a certain mapping T that transforms points or vectors from $\mathbb{R}^n$ into some new points or vectors in $\mathbb{R}^m$. Mathematically, this sentence can be written as

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m,$$

where T is the name of the transformation, $\mathbb{R}^n$ is the input space and $\mathbb{R}^m$ is the output space.

More precisely we define

Definition 4.1 A mapping T

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

is called from $\mathbb{R}^n$ into $\mathbb{R}^m$ if for any two vectors $\vec{u}$ and $\vec{w}$ from $\mathbb{R}^n$ and any nonzero real number c the following conditions are satisfied

$$\begin{aligned} T(\vec{u} + \vec{w}) &= T(\vec{u}) + T(\vec{w}), \\ T(c\vec{u}) &= cT(\vec{u}), \\ T(\vec{0}) &= \vec{0}. \end{aligned} \tag{4.75}$$

Any vector $\vec{v}$ is mapped by T into a new vector $T(\vec{v})$, which is called the image of $\vec{v}$.

Example 4.1 Show that a mapping

$$T : \mathbb{R}^1 \rightarrow \mathbb{R}^1$$

defined by

$$T(x) = \alpha x,$$

with some real $\alpha \neq 0$ is a linear transformation. Such a transformation is called .

Solution

We need to show that all three conditions (4.75) are satisfied. Thus we have

$$\begin{aligned} T(x + y) &= \alpha(x + y) = \alpha x + \alpha y = T(x) + T(y), \\ T(cx) &= \alpha cx = c\alpha x = cT(x), \\ T(0) &= \alpha \times 0 = 0. \end{aligned}$$

Therefore, the mapping T is a linear transformation.

4 Linear transformations

Example 4.2 Show that a mapping

$$T : \mathbb{R}^1 \rightarrow \mathbb{R}^1$$

defined by

$$T(x) = x + \alpha,$$

with some real $\alpha \neq 0$ is not a linear transformation. Such a transformation is called .

Solution

Let us check if the third condition in (4.75) is satisfied

$$T(0) = 0 + \alpha = \alpha \neq 0.$$

Therefore, the mapping T is not a linear transformation.

Theorem 4.1 (Matrix of a linear transformation) Any linear transformation T

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m,$$

can be represented as a right-hand side product of a constant $m \times n$ matrix A with any vector $\vec{v}$ from $\mathbb{R}^n$

$$T(\vec{v}) = A\vec{v} \rightarrow \vec{w} \in \mathbb{R}^m.$$

Example 4.3 Consider the mapping T

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^1$$

such that

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 1$$

and

$$T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = 1$$

Find the matrix of this linear transformation and determine the image of vector $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ under this transformation.

Solution

The matrix A of the transformation from $\mathbb{R}^2$ to $\mathbb{R}^1$ is a 1×2 matrix

$$A = \begin{bmatrix} a_1 & a_2 \end{bmatrix}$$

In order to find a_1 and a_2 we note that

$$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} a_1 & a_2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = a_1 = 1$$

4 Linear transformations

and

$$T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} a_1 & a_2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = a_1 + a_2 = 1$$

Therefore, $a_1 = 1$ and $a_2 = 0$.

The image of $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ under the transformation T is

$$T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2.$$

Example 4.4 Find the matrix A of the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$T\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad T\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Solution In order to find the 2×2 transformation matrix A , we represent an arbitrary vector

$$\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$$

as a linear combination of the vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} x \\ y \end{bmatrix} = a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \end{bmatrix},$$

where a and b must be solved for.

Thus we have

$$a - b = x, \quad 2a + b = y.$$

To solve for a and b we use Gaussian elimination and row reduced the augmented matrix to obtain

$$\left[\begin{array}{cc|c} 1 & -1 & x \\ 2 & 1 & y \end{array} \right] \Rightarrow \left[\begin{array}{cc|c} 1 & -1 & x \\ 0 & 3 & y - 2x \end{array} \right].$$

The solution is given by

$$b = \frac{y - 2x}{3}, \quad a = x + b = x + \frac{y - 2x}{3} = \frac{x + y}{3}.$$

Next, we find the image of $\begin{bmatrix} x \\ y \end{bmatrix}$ under T . Taking into account the properties of linear transformations

$$\begin{aligned} T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) &= T\left(a \begin{bmatrix} 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) = aT\left(\begin{bmatrix} 1 \\ 2 \end{bmatrix}\right) + bT\left(\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) \\ &= \frac{x + y}{3} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \frac{y - 2x}{3} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3x \\ 2y - x \end{bmatrix}. \end{aligned}$$

4 Linear transformations

Finally, we find the transformation matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

According to the definition of A , the following identity holds for any x and y

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = A \begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3x \\ 2y - x \end{bmatrix}.$$

Consequently

$$A = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ -1 & 2 \end{bmatrix}.$$

Alternative solution is obtained by applying T with some unknown transformation matrix A

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

on vectors $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

Thus we have

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} a_{11} + 2a_{12} = 1 \\ a_{21} + 2a_{22} = 1 \end{bmatrix}$$

and

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -a_{11} + a_{12} = -1 \\ -a_{21} + a_{22} = 1 \end{bmatrix}.$$

Combining the equations for a_{11}, a_{12} and for a_{21}, a_{22} we obtain two linear systems

$$\begin{aligned} a_{11} + 2a_{12} &= 1 \\ -a_{11} + a_{12} &= -1 \end{aligned}$$

and

$$\begin{aligned} a_{21} + 2a_{22} &= 1 \\ -a_{21} + a_{22} &= 1. \end{aligned}$$

These systems can be solved using a method of your choice. For example, one could use Cramer's rule to obtain

$$a_{11} = \frac{\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix}} = 1, \quad a_{12} = \frac{\begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}} = 0, \quad a_{21} = \frac{\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}} = -\frac{1}{3}, \quad a_{22} = \frac{\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix}}{\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}} = \frac{2}{3}.$$

Note that the $n \times n$ matrix A of a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is uniquely determined if the action of T on any n linearly independent vectors is known.

Example 4.5 Find the matrix of a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by

$$T\left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad T\left(\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad T\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Solution First, we represent an arbitrary vector $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ as a linear combination

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

This implies

$$x = a + c, \quad y = b, \quad z = c - b - a.$$

We can now express a, b, c in terms of x, y, z

$$a = \frac{x - y - z}{2}, \quad b = y, \quad c = \frac{x + y + z}{2}.$$

Using the properties of linear transformations, we obtain

$$\begin{aligned} T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) &= T\left(a \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right) \\ &= aT\left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}\right) + bT\left(\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}\right) + cT\left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}\right) \\ &= \frac{x - y - z}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \frac{x + y + z}{2} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} x/2 - y/2 - z/2 \\ x/2 + y/2 - z/2 \\ x + y \end{bmatrix}. \end{aligned}$$

The transformation matrix A is given by

$$A = \begin{bmatrix} 1/2 & -1/2 & -1/2 \\ 1/2 & 1/2 & -1/2 \\ 1 & 1 & 0 \end{bmatrix}.$$

4 Linear transformations

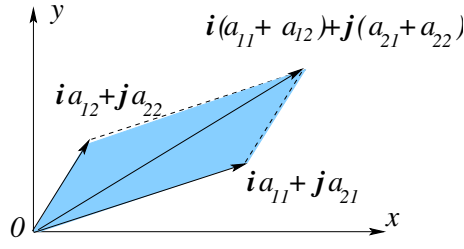


Figure 4.1: Image of a square with vertices in $(0,0)$, $(0,1)$, $(1,0)$ and $(1,1)$ under the linear transformation T , with the transformation matrix A from Eq. (4.76). The area of the parallelogram is $|\det(A)|$.

Theorem 4.2 (Geometric meaning of a linear transformation) Under the linear transformation T with matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad (4.76)$$

the rectangle (square) with the vertices in $(0,0)$, $(0,1)$, $(1,0)$ and $(1,1)$ transforms to a parallelogram, built on the vectors

$$\begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix}, \text{ and } \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix}.$$

The area of the parallelogram is given by

$$S' = |\det(A)|. \quad (4.77)$$

In order to illustrate the theorem, we calculate the images of the vertices of the rectangle under T

$$\begin{aligned} T \begin{pmatrix} \vec{j} \end{pmatrix} &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = a_{12}\vec{i} + a_{22}\vec{j}, \\ T \begin{pmatrix} \vec{i} \end{pmatrix} &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = a_{11}\vec{i} + a_{21}\vec{j}, \\ T \begin{pmatrix} \vec{i} + \vec{j} \end{pmatrix} &= \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} + a_{12} \\ a_{21} + a_{22} \end{bmatrix} = (a_{11} + a_{12})\vec{i} + (a_{21} + a_{22})\vec{j}. \end{aligned}$$

The relative position of the images $T \begin{pmatrix} \vec{i} \end{pmatrix}$, $T \begin{pmatrix} \vec{j} \end{pmatrix}$ and $T \begin{pmatrix} \vec{i} + \vec{j} \end{pmatrix}$ is schematically shown in Figure 4.1. Clearly, the image of the square S is a parallelogram, because

$$T \begin{pmatrix} \vec{i} + \vec{j} \end{pmatrix} = T \begin{pmatrix} \vec{i} \end{pmatrix} + T \begin{pmatrix} \vec{j} \end{pmatrix}.$$

4 Linear transformations

The area of the parallelogram S' can be found using the cross product of $\vec{v} = T\left(\begin{smallmatrix} \vec{i} \\ \vec{j} \end{smallmatrix}\right)$ and $\vec{w} = T\left(\begin{smallmatrix} \vec{j} \\ \vec{k} \end{smallmatrix}\right)$

$$S' = |\vec{v} \times \vec{w}| = \left| \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_{11} & a_{12} & 0 \\ a_{21} & a_{22} & 0 \end{vmatrix} \right| = |\vec{k}(a_{11}a_{22} - a_{12}a_{21})| = |\det(A)|.$$

Theorem 4.3 Let S be a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$, given by $m \times n$ matrix A and let T be a linear transformation from $\mathbb{R}^m$ to $\mathbb{R}^k$, given by $k \times m$ matrix B . Then TS is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^k$, given by the matrix C such that

$$C = BA.$$

Example 4.6 Linear transformation T is defined as the operation of interchanging the coordinates of a two-dimensional vector $\vec{u} \in \mathbb{R}^2$, i.e.

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} y \\ x \end{bmatrix}.$$

Linear transformation S maps a two-dimensional vector into a scalar by taking its projection onto the y -axis. Determine the matrix of the linear transformation ST .

Solution

First let us determine the matrix A of the transformation T

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} y \\ x \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix},$$

so that

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

The matrix B that corresponds to the projection onto the y -axis is

$$B = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

Consequently, the matrix C of the transformation ST is given by

$$C = BA = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

4 Linear transformations

Theorem 4.4 Let T be a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$, given by the 2×2 matrix A .

Under the linear transformation T , the image of any parallelogram of area S is a new parallelogram. The area S' of the image is

$$S' = S |\det(A)|.$$

Proof: Any parallelogram can be seen as an image of a square with vertices in $(0,0)$, $(0,1)$, $(1,0)$ and $(1,1)$ under a certain linear transformation. Let A_1 be the linear transformation matrix that transforms a square into a parallelogram of the area S . Then according to Eq.(4.77)

$$S = |\det(A_1)|.$$

The sequence of two linear transformations with matrices A_1 and A (A_1 followed by A) transforms the square $(0,0)$, $(0,1)$, $(1,0)$ and $(1,1)$ into the parallelogram of the area S' . Therefore, using the property of determinants ($\det(AB) = \det(A)\det(B)$), we obtain

$$S' = |\det(AA_1)| = |\det(A)| |\det(A_1)| = S |\det(A)|.$$

Example 4.7 A linear transformation T is given by the matrix

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}.$$

Let B be a rectangle bounded by $x = 1$, $x = 2$, $y = 1$ and $y = 5$. What is the area of B ? What is the image of B under T ? What is the area of the image of B under T ?

Solution

The area of B is

$$S = (2 - 1) \times (5 - 1) = 4.$$

Under T , the rectangle transforms to a parallelogram with the vertices in

$$\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \end{bmatrix} = \begin{bmatrix} 11 \\ 14 \end{bmatrix},$$

$$\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 \\ 13 \end{bmatrix}.$$

The area of the image is

$$S' = S \times |\det(A)| = 4 \times 5 = 20.$$

4 Linear transformations

Definition 4.2 Let T be a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$, given by the $n \times n$ matrix A

$$T(\vec{v}) = A\vec{v} = \vec{w}$$

The transformation T is called *if* the matrix A is invertible, i.e if A^{-1} exists. The transformation T^{-1} is then given by

$$T^{-1}(\vec{w}) = A^{-1}\vec{w} = \vec{v}.$$

Example 4.8 Linear transformation T is defined as the operation of interchanging the coordinates of a two-dimensional vector $\vec{u} \in \mathbb{R}^2$, i.e.

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} y \\ x \end{bmatrix}.$$

Determine if the transformation is invertible and if it is, find the matrix of the inverse transformation.

Solution

The matrix A of the forward transformation is

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Because $\det(A) = -1 \neq 0$, the inverse matrix exists and is given by

$$A^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = A.$$

Note that

Under linear transformations, the image of a line is a new line and the image of a plane is a new plane.

The following examples demonstrate how the the image of a line or the image of the plane can be found, if the transformation matrix is known.

Example 4.9 Find the image of the line $y = x$ under the linear transformation T from Example 4.4.

Solution

(Method of the inverse transformation). The transformation matrix A is given by

$$A = \frac{1}{3} \begin{bmatrix} 3 & 0 \\ -1 & 2 \end{bmatrix}.$$

4 Linear transformations

Because $\det(A) = \frac{6}{9} = \frac{2}{3} \neq 0$, the transformation T is invertible. The matrix of the inverse transformation is given by A^{-1}

$$A^{-1} = \frac{3}{2} \frac{1}{3} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}.$$

Let the image of $\begin{bmatrix} x \\ y \end{bmatrix}$ under T be

$$T : \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix}$$

so that

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix}$$

and the inverse transformation can be written as

$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2x' \\ x' + 3y' \end{bmatrix}.$$

Now we substitute x by x' and y by $\frac{1}{2}(x' + 3y')$ in the equation of the line $x = y$ and obtain its image under T

$$x' = \frac{x' + 3y'}{2}.$$

Or, equivalently

$$y' = \frac{x'}{3}.$$

Alternatively, the image of the line can be found using the **method of the forward transformation**. Firstly, we need to rewrite the equation of the line $y = x$ in the parametric form. Using Eq. (3.46), we obtain

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad t \in (-\infty, \infty).$$

Now we are ready to find the image of an arbitrary point $\begin{bmatrix} x \\ y \end{bmatrix}$ on the line $y = x$

$$\begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix} = \frac{t}{3} \begin{bmatrix} 3 & 0 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{t}{3} \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Finally, eliminating the parameter t , we find

$$x' = t, \quad y' = \frac{t'}{3} = \frac{x'}{3}.$$

4 Linear transformations

Example 4.10 Find the image of the plane $x + y + z = 1$ under the transformation T with the matrix

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution

(Method of the inverse transformation). Because $\det(A) = 1 \neq 0$, the transformation T is invertible. The matrix of the inverse transformation is

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -1 \\ 1 & 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}.$$

Let the image of $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ under T be

$$T : \begin{bmatrix} x \\ y \\ z \end{bmatrix} \Rightarrow \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}$$

so that

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = A \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

and the inverse transformation can be written as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} x' - y' + z' \\ -x' + y' \\ -y' + z' \end{bmatrix}.$$

Now we substitute x by $x' - y' + z'$, y by $-x' + y'$ and z by $-y' + z'$ in the equation of the plane $x + y + z = 1$ and obtain its image under T

$$x' - y' + z' - x' + y' + z' - y' = 2z' - y' = 1.$$

Alternatively, the image of the plane can be found using the **method of the forward transformation**. First, we need to rewrite the equation of the plane $x + y + z = 1$ in the parametric form. According to Eq. (3.72), we require an arbitrary point E on the plane and two linearly independent vectors $\vec{v}$ and $\vec{w}$ that are parallel to the plane. The vectors $\vec{v}$ and $\vec{w}$ can be found using any three points on the plane that do not fall onto one line. It is convenient to choose the following points: $E = (0, 0, 1)$, $A = (1, 0, 0)$ and $B = (0, 1, 0)$. Therefore, we may choose

$$\vec{v} = \overrightarrow{EA} = \vec{i} - \vec{k}, \quad \vec{w} = \overrightarrow{EB} = \vec{j} - \vec{k}.$$

4 Linear transformations

The parametric equation of the plane can now be written as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

The image of an arbitrary point on the plane is found according to

$$\begin{aligned} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &\Rightarrow \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 1 & 1 & 0 \end{bmatrix} \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + s \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right) \\ &= \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} + s \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}. \end{aligned}$$

The final step is to transform the parametric equation of the image back into the Cartesian form. This is done by calculating the normal vector to the image

$$\vec{n} = (2\vec{i} + 2\vec{j} + \vec{k}) \times (\vec{i} + 2\vec{j} + \vec{k}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 2 & 1 \\ 1 & 2 & 1 \end{vmatrix} = -\vec{j} + 2\vec{k}.$$

Now the Cartesian equation of the image can be written as

$$-y' + 2z' = d.$$

Substituting $(-1, -1, 0)$, we finally obtain $d = 1$. The equation of the image of $x + y + z = 1$ is

$$2z' - y' = 1.$$

In what follows we discuss some special cases of linear transformation in two dimensional space.

4.1 Rotation

Consider the rotation of a point $P = (x, y)$ clockwise by angle α , as shown in Fig. 4.2.

The old coordinates of $P = (x, y)$ can be related to the length of the vector $\overline{OP}$ and the angle Ψ as follows

$$x = \rho \cos \Psi, \quad y = \rho \sin \Psi.$$

Since the rotation preserves the length of the vector $\overline{OP}$, the new coordinates of the point $P' = (x', y')$ are

$$\begin{aligned} x' &= \rho \cos(\Psi - \alpha) = \rho \cos(\Psi) \cos(\alpha) + \rho \sin(\Psi) \sin(\alpha) = x \cos \alpha + y \sin \alpha, \\ y' &= \rho \sin(\Psi - \alpha) = \rho \sin(\Psi) \cos(\alpha) - \rho \cos(\Psi) \sin(\alpha) = y \cos \alpha - x \sin \alpha. \end{aligned}$$

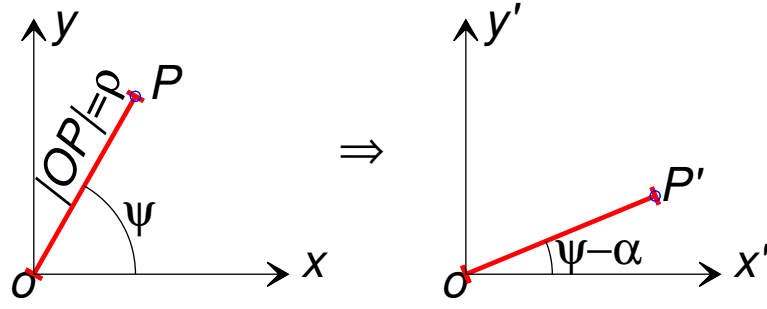


Figure 4.2: Rotation clockwise by angle α .

Rewriting this equation in the matrix form, we obtain

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Consequently, the matrix R_α that corresponds to this linear transformation is given by

$$R_\alpha = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \quad (4.78)$$

Note that $\det(R_\alpha) = \cos^2 \alpha + \sin^2 \alpha = 1$. This implies that the rotation is an invertible transformation. The inverse transformation is given by R_α^{-1}

$$R_\alpha^{-1} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}.$$

Thus, we conclude that

$$R_\alpha^{-1} = R_{(-\alpha)} = R_\alpha^T.$$

Definition 4.3 Linear transformation, given by a real matrix A is called if

$$A^{-1} = A^T.$$

Note that in some text books, the rotation is applied on to the coordinate axes, while the points in space remain fixed. Clearly, the rotation of the coordinate axes by angle α clockwise corresponds to the rotation of points by α anti-clockwise, with the coordinates axes remaining fixed.

- Any orthogonal transformation T preserves the distance between two points.
- If A is the matrix of an orthogonal transformation, then $\det(A) = \pm 1$. (Note that the converse is not true).

Indeed, let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be represented by the $n \times n$ orthogonal matrix A . Let $\vec{v}$ be associated with two given points in $\mathbb{R}^n$, so that $|\vec{v}| = d$ is the distance between the points. Under T the vector $\vec{v}$ transforms according to

$$\vec{v} \rightarrow \vec{v}' = A\vec{v}.$$

The square magnitude of the vector $\vec{v}$ can be written as

$$|\vec{v}|^2 = \vec{v}^T \vec{v}.$$

The following line proves that the distance $|\vec{v}| = d$ is preserved under T

$$d^2 = \vec{v}^T \vec{v} = \vec{v}^T (A^{-1} A) \vec{v} = \vec{v}^T (A^T A) \vec{v} = (A\vec{v})^T \vec{v}' = (\vec{v}')^T \vec{v}' = d'^2.$$

Example 4.11 Find the image of the ellipse

$$x^2 + 2y^2 = 1$$

after rotating it by $\pi/4$ anti-clockwise.

Solution

First, we need to determine the rotation matrix A . Clearly, A is given by (4.78) with $\alpha = -\pi/4$

$$A = \begin{bmatrix} \cos(-\pi/4) & \sin(-\pi/4) \\ -\sin(-\pi/4) & \cos(-\pi/4) \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

The forward transformation is

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} (x - y)/\sqrt{2} \\ (x + y)/\sqrt{2} \end{bmatrix}.$$

In order to find the image of the ellipse $x^2 + 2y^2 = 1$, we need to find the inverse transformation, i.e. we need to express the old coordinates (x, y) in terms of the new coordinates (x', y') . Thus we have

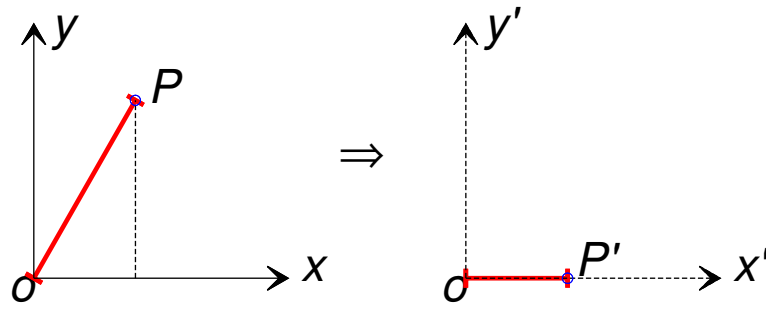
$$\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} (x' + y')/\sqrt{2} \\ (y' - x')/\sqrt{2} \end{bmatrix}.$$

Finally, replacing x by $(x' + y')/\sqrt{2}$ and y by $(y' - x')/\sqrt{2}$ in $x^2 + 2y^2 = 1$, we obtain the image of the ellipse

$$T : x^2 + 2y^2 = 1 \Rightarrow ((x' + y')/\sqrt{2})^2 + 2((y' - x')/\sqrt{2})^2 = 1$$

After some simplifications the new equation becomes

$$(x' + y')^2 + 2(y' - x')^2 = 2.$$

Figure 4.3: Projection onto the x -axis.

4.2 Projection

Consider the operation of projection of a point $P = (x, y)$ onto the x -axis. Schematically, the result of this operation is shown in Fig. 4.3. Projection onto the x -axis can be regarded as a linear transformation T

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

given by

$$T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x \\ 0 \end{bmatrix}$$

The matrix P_x that corresponds to this transformation is given by

$$P_x = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Projection is not a reversible transformation, because $\det(P_x) = 0$. Geometrically, this means that one cannot *undo* the projection, because when the point $P = (x, y)$ is projected onto the x -axis, the information about the y coordinate is lost and cannot be recovered.

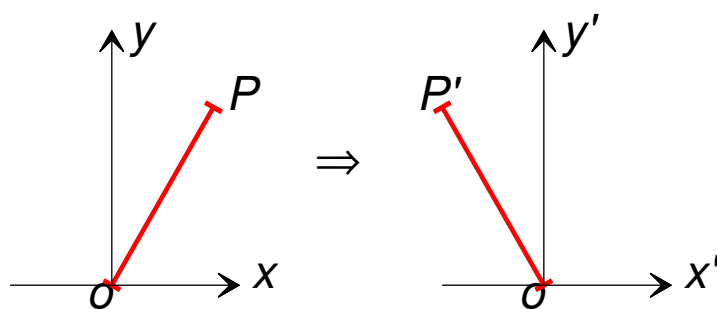
Example 4.12 Find the image of line $y = x$ under the projection onto the x -axis.

Solution

The forward transformation is written as

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = P_x \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix}$$

In order to find the image of the line $y = x$ under this transformation, we observe that each point on the line $(x, y = x)$ is transformed to $(x, 0)$. Therefore, the new equation of the line is $y' = 0$.

Figure 4.4: Reflection through y -axis.

4.3 Reflection

Consider the operation of taking a mirror image of a point $P = (x, y)$ through the y -axis (y). Schematically, the result of this operation is shown in Fig. 4.4. Reflection through the y -axis can be regarded as a linear transformation T

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

given by

$$T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} -x \\ y \end{bmatrix}$$

The matrix M_y that corresponds to this transformation is given by

$$M_y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Since $\det(M_y) = -1 \neq 0$, the operation of reflection is invertible. The inverse transformation matrix M_y^{-1} coincides with M_y

$$M_y^{-1} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Because M_y is also a symmetric matrix, i.e. $M_y = M_y^T$, we conclude that the transformation of reflection is an orthogonal transformation.

Example 4.13 Find the image of the parabola $y = (x - 1)^2$ under the transformation of reflection through the y -axis.

Solution

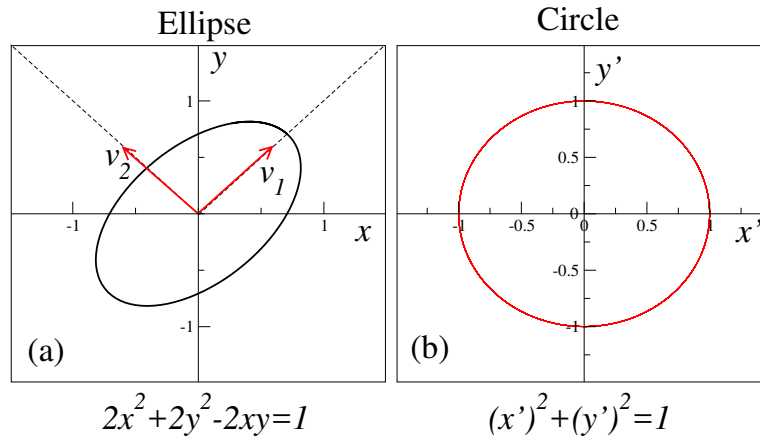


Figure 4.5: Transformation of the ellipse $2x^2 + 2y^2 - 2xy = 1$ (a) under the sequence of transformations in Example 4.14. The image of the ellipse is a circle $x'^2 + y'^2 = 1$ (b).

In order to find the image of the parabola, we need to express the old coordinates (x, y) in terms of the new coordinates (x', y') . This is done by writing down the inverse transformation

$$x = -x', \quad y = y'$$

Therefore, the image of the parabola $y = (x - 1)^2$ is

$$y' = (-x' - 1)^2 = (x' + 1)^2.$$

Example 4.14 Determine the image of the ellipse $2x^2 + 2y^2 - 2xy = 1$ under the following sequence of transformations: rotation by $\pi/4$ clockwise, followed by the reflection with respect to the y -axis, followed by the stretch along the y -axis by the factor of $\sqrt{3}$.

Solution

Rotation by $\pi/4$ clockwise is given by

$$R = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}.$$

Reflection with respect to the y -axis is given by

$$M = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Stretch along the y -axis by the factor of $\sqrt{3}$ is

$$S = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{bmatrix}.$$

4 Linear transformations

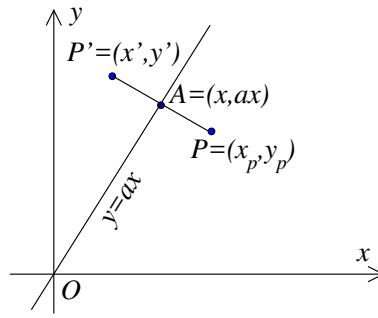


Figure 4.6: Reflection through the line $y = ax$, in Example 4.15. The image of point $P = (x_p, y_p)$ is a new point $P' = (x', y')$ such that $AP = AP'$ and $\overline{PP'}$ is orthogonal to $y = ax$.

Thus, the action of T onto an arbitrary point (x, y) is given by

$$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \frac{-1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ \sqrt{3} & -\sqrt{3} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

The inverse transformation is written as

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{-1}{\sqrt{6}} \begin{bmatrix} \sqrt{3} & 1 \\ \sqrt{3} & -1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}.$$

In order to find the image of the ellipse $2x^2 + 2y^2 - 2xy = 1$, we substitute

$$x = \frac{-1}{\sqrt{6}}(\sqrt{3}x' + y'), \text{ and } y = \frac{-1}{\sqrt{6}}(\sqrt{3}x' - y')$$

to obtain

$$\frac{2}{6} \left((\sqrt{3}x' + y')^2 + (\sqrt{3}x' - y')^2 - (\sqrt{3}x' + y')(\sqrt{3}x' - y') \right) = x'^2 + y'^2 = 1.$$

This is the equation for a circle with radius 1 and centre at the origin. This result is illustrated in Fig. 4.5.

Example 4.15 Determine the transformation matrix of the reflection through the line $y = ax$ that passes through the origin.

Solution As shown in Fig. 4.6, the image of an arbitrary point $P = (x_p, y_p)$ after the reflection through $y = ax$ is a new point $P' = (x', y')$ such that $AP = AP'$ and $\overline{PP'}$ is orthogonal to $y = ax$. First, we find the coordinates of point $A = (x, ax)$ that corresponds to the projection of P onto the line $y = ax$. The vector $\overline{AP} = (x_p - x)\vec{i} + (y_p - ax)\vec{j}$ is orthogonal to the direction of the line $\vec{i} + a\vec{j}$. This gives

$$(x_p - x) + a(y_p - ax) = 0, \rightarrow x = \frac{x_p + ay_p}{1 + a^2}.$$

4 Linear transformations

Next, we notice that A is a mid-point of PP' so that

$$\frac{x' + x_p}{2} = x = \frac{x_p + ay_p}{1 + a^2}, \rightarrow x' = \frac{(1 - a^2)x_p + 2ay_p}{1 + a^2}.$$

Similarly

$$\frac{y' + y_p}{2} = ax = \frac{a(x_p + ay_p)}{1 + a^2}, \rightarrow y' = \frac{2ax_p + (a^2 - 1)y_p}{1 + a^2}.$$

Therefore, the transformation matrix A is given by

$$A = \frac{1}{1 + a^2} \begin{bmatrix} (1 - a^2) & 2a \\ 2a & (a^2 - 1) \end{bmatrix}.$$

The special case of this transformation is the reflection through $y = x$, i.e. $a = 1$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

In this case, the image of $P = (x, y)$ is $P' = (y, x)$.

Note that the reflection through a line $y = ax + b$ that does not pass through the origin, is not a linear transformation. Indeed, consider the reflection through $y = 1$. Clearly, in this case

$$\begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ 2 - y \end{bmatrix}, \text{ if } y < 1.$$

Therefore, the image of $\vec{0}$ is

$$T(\vec{0}) = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \neq \vec{0}.$$

This shows that T is not a linear transformation.

Exercises

1. Determine for which values of a , the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^1$

$$T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = x + y + a$$

is linear.

2. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the transformation defined by

$$T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} 2x \\ 3y \end{bmatrix}$$

- a) Show that T is a linear map.
- b) Write down the 2×2 matrix that represents T .

4 Linear transformations

c) What is the image under T of the square with vertices at $(0,0)$, $(1,0)$, $(1,1)$ and $(0,1)$?

d) Find the image under T of the circle $x^2 + y^2 = 1$.

3. Let T be a linear transformation, represented by the matrix

$$A = \begin{bmatrix} 2 & -5 \\ 1 & -2 \end{bmatrix}.$$

Let B be a rectangle bounded by $x = 2$, $x = 4$, $y = 2$ and $y = 4$. What is the area of B ? What is the image of B under T ? What is the area of the image of B under T ?

4. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by

$$T \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \begin{bmatrix} x + 3y \\ 3x - y \end{bmatrix}$$

Let R be the triangular region in the (x,y) -plane with vertices at $(0,0)$, $(1,0)$ and $(0,2)$.

a) Find the image R' of R under T .

b) Calculate the area of R' .

5. Suppose that a linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is such that

$$T \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad T \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix} \quad \text{and} \quad T \left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$$

What is the image of the plane $x - 3y - 4z = 1$ under T ?

6. Find the image of the ellipse $3x^2 + y^2 = 1$ under the sequence of transformations given by: rotation by $\pi/3$ clockwise, followed by the reflection through the y -axis, followed by the rotation by $2\pi/3$ anti-clockwise.

Answers

1. $a = 0$

2. a) Follow the standard procedure.

b)

$$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

c) Rectangle with the vertices in $(0,0)$, $(2,0)$, $(0,3)$ and $(2,3)$.

d) Ellipse

$$\frac{x'^2}{4} + \frac{y'^2}{9} = 1.$$

3. Area of B is 4. The image of B is a parallelogram with vertices in $(-6, -2)$, $(-16, -6)$, $(-2, 0)$ and $(-12, -4)$. Area of the image is $S = 4 \times |\det(A)| = 4$.

4. a) The image is the triangular region R' with vertices at $(0, 0)$, $(1, 3)$ and $(6, -2)$.

b) The area of the image is 10 square units.

5. $3x' - 2z' = 1$

6. $3x'^2 + y'^2 = 1$

4.4 Stretching without rotation: the eigenvalues and eigenvectors

As we have established in the previous sections of this chapter, any constant matrix A can be associated with a linear transformation of vectors. Namely any vector v is mapped into a new vector w according to $v \rightarrow w = Av$. From the geometrical point of view, the transformation of any given vector v can be represented as a combination of stretching followed by rotation or vice versa. Indeed, consider a constant transformation matrix

$$A = \begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix}.$$

The vector $v = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is mapped into

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}.$$

On the other hand, the vector $\begin{bmatrix} 5 \\ 2 \end{bmatrix}$ can also be represented as the rotation of v by 68.2° clockwise, followed by isotropic stretching by factor $\sqrt{5^2 + 2^2}$

$$\begin{bmatrix} 5 \\ 2 \end{bmatrix} = \sqrt{5^2 + 2^2} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

where $\theta = \cos^{-1} \left(2/\sqrt{2^2 + 5^2} \right) \approx 68.2^\circ$. It should be emphasized that in this simple example, the stretching factor $\sqrt{2^2 + 5^2}$ and the rotation angle 68.2° are not unique and depend on the original vector v .

4 Linear transformations

On the other hand, one can find such vectors that only stretch under A without rotating. Thus consider the transformation of the vector $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$:

$$v \rightarrow A \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -v.$$

As we see, the multiplication of A by v from the right is equivalent to reversing the direction of v , or, alternatively, stretching by factor -1 . Such vectors that stretch without rotation under multiplication by A are called of A .

Definition 4.4 Let A be an $n \times n$ real matrix. The $n \times 1$ column vector v such that

$$Av = \lambda v,$$

where λ is a scalar, is called the of A and the corresponding λ is called the of A .

In order to find all such pairs (λ, v) we rearrange $Av = \lambda v$ according to

$$Av - \lambda v = (A - \lambda I)v = 0, \quad (4.79)$$

where I is the identity matrix.

Eq. (4.79) is a homogeneous system of linear equations for the coordinates of the vector v . Using the fundamental results of Chapter 2, we conclude that Eq. (4.79) has a non-trivial solution only if

$$\det(A - \lambda I) = 0. \quad (4.80)$$

Eq. (4.80) is called the for the eigenvalues λ of the matrix A . For an $n \times n$ matrix A , the left-hand-side of Eq. (4.80) is a polynomial of the degree n of λ , i.e. $P_n(\lambda) = 0$. According to the fundamental theorem of algebra, any polynomial of the degree n has exactly n distinct roots, counting the multiplicity. Consequently, any $n \times n$ matrix A has up to n distinct eigenvalues λ . For each specific eigenvalue λ , the corresponding eigenvector v can be found from Eq. (4.79).

We note that if v is an eigenvector that corresponds to eigenvalue λ , then any multiple of v is also an eigenvector with the same eigenvalue. Indeed, let $w = cv$, with some $c \neq 0$, then

$$Aw = Acv = c\lambda v = \lambda w.$$

Definition 4.5 The set of all eigenvalues of A is called the of A . The is determined as $\max(|\lambda_1|, |\lambda_2|, \dots)$.

Let A be an $n \times n$ real matrix. The following properties of the eigenvalues are in order:

4 Linear transformations

- The eigenvalues of matrices A^T and A are identical.

Proof:

$$\det(A - \lambda I) = \det(A - \lambda I)^T = \det(A^T - \lambda I).$$

Note that the eigenvectors of A^T and A may be different.

- The inverse A^{-1} exists only if $\lambda = 0$ is not the eigenvalue of A .

Proof: assume that $\lambda = 0$ is an eigenvalue of A then

$$\det(A - 0I) = \det(A) = 0.$$

Consequently, A is not invertible.

- If λ is an eigenvalue of A , then λ^{-1} is the eigenvalue of the inverse matrix A^{-1} .

Proof: multiply $Av = \lambda v$ by $\lambda^{-1}A^{-1}$ from the left

$$\lambda^{-1}A^{-1}Av = \lambda^{-1}A^{-1}\lambda v \rightarrow A^{-1}v = \lambda^{-1}v.$$

- If λ is an eigenvalue of A , then λ^m is the eigenvalues of A^m , where m is some integer.

Proof:

$$A^m v = A^{m-1}Av = \lambda A^{m-1}v = \lambda^2 A^{m-2}v = \dots = \lambda^m v.$$

- Let $\lambda_1, \lambda_2, \dots$ be n eigenvalues of A counting the multiplicity. Then $\det(A) = \lambda_1 \lambda_2 \dots$
- Let $\lambda_1, \lambda_2, \dots$ be n eigenvalues of A counting the multiplicity. Then the trace of the matrix, determined by $\text{Tr}(A) = a_{11} + a_{22} + \dots + a_{nn}$ is also $\text{Tr}(A) = \lambda_1 + \lambda_2 + \dots$

Example 4.16 In order to find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix},$$

we proceed as follows. Firstly, we solve the characteristic equation Eq. (4.80)

$$\det\left(\begin{bmatrix} 4 & 5 \\ 3 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0$$

$$0 = \det\left(\begin{bmatrix} 4-\lambda & 5 \\ 3 & 2-\lambda \end{bmatrix}\right) = (4-\lambda)(2-\lambda) - 15$$

$$0 = \lambda^2 - 6\lambda - 7$$

$$\lambda = 7, -1.$$

Next we find the eigenvector $v_{\lambda=7} = \begin{bmatrix} x \\ y \end{bmatrix}$ that corresponds to $\lambda = 7$. From Eq. (4.79) we obtain

$$\begin{bmatrix} 4-7 & 5 \\ 3 & 2-7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

4 Linear transformations

Gauss-reducing the augmented matrix

$$\left[\begin{array}{cc|c} -3 & 5 & 0 \\ 3 & -5 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 + R_1} \left[\begin{array}{cc|c} -3 & 5 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

we observe that y can be used as a free variable and write the solution in the vector form

$$\mathbf{v}_{\lambda=7} = \begin{bmatrix} x \\ y \end{bmatrix} = y \begin{bmatrix} 5/3 \\ 1 \end{bmatrix}, y \in (-\infty, \infty).$$

Alternatively, we may rescale the free parameter to eliminate the fractions

$$\mathbf{v}_{\lambda=7} = t \begin{bmatrix} 5 \\ 3 \end{bmatrix}, t \in (-\infty, \infty).$$

Similarly, we find the second eigenvector $\mathbf{v}_{\lambda=-1} = \begin{bmatrix} x \\ y \end{bmatrix}$ that corresponds to $\lambda = -1$. From Eq. (4.79) we obtain

$$\left[\begin{array}{cc} 4 - (-1) & 5 \\ 3 & 2 - (-1) \end{array} \right] \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Gaussian reduction

$$\left[\begin{array}{cc|c} 5 & 5 & 0 \\ 3 & 3 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 5/3R_1} \left[\begin{array}{cc|c} 5 & 5 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

yields the solution in the vector form

$$\mathbf{v}_{\lambda=-1} = \begin{bmatrix} x \\ y \end{bmatrix} = y \begin{bmatrix} -1 \\ 1 \end{bmatrix}, y \in (-\infty, \infty).$$

The above example allows us to discuss geometric interpretation of the eigenvectors. Thus, any eigenvector $\mathbf{v}_{\lambda=7}$ that corresponds to $\lambda = 7$ can be associated with an arbitrary point on the line $y = 3/5x$. Similarly, any eigenvector $\mathbf{v}_{\lambda=-1}$ that corresponds to $\lambda = -1$ is an arbitrary point on the line $y = -x$, as shown in Fig.4.7. Observing that the lines $y = -x$ and $y = 3/5x$ are not parallel, we conclude that the eigenvectors $\mathbf{v}_{\lambda=7}$ and $\mathbf{v}_{\lambda=-1}$ are linearly independent. This is a general result that can be proved as follows. Let $\mathbf{v}$ and $\mathbf{w}$ be any two eigenvectors of A that correspond to two distinct eigenvalues λ and μ , ($\lambda \neq \mu$) respectively. Assume that $\mathbf{v}$ and $\mathbf{w}$ are linearly dependent, which implies that $\mathbf{v} = c\mathbf{w}$, with some scalar $c \neq 0$. Then

$$A\mathbf{v} = Ac\mathbf{w} = cA\mathbf{w} = c\mu\mathbf{w}.$$

On the other hand $A\mathbf{v} = \lambda\mathbf{v}$ so that

$$A\mathbf{v} = \lambda\mathbf{v} = c\mu\mathbf{w} \rightarrow \mathbf{v} = c\frac{\mu}{\lambda}\mathbf{w} \neq c\mathbf{w}.$$

The last result is in contradiction with the assumption $\mathbf{v} = c\mathbf{w}$, which shows that $\mathbf{v}$ and $\mathbf{w}$ are linearly independent.

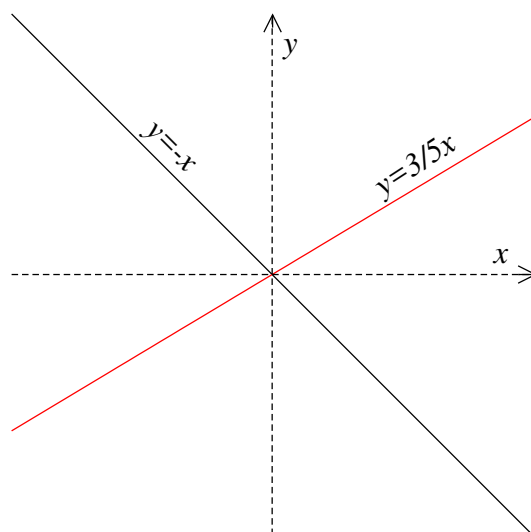


Figure 4.7: Eigenvectors $v_{\lambda=7}$ and $v_{\lambda=-1}$ of A from the example form two lines $y = -x$ and $y = 3/5x$.

4.4.1 Eigenvalues of a symmetric matrix and the spectral theorem

Theorem 4.5 Let A be a symmetric real matrix $A^T = A$.

- The eigenvalues of A are real.
- The eigenvectors that correspond to distinct eigenvalues are orthogonal.
- Any symmetric $n \times n$ matrix has n linearly independent eigenvectors.

We will prove the second statement of the theorem. Let v and w be the eigenvectors of A that correspond to distinct eigenvalues λ and μ , respectively ($\lambda \neq \mu$). Then

$$Av = \lambda v, \quad Aw = \mu w$$

Let us now multiply the first equation by w^T from the left and the second equation by v^T from the left

$$w^T Av = \lambda w^T v, \quad v^T Aw = \mu v^T w.$$

Using the property of transposition $(AB)^T = B^T A^T$, we obtain

$$\mu(v^T w)^T = (v^T Aw)^T = w^T A^T (v^T)^T = w^T Av = \lambda w^T v$$

Since $(v^T w)^T = w^T v$, we see that the last equation is only true for some $\lambda \neq \mu$ if $w^T v = v \cdot w = 0$. This shows that the vectors v and w are orthogonal.

4 Linear transformations

Example 4.17 Determine the eigenvalues and the eigenvectors of the symmetric matrix

$$A = \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}.$$

and demonstrate that the eigenvectors that correspond to distinct eigenvalues are orthogonal.

Solution

Firstly, we solve the characteristic equation Eq. (4.80) to find the eigenvalues

$$\begin{aligned} 0 &= \det \left(\begin{bmatrix} 2-\lambda & 2 \\ 2 & 5-\lambda \end{bmatrix} \right) \\ 0 &= (2-\lambda)(5-\lambda) - 4 = \lambda^2 - 7\lambda + 6 = 0 \\ \lambda &= 6, 1. \end{aligned}$$

Next we find the eigenvector $\mathbf{v}_{\lambda=6} = \begin{bmatrix} x \\ y \end{bmatrix}$ that corresponds to $\lambda = 6$. From Eq. (4.79) we obtain

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Gauss-reducing the augmented matrix

$$\left[\begin{array}{cc|c} -4 & 2 & 0 \\ 2 & -1 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow 2R_2 + R_1} \left[\begin{array}{cc|c} -4 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

we observe that y can be used as a free variable and write the solution in the vector form

$$\mathbf{v}_{\lambda=6} = \begin{bmatrix} x \\ y \end{bmatrix} = y \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}, y \in (-\infty, \infty).$$

Similar we obtain the second eigenvector $\mathbf{v}_{\lambda=1} = \begin{bmatrix} x \\ y \end{bmatrix}$ that corresponds to $\lambda = 1$. From Eq. (4.79) we obtain

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Gauss-reducing the augmented matrix

$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 2 & 4 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

we observe that y can be used as a free variable and write the solution in the vector form

$$\mathbf{v}_{\lambda=1} = \begin{bmatrix} x \\ y \end{bmatrix} = y \begin{bmatrix} -2 \\ 1 \end{bmatrix}, y \in (-\infty, \infty).$$

Finally, we observe that

$$\mathbf{v}_{\lambda=1} \cdot \mathbf{v}_{\lambda=6} = (\mathbf{v}_{\lambda=1})^T \mathbf{v}_{\lambda=6} = \begin{bmatrix} -2 & 1 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1 \end{bmatrix} = 0.$$

4.4.2 Diagonalization of symmetric matrices

In this section we will use the properties of the eigenvectors of a symmetric matrix to transform it to a new diagonal matrix.

Definition 4.6 Square matrices A and B are called *similar* if there exists such an invertible matrix M so that

$$A = M^{-1}BM.$$

The corresponding transformation of matrix B : $B \rightarrow A = M^{-1}BM$ is called the *similarity transformation*.

Let us now introduce the so-called matrix that corresponds to a given symmetric real matrix A .

Definition 4.7 Let A be an $n \times n$ symmetric real matrix, i.e. $A^T = A$. Let $e_1, e_2, \dots, e_n$ be n **normalized** and linearly independent eigenvectors of A that correspond to n distinct eigenvalues $\lambda_i, i = 1, 2, \dots, n$. Then the matrix $M = [e_1, e_2, \dots, e_n]$, whose **columns** are the eigenvectors of A is called for A .

Using the properties of the eigenvectors of a symmetric real matrix, it is possible to show that the modal matrix M is orthogonal 4.3, namely

$$M^{-1} = M^T$$

Indeed, taking into account that the eigenvectors e_i are normalized, i.e. $|e_i| = 1$ and recalling that the eigenvectors e_i that correspond to distinct eigenvalues are orthogonal 4.5, we obtain

$$e_i \cdot e_k = \begin{cases} 0 & i \neq k \\ 1 & i = k \end{cases}.$$

This result allows us to formulate the following important diagonalization theorem

Theorem 4.6 Let A be an $n \times n$ symmetric real matrix with n distinct eigenvalues $\lambda_i, i = 1, 2, \dots, n$. The matrix A is similar to the diagonal matrix $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, formed by the eigenvalues $\lambda_i, i = 1, 2, \dots, n$

$$\Lambda = M^T A M,$$

where M is the modal matrix for A .

To prove this theorem we note that the matrix AM can be represented in the form $AM = [\lambda_1 e_1, \lambda_1 e_1, \dots, \lambda_n e_n]$, i.e. the columns of AM are given by $\lambda_i e_i$. Then

$$M^T A M = M^T [\lambda_1 e_1, \lambda_1 e_1, \dots, \lambda_n e_n] = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n).$$

4 Linear transformations

Example 4.18 The application of the theorem 4.6 is illustrated on the example of a 2×2 symmetric matrix from 4.17.

$$A = \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix},$$

whose eigenvalues are $\lambda_1 = 1$, $\lambda_2 = 6$ and the corresponding eigenvectors are

$$v_1 = \begin{bmatrix} x \\ y \end{bmatrix} = y \begin{bmatrix} -2 \\ 1 \end{bmatrix}, y \in (-\infty, \infty), \quad v_2 = \begin{bmatrix} x \\ y \end{bmatrix} = y \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}, y \in (-\infty, \infty).$$

First, we normalize the eigenvectors

$$e_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad e_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Now we form the modal matrix M that has the normalized eigenvectors e_i as its columns

$$M = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}$$

Finally we observe

$$\begin{aligned} M^T A M &= \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & 6 \\ 1 & 12 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 30 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \Lambda. \end{aligned}$$

Exercises

1. Determine the eigenvalues and eigenvectors of the following matrices

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 5 & 8 & 16 \\ 4 & 1 & 8 \\ -4 & -4 & -11 \end{bmatrix}.$$

2. Determine the eigenvalues λ_i and the corresponding eigenvectors of the symmetric matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

Construct the modal matrix M and show that $M^T A M = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$.

Answers

4 Linear transformations

1. For the matrix A : $\lambda = -1, -2$, $\mathbf{v}_{\lambda=-1} = t \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $t \in (-\infty, \infty)$, $\mathbf{v}_{\lambda=-2} = t \begin{bmatrix} 1 \\ -2 \end{bmatrix}$, $t \in (-\infty, \infty)$. For the matrix B : $\lambda = 3, -2$, $\mathbf{v}_{\lambda=3} = t \begin{bmatrix} -4 \\ 1 \end{bmatrix}$, $t \in (-\infty, \infty)$, $\mathbf{v}_{\lambda=-2} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $t \in (-\infty, \infty)$. For the matrix C : $\lambda = 1, -3, -3$, $\mathbf{v}_{\lambda=1} = t \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}$, $t \in (-\infty, \infty)$, $\mathbf{v}_{\lambda=-3} = t \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$, $t, s \in (-\infty, \infty)$.

2. The eigenvalues are $\lambda = 3, -1$. The modal matrix is $M = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$.

5 Curvilinear coordinate systems and curves

5.1 Polar coordinates

In two dimensions any point $P(x, y)$ can be determined by the distance from the origin r and the angle θ , measured anticlockwise from the x -axis to the position vector $\overline{OP}$, as shown in Fig. 5.1.

The relation between r and θ and the Cartesian coordinates (x, y) of the point P can be derived from Fig. 5.1

$$x = r \cos \theta \text{ and } y = r \sin \theta.$$

Since $\cos \theta$ and $\sin \theta$ are both 2π -periodic functions, it is convenient to limit the range of possible values of θ to $\theta \in [0, 2\pi)$.

Definition 5.1 *The pair of real numbers (r, θ) with $r \geq 0$ and $\theta \in [0, 2\pi)$ is called .*

Any such pair (r, θ) uniquely determines the position of a point in a two dimensional space.

The conversion from (x, y) to (r, θ) occurs according to

$$r = \sqrt{x^2 + y^2} \text{ and } \theta = \begin{cases} \arctan y/x & x, y > 0 \\ \arctan y/x + \pi & x < 0, y \geq 0 \\ \arctan y/x + \pi & x < 0, y \leq 0 \\ \arctan y/x + 2\pi & x > 0, y \leq 0 \\ \text{undefined} & x = y = 0. \end{cases} \quad (5.81)$$

There exist several different conventions in the literature, according to which the range of θ and r is specified. Thus, sometimes r can be taken as negative, in which case the distance from the origin is taken in the opposite direction. For instance, the point with polar coordinates $(-1, \pi/4)$ is equivalent to $(1, \pi/4 + \pi = 5\pi/4)$. Also in some textbooks, the angle θ is restricted to $(-\pi, \pi]$.

For the remainder of this Section, we use $r \geq 0$ and $\theta \in [0, 2\pi)$.

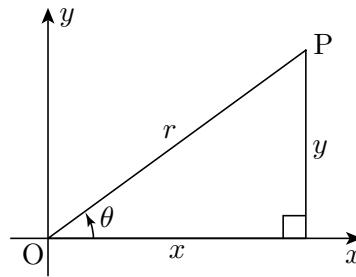


Figure 5.1: Point P in polar coordinates.

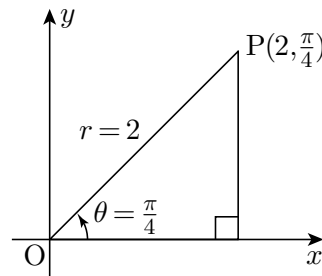


Figure 5.2: Point P with polar coordinates $(2, \pi/4)$

Example 5.1 Plot the point P , which has polar coordinates $(2, \pi/4)$

Solution. See Fig. 5.2.

Example 5.2 Point P has rectangular (Cartesian) coordinates $(-3, -7)$. Find the polar coordinates for P .

Solution. Point P is shown in Fig. 5.3.

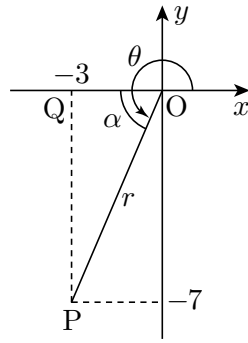
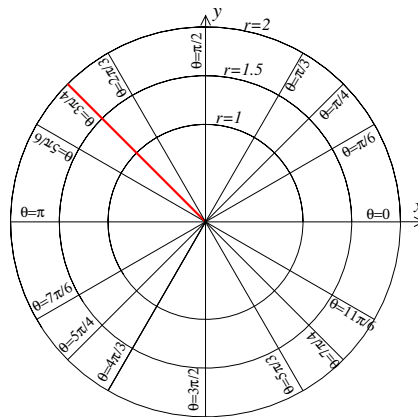
From (5.81) we find $r = \sqrt{(-3)^2 + (-7)^2} = \sqrt{58} \approx 7.62$. To find θ , first find $\alpha = \arctan 7/3 = 1.166$ and then θ , using $\theta = \pi + \alpha = 4.307$ (radians) ($\approx 246^\circ 48'$).

Hence, the polar coordinates of P are approximately $(7.62, 4.307)$.

5.1.1 Drawing polar graphs

A curve of the form $y = f(x)$ can be drawn using Cartesian coordinates. Similarly, a curve of the form $r = f(\theta)$ can be drawn using polar coordinates.

Recall that in Cartesian coordinates, the equations $x = x_0$ and $y = y_0$, where x_0 and y_0 are some constants, represent lines that are parallel to the coordinate axes: $x = x_0$ is


 Figure 5.3: Point P with Cartesian coordinates $(-3, -7)$.

 Figure 5.4: Curves $r = \text{const}$ and $\theta = \text{const}$ in polar coordinates.

parallel to the y -axis and $y = y_0$ is parallel to the x -axis. Such lines are sometimes called **coordinate lines**.

In polar coordinates the equations $r = r_0$ with $\theta \in [0, 2\pi)$ and $\theta = \theta_0$ with $r \geq 0$, where r_0 and θ_0 are some constants, represent a circle and a straight line segment, as shown in Fig.

Indeed, $r = r_0$ with $\theta \in [0, 2\pi)$ corresponds to $\sqrt{x^2 + y^2} = r_0$, or, equivalently $x^2 + y^2 = r_0^2$, which is the standard equation for a circle of a radius r_0 with the center in the origin.

On the other hand, consider, for instance, the equation $\theta = 3\pi/4$ with $r \geq 0$. This equation can be written in terms of (x, y) as $\tan \theta = y/x = \tan 3\pi/4 = \tan(\pi - \pi/4) = -\tan \pi/4 = -1$ with $x \leq 0$. This gives a segment of a straight line that passes through the origin $y = -x$ with $x \leq 0$, as shown by the red solid line in Fig.5.4.

Note that the entire line $y = -x$ can be represented in polar coordinates as $\theta = 3\pi/4 \cup \theta = 2\pi - \pi/4 = 7\pi/4$ with $r \geq 0$.

Example 5.3 A graph is defined by the polar equation $r = 3 \cos \theta$, $\theta \in [0, \pi/2] \cup [3\pi/2, 2\pi)$. Find its Cartesian equation and identify the type of the curve.

Solution. $r = \sqrt{x^2 + y^2}$, $\cos \theta = \frac{x}{r} = \frac{x}{\sqrt{x^2 + y^2}}$. Hence the polar equation is transformed into the equation

$$\sqrt{x^2 + y^2} = \frac{3x}{\sqrt{x^2 + y^2}}.$$

Multiplying both sides by $\sqrt{x^2 + y^2}$ gives

$$\begin{aligned} x^2 + y^2 &= 3x \\ \implies x^2 - 3x + y^2 &= 0 \end{aligned}$$

Completing the square, we obtain

$$\begin{aligned} \implies x^2 - 3x + \frac{9}{4} + y^2 &= \frac{9}{4} \\ \implies \left(x - \frac{3}{2}\right)^2 + y^2 &= \left(\frac{3}{2}\right)^2 \end{aligned}$$

This is a circle, centre $(\frac{3}{2}, 0)$, and radius $\frac{3}{2}$.

5.1.2 Cartesian graphs in polar form

Example 5.4 Describe the region inside a circle $x^2 + (y - 2)^2 = 4$ in polar coordinates

Solution Substituting

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta. \end{aligned}$$

into the equation for the circle, gives

$$\begin{aligned} 4 &= r^2 \cos^2 \theta + (r \sin \theta - 2)^2, \\ 4 &= r^2 \cos^2 \theta + r^2 \sin^2 \theta - 4r \sin \theta + 4, \\ 0 &= r(r - 4 \sin \theta). \end{aligned}$$

Therefore, for the points on the circle, we have

$$r = 0, \text{ or } r = 4 \sin \theta.$$

Clearly, the region inside a circle corresponds to the values of r , smaller than $4 \sin \theta$

$$r \leq 4 \sin \theta.$$

5 Curvilinear coordinate systems and curves

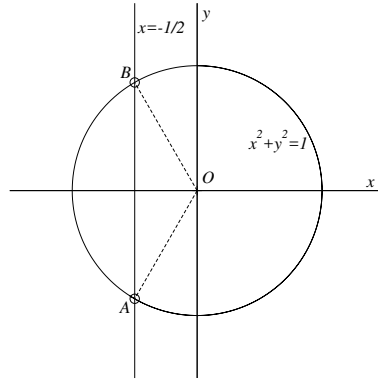


Figure 5.5: Region in Example 5.5.

This inequality must be supplemented with the appropriate limits for the angle θ . Since the origin lies on the circle, the value of $r = 0$ is reached, when $\sin \theta = 0$

$$\theta = 0, \text{ and } \theta = \pi.$$

Finally, in polar coordinates, the region inside a circle is given by

$$r \leq 4 \sin \theta, \quad 0 \leq \theta \leq \pi.$$

Example 5.5 Describe the region inside a circle $x^2 + y^2 = 1$ with $x \leq -1/2$ in polar coordinates.

Solution

Substituting

$$x = r \cos \theta$$

$$y = r \sin \theta.$$

into $x \leq -1/2$ and $x^2 + y^2 \leq 1$, gives

$$r \cos \theta \leq -1/2, \quad r^2 = x^2 + y^2 \leq 1.$$

Since the region lies to the left from the line $x = -1/2$, the angle θ is in the second and third quadrants as shown in Fig.5.5. In fact

$$\theta \in [\pi - \cos^{-1}(1/2), \pi + \cos^{-1}(1/2)] = [2\pi/3, 4\pi/3].$$

Consequently

$$\cos \theta < 0.$$

Therefore, the values of r satisfy the inequality

$$r \geq -\frac{1}{2 \cos \theta}$$

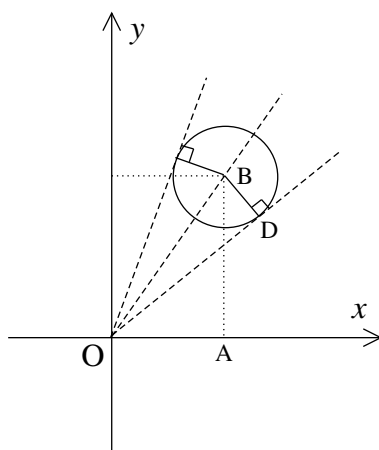


Figure 5.6: Illustration to Example 5.6.

(note the reversed sign of the inequality).

Finally, we obtain

$$r \leq 1, \quad r \geq -\frac{1}{2\cos\theta}, \quad \theta \in [2\pi/3, 4\pi/3].$$

Example 5.6 A graph of a circle is given by the Cartesian (rectangular) equation

$$(x-2)^2 + (y-3)^2 = 1 \quad (5.82)$$

Find its polar equation.

Solution.

$$x = r \cos \theta \quad (5.83)$$

$$y = r \sin \theta. \quad (5.84)$$

Substituting (5.83) and (5.84) into (5.82) gives

$$(r \cos \theta - 2)^2 + (r \sin \theta - 3)^2 = 1.$$

Expanding gives

$$r^2 \cos^2 \theta + 4 - 4r \cos \theta + r^2 \sin^2 \theta - 6r \sin \theta + 9 = 1$$

and hence

$$r^2(\cos^2 \theta + \sin^2 \theta) + 13 - 4r \cos \theta - 6r \sin \theta = 1$$

 SO

$$r^2 - r(4 \cos \theta + 6 \sin \theta) + 12 = 0.$$

5 Curvilinear coordinate systems and curves

This is a quadratic equation for r . Solving for r yields

$$r = \frac{1}{2} \left(4 \cos \theta + 6 \sin \theta \pm \sqrt{(4 \cos \theta + 6 \sin \theta)^2 - 48} \right).$$

Finally, we need to determine the limits in which the angle θ is changing. From Fig.5.6 it is clear that

$$\theta \in [\beta - \alpha, \beta + \alpha],$$

where β is the angle BOA and α is the angle BOD. From the right-angled triangles BOA and BOD we obtain $\beta = \tan^{-1}(3/2)$ and $\alpha = \sin^{-1}(1/\sqrt{13})$.

Exercises

1. In each of the following problems, a point's coordinates are given in rectangular (Cartesian) form. Give the polar coordinates for each point.
 (a) $(1, 1)$, (b) $(-2, 2)$ (c) $(\sqrt{3}, 1)$ (d) $(-1, \sqrt{3})$ (e) $(3, 0)$ (f) $(0, 1)$
2. Plot the following polar curves:
 - a) $r = \cos \theta$
 - b) $r = \sin \theta$
 - c) $r = \sqrt{\sin 2\theta}$ (this curve is called the lemniscate and is sometimes used as a horizontal road curve)
 - d) $r = \sqrt{-\sin 2\theta}$
 - e) $r = \sin^2 \theta$
3. In exercises (a)–(d), change the given equation in r and θ to an equivalent equation in x and y .
 (a) $r = 3 \sin \theta$ (b) $r = \sin 2\theta$ (c) $r = \tan \theta$ (d) $r = 4 + \cos \theta$
4. In exercises (a)–(d), change the given equation in x and y to an equivalent equation in r and θ .
 (a) $x^2 + y^2 = 4$ (b) $x = 2$ (c) $y = 3x$ (d) $(x - 2)^2 + y^2 = 4$

Answers

1. (a) $\left(\sqrt{2}, \frac{\pi}{4}\right)$ (b) $\left(2\sqrt{2}, \frac{3\pi}{4}\right)$ (c) $\left(2, \frac{\pi}{6}\right)$
 (d) $\left(2, \frac{2\pi}{3}\right)$ (e) $(3, 0)$ (f) $\left(1, \frac{\pi}{2}\right)$

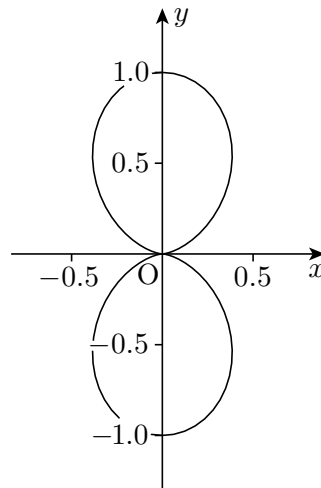


Figure 5.7: Curve from part (e) of question 2.

2. Selected answer: Curve from part (e) is shown in Fig. 5.7.

3.(a) $x^2 + (y - \frac{3}{2})^2 = (\frac{3}{2})^2$

(b) $(x^2 + y^2)^3 = 4x^2y^2$

(c) $x^4 + x^2y^2 = y^2$

(d) $(x^2 + y^2 - x)^2 = 16(x^2 + y^2)$

4.(a) $r = 2$, (b) $r = 2 \sec \theta$, (c) $\tan \theta = 3$, (d) $r = 4 \cos \theta$

5.2 Curvilinear coordinate systems in three dimensions

The position of an arbitrary point in three dimensions can be described using the *cylindrical* or *spherical* coordinates.

5.2.1 Cylindrical coordinates

Cylindrical coordinates system is a three-dimensional extension of the polar coordinate system. The position of a point P with Cartesian coordinates $P = (x, y, z)$ is specified by the polar coordinates (r, θ) of the projection of point P onto the xy plane and the

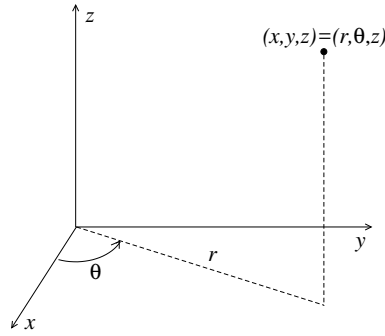


Figure 5.8: Cylindrical coordinates (r, θ, z) of a point with Cartesian coordinates (x, y, z) .

z -coordinate of P , as shown in Fig. 5.8. The transition from the cylindrical coordinates (r, θ, z) to Cartesian coordinates (x, y, z) occurs according to

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

Example 5.7 Express the intersection of the sphere $x^2 + y^2 + z^2 = 10$ with the plane $z = 1$ in cylindrical coordinates.

Solution

Substituting $z = 1$ into the equation of the sphere $x^2 + y^2 + z^2 = 10$, we obtain

$$x^2 + y^2 = 9.$$

In cylindrical coordinates $x = r \cos \theta$ and $y = r \sin \theta$, which yields

$$r^2 = 9, \quad \rightarrow r = 3.$$

Therefore, the intersection of the sphere $x^2 + y^2 + z^2 = 10$ with the plane $z = 1$ in cylindrical coordinates is given by

$$z = 1, \quad r = 3, \quad \theta = [0, 2\pi).$$

5.2.2 Spherical coordinates

In spherical coordinates, the position of a point $P = (x, y, z)$ is specified by the distance $\rho = \sqrt{x^2 + y^2 + z^2}$ from the origin, the angle θ between the x -axis and the projection of P onto the xy -plane and the angle ϕ between the z -axis and the position vector of point P , as shown in Fig. 5.9. The transition from (ρ, θ, ϕ) occurs according to

$$x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

The angles θ and ϕ are naturally restricted to

$$\theta \in [0, 2\pi), \quad \phi \in [0, \pi].$$

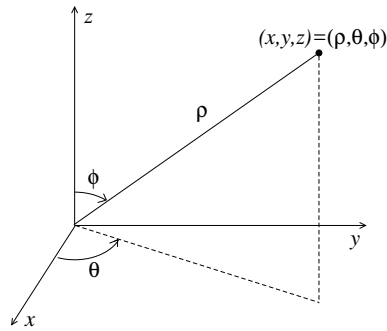


Figure 5.9: Spherical coordinates (ρ, θ, ϕ) of a point with Cartesian coordinates (x, y, z) .

In spherical coordinates

- the angle ϕ between the z -axis and the position vector of point P is called the *polar angle*
- the angle θ between the x -axis and the projection of the position vector of point P onto the xy -plane is called the *azimuthal angle*.

Note that in some textbooks the polar angle is denoted by θ and the azimuthal angle is denoted by ϕ . It is also advisable not to confuse the polar angle ϕ with the angle θ in polar or cylindrical coordinates.

Example 5.8 A point P has spherical coordinates $(\sqrt{8}, \pi/3, \pi/4)$. Find both, the Cartesian and the cylindrical coordinates of P .

Solution

The Cartesian coordinates of P are obtained by

$$\begin{aligned} x &= \sqrt{8} \sin(\pi/4) \cos(\pi/3) = \frac{\sqrt{8}}{2\sqrt{2}} = 1, \\ y &= \sqrt{8} \sin(\pi/4) \sin(\pi/3) = \frac{\sqrt{8}\sqrt{3}}{\sqrt{22}} = \sqrt{3}, \\ z &= \sqrt{8} \cos(\pi/4) = 2. \end{aligned}$$

The cylindrical coordinates (r, θ, z) of P are found as

$$r = \sqrt{x^2 + y^2} = \sqrt{1^2 + 3} = 2, \quad \theta = \pi/3, \quad z = 2.$$

Example 5.9 Express the circle $x^2 + y^2 + z^2 = 32, z = 4$ in spherical coordinates.

Solution

Substituting $z = 4$ into $x^2 + y^2 + z^2 = 32$, we obtain the equation of the circle

$$x^2 + y^2 = 16.$$

The radius of the circle is $R = 4$. All points on the circle have identical polar angle ϕ given by

$$\tan \phi = \frac{R}{4} = 1.$$

Therefore, $\phi = \tan^{-1}(1) = \pi/4$.

Finally, the equation of the circle in spherical coordinates can be written as

$$\rho = \sqrt{32}, \quad \theta \in [0, 2\pi), \quad \phi = \pi/4.$$

Example 5.10 Convert from cylindrical $(1, 3\pi/4, -1)$ to Cartesian and spherical coordinates.

Solution

Cylindrical coordinates are $r = 1, \theta = 3\pi/4, z = -1$.

The Cartesian coordinates are found as

$$\begin{aligned} x &= r \cos \theta = \cos 3\pi/4 = \cos(\pi - \pi/4) = -\cos \pi/4 = -\frac{1}{\sqrt{2}}, \\ y &= r \sin \theta = \sin 3\pi/4 = \sin(\pi - \pi/4) = \sin \pi/4 = \frac{1}{\sqrt{2}}, \quad z = -1, \end{aligned}$$

where we have used the trigonometric identities $\cos(a+b) = \cos a \cos b - \sin a \sin b$ and $\sin(a+b) = \sin a \cos b + \cos a \sin b$.

The distance from the origin ρ and the azimuthal angle θ in the spherical coordinates are

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{1/2 + 1/2 + 1} = \sqrt{2}, \quad \theta = 3\pi/4.$$

The polar angle ϕ can be found from

$$\tan \phi = \frac{\sqrt{x^2 + y^2}}{z} = -1, \quad \phi \in [0, \pi].$$

Therefore, $\phi = \pi - \pi/4 = 3\pi/4$.

Example 5.11 Rewrite the equation of the cone $x^2 + y^2 = z^2$ in spherical coordinates.

Solution

Substituting $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$ and $z = \rho \cos \phi$ into $x^2 + y^2 = z^2$, we obtain

$$\rho^2 \sin^2 \phi (\cos^2 \theta + \sin^2 \theta) = \rho^2 \cos^2 \phi.$$

Or, equivalently

$$\tan \phi = \pm 1.$$

This yields

$$\phi = \frac{\pi}{4}, \text{ or } \phi = \pi - \frac{\pi}{4} = \frac{3\pi}{4}.$$

Example 5.12 Rewrite the equation of a plane $x + y = 0$ in spherical coordinates.

Solution

The equation of the plane is

$$\rho \sin \phi \cos \theta + \rho \sin \phi \sin \theta = \rho \sin \phi (\cos \theta + \sin \theta) = 0$$

Thus we obtain

$$\cos \theta + \sin \theta = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos \theta + \frac{1}{\sqrt{2}} \sin \theta \right) = 0.$$

Recalling that $\cos \pi/4 = \sin \pi/4 = 1/\sqrt{2}$ and using $\cos(a - b) = \cos a \cos b + \sin a \sin b$, we obtain

$$\sqrt{2} \cos \left(\theta - \frac{\pi}{4} \right) = 0.$$

Consequently, the equation of the plane in spherical coordinates is

$$\theta - \frac{\pi}{4} = \frac{\pi}{2} + \pi k, \quad k = 0, \pm 1, \pm 2, \pm 3, \dots$$

Taking into account that $\theta \in [0, 2\pi)$, we finally obtain

$$\theta = \frac{3\pi}{4}, \text{ or } \theta = \pi + \frac{3\pi}{4} = \frac{7\pi}{4}.$$

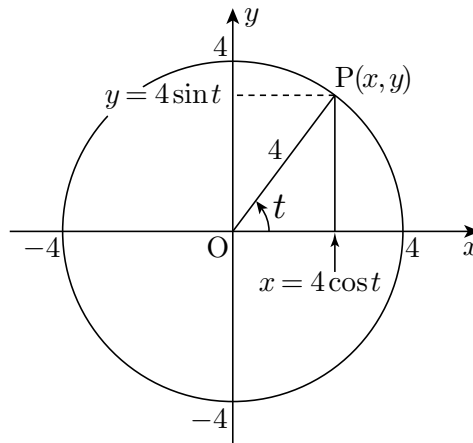


Figure 5.10: Circle given by the parametric equation $x = 4 \cos t$, $y = 4 \sin t$, $t \in [0, 2\pi)$.

5.3 Two-dimensional (2D) parametric curves

If a curve is defined by two equations of the form

$$x = f(t) \quad (5.85)$$

$$y = g(t) \quad (5.86)$$

(i.e. x and y are functions of t), then the variable t is said to be a parameter and (5.85) and (5.86) are said to be parametric equations.

Example 5.13 A curve is defined by $x = 4 \cos t$, $y = 4 \sin t$, $t \in [0, 2\pi)$.

1. Find a relation between y and x .
2. Plot the graph for $0 \leq t \leq 2\pi$.

Solution. (a) The equations imply that $\cos t = \frac{x}{4}$ and $\sin t = \frac{y}{4}$. As $\sin^2 t + \cos^2 t = 1$,

$$\begin{aligned} \left(\frac{y}{4}\right)^2 + \left(\frac{x}{4}\right)^2 &= 1. \\ \implies x^2 + y^2 &= 16. \end{aligned} \quad (5.87)$$

This is the equation of a circle with radius 4 and centre the origin.

(b) The circle can be plotted as shown in Fig. 5.10. It can be seen from Fig. 5.10 that the parameter t is the angle between the x -axis and OP .

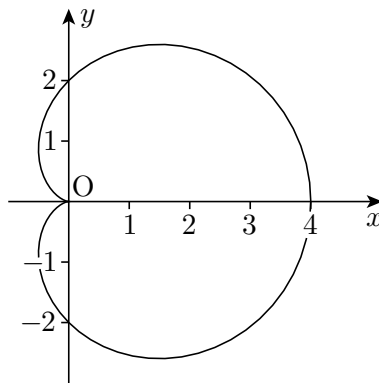


Figure 5.11: Curve given by $r = 2 + 2 \cos \theta$, for $0 \leq \theta \leq 2\pi$.

5.3.1 Polar curves in parametric form

Example 5.14 Given that a polar curve is defined by the equation $r = 2 + 2 \cos \theta$, for $0 \leq \theta \leq 2\pi$, represent the Cartesian coordinates (x, y) in terms of the angle θ and thus, derive the parametric equation of the curve.

Solution.

$$r = 2 + 2 \cos \theta. \quad (5.88)$$

As for polar curves

$$x = r \cos \theta \quad (5.89)$$

and

$$y = r \sin \theta. \quad (5.90)$$

Substituting (5.88) into (5.89), we obtain

$$x = (2 + 2 \cos \theta) \cos \theta.$$

Substituting (5.88) into (5.90) we obtain

$$y = (2 + 2 \cos \theta) \sin \theta.$$

The graph of the curve is shown in Fig. 5.11.

Exercises

1. The following are parametric relations between x and y .

a) $x = t^2, \quad y = t^3, \quad -3 \leq t \leq 3$

b) $x = \cos t, \quad y = \sin t$

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- c) $x = 2 \cos \theta, \quad y = \sin \theta$
- d) $x = t \cos t, \quad y = t \sin t \quad 0 \leq t \leq 4\pi$
- e) $x = \ln U, \quad y = \ln(2/U)$
- Eliminate the parameter to give a direct relation between x and y .
 - Plot the graphs of the curves.
2. For each polar curve below represent the Cartesian coordinates (x, y) in terms of the angle θ .
- the circle $r = 1$, for $0 \leq \theta \leq 2\pi$
 - the spiral $r = \theta$, for $0 \leq \theta \leq 4\pi$
 - the cardioid $r = 1 + \cos \theta$, for $0 \leq \theta \leq 4\pi$

Answers

1. a) i. $y = \pm x^{1.5}$
- b) i. $x^2 + y^2 = 1$
- c) i. $\frac{x^2}{4} + y^2 = 1$
- d) i. $y = x \tan(\sqrt{x^2 + y^2})$
- e) i. $y = \ln 2 - x$
2. a) $x = \cos \theta, \quad y = \sin \theta$
- b) $x = \theta \cos \theta, \quad y = \theta \sin \theta$
- c) $x = \cos \theta + \cos^2 \theta, \quad y = \sin \theta + \sin \theta \cos \theta$

5.3.2 Lissajous curves

An example of parametric curves with a direct engineering application are the , given by

$$\begin{aligned}x &= a \sin(mt + \varepsilon_1) \\y &= b \sin(nt + \varepsilon_2).\end{aligned}$$

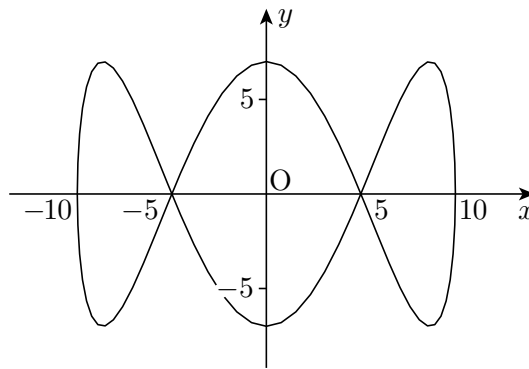


Figure 5.12: A Lissajous curve $x = 10 \sin t$, $y = 7 \cos 3t$.

As a sine curve can also be written as a cosine curve, often one or both of the x and y expressions is written in cosine form. Both coordinates x and y oscillate with frequencies m and n , respectively. The frequency rotation is determined as $m : n$.

An example of a Lissajous curve is the Australian Broadcasting Corporation symbol, which has the parametric equations

$$\begin{aligned}x &= 10 \sin t \\y &= 7 \cos 3t.\end{aligned}$$

A plot of this curve is shown in Fig. 5.12.

Lissajous curves are important in electronics when comparing the frequencies of two voltages using a cathode ray oscilloscope (CRO). The CRO has the vertical deflection of an electron beam controlled by one voltage $y(t)$ (where t is time in seconds) and the horizontal deflection controlled by the other voltage $x(t)$. The beam traces out on the screen a curve specified parametrically by $x = x(t)$ and $y = y(t)$, as illustrated in Fig. 5.13.

5.4 Curves in space

Any curve in a three dimensional space can be represented by a vector parametric equation

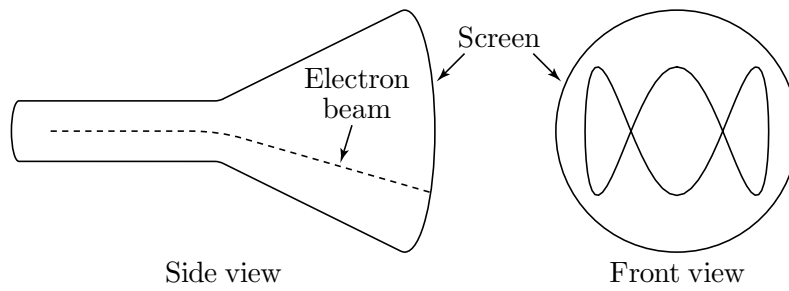


Figure 5.13: Views of a CRO

Table 5.1:

t	x	y	z	Point
0	2	0	0	A
$\frac{\pi}{2}$	0	2	$\frac{\pi}{2}$	B
π	-2	0	π	C
$\frac{3\pi}{2}$	0	-2	$\frac{3\pi}{2}$	D
2π	2	0	2π	E
$\frac{5\pi}{2}$	0	2	$\frac{5\pi}{2}$	F

$$\vec{r} = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k},$$

where $\vec{r}$ is the position vector of a point on the curve.

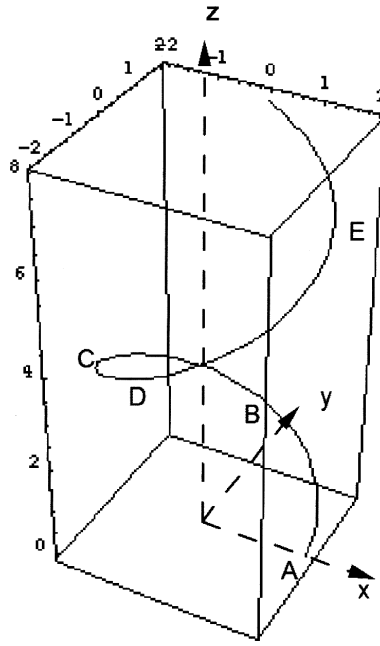
Example 5.15 A particle travels along a path defined by

$$\vec{r} = (2 \cos t)\vec{i} + (2 \sin t)\vec{j} + t\vec{k}.$$

Graph the path for $0 \leq t \leq \frac{5\pi}{2}$.

Solution. Table 5.1 shows values for t , x , y , and z .

Draw 3D axes, roughly plot the points, and join them up with a smooth curve, as shown in Fig. 5.14. This curve is called a (a spiralling 3D curve). It is a curve whose tangent makes a constant angle with the horizontal. The curve can be thought of as spiralling around a cylinder. A thread on a screw and the bannisters of a spiral staircase are both helical curves.

Figure 5.14: Graph of a helix $\vec{r} = (2 \cos t)\vec{i} + (2 \sin t)\vec{j} + t\vec{k}$.

5.4.1 Velocity and acceleration

Suppose a particle (e.g. a fluid particle) is at point P with position vector $\vec{r}$ at time t , and after a short interval of time δt (e.g. 0.01 seconds), it is at position Q, with position vector $\vec{r} + \delta\vec{r}$ (see Fig. 5.15). The average velocity over this time interval is $\frac{\overrightarrow{PQ}}{\delta t}$, i.e. $\frac{\delta\vec{r}}{\delta t}$. The $\vec{v}$ at time t (and hence at point P) is defined to be the limiting value of $\frac{\delta\vec{r}}{\delta t}$ as $\delta t \rightarrow 0$, i.e.

$$\begin{aligned}\vec{v} &= \lim_{\delta t \rightarrow 0} \frac{\delta\vec{r}}{\delta t} \\ &= \frac{d\vec{r}}{dt}.\end{aligned}$$

The velocity $\vec{v}$ at P is tangential to the curve $\vec{r} = \vec{r}(t)$ at P.

The $\vec{a}$ is defined by $\vec{a} = \frac{d\vec{v}}{dt}$.

Example 5.16 A particle moves with position vector $\vec{r}$, defined by

$$\vec{r} = (\sin t)\vec{i} + (\cos t)\vec{j} + t\vec{k}.$$

Find its velocity $\vec{v}$ and acceleration $\vec{a}$ at time t .

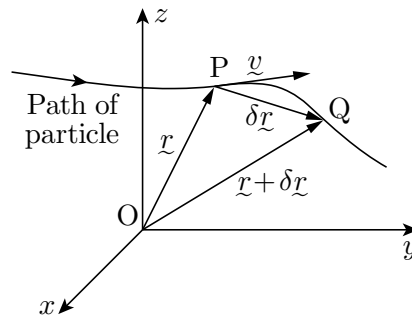


Figure 5.15: Velocity of a particle moving along the curve.

Solution.

$$\vec{v} = \frac{d\vec{r}}{dt} = (\cos t)\vec{i} - (\sin t)\vec{j} + 1\vec{k}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = -(\sin t)\vec{i} - (\cos t)\vec{j} + 0\vec{k} = -(\sin t)\vec{i} - (\cos t)\vec{j}.$$

Exercises

- For each case below, the position vector $\vec{r}$ of a particle is specified for any time t . Find the velocity and acceleration at time t .

a) $\vec{r} = (\cos t)\vec{i} + (\sin t)\vec{j} + t\vec{k}$

b) $\vec{r} = 2(\cos t)\vec{i} + (\sin t)\vec{j} + t\vec{k}$

c) $\vec{r} = t^2\vec{i} + (\cos t)\vec{j} + (\sin t)\vec{k}$

d) $\vec{r} = (\cos t)\vec{i} + (\sin t)\vec{j} + 1\vec{k}$

e) $\vec{r} = t\vec{i} + (1 + t)\vec{j} + (2 - t)\vec{k}$

f) $\vec{r} = t\vec{i} + t\vec{j} + t^2\vec{k}$

- The position vector of the centre of gravity (CG) of a person on a roller-coaster ride is $\vec{r} = (\cos t)\vec{i} + (\sin t)\vec{j} + (\sin 4t)\vec{k}$. Find the velocity and acceleration of the CG at time t and graph the three-dimensional curve traversed by the CG.
- The CG of a football that has been kicked has position vector

$$\vec{r} = 10t\vec{i} + (25t - 4.9t^2)\vec{j}.$$

- Find the velocity and acceleration of the CG at time t
- Find the Cartesian equation of the curve the CG travels along

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- c) Taking the x -axis to lie along the football ground, at what time does the football land?
4. An electron of mass m kilograms and charge q coulombs moves under the sole influence of a magnetic field $\vec{B} = B\vec{k}$. Its position vector at time t is given by $\vec{r} = (\cos wt)\vec{i} + (\sin wt)\vec{j} + t\vec{k}$.
- a) Name the three-dimensional curve that the particle travels along.
- b) Find the velocity $\vec{v}$ and acceleration $\vec{a}$ of the particle.
- c) Find $\frac{q}{m}$ (the ratio of charge to mass for the electron) in coulombs given that $m\vec{a} = q\vec{v} \times \vec{B}$ and that $B = 10^{-6}$ weber/m² and $w = 1.7 \times 10^5$ radians/s.

Answers

1. velocity, acceleration:

a) $-(\sin t)\vec{i} + (\cos t)\vec{j} + \vec{k}, \quad -(\cos t)\vec{i} - (\sin t)\vec{j}$

b) $-2(\sin t)\vec{i} + (\cos t)\vec{j} + \vec{k}, \quad -2(\cos t)\vec{i} - (\sin t)\vec{j}$

c) $2t\vec{i} - (\sin t)\vec{j} + (\cos t)\vec{k}, \quad 2\vec{i} - (\cos t)\vec{j} - (\sin t)\vec{k}$

d) $-(\sin t)\vec{i} + (\cos t)\vec{j}, \quad -(\cos t)\vec{i} - (\sin t)\vec{j}$

e) $\vec{i} + \vec{j} - \vec{k}, \quad \vec{0}$

f) $\vec{i} + \vec{j} + 2t\vec{k}, \quad 2\vec{k}$

2. $-(\sin t)\vec{i} + (\cos t)\vec{j} + (4 \cos 4t)\vec{k}, \quad -(\cos t)\vec{i} - (\sin t)\vec{j} - (16 \sin 4t)\vec{k}$

3. a) $10\vec{i} + (25 - 9.8t)\vec{j}, \quad -9.8$

b) $y = 2.5x - 0.049x^2$

c) 5.1 s

4. a) helix

b) $\vec{v} = -(w \sin wt)\vec{i} + (w \cos wt)\vec{j} + \vec{k}$

$\vec{a} = -(w^2 \cos wt)\vec{i} - (w^2 \sin wt)\vec{j}$

$\frac{q}{m} = 1.7 \times 10^{11} \text{ c/kg}$

5.4.2 Ballistic trajectories

Any object (projectile), launched at some initial velocity in a certain direction, follows a . The simplest model for describing the motion of projectiles is one that assumes that the only force acting on the projectile is the force of gravity. More complicated (and more realistic) models take into account other forces such as air resistance and the effect of wind. In the example given below it is assumed that gravity is the only force acting, and hence the acceleration is due only to gravity. Taking the vertical axis to be the y -axis and the x -axis to be horizontal, the acceleration is given by $\vec{a} = -g\vec{j}$, where g is the acceleration due to gravity. At the surface of the earth, $g = 9.8 \text{ ms}^{-2}$. In the models, the projectile is regarded as a particle (i.e. a point).

Example 5.17 A bullet is fired at ground-level from $(0,0)$ at time $t = 0$. Its initial velocity is 700 m/s in a direction at 25° to the horizontal.

1. Find its velocity ($\vec{v}$) at any time t seconds.

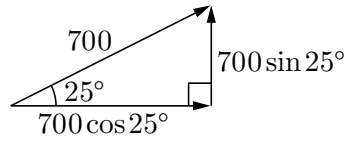


Figure 5.16: The components of the velocity when $t = 0$.

2. Find its position vector ($\vec{r}$) at any time t seconds.
3. Find the equation of its trajectory (path).
4. When does it hit the ground again?
5. Find its maximum height and the time it reaches this height.

Solution.

1.

$$\vec{a} = -g\vec{j}. \quad (5.91)$$

Integrating (5.91) with respect to t implies that

$$\vec{v} = \int \vec{a} dt = c_1\vec{i} + (-gt + c_2)\vec{j}, \quad (5.92)$$

where c_1 and c_2 are constants.

From Fig. 5.16, it can be seen that at $t = 0$, $\vec{v}$ is given by

$$\vec{v}(0) = 700 \cos 25^\circ \vec{i} + 700 \sin 25^\circ \vec{j}. \quad (5.93)$$

Substituting $t = 0$ into (5.92) gives

$$\vec{v}(0) = c_1\vec{i} + (-g \times 0 + c_2)\vec{j} = c_1\vec{i} + c_2\vec{j}. \quad (5.94)$$

Equating the respective components in (5.93) and (5.94) gives

$$\begin{aligned} c_1 &= 700 \cos 25^\circ \approx 634.42 \\ c_2 &= 700 \sin 25^\circ \approx 295.83 \\ \text{so } \vec{v} &\approx 634.42\vec{i} + (295.83 - 9.8t)\vec{j}. \end{aligned} \quad (5.95)$$

2. Integrating (5.95) with respect to t results in

$$\vec{r} = \int \vec{v} dt \approx (634.42t + d_1)\vec{i} + (295.83t - 4.9t^2 + d_2)\vec{j}, \quad (5.96)$$

5 Curvilinear coordinate systems and curves

where d_1 and d_2 are constants. Substituting $t = 0$ into (5.96) tells us the initial position vector is equal to

$$\begin{aligned}\vec{r}(0) &\approx (634.42 \times 0 + d_1)\vec{i} + (295.83 \times 0 - 4.9 \times 0^2 + d_2)\vec{j} \\ &= (0 + d_1)\vec{i} + (0 + 0 + d_2)\vec{j} \\ &= d_1\vec{i} + d_2\vec{j}.\end{aligned}$$

But it is given that initially the bullet is at the origin. Hence, the initial position vector is equal to

$$\vec{r}(0) = 0\vec{i} + 0\vec{j}.$$

Equating the respective components in (5.97) and (5.98) then gives

$$d_1 = 0 \quad \text{and} \quad d_2 = 0 \quad (5.97)$$

Hence, substituting (5.97) into (5.96) gives

$$\vec{r} \approx 634.42t\vec{i} + (295.83t - 4.9t^2)\vec{j}. \quad (5.98)$$

3. As

$$\vec{r} = x\vec{i} + y\vec{j}, \quad (5.99)$$

it follows from equating the respective components of (5.98) and (5.99) that

$$x \approx 634.42t \quad (5.100)$$

and

$$y \approx 295.83t - 4.9t^2. \quad (5.101)$$

To obtain the equation of the curve (path) that the bullet travels along, i.e. y as a function of x , first express t in terms of x , using (5.100), which gives

$$t \approx \frac{x}{634.42}. \quad (5.102)$$

Then, substituting (5.102) into (5.101) gives

$$y \approx 295.83\left(\frac{x}{634.42}\right) - 4.9\left(\frac{x}{634.42}\right)^2,$$

which is the equation of a parabola. The graph of this function is shown in Fig. 5.17.

4. The bullet hits the ground again when $y = 0$. Substituting $y = 0$ into (5.101), we find that

$$y \approx 295.83t - 4.9t^2 = 0$$

so solve

$$295.83t - 4.9t^2 = 0.$$

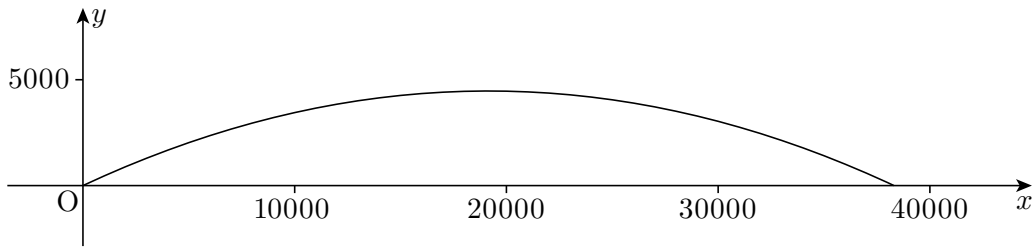


Figure 5.17: Ballistic trajectory of a projectile

Factorising gives

$$t(295.83 - 4.9t) = 0$$

from which

$$t = 0 \quad \text{or} \quad 295.83 - 4.9t \approx 0.$$

Hence

$$t = 0 \text{ seconds} \quad \text{or} \quad t \approx \frac{295.83}{4.9} \approx 60.4 \text{ seconds}.$$

The bullet is fired at ground level when $t = 0$. It is at ground level again when $t \approx 60.4$ s.

5. When the bullet is at its maximum height, the velocity is horizontal, and hence the $\vec{j}$ component of the velocity is 0. Hence, as $\vec{v} \approx 634.42\vec{i} + (295.83 - 9.8t)\vec{j}$, it follows that

$$\begin{aligned} 295.83 - 9.8t &\approx 0 \\ \implies t &\approx \frac{295.83}{9.8} \approx 30.2 \text{ seconds.} \end{aligned} \tag{5.103}$$

Substituting (5.103) into (5.101) shows that the maximum height is given by

$$y \approx 295.83 \times 30.2 - 4.9 \times 30.2^2 \approx 4465 \text{ metres}.$$

Exercises

1. A bullet is fired with a velocity of 500 m/s in a direction 20° above the horizontal. Given that its acceleration is $\vec{a} = -g\vec{j}$, where $g = 9.8 \text{ m/s}^2$, and it is initially at $(0,0)$, find its velocity ($\vec{v}$) at any time t seconds.
2. A bullet is fired with a velocity of 400 m/s in a direction 30° above the horizontal. Given that its acceleration is $\vec{a} = -g\vec{j}$, where $g = 9.8 \text{ m/s}^2$, and it is initially at $(0,0)$, find its velocity ($\vec{v}$) at any time t seconds.

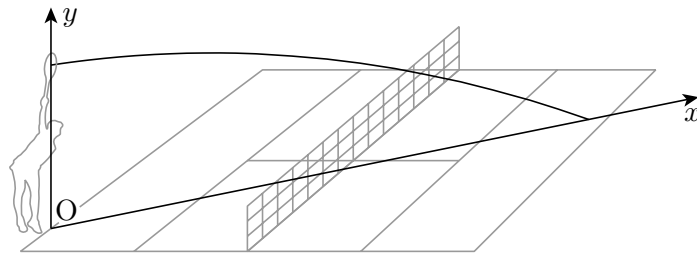


Figure 5.18: Tennis player, serving the ball horizontally.

3. A tennis ball is served horizontally from 2.5 m above the ground at 30 m/s, as shown in Fig. 5.18. The net is 12 m away and 0.9 m high.
 - a) Find its velocity ($\vec{v}$) at any time t seconds
 - b) Find its position vector ($\vec{r}$) at any time t seconds
 - c) Will the ball clear the net? If so, by how much?
 - d) When will the ball land?
 - e) Where will the ball land?

Answers

1. $469.9\vec{i} + (-9.8t + 171.0)\vec{j}$
2. $346.4\vec{i} + (-9.8t + 200.0)\vec{j}$
3.
 - a) $30\vec{i} - 9.8t\vec{j}$
 - b) $30t\vec{i} + (2.5 - 4.9t^2)\vec{j}$
 - c) Yes, 0.82 m
 - d) 0.714 s after it is hit
 - e) 21.42 m from its starting point

5.5 Curvature (bending)

Given a curve $y = f(x)$, the K at any point $P(x, y)$ on the curve, is defined as the rate of change of the direction of the curve with respect to arc length. As a straight line does not change direction, it has zero curvature.

To illustrate the concept of curvature consider Fig. 5.19.

Let θ be the angle (in radians) that the tangent at P makes with the x -axis. Let Q be a nearby point, a distance δs along the curve from P, and let $\theta + \delta\theta$ be the angle that the tangent at Q makes with the x -axis.

Definition 5.2 The curvature K at point P is given by

$$K = \lim_{\delta s \rightarrow 0} \frac{\delta\theta}{\delta s} = \frac{d\theta}{ds}.$$

Hence, K measures how fast θ (the direction) changes with distance s along the curve. If K is positive, then θ increases with distance and the curve is concave upwards, e.g. the curve in Fig. 5.19. If K is negative, then θ decreases with distance and the curve is concave downwards.

Let the normals (perpendiculars) at P and Q meet at C. As P and Q are close, $CQ \approx CP = R$ and approximately δs is the arc of a circle of radius R . The angle $PCQ = \delta\theta$ because if the tangent changes direction by $\delta\theta$, then the normals, which are perpendicular to the tangent, also change direction by $\delta\theta$.

Hence, as the arc PQ is approximately the arc of a circle of radius R , then

$$\delta s \approx R\delta\theta \implies R \approx \frac{\delta s}{\delta\theta}.$$

Therefore $R = \frac{1}{K}$.

The circle of which C is the centre is called the at P, and C is called the .

Definition 5.3 The radius of the circle of curvature is given by

$$|R| = |K|^{-1}.$$

Since, usually, curves are described by their Cartesian equation, it would be useful to have a formula relating K to the variables x, y and derivatives of y .

Since θ is the angle the tangent makes with the x -axis, it should be clear that θ is a function of the gradient y' , $\theta = \theta(y')$. Clearly also $y' = y'(x)$, and finally, there is certainly a relation between the coordinate x and the distance s travelled along the curve, $x = x(s)$.

Hence we can write a chain of composite functions for θ : $\theta = \theta(y'(x(s)))$.

The chain rule then tells us

$$K = \frac{d\theta}{dy'} \frac{dy'}{dx} \frac{dx}{ds}.$$

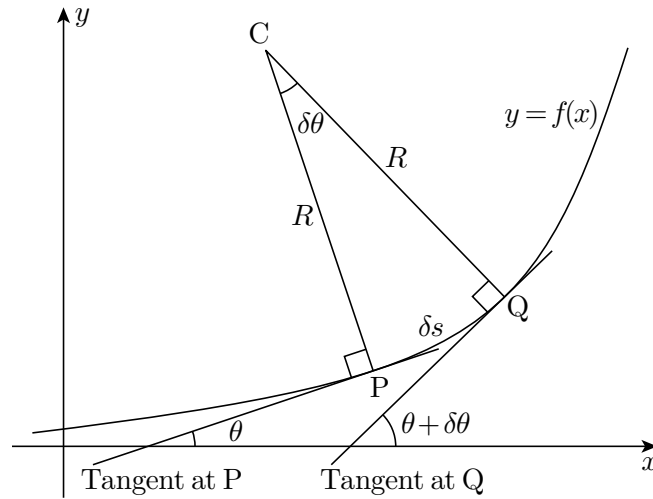


Figure 5.19: Curvature and the circle of curvature

All three of these derivatives can be rewritten in terms of derivatives of y .

Obviously, the middle one is trivial:

$$\frac{dy'}{dx} = y''.$$

For the first one, we note that the $\tan$ of the angle θ is just the gradient y' , $y' = \tan(\theta)$, and so $\theta = \tan^{-1}(y')$. This gives

$$\frac{d\theta}{dy'} = \frac{1}{1 + y'^2}.$$

Finally, we use the relation between infinitesimals, $ds^2 = dx^2 + dy^2$, divide through by dx^2 , take square roots both sides then the reciprocal to see that

$$\frac{dx}{ds} = \frac{1}{\sqrt{1 + y'^2}}.$$

Putting all three derivatives together yields a compact formula for K and R

$$K = \frac{y''}{(1 + y'^2)^{3/2}}, \quad R = |K|^{-1}. \quad (5.104)$$

Example 5.18 Find the curvature K , radius of curvature R , and the centre of curvature, for the point $(1, 0.5)$ on the curve $y = \frac{x^2}{2}$.

Solution.

$$y = \frac{x^2}{2}. \quad (5.105)$$

$$y' = x. \quad (5.106)$$

At (1,0.5),

$$y' = 1. \quad (5.107)$$

$$y'' = 1. \quad (5.108)$$

Using (5.104) we obtain

$$K = \frac{1}{(1 + 1^2)^{3/2}} = \frac{1}{2\sqrt{2}}$$

and

$$R = \frac{1}{K} = \frac{1}{1/(2\sqrt{2})} = 2\sqrt{2}.$$

The tangent line to the graph $y = x^2/2$ at point $x = 1$ has the slope of $y'(x = 1) = 1$ and the angle $\theta = \pi/4$ with the x -axis. Therefore, the centre of curvature is located at point (x_c, y_c) with

$$x_c = 1 - R \sin \theta = 1 - \frac{2\sqrt{2}}{\sqrt{2}} = -1, \quad y_c = 1/2 + R \cos \theta = 1/2 + 2 = 5/2.$$

5.5.1 Engineering application of curvature: transition curves

Most railway lines and road centrelines are either straight lines or arcs of circular curves. However, with both types of lines a transition (or easement) curve is introduced between a straight and a circular curve or between two circular curves of different curvature.

The function of a transition curve in railway lines is twofold:

1. It eliminates the sudden change in curvature in passing from a straight to a curve or from one arc to another of different curvature. The centripetal force (the force causing the train to travel along a curved line) acting on a vehicle of mass m travelling at speed v along a curve of radius of curvature R is $\frac{mv^2}{R}$. As the radius of the straight is infinite, the centripetal force would increase instantaneously from 0 to $\frac{mv^2}{r}$ when changing to a circular curve of radius r if no transition curve were inserted. Passengers in the train would experience a sideways shock and there would be increased wear of the outer rail. A curve of variable radius of curvature is inserted between the straight and circular curves so that the centripetal force can build up in a gradual manner.

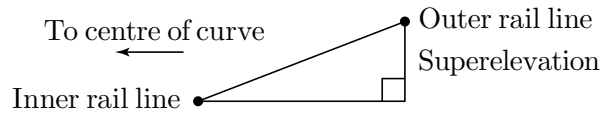


Figure 5.20: Superelevation (banking) of the outer rail.

2. It provides for the gradual introduction of the superelevation (banking) of the outer rail. The superelevation (see Fig. 5.20) is used to equalise the wear of the two rails and to reduce the possibility of the train lifting due to the moment of the centripetal force.

Similar considerations apply to the design of transition curves on roads. Usually road curves designed for a maximum speed of 60 km/h or more include transition curves.

Example 5.19 A railway line runs along the negative x -axis to the origin O , then it follows the cubic parabola transition curve $y = \frac{x^3}{500\,000}$ up to the point S , where $x = 170$ m, and then, without a change in direction, starts to follow a circular curve.

Given that the radius of curvature of the cubic parabola at S equals that of the circular arc, find

1. R , the radius of the circular arc,
2. θ , the angle that the tangent to the transition curve at S makes with the x -axis, and
3. (h, k) , the x - and y -coordinates of the centre of the circular curve.

Solution.

1. The coordinates of point S are $x_S = 170$ and $y_S = 9.82600$ m.

Using $y = \frac{x^3}{500\,000}$, we obtain

$$y' = \frac{3x^2}{500\,000}. \quad (5.109)$$

and

$$y'(x = x_S) = 0.17340. \quad (5.110)$$

Similarly,

$$y'' = \frac{3x}{250\,000}. \quad (5.111)$$

and

$$y''(x = x_S) = 0.00204. \quad (5.112)$$

The radius of curvature at S is

$$R = \frac{(1 + (y'_1)^2)^{3/2}}{y''_1} = 512.46994 \text{ m}. \quad (5.113)$$

2. At S, the angle θ of the slope can be found from

$$\tan \theta = y'(x = x_S). \quad (5.114)$$

This gives

$$\theta = 9^\circ 50' 14''.$$

3. The coordinates (h, k) of the centre of the circle of curvature are found as

$$h = x_1 - R \sin \theta \quad (5.115)$$

$$k = y_1 + R \cos \theta. \quad (5.116)$$

Substituting $x_1 = 170$ and $y_1 = 9.82600$, we obtain $h \approx 82.44470$ m and $k \approx 514.76115$ m.

5.5.2 Curvature of curve given by a parametric equation

For any curve on a plane or in space which is parametrised by time t , the position of a particle on the curve at time t , is given by the vector function $\vec{r} = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$.

The curve can also be parametrised by the arc length s measured from some arbitrary fixed point on the curve.

As the velocity vector $\vec{v} = \dot{\vec{r}}(t)$ and $\vec{v}$ is tangential to the curve, a unit tangent vector to a curve at a point given by time t , is $\vec{T}(t) = \dot{\vec{r}}(t)/|\dot{\vec{r}}(t)|$.

The curvature, K , at the point $P(t)$ can be shown to be given by

$$K = |\dot{\vec{r}}(t) \times \ddot{\vec{r}}(t)|/|\dot{\vec{r}}(t)|^3 = |\vec{v}(t) \times \vec{a}(t)|/|\vec{v}(t)|^3.$$

Example 5.20 Find the unit tangent vector $\vec{T}(t)$, and the curvature, K , of the twisted cubic curve, which has equation $\vec{r}(t) = t\vec{i} + t^2\vec{j} + t^3\vec{k}$.

5 Curvilinear coordinate systems and curves

Solution. Differentiating with respect to t gives $\vec{r}(t) = \vec{i} + 2t\vec{j} + 3t^2\vec{k}$ and $\vec{v}(t) = 2\vec{j} + 6t\vec{k}$.
Then

$$|\vec{r}(t)| = \sqrt{1 + 4t^2 + 9t^4},$$

$$\vec{T} = \frac{1}{\sqrt{1 + 4t^2 + 9t^4}}(\vec{i} + 2t\vec{j} + 3t^2\vec{k}),$$

and

$$\vec{r}(t) \times \vec{v}(t) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2t & 3t^2 \\ 0 & 2 & 6t \end{vmatrix} = 6t^2\vec{i} - 6t\vec{j} + 2\vec{k}.$$

Hence,

$$|\vec{r}(t) \times \vec{v}(t)| = \sqrt{4 + 36t^2 + 36t^4} = 2\sqrt{1 + 9t^2 + 9t^4}.$$

Thus,

$$K = \frac{2\sqrt{1 + 9t^2 + 9t^4}}{(\sqrt{1 + 4t^2 + 9t^4})^3}.$$

If x and y are expressed in parametric form, then the curvature formula in terms of the parameter t becomes

$$K = \frac{\dot{x}\ddot{y} - \ddot{x}\dot{y}}{(\dot{x}^2 + \dot{y}^2)^{3/2}}.$$

It is left as an exercise to prove this result starting with the curvature formula written in terms of y' and y'' (see (5.104)) and substituting for y' and y'' using their expressions derived from parametric differentiation:

$$y' = \frac{\dot{y}}{\dot{x}}y'' = \frac{\dot{x}\ddot{y} - \ddot{x}\dot{y}}{\dot{x}^3}. \quad (5.117)$$

Example 5.21 The position vector $\vec{r}$ of a particle changes with time according to

$$\vec{r}(t) = t \cos t \vec{i} + t \sin t \vec{j} + t \vec{k}, \quad t \geq 0.$$

1. At what time does the particle reach the point $B = (0, \pi/2, \pi/2)$?
2. Find the velocity and the acceleration vectors of the particle.
3. Calculate the angle between $\vec{r}'(0)$ and $\vec{r}''(0)$.
4. Write down the vector equation of the line, tangent to the trajectory of the particle at point $B = (0, \pi/2, \pi/2)$.
5. Calculate

$$\vec{r}'(\pi/2) \times \vec{r}''(\pi/2)$$

6. Calculate the curvature of the trajectory at point $B = (0, \pi/2, \pi/2)$.

Solution

1. The particle reaches point $B = (0, \pi/2, \pi/2)$ at $t = \pi/2$. Indeed

$$x = t \cos \pi/2 = 0, \quad y = \pi/2 \sin \pi/2 = \pi/2, \quad z = t = \pi/2.$$

2. The velocity vector at time t is

$$\vec{r}(t) = (\cos t - t \sin t)\vec{i} + (\sin t + t \cos t)\vec{j} + \vec{k}.$$

The acceleration vector at time t is

$$\vec{r}'(t) = (-2 \sin t - t \cos t)\vec{i} + (2 \cos t - t \sin t)\vec{j}.$$

3. The angle between $\vec{r}(0) = \vec{i} + \vec{k}$ and $\vec{r}'(0) = 2\vec{j}$ is found using the scalar product

$$\vec{r}(0) \cdot \vec{r}'(0) = (\vec{i} + \vec{k}) \cdot 2\vec{j} = 0, \quad |\vec{r}(0)| = \sqrt{2}, \quad |\vec{r}'(0)| = 2.$$

Consequently, the angle α between $\vec{r}(0) = \vec{i} + \vec{k}$ and $\vec{r}'(0) = 2\vec{j}$ is

$$\cos \alpha = 0, \rightarrow \alpha = \pi/2.$$

4. At point $B = (0, \pi/2, \pi/2)$, the tangent to the trajectory of the particle is given by the velocity vector

$$\vec{r}'(\pi/2) = -\frac{\pi}{2}\vec{i} + \vec{j} + \vec{k}.$$

The vector equation of the line, tangent to the trajectory can be written as

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ \pi/2 \\ \pi/2 \end{bmatrix} + s \begin{bmatrix} -\pi/2 \\ 1 \\ 1 \end{bmatrix}, \quad s \in (-\infty, \infty).$$

5. At point $B = (0, \pi/2, \pi/2)$, the acceleration vector is $\vec{r}''(\pi/2) = -2\vec{i} - \pi/2\vec{j}$. Therefore

$$\vec{r}'(\pi/2) \times \vec{r}''(\pi/2) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -\pi/2 & 1 & 1 \\ -2 & -\pi/2 & 0 \end{vmatrix} = \pi/2\vec{i} - 2\vec{j} + (2 + \pi^2/4)\vec{k}.$$

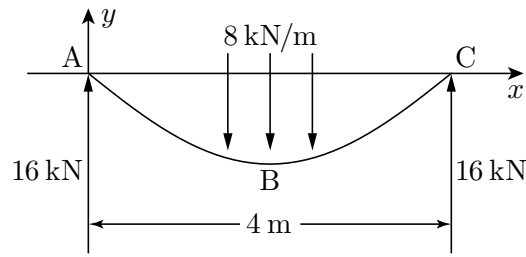


Figure 5.21: A simply supported beam bends under a uniform load.

6. The curvature at point $B = (0, \pi/2, \pi/2)$ is found as

$$\kappa = \frac{|\vec{r}'(\pi/2) \times \vec{r}''(\pi/2)|}{|\vec{r}'(\pi/2)|^3} = \frac{\sqrt{\pi^2/4 + 4 + (2 + \pi^2/4)^2}}{(\pi^2/4 + 2)^{3/2}} = 0.544$$

Exercises

- Find the curvature, radius of curvature, and centre of curvature
 - at $(1, 1)$ on the curve $y = x^2$
 - at $(2, 4)$ on the curve $y^2 = 2x^3$
 - at $\left(\frac{\pi}{6}, \frac{1}{2}\right)$ on the curve $y = \sin x$
- A simply supported beam of length 4 m has a uniform load of 8 k N/m (see Fig. 5.21).

The equation of the curve (ABC) formed by the beam is

$$y = \frac{1}{14.55 \times 10^3} \left(\frac{8x^3}{3} - \frac{x^4}{3} - \frac{64x}{3} \right).$$

Find

- R (the radius of curvature) at the midpoint B (where $x = 2$)
 - M (the bending moment) at B, given that $M = \frac{14.55 \times 10^6}{R}$ Newton metres
- A railway runs along the negative x -axis up to the origin O, then it follows the cubic parabola transition curve $y = \frac{x^3}{300000}$ up to the point S, where $x = 100$ m and then, without any change in direction, it follows a circular curve.

5 Curvilinear coordinate systems and curves

Given that the radius of curvature of the cubic parabola at S equals that of the circular arc, find (correct to 2 decimal places) the radius of the circular arc and the coordinates of the centre of the arc.

The cant or superelevation c is the amount by which the outer rail is raised above the inner rail to counter the effects of the centripetal force. At S, c is given by

$$c = \frac{Gv^2}{gR} \text{ metres,}$$

where G is the gauge of the railway in metres, v is the maximum velocity of a train, R is the radius of curvature of the railway line, and g is the acceleration due to gravity. Find c given that $G = 2$ m, $v = 60$ km/h, and $g = 9.81$ m/s².

4. A curve has vector equation $\vec{r} = (\cos t)\vec{i} + (\sin t)\vec{j} + t\vec{k}$.
Find
 - a) its unit tangent vector $\vec{T}$
 - b) its curvature, K .
5. A curve has vector equation $\vec{r} = (3t - t^3)\vec{i} + 3t^2\vec{j} + (3t + t^3)\vec{k}$.
Find
 - a) its unit tangent vector $\vec{T}$
 - b) its curvature, K .
6. A particle moves with position vector $\vec{r} = (t + \cos t)\vec{i} + (t - \cos t)\vec{j} + (\sqrt{2} \sin t)\vec{k}$.
Find its velocity $\vec{v}$, acceleration $\vec{a}$, and curvature K .
7. A curve has vector equation $\vec{r} = \left(\frac{4}{5} \cos t\right)\vec{i} + (1 - \sin t)\vec{j} - \left(\frac{3}{5} \cos t\right)\vec{k}$.
Find its curvature, K .

Answers

1. a) $\frac{2\sqrt{5}}{25}, \frac{5\sqrt{5}}{2}, (-4, 3.5)$
 b) $\frac{3}{40\sqrt{10}}, \frac{40\sqrt{10}}{3}, (-38, 17\frac{1}{3})$
 c) $-\frac{4}{7\sqrt{7}}, -\frac{7\sqrt{7}}{4}, \left(\frac{\pi}{6} + \frac{7\sqrt{3}}{4}, -3\right)$
2. a) 909 m
 b) 16 000 Nm
3. 507.52 m, (49.50, 508.33), 11.16 cm
4. a) $\vec{T} = \frac{1}{\sqrt{2}}(-(\sin t)\vec{i} + (\cos t)\vec{j} + \vec{k})$
 b) $K = \frac{1}{2}$
5. a) $\vec{T} = \frac{1}{\sqrt{2}(t^2 + 1)}((1 - t^2)\vec{i} + 2t\vec{j} + (1 + t^2)\vec{k})$
 b) $K = \frac{1}{3(1 + t^2)^2}$
6. $(1 - \sin t)\vec{i} + (1 + \sin t)\vec{j} + \sqrt{2}(\cos t)\vec{k}, -(\cos t)\vec{i} + (\cos t)\vec{j} - \sqrt{2}(\sin t)\vec{k}, \frac{\sqrt{2}}{4}$
7. 1

6 Complex Numbers

The equation

$$x^2 + 1 = 0$$

is not satisfied by any real number. For this reason mathematicians have extended the set of real numbers by introducing a new quantity, that is called .

Definition 6.1 *The imaginary unit, written j , satisfies the equation*

$$j^2 + 1 = 0,$$

that is, $j^2 = -1$.

Given this new quantity, we can develop the algebra of quantities involving j by applying all of the usual rules of algebra with the single additional rule $j^2 = -1$.

Definition 6.2 *We call any number of the form $x + jy$, where x and y are real numbers, a . We call x the real part of the complex number and y the imaginary part of the complex number, and if $z = x + jy$, we write $\Re(z) = x$ and $\Im(z) = y$.*

Here are examples of complex numbers with their real and imaginary parts:

$z_1 = -2 + 3j,$	$\Re(z_1) = -2,$	$\Im(z_1) = 3$
$z_2 = -5.324 - j,$	$\Re(z_2) = -5.324,$	$\Im(z_2) = -1$
$z_3 = \sqrt{3} + \frac{4}{7}j,$	$\Re(z_3) = \sqrt{3},$	$\Im(z_3) = \frac{4}{7}$
$z_4 = 8e + \pi j,$	$\Re(z_4) = 8e,$	$\Im(z_4) = \pi$
$z_5 = 457.92 + 106.1j,$	$\Re(z_5) = 457.92,$	$\Im(z_5) = 106.1$

6.1 Arithmetic operations with complex numbers

To state the rules for operating with complex numbers we will use the convention that $z = x + jy$ is a complex number and if z_1 and z_2 are complex then $z_1 = x_1 + jy_1$ and $z_2 = x_2 + jy_2$.

6 Complex Numbers

- Equality

Two complex numbers z_1 and z_2 are equal if and only if

$$\Re(z_1) = \Re(z_2) \text{ and } \Im(z_1) = \Im(z_2)$$

Example 6.1 Solve the equation $3z = 6 - 9j$.

Solution. Letting $z = x + jy$ we have $3(x + jy) = 6 - 9j$ so $3x + j3y = 6 - 9j$.

Equating real and imaginary parts gives $3x = 6$ and $3y = -9$. Thus, $x = 2$ and $y = -3$ so $z = 2 - 3j$.

- Addition and subtraction

$$\begin{aligned} z_1 + z_2 &= (x_1 + jy_1) + (x_2 + jy_2) = (x_1 + x_2) + j(y_1 + y_2) \\ z_1 - z_2 &= (x_1 + jy_1) - (x_2 + jy_2) = (x_1 - x_2) + j(y_1 - y_2) \end{aligned}$$

Example 6.2 $(2 - 4j) + (-4 + j) = (2 - 4) + j(-5 + 1) = -2 - 4j$

(b) $(-3 + 0.7j) - (4 - 2j) = (-3 - 4) + j(0.7 - (-2)) = -7 + 2.7j$

- Multiplication

The important new element here is $j^2 = -1$. When we apply this to multiplication of complex numbers, we obtain

$$\begin{aligned} z_1 z_2 &= (x_1 + jy_1)(x_2 + jy_2) = x_1 x_2 + jx_1 y_2 + jy_1 x_2 + j^2 y_1 y_2 \\ z_1 z_2 &= x_1 x_2 - y_1 y_2 + j(x_1 y_2 + x_2 y_1). \end{aligned}$$

Example 6.3 $(3 + 6j)(2 - 5j) = 6 - 10j^2 + j(4 - 15) = 6 + 10 - 11j = 16 - 11j$

(b) $(\sqrt{2} + j)(\sqrt{2} - j) = (\sqrt{2})^2 - j^2 + j(\sqrt{2} - \sqrt{2}) = 2 + 1 = 3$

Note that we can compute the successive powers of j :

$$\begin{aligned} j^3 &= j^2 j = -1 \cdot j = -j, \\ j^4 &= j^2 j^2 = -1 \cdot -1 = 1, \\ j^5 &= j, \end{aligned}$$

and so on.

Note also that

$$(x + jy)(x - jy) = x^2 - j^2 y^2 = x^2 + y^2.$$

From this we have such remarkable results as

$$(3 + 2j)(3 - 2j) = 3^2 + 2^2 = 9 + 4 = 13.$$

Thus the number 13 can be factorised if we use complex numbers.

3.2. Geometrical representation of complex numbers

- Conjugation

Definition 6.3 If $z = x + jy$ then we call the number $\bar{z} = x - jy$ the of z .

For example, $\overline{3 - 2j} = 3 + 2j$. We always have $z\bar{z} = x^2 + y^2$, i.e. the product of a number and its conjugate is a real number. Note that sometimes the conjugate of z is written z^* , i.e. $(x + jy)^* = x - jy$. It is clear that $(z^*)^* = z$.

- Division

To express $\frac{z_1}{z_2} = \frac{x_1 + jy_1}{x_2 + jy_2}$ in the form $z = x + jy$, we multiply both numerator and denominator by $\bar{z}_2$

$$\begin{aligned}\frac{x_1 + jy_1}{x_2 + jy_2} &= \frac{(x_1 + jy_1)(x_2 - jy_2)}{(x_2 + jy_2)(x_2 - jy_2)} \\ &= \frac{(x_1x_2 + y_1y_2) + j(x_2y_1 - x_1y_2)}{(x_2^2 + y_2^2)}.\end{aligned}$$

Example 6.4

$$\frac{(5 + 3j)}{(4 - j)} = \frac{(5 + 3j)(4 + j)}{(4 - j)(4 + j)} = \frac{20 + 5j + 12j + 3j^2}{4^2 + 1^2} = \frac{17 + 17j}{17} = 1 + j.$$

Example 6.5

$$\frac{1}{2 - 3j} = \frac{1}{(2 - 3j)(2 + 3j)} = \frac{2 + 3j}{2^2 + 3^2} = \frac{2}{13} + \frac{3j}{13}.$$

6.2 Geometrical representation of complex numbers

A complex number $z = x + jy$ can be represented by the point P with coordinates (x, y) in the Cartesian plane. We then call the x -axis the and the y -axis the . A representation of complex numbers in this way is called an *Argand diagram*. The Cartesian plane is referred to as the complex plane, when used to represent the set of complex numbers. Addition of complex numbers is analogous to head-to-tail addition of vectors with components x and y .

6.2.1 Polar form

Given a complex number $z = x + jy$ we can convert to polar coordinates in the plane using the relations $x = r \cos \theta$, $y = r \sin \theta$. Then we have $z = r \cos \theta + jr \sin \theta$ or

3.2. Geometrical representation of complex numbers

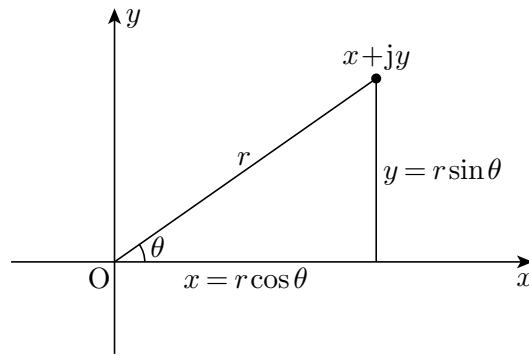


Figure 6.1: Complex number $z = x + jy$ in the polar form.

$$z = r(\cos \theta + j \sin \theta),$$

as shown in Fig. 6.1.

The quantity $r = \sqrt{x^2 + y^2}$ is called the *magnitude* or *modulus* of z . It is written as $|z|$,

$$r = |z| = |x + jy| = \sqrt{x^2 + y^2}.$$

The quantity

$$|z_1 - z_2|$$

can be regarded as the distance between the complex numbers z_1 and z_2 in the Argand plane.

Note the relation $z\bar{z} = |z|^2 = r^2$

The angle θ is called the *argument* or *angle* of the complex number z .

Example 6.6 $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

$$|1 - j| = \sqrt{1 + 1} = \sqrt{2}$$

$$|-1 + \sqrt{3}j| = \sqrt{1 + 3} = \sqrt{4} = 2$$

Now there are infinitely many possible values for θ , for if $z = r(\cos \theta + j \sin \theta)$ then

$$z = r(\cos(\theta + 2n\pi) + j \sin(\theta + 2n\pi))$$

for any integer n . For this reason we define the **principal argument** of z to be that θ such that $z = r(\cos \theta + j \sin \theta)$ where r is positive and θ satisfies the condition $-\pi < \theta \leq \pi$. This is denoted by $\arg(z)$.

Example 6.7 Express $1 + j$ in polar form.

3.2. Geometrical representation of complex numbers

Solution. $|1 + j| = \sqrt{2}$, and $\arg(1 + j) = \arctan(1) = \frac{\pi}{4}$, so $1 + j = \sqrt{2}(\cos \pi/4 + j \sin \pi/4)$.

Example 6.8 Express $2\sqrt{3} - 2j$ in polar form.

Solution. $|2\sqrt{3} - 2j| = \sqrt{(2\sqrt{3})^2 + (-2)^2} = \sqrt{16} = 4$.

To calculate the argument, note that $2\sqrt{3} - 2j$ lies in the fourth quadrant so

$$\arg(2\sqrt{3} - 2j) = \arctan\left(\frac{-2}{2\sqrt{3}}\right) = -\arctan\frac{2}{2\sqrt{3}} = -\frac{\pi}{6}.$$

Thus, $2\sqrt{3} - 2j = 4(\cos(-\pi/6) + j \sin(-\pi/6))$.

Generally, we have, for $z = x + jy$,

$$\begin{aligned}\arg(z) &= \arctan \frac{y}{x} \quad \text{if } x > 0 \text{ (1st and 4th quadrants)} \\ \arg(z) &= \arctan \frac{y}{x} + \pi \quad \text{if } x < 0, y \geq 0 \text{ (2nd quadrant)} \\ \arg(z) &= \arctan \frac{y}{x} - \pi \quad \text{if } x < 0, y < 0 \text{ (3rd quadrant)}.\end{aligned}$$

In fact, when calculating the polar form of a complex number, it is simplest to draw a diagram locating the complex number on an Argand diagram and then it should be clear which value to give to the argument.

6.2.2 Multiplication in polar form

Consider first those complex numbers which have the form $z = \cos \theta + j \sin \theta$; we have, for these numbers, $|z| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$. Thus, these numbers are represented by the points on the circle $x^2 + y^2 = 1$.

If we multiply two such numbers, $\cos \theta + j \sin \theta$ and $\cos \phi + j \sin \phi$, we have

$$\begin{aligned}(\cos \theta + j \sin \theta)(\cos \phi + j \sin \phi) &= (\cos \theta \cos \phi - \sin \theta \sin \phi) + j(\sin \theta \cos \phi + \sin \phi \cos \theta) \\ &= \cos(\theta + \phi) + j \sin(\theta + \phi) \\ &= \cos(\theta + \phi) + j \sin(\theta + \phi).\end{aligned}$$

Thus, when we multiply these numbers, we simply add their arguments. Now to multiply *any* two complex numbers z_1, z_2 , we can bring them into polar form $z_1 = r_1 \cos \theta_1 + jr_1 \sin \theta_1, z_2 = r_2 \cos \theta_2 + jr_2 \sin \theta_2$. Then we can multiply them as follows

3.2. Geometrical representation of complex numbers

$$z_1 z_2 = r_1 r_2 [\cos (\theta_1 + \theta_2) + j \sin (\theta_1 + \theta_2)].$$

That is, we multiply the magnitudes and *add* the arguments. (We may need to add or subtract π to get the principal argument of the product.)

Example 6.9 Let $z_1 = 3j$, $z_2 = 4 + 4j$. Then in Cartesian form we have

$$z = z_1 z_2 = 3j(4 + 4j) = 12j + 12j^2 = -12 + 12j;$$

in polar form we have

$$|z_1| = 3 \quad \text{and} \quad \arg(z_1) = \frac{\pi}{2}$$

and

$$|z_2| = \sqrt{4^2 + 4^2} = 4\sqrt{2} \quad \text{and} \quad \arg(z_2) = \frac{\pi}{4},$$

so

$$z_1 = 3(\cos \frac{\pi}{2} + j \sin \frac{\pi}{2}) \quad \text{and} \quad z_2 = 4\sqrt{2}(\cos \frac{\pi}{4} + j \sin \frac{\pi}{4}),$$

and

$$\begin{aligned} z_1 z_2 &= 12\sqrt{2} \left(\cos \frac{3\pi}{4} + j \sin \frac{3\pi}{4} \right) \\ &= 12\sqrt{2} \left(-\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} j \right) \\ &= -12 + 12j. \end{aligned}$$

6.2.3 Division in polar form

We can treat division of complex numbers as multiplication by the reciprocal:

$$\frac{z_1}{z_2} = z_1 \frac{1}{z_2}.$$

Now if we calculate $\frac{1}{(\cos \theta + j \sin \theta)}$, we have

$$\begin{aligned} \frac{1}{(\cos \theta + j \sin \theta)} &= \frac{1}{\cos \theta + j \sin \theta} \\ &= \frac{1}{\cos \theta + j \sin \theta} \frac{\cos \theta - j \sin \theta}{\cos \theta - j \sin \theta} \\ &= \frac{\cos \theta - j \sin \theta}{\cos^2 \theta + \sin^2 \theta} \\ &= \cos \theta - j \sin \theta. \end{aligned}$$

i.e. $\frac{1}{(\cos \theta + j \sin \theta)} = (\cos \theta + j \sin \theta)^*$. Thus we have

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos (\theta_1 - \theta_2) + j \sin (\theta_1 - \theta_2)].$$

Thus, to divide complex numbers, we divide their magnitudes and subtract their arguments.

6.3 Powers and roots of complex numbers

We now consider powers of $r(\cos \theta + j \sin \theta)$. Using the results from the previous section, we can show that the following is true

Theorem 6.1 (De Moivre's Theorem)

$$\begin{aligned} r^n (\cos \theta + j \sin \theta)^n &= r^n \underbrace{(\cos \theta + j \sin \theta)(\cos \theta + j \sin \theta) \dots (\cos \theta + j \sin \theta)}_n \\ &= r^n \cos (n\theta) + j \sin (n\theta) \end{aligned} \quad (6.118)$$

Example 6.10 By first converting to polar form, find $(1 + j)^{10}$.

Solution. We have

$$|z| = \sqrt{1+1} = \sqrt{2} \quad \text{and} \quad \arg(z) = \frac{\pi}{4},$$

thus

$$z = \sqrt{2} \left(\cos \frac{\pi}{4} + j \sin \frac{\pi}{4} \right)$$

so

$$z^{10} = 32 \left(\cos \frac{5\pi}{2} + j \sin \frac{5\pi}{2} \right) = 32 \left(\cos \frac{\pi}{2} + j \sin \frac{\pi}{2} \right) = 32j.$$

6.3.1 Roots of complex numbers

By $z^{1/n}$ we mean any number which, when raised to the power n , gives z . Thus $w = z^{1/n}$ if and only if $w^n = z$. To find an expression for $z^{1/n}$ given z , we write z in polar form $z = r(\cos \theta + j \sin \theta)$; then we suppose that $w = \rho(\cos \phi + j \sin \phi)$ satisfies $w = z^{1/n}$. Thus $w^n = z$, so $\rho^n(\cos \phi + j \sin \phi)^n = r(\cos \theta + j \sin \theta)$.

Then we must have

$$\rho^n (\cos (n\phi) + j \sin (n\phi)) = r(\cos \theta + j \sin \theta).$$

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But note also that

$$\rho^n (\cos (n\phi) + j \sin (n\phi)) = \rho^n (\cos (n\phi + 2\pi k) + j \sin (n\phi + 2\pi k)),$$

where $k = 0, \pm 1, \pm 2, \dots$

so that

$$\rho = r^{1/n} \quad \text{and} \quad \phi = \frac{\theta + 2\pi k}{n}.$$

Thus there are n different complex numbers which when raised to the power of n give $z = r(\cos \theta + j \sin \theta)$. These are

$$\begin{aligned} & r^{1/n} \left(\cos \frac{\theta}{n} + j \sin \frac{\theta}{n} \right), \\ & r^{1/n} \left(\cos \frac{\theta + 2\pi}{n} + j \sin \frac{\theta + 2\pi}{n} \right), \\ & r^{1/n} \left(\cos \frac{\theta + 4\pi}{n} + j \sin \frac{\theta + 4\pi}{n} \right), \\ & \dots \\ & r^{1/n} \left(\cos \frac{\theta + (n-1)2\pi}{n} + j \sin \frac{\theta + (n-1)2\pi}{n} \right). \end{aligned} \quad (6.119)$$

Example 6.11 Find $(-8)^{1/3}$.

Solution. We express -8 in polar form: $-8 = 8(\cos \pi + j \sin \pi)$. Using (6.119) we obtain

$$\begin{aligned} (-8)^{1/3} = & \\ & 2 \left(\cos \frac{\pi}{3} + j \sin \frac{\pi}{3} \right), \\ & 2(\cos \pi + j \sin \pi), \\ & 2 \left(\cos \frac{5\pi}{3} + j \sin \frac{5\pi}{3} \right). \end{aligned}$$

That is,

$$(-8)^{1/3} = 1 + j\sqrt{3}, \quad -2, \quad 1 - j\sqrt{3}.$$

If we plot them on an Argand diagram these roots are equally spaced on the circle of radius 2 centred at 0, as shown in Fig. 6.2.

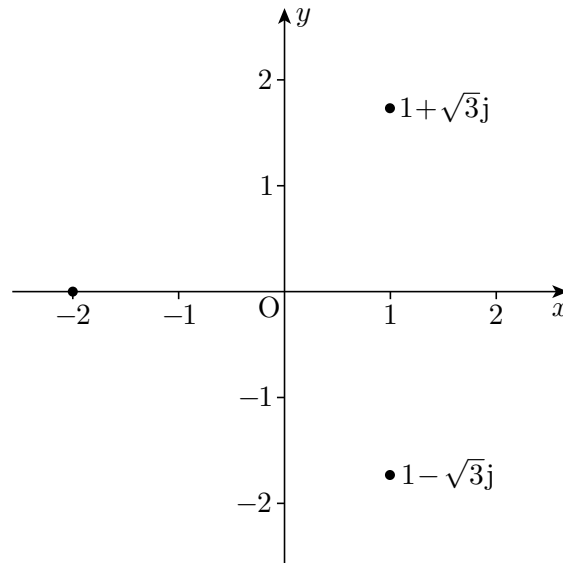


Figure 6.2: Three different values of $(-8)^{1/3}$

6.4 Exponential form of $(\cos \theta + j \sin \theta)$ and Euler's formula

The following power series can be established for the functions $\sin x$, $\cos x$, and e^x :

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \frac{x^7}{7!} + \dots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \\ \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \end{aligned}$$

If we replace x by $j\theta$ in the series for e^x , we obtain

$$e^{j\theta} = 1 + j\theta + \frac{(j\theta)^2}{2!} + \frac{(j\theta)^3}{3!} + \frac{(j\theta)^4}{4!} + \frac{(j\theta)^5}{5!} + \frac{(j\theta)^6}{6!} + \frac{(j\theta)^7}{7!} + \dots$$

We can rearrange this gathering real and imaginary parts to obtain the result

$$\begin{aligned} e^{j\theta} &= 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} \dots + j\theta + \frac{(j\theta)^3}{3!} + \frac{(j\theta)^5}{5!} - \frac{(j\theta)^7}{7!} + \dots \\ e^{j\theta} &= \underbrace{1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} \dots}_{\cos \theta} + j \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots \right)}_{\sin \theta} \\ e^{j\theta} &= \cos \theta + j \sin \theta. \end{aligned}$$

3.2. Geometrical representation of complex numbers

Definition 6.4 The exponential function e^z of a complex argument $z = x + jy$ is defined by

$$e^z = e^{x+jy} = e^x e^{jy} = e^x (\cos y + j \sin y). \quad (6.120)$$

Thus $|e^z| = e^{\Re(z)}$.

(6.120) is known as for complex numbers.

We thus have another way of writing complex numbers in polar form, namely

$$z = re^{j\theta},$$

where $|z| = r$ and $\arg(z) = \theta$.

For example

$$1 + j = \sqrt{2}e^{j\frac{\pi}{4}}, \quad j = e^{j\frac{\pi}{2}}$$

and note the remarkable result

$$e^{j\pi} = -1.$$

Example 6.12 Express $z = e^{3-j}$ in Cartesian form.

Solution.

$$z = e^3 e^{-j} = e^3 (\cos(-1) + j \sin(-1)) = 10.85226 - 16.90140j.$$

It is often clearer to use the notation $\exp(x)$ for the exponential function e^x ; that is, $\exp(x) \equiv e^x$. Thus,

$$\exp\left(\frac{\pi}{4}j\right) = \frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}.$$

6.4.1 The fundamental theorem of algebra

The quantity j was introduced to solve the equation

$$x^2 + 1 = 0.$$

We thus enlarge the system of real numbers to the set of complex numbers, that is, the set of numbers of the form $x + iy$. It is natural to ask whether there are algebraic equations which can be written involving complex numbers, which do not have a solution. In fact, this is not the case and the states that

3.2. Geometrical representation of complex numbers

Theorem 6.2 *any algebraic equation*

$$a_0 + a_1z + a_2z^2 + \cdots + a_nz^n = 0,$$

where the coefficients may be any complex numbers, has exactly n possibly complex solutions, counting their respective multiplicities.

Thus the set of complex numbers is *complete* in the sense that any algebraic equation whose coefficients are complex numbers has a solution in the set of complex numbers. We illustrate how the roots of equations involving complex numbers may be found.

Example 6.13 *Find all solutions of the equation*

$$z^2 + 4j = 0.$$

Solution. We have $z^2 = -4j$, so $z^2 = 4\left(\cos\left(-\frac{\pi}{2}\right) + j\sin\left(-\frac{\pi}{2}\right)\right)$. Hence, $z = 2\left(\cos\left(-\frac{\pi}{4}\right) + j\sin\left(-\frac{\pi}{4}\right)\right)$ and $z = 2\left(\cos\left(-\frac{\pi}{4} + \pi\right) + j\sin\left(-\frac{\pi}{4} + \pi\right)\right)$ That is, the roots are

$$z_1 = 2\left(\cos\left(-\frac{\pi}{4}\right) + j\sin\left(-\frac{\pi}{4}\right)\right) = \sqrt{2} - j\sqrt{2}$$

and

$$z_2 = 2\left(\cos\left(\frac{3\pi}{4}\right) + j\sin\left(\frac{3\pi}{4}\right)\right) = -\sqrt{2} + j\sqrt{2}.$$

(You can check this by squaring them.)

Example 6.14 *Find all solutions of the equation*

$$z^4 + 4z^2 + 16 = 0.$$

Solution. We first make the substitution $u = z^2$; then the equation can be written

$$u^2 + 4u + 16 = 0.$$

The quadratic formula gives the two roots of this equation

$$u = \frac{-4 \pm \sqrt{-48}}{2} = -2 \pm 2\sqrt{-3} = -2 \pm 2\sqrt{3}j.$$

We then have

$$z^2 = -2 + 2\sqrt{3}j \quad \text{and} \quad z^2 = -2 - 2\sqrt{3}j.$$

We now solve these equations by first putting them in polar form. Firstly we consider

$$z^2 = -2 + 2\sqrt{3}j = 4\left(\cos\frac{2\pi}{3} + j\sin\frac{2\pi}{3}\right).$$

3.2. Geometrical representation of complex numbers

This gives

$$z = 2\left(\cos \frac{2\pi}{6} + j \sin \frac{2\pi}{6}\right) \quad \text{and} \quad z = 2\left(\cos \frac{8\pi}{6} + j \sin \frac{8\pi}{6}\right),$$

while

$$z^2 = -2 - 2\sqrt{3}j = 4\left(\cos \frac{-2\pi}{3} + j \sin \frac{-2\pi}{3}\right) = 4\left(\cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3}\right)$$

gives

$$z = 2\left(\cos \frac{-\pi}{3} + j \sin \frac{-\pi}{3}\right) \quad \text{and} \quad z = 2\left(\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3}\right).$$

Thus the four roots of the equation are

$$2\left(\cos \frac{\pi}{3} + j \sin \frac{\pi}{3}\right), \quad 2\left(\cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3}\right), \quad 2\left(\cos \left(-\frac{\pi}{3}\right) + j \sin \left(-\frac{\pi}{3}\right)\right)$$

and

$$2\left(\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3}\right).$$

Exercises

1. Express in simplest Cartesian form:

- | | | |
|---|---|----------------------------|
| (a) $(2 + j) + (3 - 2j)$ | (b) $(3 - 5j) + (-2 + 6j)$ | (c) $(3 + 4j)(2 + j)$ |
| (d) $(-2 + j) - (-5 - 3j)$ | (e) $j(2 - 3j)$ | (f) $(1 + j)^2$ |
| (g) $(1 + \sqrt{3}j)^3$ | (h) $(1 + j)^{-1}$ | (j) $\frac{25}{(4 + 3j)}$ |
| (i) $(-4 + 3j)^{-2}$ | (k) $(1 - j)^3$ | (l) $\frac{1 + j}{1 - j}$ |
| (m) $\frac{2 + 3j}{3 - 2j}$ | (n) $\left(\frac{1 - j}{1 + j}\right)^2$ | (o) $\frac{2 + j}{4 + 3j}$ |
| (p) $\frac{1}{(1 - j)^2} - \frac{1}{(1 + j)^2}$ | (q) $\frac{a + bj}{a - bj} - \frac{a - bj}{a + bj}$ | |

2. Taking $z = x + jy$ express in Cartesian form

- (a) $\frac{1}{z}$ (b) $\frac{1}{z + 1}$ (c) $\frac{\bar{z}}{z}$ (d) $\frac{-(\bar{z})}{-z}$

3. If $z = x + jy$ we write $\Re(z) = x$ and $\Im(z) = y$. For the following complex numbers, find $\Re(z)$ and $\Im(z)$.

- (a) $z = (x + jy)^3$ (b) $z = (x + 1 + jy)^2$ (c) $z = ((x - j) + jy)j$ (d) $\frac{5 + j}{5 - 3j}$

4. Find real numbers x, y such that

- (a) $x(3 - j) + y(5 + 2j) = 7 + 5j$ (b) $(3 - 2j)(x + jy) = 2(x - 2jy) + 2j - 1.$

3.2. Geometrical representation of complex numbers

5. Express the following numbers in polar form $z = r(\cos \theta + j \sin \theta)$
 - (a) $4 - 4j$
 - (b) $-1 + \sqrt{3}j$
 - (c) $\frac{1 - 5j}{2 + 3j}$
 - (d) $\frac{3j}{1 + j}$
6. Find the absolute value, and principal argument of the following complex numbers:
 - (a) $5 + 12j$
 - (b) -4
 - (c) $2j$
 - (d) $\sqrt{3} - j$
 - (e) $-1 - \sqrt{3}j$
 - (f) $-3 + 3j$
7. Find the modulus of
 - (a) $x + 2j$
 - (b) $\frac{1}{x + jy}$
 - (c) $\cos \theta + j \sin \theta$
 - (d) $(x - 1) + (y - 2)j$
 - (e) $j(1 + j \cos \theta + \sin \theta)$
8. Find the modulus and argument, correct to five decimal places, of
 - (a) $4 - 3j$
 - (b) $-5 + 3j$
 - (c) $-2 - 3j$
 - (d) $-0.5 - 6j$
9. Write in simplest Cartesian form
 - (a) $2 + 3j$
 - (b) $\frac{1}{2 + 3j}$
 - (c) $\frac{1 + j}{1 - j}$
 - (d) $\left(\frac{1 + j}{1 - j}\right)^2$
10. Write in simplest Cartesian form:
 - (a) $1 + j$
 - (b) $(1 + j)^2$
 - (c) $(1 + j)^3$
 - (d) $(1 + j)^4$
11. Specify the set of points in the complex plane which satisfy $|z + 1 + j| = 2$
12. Plot the set of points in the complex plane for which $|2z + 3| \leq 1$
13. Express the complex number $2 + 2\sqrt{3}j$ in the form $re^{j\theta}$, hence obtaining the Cartesian form of $(2 + 2\sqrt{3}j)^6$
14. By first converting to polar form, evaluate $(-1 + j)^8$
15. By first converting to polar form, evaluate
 - (a) $\frac{-1 + j}{1 + j}$
 - (b) $\frac{(-1 - j)^6}{(1 + j)^4}$
 - (c) $\frac{1 + j\sqrt{3}}{1 - j\sqrt{3}}$
 - (d) $\frac{(-1 + j)^5}{(1 - j)^4}$
16. In each case below, give your answers in polar form with Principal Argument then show the roots on an Argand diagram.
 - a) Find the three cube roots of -1 .
 - b) Find all of the fourth roots of $16j$.
 - c) Find all of the cube roots of $-8j$.
 - d) Find all cube roots of $\sqrt{3} + j$.

3.2. Geometrical representation of complex numbers

17. Find all roots of the equations

a) $z^4 + 2z^2 + 4 = 0$

b) $w^4 - w^2 + 1 = 0$

c) $z^4 + 1 = 0$

d) $z^4 + 2z^3 + 5z^2 + 8z + 4 = 0$

e) $z^3 + 3z^2 + 12z + 10 = 0$

18. Express in Cartesian form

(a) e^{2+j} (b) $e^{-1+\frac{\pi j}{3}}$ (c) $\sqrt{2}e^{-\frac{\pi j}{4}}$ (d) $5e^{1-\frac{\pi j}{3}}$

Answers

1. (a) $5 - j$ (b) $1 + j$ (c) $2 + 11j$ (d) $3 + 4j$
 (e) $3 + 2j$ (f) $2j$ (g) -8 (h) $\frac{1}{2} - \frac{1}{2}j$
 (i) $4 - 3j$ (j) $\frac{7}{625} + \frac{24j}{625}$ (k) $-2 - 2j$ (l) j
 (m) j (n) -1 (o) $\frac{11}{25} - \frac{2j}{25}$ (p) j
 (q) $\frac{4abj}{a^2 + b^2}$

2. (a) $\frac{x - yj}{x^2 + y^2}$ (b) $\frac{(x + 1) - yj}{(x + 1)^2 + y^2}$ (c) $\frac{(x^2 - y^2 - 2xyj)}{x^2 + y^2}$ (d) 1

3. a) $\Re((x + jy)^3) = x^3 - 3xy^2, \quad \Im((x + jy)^3) = 3x^2y - y^3$
 b) $\Re((x + 1 + jy)^2) = (x + 1)^2 - y^2, \quad \Im((x + 1 + jy)^2) = 2(x + 1)y$
 c) $\Re(((x - j) + jy)j) = -y + 1, \quad \Im(((x - j) + jy)j) = x$
 d) $\Re\left(\frac{5 + j}{5 - 3j}\right) = \frac{11}{17}, \quad \Im\left(\frac{5 + j}{5 - 3j}\right) = \frac{10}{17}.$

4. (a) $x = -1, y = 2$ (b) $x = -1, y = 0$

5. (a) $4 - 4j = 4\sqrt{2}\left(\cos \frac{\pi}{4} - j \sin \frac{\pi}{4}\right)$ (b) $-1 + \sqrt{3}j = 2\left(\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3}\right)$
 (c) $-1 - j = \sqrt{2}\left(\cos \frac{3\pi}{4} - j \sin \frac{2\pi}{4}\right)$ (d) $\frac{3}{2}(1 + j) = \frac{3}{2}\sqrt{2}\left(\cos \frac{\pi}{4} + j \sin \frac{\pi}{4}\right)$

6. a) $|5 + 12j| = 13, \quad \arg(5 + 12j) = 1.17601$

3.2. Geometrical representation of complex numbers

- b) $|-4| = 4, \quad \arg(-4) = \pi$
- c) $|2j| = 2, \quad \arg(2j) = \frac{\pi}{2}$
- d) $|\sqrt{3} - j| = 2, \quad \arg(\sqrt{3} - j) = -\frac{\pi}{6}$
- e) $|-1 - \sqrt{3}j| = 2, \quad \arg(-1 - \sqrt{3}j) = -\frac{2\pi}{3}$
- f) $|-3 + 3j| = 3\sqrt{2}, \quad \arg(-3 + 3j) = \frac{3\pi}{4}$
7. a) $|x + 2j| = \sqrt{x^2 + 4}$
- b) $\left| \frac{1}{x + jy} \right| = \frac{1}{\sqrt{x^2 + y^2}}$
- c) 1
- d) $\sqrt{(x-1)^2 + (y-2)^2}$
- e) $\sqrt{2(1 + \sin \theta)}$
8. a) $|4 - 3j| = 5, \arg(4 - 3j) = -0.643501$
- b) $|-5 + 3j| = 5.83095, \arg(-5 + 3j) = 2.60117$
- c) $|-2 - 3j| = 3.60555, \arg(-2 - 3j) = -2.1588$
- d) $|-0.5 - 6j| = 6.0208, \arg(-0.5 - 6j) = -1.65394$
9. (a) $2 + 3j$ (b) $\frac{2}{13} - \frac{3j}{13}$ (c) j (d) -1
10. (a) $1 + j$ (b) $2j$ (c) $-2 + 2j$ (d) -4
11. The circumference of the circle of radius 2 and centre the point $-1 - j$, as shown in Fig. 6.3.

3.2. Geometrical representation of complex numbers

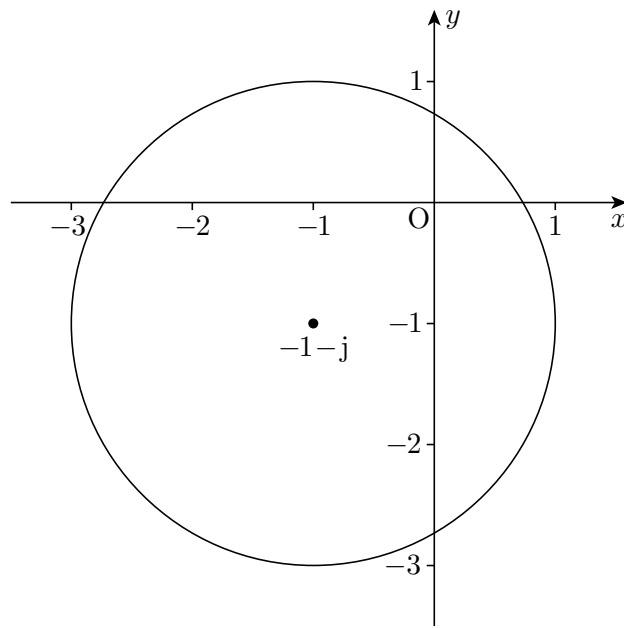


Figure 6.3: Set of points $|z + 1 + j| = 2$

12. See Fig. 6.4.

13. $2 + 2\sqrt{3}j = 4e^{j\frac{\pi}{3}}, \quad (2 + 2\sqrt{3}j)^6 = 4096$

14. $-1 + j = \sqrt{2}\left(\cos \frac{3\pi}{4} + j \sin \frac{3\pi}{4}\right), \quad (-1 + j)^8 = 16$

15. (a) j (b) $2j$ (c) $-\frac{1}{2} + \frac{\sqrt{3}}{2}j$ (d) $-1 + j$

16. a) $\left(\cos \frac{\pi}{3} + j \sin \frac{\pi}{3}\right), \quad (\cos \pi + j \sin \pi), \quad \left(\cos \frac{\pi}{3} - j \sin \frac{\pi}{3}\right),$ as shown in Fig. 6.5

3.2. Geometrical representation of complex numbers

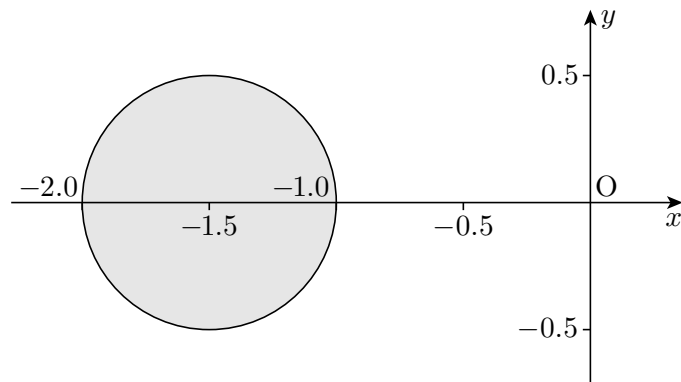


Figure 6.4: Inside of the circle $|2z + 3| \leq 1$.

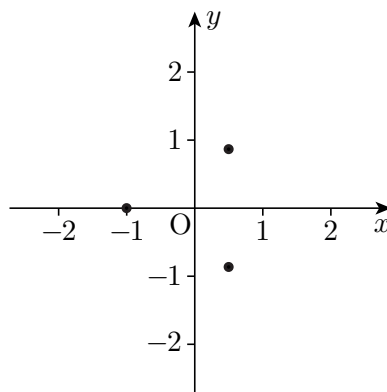


Figure 6.5: Three values of $(-1)^{1/3}$.

b) $2\left(\cos \frac{\pi}{8} + j \sin \frac{\pi}{8}\right)$, $2\left(\cos \frac{5\pi}{8} + j \sin \frac{5\pi}{8}\right)$, $2\left(\cos \frac{7\pi}{8} - j \sin \frac{7\pi}{8}\right)$, $2\left(\cos \frac{3\pi}{8} - j \sin \frac{3\pi}{8}\right)$, as shown in Fig. 6.6

c) $2\left(\cos \frac{\pi}{2}\right)$, $2\left(\cos \frac{5\pi}{6} - j \sin \frac{5\pi}{6}\right)$, $2\left(\cos \frac{\pi}{6} - j \sin \frac{\pi}{6}\right)$, as shown in Fig. 6.7

3.2. Geometrical representation of complex numbers

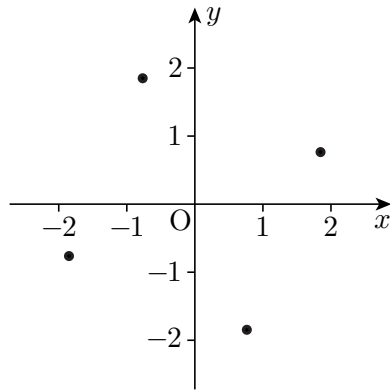


Figure 6.6: Four values of $(16j)^{1/4}$.

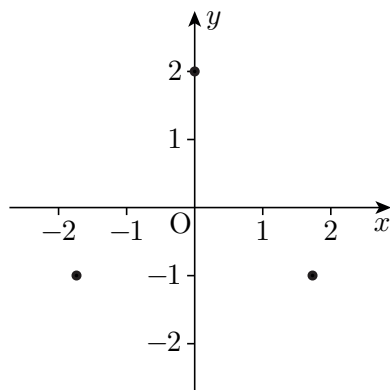


Figure 6.7: Three values of $(-8j)^{1/3}$

3.2. Geometrical representation of complex numbers

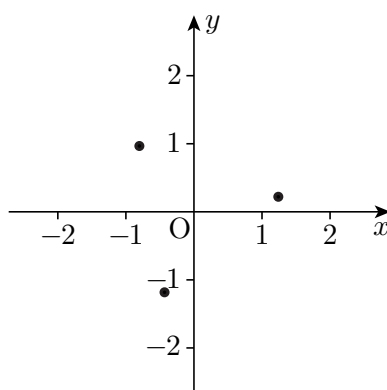


Figure 6.8: Three values of $(\sqrt{3} + j)^{1/3}$

d) $2^{1/3} \left(\cos \frac{\pi}{18} + j \sin \frac{\pi}{18} \right), \quad 2^{1/3} \left(\cos \frac{13\pi}{18} + j \sin \frac{13\pi}{18} \right), \quad 2^{1/3} \left(\cos \frac{11\pi}{18} - j \sin \frac{11\pi}{18} \right),$
as shown in Fig. 6.8

17. a) $\frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2}j, \quad -\frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2}j, \quad \frac{\sqrt{2}}{2} - \frac{\sqrt{6}}{2}j, \quad -\frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2}j$

b) $-\frac{\sqrt{3}}{2} - \frac{1}{2}j, \quad \frac{\sqrt{3}}{2} + \frac{1}{2}j, \quad \frac{\sqrt{3}}{2} - \frac{1}{2}j, \quad -\frac{\sqrt{3}}{2} + \frac{1}{2}j$

c) $\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j, \quad -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j, \quad \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}j, \quad -\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}j$

d) $-1, \quad -2j, \quad 2j$

e) $-1, \quad -1 - 3j, \quad -1 + 3j$

18. (a) $e^{2+j} = e^2(\cos 1 + j \sin 1),$ (b) $\frac{1}{e} \left(\frac{1}{2} + \frac{\sqrt{3}}{2}j \right),$ (c) $1 - j,$ (d) $\frac{5e}{2} - \frac{5\sqrt{3}e}{2}i$

6.5 Engineering application: impedance

One application of complex numbers is in the analysis of circuits in which there is an alternating current (ac) power supply. For voltage supplies which vary sinusoidally, it is often most efficient to analyse circuits using complex numbers.

A typical ac circuit will consist of a voltage source, a resistance R , an inductance L and a capacitance C in series, with a current I amps flowing through the circuit (called an RLC circuit), as shown in Fig. 6.9.

3.2. Geometrical representation of complex numbers

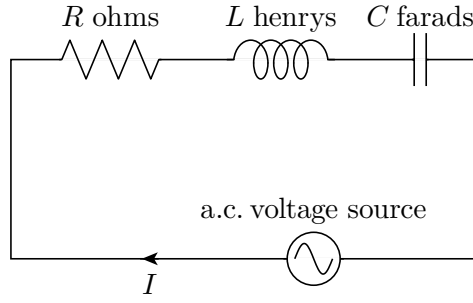


Figure 6.9: Complex impedance of an electric circuit.

The voltage across the resistance is given by $V_R = IR$ (Ohm's law). Across the inductance the voltage drop is given by

$$V_L = L \frac{dI}{dt},$$

where t = time in seconds and across the capacitance C the voltage satisfies the relation

$$\frac{dV_C}{dt} = \frac{I}{C}.$$

If the current in the circuit is given by $I = I_0 \sin \omega t$, then the equations allow us to calculate

$$V_R = RI_0 \sin \omega t, \quad V_L = \omega LI_0 \cos \omega t, \quad V_C = -\frac{1}{\omega C} I_0 \cos \omega t.$$

We then obtain an expression for the total voltage using

$$V = V_R + V_L + V_C.$$

It is, in fact, simpler to express the relation between the current and voltage by introducing the *complex current* I_c and the *complex voltage* V_c . $I_c = I_0 e^{j\omega t}$, where the actual current is given by the *imaginary* part of I_c . We then calculate

$$\begin{aligned} V_R &= RI_0 e^{j\omega t} = RI_c, \\ V_L &= j\omega LI_0 e^{j\omega t} = j\omega LI_c, \\ V_C &= \frac{1}{j\omega C} I_0 e^{j\omega t} = \frac{1}{j\omega C} I_c. \end{aligned}$$

The total actual voltage is given by the imaginary part of V_c where

$$V_c = V_R + V_L + V_C = \left[R + j \left(\omega L - \frac{1}{\omega C} \right) \right] I_c.$$

If we define the complex quantity, Z , called the impedance, by

$$Z = R + j \left(\omega L - \frac{1}{\omega C} \right),$$

3.2. Geometrical representation of complex numbers

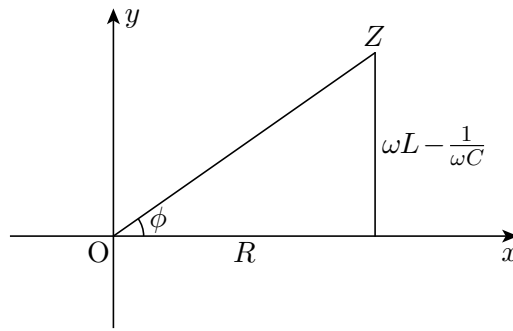


Figure 6.10: Complex impedance

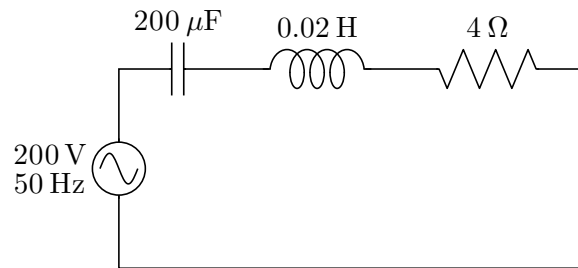


Figure 6.11: RLC circuit in Example (6.15).

then we can write the equation $V_c = ZI_c$, which is of the same form as Ohm's law. The quantity Z , the impedance corresponds to the resistance in a dc circuit. The impedance can be shown on an Argand diagram as shown in Fig. 6.10.

Bearing in mind that in the product ZI_c the arguments are added, we see that the angle ϕ gives the lag of the current behind the voltage, with $V = V_0 \sin(\omega t + \phi)$. (Note: ϕ can be positive or negative.) The amplitude of the voltage V_0 is given by $V_0 = |Z|I_0$, which sometimes needs to be rearranged to give $I_0 = \frac{V_0}{|Z|}$ if the voltage is given and the current not known.

Example 6.15 For the RLC circuit shown in Figure 6.11 which has a source of voltage of amplitude 200 volt at 50 hertz (i.e. cycles per second), find

1. the complex impedance in Cartesian and polar form,
2. the amplitude of the current,
3. the phase shift between the voltage and the current.

3.2. Geometrical representation of complex numbers

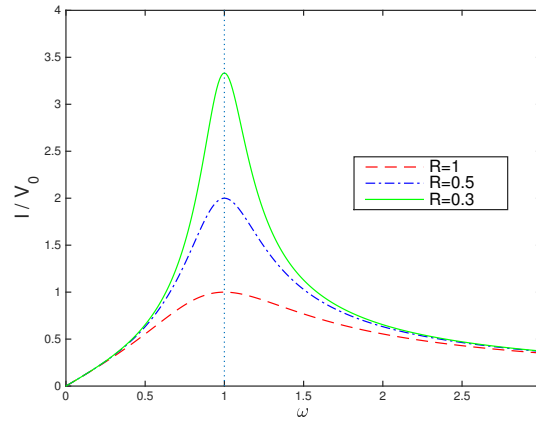


Figure 6.12: Resonance of the current in an RLC electric circuit. The amplitude of the current I/V_0 vs ω for $L = C = 1$ and different R as in the legend.

Solution. We have $Z = R + j(\omega L - \frac{1}{\omega C})$ and $\omega = 2\pi f = 100\pi$ so

$$\begin{aligned} Z &= 4 + j\left(100\pi \times 0.02 - \frac{10^6}{100\pi \times 200}\right) \\ &= 4 + j(6.28 - 15.92) = 4 - 9.64j \quad (\text{Cartesian form}) \\ Z &= 10.43(\cos(1.178) + j\sin(-1.178)) \quad (\text{polar form}) \\ |Z| &= \sqrt{4^2 + 9.64^2} = 10.43; \end{aligned}$$

If the complex voltage is $V = 200e^{j100\pi t}$, then the complex current is given by

$$I = \frac{200}{10.43}e^{j(100\pi t + 1.178)}$$

Consequently, the amplitude of the current is

$$\frac{200}{10.43} = 19.16$$

and the phase shift between the voltage and the current is 1.178.

More precisely, if we take the voltage as $V = 200 \sin(100\pi t)$, then the current in the circuit is

$$I = 19.16 \sin(100\pi t + 1.178).$$

6.5.1 Resonance in RLC circuit.

Consider the dependence of the amplitude of the current I in an RLC circuit on the frequency of the applied voltage ω .

3.2. Geometrical representation of complex numbers

$$I = \frac{V_0}{|Z|} = \frac{V_0}{\sqrt{R^2 + (\omega L - (\omega C)^{-1})^2}}.$$

where V_0 is the amplitude of the applied voltage and $Z = R + j(\omega L - (\omega C)^{-1})$ is the impedance of the circuit.

The function $I(\omega)$ is non-monotonic. It has a maximum at a certain value of ω_c that can be found from

$$\frac{dI}{d\omega} = 0 = (-0.5) \frac{V_0}{(R^2 + (\omega L - (\omega C)^{-1})^2)^{3/2}} \left(2(\omega L - (\omega C)^{-1})(L + \frac{1}{C\omega^2}) \right).$$

Solving for ω , we obtain

$$\omega_c^2 = \frac{1}{LC}, \rightarrow \omega_c = \frac{1}{\sqrt{LC}}.$$

In order to visualize the function $|I|(\omega)$ we set $L = C = 1$ and plot $|I|(\omega)$ in the units of V_0 for three different values of the resistance $R = (1, 0.5, 0.3)$, as shown in Fig.6.12.

Each of the three lines in Fig.6.12 has a maximum at $\omega_c = 1$. As the resistance R decreases, the maximal value of the current $I(\omega_c)$ increases. In the limit, when the resistance of the circuit is negligibly small, i.e. $R \rightarrow 0$, the maximal value of the current diverges

$$\lim_{R \rightarrow 0} I(\omega_c) = \lim_{R \rightarrow 0} \frac{V_0}{R} = \infty.$$

This phenomenon is called *resonance* and the critical frequency $\omega_c = (LC)^{-1/2}$ is called the resonance frequency of the circuit.

6.6 Parallel RLC circuit

Consider an RLC circuit, where all three elements are connected in parallel, as shown in Fig.6.13.

In this case, the voltage drops across the resistor, the capacitor and the inductor are identical and equal to the voltage at the source $V(t) = V_0 e^{j\omega t}$

$$V_R = V_C = V_L = V(t).$$

3.2. Geometrical representation of complex numbers

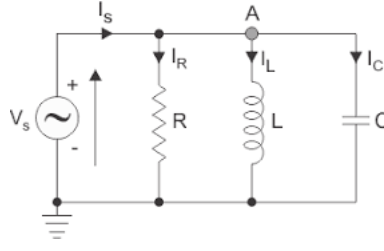


Figure 6.13: RLC circuit connected in parallel.

The respective complex currents I_R , I_C and I_L are found from

$$I_R = \frac{V}{Z_R}, \quad I_C = \frac{V}{Z_C}, \quad I_L = \frac{V}{Z_L},$$

where the complex impedances Z_R , Z_C and Z_L are given by

$$Z_R = R, \quad Z_C = \frac{1}{j\omega C}, \quad Z_L = \omega L j.$$

Conservation of current implies that the current through the source I_S is equal to the sum of I_R , I_C and I_L

$$I_S = I_R + I_C + I_L.$$

Writing I as $I = \frac{V}{Z}$, where Z is the complex impedance of the parallel circuit, we obtain

$$I_S = \frac{V}{Z} = \frac{V}{Z_R} + \frac{V}{Z_C} + \frac{V}{Z_L}.$$

Consequently

$$\frac{1}{Z} = \frac{1}{Z_R} + \frac{1}{Z_C} + \frac{1}{Z_L} = \frac{1}{R} - \frac{\omega C}{j} + \frac{1}{j\omega L} = \frac{1}{R} + j \left(\omega C - \frac{1}{\omega L} \right).$$

This shows that the current through the source I_S is

$$I_S = V_0 \sqrt{R^{-2} + (\omega C - (\omega L)^{-1})^2} e^{j(\omega t + \phi)},$$

where the phase shift ϕ between the voltage $V(t)$ and the current I_S is

3.2. Geometrical representation of complex numbers

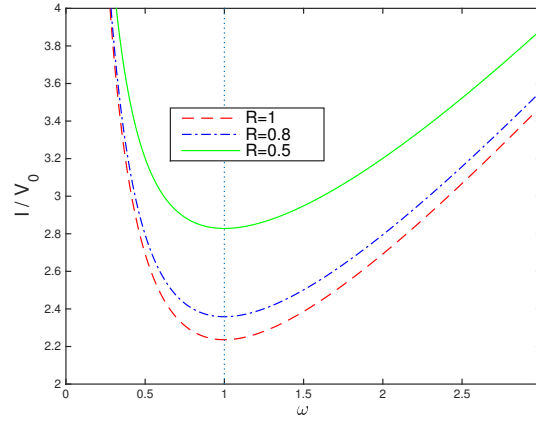


Figure 6.14: Anti-resonance in the RLC circuit connected in parallel.

$$\phi = \arg \left(1/R + j \left(\omega C - \frac{1}{\omega L} \right) \right).$$

The dependence of the amplitude of the current I_S on the frequency ω is shown in Fig.6.14.

Remarkably, the magnitude of the current I_S has a minimum, rather than a maximum at the resonance frequency $\omega_c = 1/\sqrt{CL}$. Indeed from

$$\frac{dI_S}{d\omega} = \frac{V_0}{2(R^{-2} + (\omega C - (\omega L)^{-1})^2)^{1/2}} \left(2(\omega C - (\omega L)^{-1})(C + \frac{1}{\omega^2 L}) \right) = 0$$

we obtain the frequency at the minimum of I_S

$$\omega_c = \frac{1}{\sqrt{CL}}.$$

The RLC circuit connected in parallel is called anti-resonator.

6.7 Complex rotation

Consider the operation of rotation of a complex number $z = x + jy$ by the angle α anti-clockwise. Under the rotation, the modulus $|z| = \sqrt{x^2 + y^2} = \rho$ remains unchanged, while the argument $\arg(z)$ increases by α . Using the exponential form for z we obtain

3.2. Geometrical representation of complex numbers

$$z = \rho e^{j\theta} \rightarrow z' = \rho e^{j(\theta+\alpha)} = e^{j\alpha} \rho e^{j\theta} = e^{j\alpha} z$$

Let us construct a two-dimensional real vector

$$\vec{v} = \begin{bmatrix} \Re(z) \\ \Im(z) \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

We are looking to derive the transformation rule for $\vec{v}$ under the rotation by the angle α anti-clockwise. Using the Euler's formula, we obtain

$$z' = x' + jy' = e^{j\alpha}(x + jy) = (\cos \alpha + j \sin \alpha)(x + jy),$$

therefore,

$$z' = x \cos \alpha - y \sin \alpha + j(x \sin \alpha + y \cos \alpha).$$

Therefore

$$\vec{v} = \begin{bmatrix} \Re(z) \\ \Im(z) \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow \vec{v}' = \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Consequently, under the operation of rotation by angle α anti-clockwise, the real and imaginary parts of z transform as components of a two-dimensional vector, in agreement with Eq. (4.78).

Example 6.16 Find the image of $z = 1 + j$ under the rotation by $\pi/2$ clockwise.

Solution

$$z \rightarrow z' = e^{-j\pi/2} z = (\cos \pi/2 - j \sin \pi/2)(1 + j) = -j(1 + j) = 1 - j.$$

As we see, the rotation by $\pi/2$ clockwise corresponds to the multiplication by $-j$. Similarly, rotation by $\pi/2$ anti-clockwise corresponds to the multiplication by j .

Example 6.17 Find the image of $z = \sqrt{3} + j$ under the rotation by $\pi/4$ anti-clockwise, followed by the reflection through the origin.

Solution

After the rotation by $\pi/4$ anti-clockwise

3.2. Geometrical representation of complex numbers

$$\begin{aligned} z \rightarrow z' &= e^{j\pi/4}z = (\cos \pi/4 + j \sin \pi/4)(\sqrt{3} + j) = \frac{(1+j)(\sqrt{3}+j)}{\sqrt{2}} \\ &= \frac{\sqrt{3} - 1 + j(\sqrt{3} + 1)}{\sqrt{2}}. \end{aligned}$$

Reflection through the origin can be represented as the rotation by π clock- or anti-clockwise. Therefore

$$z' \rightarrow z'' = e^{j\pi}z' = (\cos \pi + j \sin \pi) \frac{\sqrt{3} - 1 + j(\sqrt{3} + 1)}{\sqrt{2}} = -\frac{\sqrt{3} - 1 + j(\sqrt{3} + 1)}{\sqrt{2}}.$$

6.8 Differential equations: complex method of solution

In many applications in physics and engineering, one needs to solve a vector linear differential equation of the form

$$\frac{d\vec{P}}{dt} = -\frac{1}{\epsilon_0\chi}\vec{P} + \vec{E}_{\text{ext}}(t), \quad (6.121)$$

where $\vec{P} = P_x\mathbf{i} + P_y\mathbf{j}$ is a two-dimensional solution vector and $\vec{E}_{\text{ext}}(t) = E_x(t)\mathbf{i} + E_y(t)\mathbf{j}$ is an external time-dependent forcing.

Such equation appears, for example, in electrostatics, when calculating the response of a dielectric material to a time-dependent external electric field $\vec{E}_{\text{ext}}(t)$. Here $\vec{P}$ is polarization density vector of the dielectric medium, ϵ_0 is the electric permittivity of vacuum, χ is the dielectric susceptibility of the medium.

Consider the response of the medium to the so-called circularly polarized external electric field $\vec{E}_{\text{ext}}(t) = E_0 \cos \omega t \vec{i} + E_0 \sin \omega t \vec{j}$, where E_0 is the amplitude of the field and ω is the rotation frequency. From Eq. (6.121), the components of the polarization density $\vec{P}$ vector satisfy

$$\begin{aligned} \dot{P}_x &= -\frac{1}{\epsilon_0\chi}P_x + E_0 \cos \omega t, \\ \dot{P}_y &= -\frac{1}{\epsilon_0\chi}P_y + E_0 \sin \omega t. \end{aligned} \quad (6.122)$$

In order to solve Eq. (6.122), we employ the complex method of solution and introduce a complex-valued function

$$Z(t) = P_x(t) + jP_y(t). \quad (6.123)$$

3.2. Geometrical representation of complex numbers

If we add the two equations in Eq. (6.122), we obtain

$$\dot{P}_x + j\dot{P}_y = -\frac{P_x + jP_y}{\epsilon_0\chi} + E_0 e^{j\omega t}, \rightarrow \dot{Z} = -\frac{Z}{\epsilon_0\chi} + E_c(t), \quad (6.124)$$

where E_c is the complex external forcing $E_c(t) = E_0(\cos \omega t + j \sin \omega t) = E_0 e^{j\omega t}$.

Complex differential equation Eq. (6.124) can be solved using the standard method for real-valued functions. Namely, the general solution is given by

$$Z(t) = C \exp\left(-\frac{t}{\epsilon_0\chi}\right) + A e^{j\omega t}, \quad (6.125)$$

where C is an arbitrary complex number and the complex constant A can be determined by substituting Eq. (6.125) into Eq. (6.124). Thus we obtain

$$j\omega A e^{j\omega t} = -\frac{A e^{j\omega t}}{\epsilon_0\chi} + E_0 e^{j\omega t}. \quad (6.126)$$

Consequently

$$A = \frac{E_0}{j\omega + (\epsilon_0\chi)^{-1}}. \quad (6.127)$$

This shows that in the long-time limit, when $t \rightarrow \infty$, the response of the dielectric medium to the circularly polarized electric field is given by

$$\lim_{t \rightarrow \infty} Z(t) = \frac{E_0 e^{j\omega t}}{j\omega + (\epsilon_0\chi)^{-1}}. \quad (6.128)$$

The physical components of the polarization vector $\vec{P}$ can be found from Eq. (6.128) by taking the real and imaginary parts

$$\lim_{t \rightarrow \infty} P_x(t) = \frac{E_0 \cos(\omega t - \phi)}{\sqrt{\omega^2 + (\epsilon_0\chi)^{-2}}}, \quad \lim_{t \rightarrow \infty} P_y(t) = \frac{E_0 \sin(\omega t - \phi)}{\sqrt{\omega^2 + (\epsilon_0\chi)^{-2}}}, \quad (6.129)$$

where the phase shift $\phi = \tan^{-1}(\omega\epsilon_0\chi)$.

Eqs. (6.129) demonstrate that the maximal polarization amplitude $\max(P_x) = \max(P_y) = \epsilon_0\chi E_0$ corresponds to a time-independent forcing, i.e. $\omega \rightarrow 0$. Polarization amplitude decreases monotonically with ω and the vector $\vec{P}$ forms the angle $\phi = \tan^{-1}(\omega\epsilon_0\chi)$ with the external circularly polarized forcing $\vec{E}_{\text{ext}}$.

Exercises

1. For the *RLC* circuit shown in Fig. 6.15, which has a power supply of 20 volts and amplitude and frequency of 100 hertz, a resistance of 5 ohms, a capacitance of 100 microfarads, and an inductance of 4 millihenrys, find

3.2. Geometrical representation of complex numbers

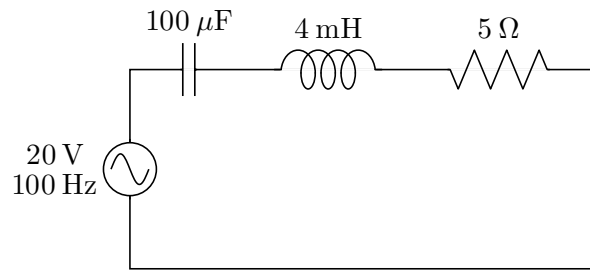


Figure 6.15: RLC circuit in Ex. (1).

- a) the complex impedance in Cartesian and polar form
 - b) the current
 - c) the angle by which the current lags or leads the applied voltage
2. A coil of resistance 5 ohms and inductance 0.04 henrys is connected in series with a capacitor of 400×10^{-6} farads across a 100-volt, 50-hertz alternating current. Find
 - a) the complex impedance in Cartesian and polar form
 - b) the current
 - c) the angle by which the current lags or leads the applied voltage
 3. A coil of resistance 100 ohms and inductance 0.1 henrys is connected in series with a capacitor of 10^{-3} farads across a 155-volt, 60-hertz alternating current. Find
 - a) the complex impedance in Cartesian and polar form
 - b) the current
 - c) the angle by which the current lags or leads the applied voltage

Answers

1.
 - a) $5 - 13.41j$, $14.31(\cos(69^\circ 33') - i \sin(69^\circ 33'))$
 - b) $1.40 \sin 200\pi t$
 - c) leads by $69^\circ 33'$
2.
 - a) $5 + 4.61j$, $6.8(\cos 42^\circ 40' + j \sin 42^\circ 40')$
 - b) $14.7 \sin 100\pi t$
 - c) lags by $42^\circ 40'$
3.
 - a) $100 + 35.04j$, $105.96(\cos 19^\circ 19' + j \sin 19^\circ 19')$
 - b) $1.46 \sin 120\pi t$
 - c) lags by $19^\circ 19'$

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